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8. ECOLOGICAL IMPACT ASSESSMENT

8.1 Introduction

8.1.1 This section presents the baseline ecological profile of the Assessment Area, and the assessment of potential ecological impacts resulting from the construction and operation of the Project. The location plan and description of the Project are presented in **60722948/A10/FIGURE 2.1** and **Chapter 2** respectively. In terms of ECR Alignment and its design, considerable thought went into selecting the most appropriate option from environmental and engineering perspectives as detailed in **Section 2.5** and **Section 2.6**. For the purpose of the ecological impact assessment, two spatial boundaries are defined as:

- Project Boundary: refers to the entire project area, including areas resumed for the Project that would not be subject to any construction works
- Construction Area of the Project: refers to all areas will be directly impacted by the project construction works.

8.1.2 According to the Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM), baseline conditions for ecological components of the environment were evaluated based on available literature and field surveys conducted for the purposes of this Environmental Impact Assessment (EIA). Measures required to mitigate any identified adverse impacts are recommended, where appropriate, and residual impacts are addressed.

8.2 Environmental Legislation, Standards and Guidelines

8.2.1 This report makes reference to the following HKSAR Government ordinances, regulations, standards, guidelines, and documents where identifying ecological importance of habitats and species, and evaluating and assessing potential ecological impacts of the Project:-

- EIA Study Brief (ESB-375/2025) issued for the proposed project, Development of the Loop – Eastern Connection Road
- Environmental Impact Assessment Ordinance (EIAO) (Cap. 499) and subsidiary legislation and guidelines
- EIAO Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM) Annexes 8 and 16
- EIAO Guidance Note No. 3/2010 - Flexibility and Enforceability of Mitigation Measures Proposed in an Environmental Impact Assessment Report
- EIAO Guidance Note No. 6/2010 - Some Observations on Ecological Assessment from the Environmental Impact Assessment Ordinance Perspective
- EIAO Guidance Note No. 7/2010 – Ecological Baseline Survey for Ecological Assessment

- EIAO Guidance Note No. 10/2010 – Methodologies for Terrestrial and Freshwater Ecological Baseline Surveys
- Forests and Countryside Ordinance (Cap. 96) and its subsidiary legislation, the Forestry Regulations
- Wild Animals Protection Ordinance (Cap. 170)
- Town Planning Board Guideline No. 12C – Application for developments within Deep Bay area under Section 16 of the Town Planning Ordinance
- Protection of Endangered Species of Animals and Plants Ordinance (Cap. 586)
- Town Planning Ordinance (Cap. 131)
- Water Pollution Control Ordinance (Cap. 358)
- Chapter 10 of the Hong Kong Planning Standard and Guidelines (HKPSG) Development Bureau (DEVB) TC(W) No. 4/2020 Tree Preservation; and
- Drainage Services Department Practice Note No. 3/2021 Guidelines on Design for Revitalisation of River Channel

8.2.2 This report also made reference to the following international conventions, national legislation and local publications:-

- The international Union of Conservation of Nature (IUCN) Red List of Threatened Species
- The List of Wild Plants and Wild Animals Under Special State Protection under the Law of the People's Republic of China on the Protection of Wildlife.
- The Convention on Biological Diversity (the CBD), and an associated city-level Biodiversity Strategy and Action Plan (BSAP) developed by Agriculture, Fisheries and Conservation Department (AFCD) under the CBD.
- The Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES);
- Category I/II/III in List of Wild Plants under State Protection;
- Rare and Precious Plants of Hong Kong
- Considered 'Rare' or 'Very Rare' plant species listed by Corlett et al. (2000) and regarded as 'Rare' plant species by Yip *et al.* (2010);
- Threatened Species List of China's Higher Plants (Qin *et al.,* 2017);
- Category I or II State Protected Wild Animals;
- PRC Wild Animal Protection Law;
- China Red Data Book of Endangered Animals and Plant

- Red List of China's Vertebrates (Jiang *et al.*, 2016)
- Fauna species considered of concern in Fellowes *et al.* (2002)
- State of Hong Kong Biodiversity (WWF HK, 2025a)
- Hong Kong Terrestrial Biodiversity Hotspot Map 2025 (WWF HK, 2025b)

8.3 Assessment Methodology

Assessment Area

8.3.1 In accordance with the EIA Study Brief No. ESB-375/2025, the Assessment Area for the purpose of the terrestrial and aquatic ecological impact assessment includes areas within 500m from the Project Boundary of the Project and associated works, as well as other areas likely to be impacted by the Project (**60722948/A10/FIGURE 8.1** refers).

Literature Review

8.3.2 A literature review was conducted including ecological information from previous EIA reports, published academic studies, relevant grey literature and relevant EM&A data. These sources provided a basis for understanding the ecological context of the Assessment Area.

Ecological Survey Methodology

8.3.3 In agreement with relevant authorities and following the ecological survey requirements described in Appendix G in the EIA Study Brief, ecological surveys were undertaken from July 2022 to December 2023 (**Table 8.1** refers), covering both the wet and dry seasons. The purpose of the ecological surveys was to verify the information from literature review, collect up-to-date ecological baseline information and fill information gaps. This facilitates subsequent assessment of ecological value, evaluation of the potential ecological impacts resulting from the Project, and provision of appropriate mitigation measures. Survey transects for habitat, vegetation, mammals, avifauna, herpetofauna, butterflies, odonates and fireflies; and sampling points for aquatic fauna, camera traps and flight-line surveys; are shown in **60722948/A10/FIGURE 8.2**.

8.3.4 In response to the amendments to EIAO and the revision of the EIAO-TM (introduced in April 2023) during the 12-month ecological survey programme undertaken for this Project, additional freshwater fish and firefly surveys (**Table 8.1**) were conducted to meet the latest EIAO-TM requirements. The ecologists and other personnel with relevant experience who led and conducted the surveys are summarised in **Appendix 8.1.**

8.3.5 The recommended timing and methodology of surveys for specific taxa followed the EIAO GN No. 7/2023 and No. 10/2023. Survey methodologies for habitats/vegetation and different faunal groups are described in the following sections.

Habitat and Vegetation

8.3.6 Habitat and vegetation surveys of the Assessment Area were undertaken in July and November (2022) and in March and June (2023), covering both dry and wet seasons.

8.3.7 Habitats were identified, sized and mapped based on recent aerial photos and field ground-truthing. Habitats were characterised and defined with reference to size, vegetation type, flora species present, dominant species, species diversity and abundance, community structure and seasonality, as well as the presence of any features of ecological importance. Representative colour photos were taken for each habitat type and any important ecological features identified. Habitat maps of suitable scale were prepared.

8.3.8 Vegetation surveys were conducted within each habitat type by direct observation. Plant species from each habitat type were identified and their relative abundance recorded. The location of any plant species of conservation importance (e.g. Incense Tree, *Aquilaria sinensis*) was recorded. Nomenclature of plant species follows the latest Hong Kong Plant Database (HKPD) available from the website of the Hong Kong Herbarium (AFCD, 2021).

Avifauna

8.3.9 Daytime and nighttime bird surveys were carried out on a monthly basis from July 2022 to June 2023, covering dry and wet seasons. Daytime surveys were carried out in the early morning at the period of peak bird activity, while night surveys were conducted after dusk to record nocturnal avifauna. The bird communities of each habitat type within the Assessment Area were surveyed along survey transects (**60722948/A10/FIGURE 8.2** refers). All birds seen or heard along the survey transects were counted and identified to species where possible. The location and nature of any substantial waterbird foraging and roosting sites were recorded. Signs of breeding (e.g. nests and/ or recently fledged juveniles) within the Assessment Area, especially within the Construction Area of the Project, were also recorded and mapped. Surveyors used 7X to 10X binoculars and/or a 30X to 65X spotting scope mounted on tripod for the surveys and photographic records. Ornithological nomenclature follows the latest List of Hong Kong Birds (HKBWS, 2024).

8.3.10 Specific surveys of the flight-line and flight behaviour of waterbirds and other larger avifauna species of conservation importance within the Hoo Hok Wai (HHW) wetland areas were undertaken in early morning during peak periods of bird activity (from November 2022 to February 2023; and from May to June 2023). The flight-line surveys were undertaken at two vantage points simultaneously and continuously for two hours. The bird species, direction and height of flight adopted were recorded. The location of the vantage points is shown in **60722948/A10/FIGURE 8.2**.

Mammals

8.3.11 A total of seven camera traps were deployed throughout the Assessment Area at secure and appropriate locations subject to the agreement of the lease owners (**60722948/A10/FIGURE 8.2** refers) to survey cryptic non-flying terrestrial mammals, focusing on areas with potential mammal usage. Monthly mammal surveys (including day and night-time surveys, covering the dry and wet seasons) were carried out from July 2022 to June 2023. As most mammals occur in low densities, tracks and signs of mammals (e.g. droppings) were actively searched for (as well as direct sightings) along the survey transects (**60722948/A10/FIGURE 8.2**). Dusk and night surveys were conducted to survey nocturnal mammal species (e.g. rodents and bats).

8.3.12 Particular attention was given to searching for Eurasian Otter (*Lutra lutra*)'s holts in potential suitable habitats such as terrestrial areas adjacent to water bodies (Shek, 2006). Active search for field signs of otter (including but not limited to spraints, footprints, scratch marks, etc) across the Assessment Area were conducted. In addition, camera traps were deployed at locations considered to be potential Eurasian Otter activity areas.

8.3.13 Bat surveys were conducted at dusk through direct observations and using a bat detector. Particular attention was given to potential foraging and drinking sites such as fruiting trees and freshwater ponds (Tong, 2016). All bat calls recorded were identified according to species-specific echolocation call structure (Tong, 2016).

8.3.14 All mammals seen during the survey were counted and identified to species where possible. Nomenclature for mammals follows that the Hong Kong Biodiversity Information Hub (AFCD, 2024c).

Herpetofauna

8.3.15 Daytime and nighttime herpetofauna surveys were carried out from July to October 2022 and from March to June 2023, covering the dry and wet seasons. Herpetofauna surveys were conducted through direct observation in all habitat types along the survey transects (**60722948/A10/FIGURE 8.2** refers) and active searching of hiding places such as among leaf litter, inside holes, under stones and logs. Particular attention was given to streams and other water features. In addition to visual surveys, auditory detection of species-specific calls was used to survey frogs and toads. During the surveys, all reptiles and amphibians sighted and heard were counted and identified, with the location of herpetofauna species with conservation importance mapped. Special attention was paid to streams and riparian areas in Ma Tso Lung, where the critically endangered Three-banded Box Turtle (*Cuora trifasciata*) was recorded in the reviewed literature, and Tse Koo Hang, which has been identified as a potential habitat for the Three-banded Box Turtle (Ove Arup, 2013a). Nomenclature for herpetofauna follows that available from the Hong Kong Biodiversity Information Hub (HKBIH) (AFCD, 2024c).

Butterflies and Odonates

8.3.16 Daytime butterfly and odonate surveys were carried out from July to November 2022 and from March to June 2023, while odonate surveys were carried out from July to October 2022 and from April to June 2023. Butterflies and odonates were identified and counted by direct observation in different habitats within the Assessment Area along the survey transects (**60722948/A10/FIGURE 8.2**). Nomenclature for butterflies and odonates follows HKBIH (AFCD, 2024).

Fireflies

8.3.17 Monthly firefly surveys were carried out along the survey transects at dusk and night from April to June 2023 and from October to December 2023. Firefly larvae and adults were identified to species level where possible. The location of firefly species of conservation importance or any notable behaviour (e.g. breeding) were recorded. Nomenclature and conservation status of fireflies follow Yiu (2023).

Aquatic Fauna

8.3.18 Daytime freshwater community surveys were carried out from July to November 2022 and from March to June 2023. Bimonthly nighttime freshwater fish surveys were carried out from April to December 2023. Sampling locations are shown in **60722948/A10/FIGURE 8.2**. Aquatic fauna (including freshwater macro-invertebrates such as freshwater crabs, shrimps, freshwater molluscs and aquatic insect larvae; and fishes) in the watercourses within the Assessment Area were surveyed by direct observation and active searching. Freshwater fishes and aquatic macro-invertebrates were recorded and identified to the lowest possible taxon and counted. The location(s) of freshwater species of conservation importance were recorded, along with any notable behaviour. Nomenclature for freshwater fishes follows the HKBIH (AFCD, 2024), while that for freshwater invertebrates follows Dudgeon (1999).

Ecological Survey Programme

8.3.19 The ecological survey schedule is presented in **Table 8.1** below.

Table 8.1 Ecological Survey Programme

Types of ecological survey		2022						2023											
		Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Habitat and vegetation		D				D				D			D						
Mammal		D+N	D+N	D+N	D+N	D+N	D+N	D+N	D+N	D+N	D+N	D+N	D+N						
Bird	Transect	D+N	D+N	D+N	D+N	D+N	D+N	D+N	D+N	D+N	D+N	D+N	D+N						
	Coordinated flight line					D	D	D	D			D	D						
Herpetofauna		D+N	D+N	D+N	D+N					D+N	D+N	D+N	D+N						
Butterfly		D	D	D	D	D				D	D	D	D						
Odonate		D	D	D	D						D	D	D						
Firefly											N	N	N				N	N	N
Freshwater fish		D	D	D	D	D				D	D+N	D	D+N		N		N		N
Freshwater invertebrate		D	D	D	D	D				D	D	D	D						

Note:
1. D: Day; N: Night and D+N: Day and Night.

Assessment Approach

8.3.20 The criteria and guidelines stated in Annexes 8 and 16 of the EIAO-TM were followed for evaluating and assessing ecological impacts arising from the construction and operation of the Project. Potential direct / indirect, on-site / off-site, primary, secondary, cumulative and residual ecological impacts arising from the construction and operation of the Project were identified and evaluated where possible. Mitigation measures and monitoring and audit programme were recommended as required.

8.4 Description of the Environment

Recognised Sites of Conservation Importance

8.4.1 The recognised sites of conservation importance identified within the Assessment Area are shown in **60722948/A10/FIGURE 8.1** and are described below.

Proposed Hoo Hok Wai Wetland Conservation Park (HHW WCP)

8.4.2 The Agriculture, Fisheries and Conservation Department (AFCD) has conducted a Strategic Feasibility Study on the Development of Wetland Conservation Parks (WCPs) System (hereafter referred as the WCPs System FS), which proposed the establishment of four new Wetland Conservation Parks (Sam Po Shue WCP, Hong Kong Wetland Park Expansion Area, Hoo Hok Wai WCP (including Sha Ling / Nam Hang area), and Nam Sang Wai WCP) across the Northwest New Territories (AFCD, 2024a). The four main objectives of the WCPs System are:

- Conserving the ecological value of the wetlands and safeguarding the integrity of the wetlands system;
- Developing modernised aquaculture industry to create more job opportunities for the agriculture and fisheries industries;
- Promoting scientific research on aquaculture to facilitate the upgrading and transformation of the agriculture and fisheries industries; and
- Providing ecological education and recreational facilities for the public.

8.4.3 While the boundary of the proposed HHW WCP is subject to further review and confirmation, the proposed park boundary of the HHW WCP (including Sha Ling / Nam Hang area) is shown in **60722948/A10/FIGURE 8.1**, covering approximately 277 ha and comprising managed and unmanaged ponds, marshes and reedbeds, as well as compensatory wetlands. These habitats support diverse wetland fauna species, particularly waterbirds and Eurasian Otter. Part of the Project Boundary falls within the proposed HHW WCP (**60722948/A10/FIGURE 8.1**). AFCD (2024a) notes that the next stage of detailed studies for the WCPs System should take into account any interface issue with these developments (including the ECR Project) when more information becomes available.

Wetland Conservation Area (WCA) and Wetland Buffer Area (WBA)

8.4.4 The northwestern part of the Assessment Area falls within the Wetland Conservation Area (WCA) (**60722948/A10/FIGURE 8.1** refers). The WCA was designated by the Town Planning Board (TPB) to conserve the ecological value of the fishponds in the

Deep Bay wetland ecosystem (TPB Guideline No. 12C – Application for developments within Deep Bay area under Section 16 of the Town Planning Ordinance revised in May 2014 by TPB). The WCA comprises the existing and contiguous active or abandoned fishponds in the Deep Bay Area, which should be conserved.

8.4.5 New development within the WCA would not be allowed unless it is required to support the conservation of the ecological value of the area or the development is an essential infrastructural project with overriding public interest. Any such development should be supported by an ecological impact assessment to demonstrate that the development would not result in a net loss in wetland function and negative disturbance impact.

8.4.6 The central part of the Project Boundary and Assessment Area falls within the Wetland Buffer Area (WBA, **60722948/A10/FIGURE 8.1** refers). The WBA is approximately 500m in width and lies along the landward boundary of the WCA. The intention of WBA is to protect the ecological integrity of wetland habitats within the WCA (TPB Guideline No. 12C). Development within the WBA causing negative impacts on the ecological value of the WCA would not be supported by TPB, unless an ecological impact assessment can demonstrate that the negative impacts could be mitigated through positive measures. All works conducted within the WCA and WBA should comply with the "No-Net-Loss in Wetland" principle. Wetland compensation is required for any development involving pond filling and mitigation measures against disturbance would be necessary and would be imposed as part of the planning approval conditions.

"Conservation Area" ("CA" zone)

8.4.7 Areas zoned "CA" under the Approved LMCL Outline Zoning Plan (OZP) No. S/LMCL/2, the Approved San Tin Technopole OZP No. S/STT/2 and the Ma Tso Lung and Hoo Hok Wai Outline Zoning Plan S/NE-MTL/3 fall within the Project Boundary and Assessment Area (**60722948/A10/FIGURE 8.1**).

8.4.8 The Lok Ma Chau (LMC) Meander (also known as Old Shenzhen River Meander) to the southeast of the Lok Ma Chau Loop (hereinafter referred to as "the Loop") is designated as "CA" zone under the Approved LMCL OZP No. S/LMCL/2. The planning intention of this zone is to protect and retain the existing natural landscape and ecological features of the LMC Meander and the associated riparian vegetation, which forms an important element of the flight line corridor for birds and is used by the Eurasian Otter, for conservation, educational and research purposes, and to separate sensitive natural environment from the adverse effects of development. There is a general presumption against development in this zone. In general, only developments that are needed to support the conservation of the ecological integrity of the wetland ecosystem or the existing natural landscape or scenic quality of the LMC Meander or are essential infrastructure projects with overriding public interest may be permitted.

8.4.9 The "CA" zone under the Approved San Tin Technopole OZP No. S/STT/2 mostly comprises the existing contiguous and continuous, active or abandoned fishponds to the south of Shenzhen River. Areas of fishponds are also found to the north-east of Lok Ma Chau Control Point. The planning intention of the "CA" zone is to conserve the ecological value of wetland and fishponds which form an integral part of the wetland ecosystem in the Deep Bay Area. Any change in use within this zone should adhere to the "no-net-loss in wetland" principle. The primary intention is to discourage

new development unless it is required to support the conservation of the ecological integrity of the wetland ecosystem or the development is an essential infrastructure project with overriding public interest. The existing fishpond culture within this zone should be maintained and its continuous operation is encouraged. Conservation management activities which will enhance the overall Deep Bay wetland ecosystem are also promoted. Alternative beneficial uses to fishponds such as nature reserve or wetland habitat are permitted as of right within the zone. For those developments which may be permitted on application to the Board, such application should be supported by an Ecological Impact Assessment (EcoIA) and a management plan to demonstrate that the development would not result in any net loss in wetland function and negative disturbance impact. Such development should also be compatible with the conservation objectives of the wetland in the Deep Bay Area and should be appropriate as well as be able to enhance the visual and landscape quality of the area. Wetland compensation is required for any development involving pond filling and mitigation measures against any disturbance would be necessary. Certain uses/facilities that are of small scale or necessary to serve the local needs may also be considered on application to the Board.

8.4.10 A patch of "Conservation Area" ("CA") is gazetted under Ma Tso Lung and Hoo Hok Wai OZP No. S/NE-MTL/3 to conserve the ecological value of wetland and fishponds which form an integral part of the wetland ecosystem at Hoo Hok Wai and the rest of the Deep Bay Area (**60722948/A10/FIGURE 8.1** refers). The "no-net-loss in wetland" principle is also adopted for any change in land use within this zone. The primary intention is to discourage new development unless it is required to support the conservation of the ecological integrity of the wetland ecosystem or the development is an essential infrastructure project within overriding public interest.

Priority Sites for Enhanced Conservation

8.4.11 The New Nature Conservation Policy (NNCP) of the HKSAR Government identified twelve priority sites for enhanced conservation. Under the NNCP, two schemes, namely the Nature Conservation Management Agreement (MA) Scheme and the Public-Private Partnership (PPP) Scheme, were introduced to enhance the conservation of ecologically important sites, particularly those on land under private ownership. (AFCD, 2024d).

8.4.12 One of the twelve sites identified is the 'Deep Bay Wetland Outside Ramsar Site Priority Site for Enhanced Conservation' which falls partly within the Project Boundary and Assessment Area (**60722948/A10/FIGURE 8.1** refers). The major habitat type of the Site is pond, which are man-made and intensively modified by humans. The Site maintains certain characteristics of natural wetland that attract waterbirds and other wildlife, making it a semi-natural habitat. One hundred and sixty-six bird species, including the Black-faced Spoonbill (*Platalea minor*), were recorded therein (AFCD, 2004a). It was ranked 9th among the 12 Priority Sites for Enhanced Conservation.

Other Ecologically Sensitive Resources

Inner Deep Bay and Shenzhen River Catchment Important Bird Area (IBA)

8.4.13 The Inner Deep Bay and Shenzhen River Catchment was identified as an Important Bird Area (IBA) in 2003 (Birdlife International, 2003). The IBA is an estuarine area comprising a variety of habitats, including (1) freshwater agricultural lands at Long Valley; (2) ponds at Ma Tso Lung, Lok Ma Chau, San Tin, Lin Barn Tsuen, Mai Po, Pak Hok Chau, Lut Chau, Tai Sang Wai, Nam Sang Wai, Tin Shui Wai and Tsim Bei

Tsui; (3) Mai Po Marshes Nature Reserve; (4) Inner Deep Bay mangroves; and (5) mudflat and oyster farm at Sheung Pak Lai. The Project Boundary and Assessment Area largely fall within the Inner Deep Bay and Shenzhen River Catchment IBA (**60722948/A10/FIGURE 8.1** refers).

Compensatory Wetlands

"Ecological Area" ("EA")

8.4.14 Ecological Area (EA) is approximately 12.78ha in size, and is located along the southeastern edge of the Loop (**60722948/A10/FIGURE 8.1** refers). It was created to compensate for the loss of wetland habitats (including 10.96ha reed marsh and 0.50 ha of freshwater marsh) as part of the Development of Lok Ma Chau Loop EIA (Ove Arup, 2013a), hereafter referred to as the LMC Loop EIA. The EA aims to provide an unfragmented, managed and undisturbed wetland habitat for maximising connectivity with existing wetland habitats and protecting the flight line corridor. Furthermore, a 50m wide low-rise building buffer zone from the EA, with appropriate screen-planting, will be established between the EA and developments within the Loop to minimise disturbance and impacts on avifaunal flight paths from the Loop. The LMC Loop EIA predicted that birds flying over the Loop area and potentially affected by construction activities would re-route to fly over LMC Meander and the EA and its buffer zone. The EA was established in December 2022, and subsequent monitoring of marsh and reed marsh habitats within the EA have shown it supports a variety of fauna species of conservation importance, including Greater Painted-snipe (*Rostratula benghalensis*), Little Egret (*Egretta garzetta*), Collared Crow (*Corvus torquatus*), Yellow-breasted Bunting (*Emberiza aureola*), Leopard Cat (*Prionailurus bengalensis*), Eurasian Otter, Chinese Bullfrog (*Hoplobatrachus rugulosus)*, Chinese Cobra (*Naja atra*) and Ruby Darter (*Rhodothemis rufa*) (AFCD, unpublished data).

Off-site Wetland Compensation Areas (OWCAs)

8.4.15 Offsite Wetland Compensation Areas (OWCAs) comprise compensatory off-site wetland areas constructed to mitigate direct, indirect, permanent and temporary wetland impacts resulting from Development of the Loop, Western Connection Road (WCR), ECR and Direct Link to LMC Station. The OWCA were proposed in the LMC Loop EIA (although the ECR was excluded prior to the approval stage of this EIA report). Three areas of existing pond habitat, comprising Areas 2, 7 and 9, were proposed as OWCA, with enhancement works at these Areas completed in October 2022. While only one of these (Area 2) falls within the Assessment Area, details of all three are summarised in **Table 8.2** below.

8.4.16 Area 2 is situated to the immediate northeast of the LMC Meander. It is composed of pond habitat and marsh (**60722948/A10/FIGURE 8.1** refers) to mitigate the permanent loss of pond and marsh habitat in LMC arising from the construction of the WCR, permanent loss of pond habitat caused by the construction of the Loop, and the potential loss of marsh habitat caused by the ECR. According to EM&A monitoring data (AFCD&CEDD, 2022), Area 2 supports a variety of waterbirds, such as Chinese Pond Heron (*Ardeola bacchus*), Little Egret, Great Egret (*Ardea alba*) and Grey Heron (*Ardea cinerea*); raptors, including Black Kite (*Milvus migrans*), Eastern Buzzard (*Buteo japonicus*), Common Kestrel (*Falco tinnunculus),* and Western Osprey (*Pandion haliaetus*)*;* and open-country species, such as Buff-bellied Pipit (*Anthus rubescens*), Japanese Quail (*Coturnix japonica*) and Zitting Cisticola (*Cisticola juncidis*)*.*

8.4.17 Area 7 is situated to the southwest of the Assessment Area, to the west of LMC MTR Station, between the Frontier Closed Area (FCA) fenced road and the LMC Spur Line Mitigation Wetland. It is composed of two marshes and three ponds created to mitigate for the temporary loss of a reed cell in the wastewater reedbed area of the LMC Spur Line Mitigation Wetland during the construction phase of the Direct Link to LMC station in the Loop. It was also intended to be used for compensating the permanent loss of pond habitat caused by the construction of LMC WCR and Direct Link to LMC Station. The recorded species in Area 7 comprise wetland-dependent waterbirds such as Chinese Pond Heron, Little Egret, Great Egret, Grey Heron, Great Cormorant (*Phalacrocorax carbo*), Little Grebe (*Tachybaptus ruficollis*), and White-throated Kingfisher (*Halcyon smyrnensis*) which commonly utilise wetland-edge habitats, as well as raptors including Black Kite and Eastern Buzzard (AFCD&CEDD, 2022).

8.4.18 Area 9 is also situated to the southwest of the Assessment Area and lies to the immediate south of Area 7 and is separated by a drainage channel, with the FCA fenced road on the west and Lok Ma Chau Wetland Mitigation Area (hereinafter referred as LMC EEA) on the east. Area 9 consists of pond habitat designed to compensate the temporary loss of pond habitat due to the construction of the WCR. Following completion of the WCR in 2026, these enhanced areas would be available to compensate for impacts resulting from the ECR. Based on EM&A monitoring data (AFCD&CEDD, 2022), Area 9 supports a diverse assemblage of wetland-dependent and open-country avifauna, including species of conservation importance such as Chinese Pond Heron, Little Egret, Great Egret, Grey Heron, Great Cormorant, Northern Pintail (*Anas acuta*), Northern Shoveler (*Spatula clypeata*), Red-billed Starling (*Spodiopsar sericeus*), Red-throated Pipit (*Anthus cervinus*), Tufted Duck *(Aythya fuligula*) and White-cheeked Starling (*Spodiopsar cineraceus*).

Table 8.2 Summary of Habitat Types and Areas Created at the EA and OWCAs

Area/Location	Area size (ha)		
	Pond	**Marsh**	**Reedmarsh**
Ecological Area (EA) within the Loop			
The EA	-	12.78	
Off-site Wetland Compensation Areas (OWCAs) [1]			
Area 2	6.61	1.66	0.22
Area 7	2.28	1.19	-
Area 9	5.80	1.03	-
Sub-total	**14.69**	**3.88**	**0.22**

Notes:
1. Areas extracted from Table 1.1 of Operation Manual for the Offsite Wetland Compensation Areas under the Development of Lok Ma Chau Loop: Main Works Package 1 (CEDD, 2023b) and differ slightly from areas stated in the LMC Loop EIA (Register No.: AEIAR-176/2013) as habitats across the three OWCAs were reconfigured in response to updated site assessment.

Offsite Compensation Woodland

8.4.19 The Offsite Compensation Woodland (~1.26 ha) is situated at a hillside near Ngau Kok Shan (Horn Hill), about 40m from the Project Boundary. It was identified to compensate for the loss of approximately 0.4 ha of woodland habitat from WCR

footprint, and 0.7 ha of secondary woodland and 0.16 ha of shrubland habitat from ECR footprint as estimated under the LMC Loop EIA (Ove Arup, 2013a). As per the latest woodland compensation plan developed for the LMC Loop EIA (CEDD, 2024), only the 0.4ha of woodland habitat creation has been provided to date to compensate for impacts associated with the WCR development.

Key Dragonfly Site in Ma Tso Lung

8.4.20 Reels (2019) identified Ma Tso Lung as a key dragonfly site, referring to the mix of active and abandoned ponds in the vicinity of Ma Tso Lung and HHW, although the boundaries of the site are not delineated so it is not clear if this Site falls within the Project Boundary. Thirty-three odonate species were recorded from the Site, including species of conservation importance such as Ruby Darter (*Rhodothemis rufa*), Sapphire Flutterer (*Rhyothemis triangularis*) and Scarlet Basker (*Urothemis signata*).

8.5 Ecological Baseline Information (Literature Review)

Literature Review

Reviewed Reports

8.5.1 In accordance with Appendix G of the EIA Study Brief ESB-375/2025, the findings of relevant studies/surveys (including the approved EIA reports of the LMC Loop EIA (Ove Arup, 2013a) and Northeast New Territories New Development Areas (NENT NDA) (Ove Arup, 2013b) were reviewed. The review was conducted to collect ecological baseline information within and in the vicinity of the Assessment Area, in particular the along the Construction Area of the Project, and information on all habitat/species of conservation importance such as their occurrence, distribution and abundance. Literature and websites reviewed are summarised in **Table 8.3** below.

Table 8.3 Baseline Information on Ecological Resources in the Assessment Area

Relevant Literature	Habitat and Vegetation	Avifauna	Mammal	Herpetofauna	Butterfly and Odonate	Freshwater Community
Agreement No. CE 46/2022 (CE) Ma Tso Lung Area and Other Sites in Kwu Tung North New Development Area and North District – Feasibility Study – Ecological Baseline Desktop Study and Survey Report for Zone 1 – Ma Tso Lung (Draft) (Ref. TR17a-01) (AECOM, unpublished)	✓	✓	✓	✓	✓	✓
Agreement No. CE 20/2021 (CE). First Phase Development of the New Territories North – San Tin / Lok. Ma Chau Development Node – Investigation (AEIAR-261/2024) (CEDD & PlanD, 2024)	✓	✓	✓	✓	✓	✓

Relevant Literature	Habitat and Vegetation	Avifauna	Mammal	Herpetofauna	Butterfly and Odonate	Freshwater Community
Agreement No. CE 13/2014(CE) Development of Kwu Tung North and Fanling North New Development Areas Phase 1 – Design and Construction (CEDD, 2013)	✓	✓	✓	✓	✓	✓
Agreement No. CE 53/2008(CE) Planning and Engineering Study on Development of Lok Ma Chau Loop – Investigation. Report for Civil Engineering and Development Department, HKSAR Government (Ove Arup, 2013a)	✓	✓	✓	✓	✓	✓
Agreement No. CE 61/2007(CE) North East New Territories New Development Areas Planning and Engineering Study – Investigation (Ove Arup, 2013b)	✓	✓	✓	✓	✓	✓
Environmental Monitoring & Auditing Reports for the Development of Lok Ma Chau Loop (Black and Veatch, 2019a-e)	✓	✓	✓	✓	✓	✓
Development of Lok Ma Chau Loop: Main Works Package 1 – Environmental Monitoring and Audit (EM&A) Report in 2022 (AFCD&CEDD, 2022)		✓	✓	✓		
Agreement No. CE 19/2019(CE) Development of Kwu Tung North New Development Area, Remaining Phase – Design and Construction Compensation Event No. CE/2019-01 – Planning and Engineering Review (KTN) [P&ER(KTN)] (CEDD, 2019)	✓	✓	✓	✓	✓	✓
Study for the Construction of a Secondary Boundary Fence and New Sections of Primary Boundary Fence and Boundary Patrol Road (ArchSD, 2009)	✓	✓	✓	✓	✓	✓
Hong Kong Biodiversity – Newsletter of Agriculture, Fisheries and Conservation Department (AFCD);		✓	✓	✓	✓	✓
“Territory-wide long-term monitoring survey results from the Agriculture, Fisheries and Conservation Department” (AFCD, 2024)			✓			
Mai Po Inner Deep Bay Ramsar Site Waterbird Monitoring Programme, Monthly Waterbird Monitoring from Hong Kong Bird Watching Society (HKBWS, 2018-2023)		✓				
Memoirs of Hong Kong Natural History Society	✓	✓	✓	✓	✓	✓
Porcupine! Newsletter of Ecology & Biodiversity, The School of Biological Sciences, The University of Hong Kong	✓	✓	✓	✓	✓	
Bird Survey of Selected Farmlands in the Proposed Northern Metropolis, Hong Kong SAR: Interim Report (KFBG, 2022)		✓				

Relevant Literature	Habitat and Vegetation	Avifauna	Mammal	Herpetofauna	Butterfly and Odonate	Freshwater Community
The Hidden Oases of Hong Kong: High Ecological Value Farmlands in the Proposed Northern Metropolis (Zheng et al., 2023)		✓		✓	✓	
State of Hong Kong Biodiversity (WWF, 2025a)	✓	✓	✓	✓	✓	✓
Hong Kong Terrestrial Biodiversity Hotspot Map 2025 (WWF, 2025b)	✓	✓	✓	✓	✓	✓

Habitat and Vegetation

8.5.2 A total of 13 habitat types were previously identified within the Assessment Area, comprising pond, compensatory wetlands, marsh, reed, modified watercourse, semi-natural / natural watercourse, seasonally wet grassland, woodland, grassland / shrubland, orchard, plantation, agricultural land and developed area. Most of these recorded habitats were also identified within the Construction Area of the Project, dominated by ponds, grassland/shrubland, seasonally wet grassland and developed area.

General Description of Habitats within the Assessment Area

8.5.3 An extensive area of mostly active ponds was recorded in HHW, along the northern part of the Assessment Area. Although vegetation diversity in this habitat was low, largely limited to common herbaceous species along the pond bunds (Ove Arup, 2013a and 2013b), the ecological value of the ponds was considered of high value, as they supported a high diversity and abundance of waterbirds and other wetland-dependent faunal species. The latest published report (WWF, 2025a and 2025b; Zheng et al, 2023) have been reviewed for ecological impact assessment on the habitats and species of conservation importance. The ponds in HHW form part of the identified biodiversity hotspot Inner Deep Bay, which supports the richest birdlife in Hong Kong and is the only know local breeding site for the Eurasian Otter (WWF, 2025a; 2025b). The area is also recognised for its important ecological connectivity function (Zheng, 2023). Previous surveys recorded three ponds to the southeast of the LMC Meander within the Project Boundary, all of which were reported as being abandoned (Ove Arup, 2013a and 2013b).

8.5.4 Patches of other wetland habitats formed from abandoned ponds/agricultural land were recorded interspersed with the pond habitats and close to Ma Tso Lung Village. These included marsh, reedbeds and seasonally wet grassland.

8.5.5 Two areas of compensatory wetland were recorded in the Assessment Area, the EA and OWCA Area 2. The EA comprises an area of approximately 12.78 ha containing reed marsh and marsh habitat, support a mix of cultivated floating, emergent and submerged plants (Ove Arup, 2013a and 2013b). Following establishment and

maturation, the ecological value of the EA is considered of moderate to high value (CEDD & PlanD, 2024). The OWCA Area 2 consists of a mix of managed pond and marsh with native and brackish-tolerant plants that aims to provide suitable habitats for Eurasian Otter and waterbirds (AECOM, 2021).

8.5.6 The hillsides in the south and southeast of the Assessment Area were dominated by grassland / shrubland habitat which is generally of low species diversity and disturbed by frequent hillfires (Chan *et al.*, 2023 and 2024). Smaller patches of woodland, orchard and plantation were recorded from hillside areas, mostly around Ma Tso Lung Village.

8.5.7 Several watercourses were recorded from the Assessment Area, the largest of which is the LMC Meander which surrounds the Loop. The LMC Meander was originally part of Shenzhen River before diversion, and was restored to mitigate the loss of ponds from training of the Shenzhen River (Ove Arup, 2013a). The Ma Tso Lung Stream flows largely south to north through the Assessment Area, and is in generally good condition in the upper reaches. The mid-section of the stream (to the northwest of Ma Tso Lung Village) has been channelised with concrete. The downstream section of the stream flows into the larger wetland area in HHW.

8.5.8 The majority of the flora species recorded in the Assessment Area were common and widespread in Hong Kong, and typical of the observed habitats. For instance, the seasonally wet grassland was occupied by herb species such as *Miscanthus sinensis*. Plantation habitat was mainly comprised of exotic tress species such Taiwan Acacia (*Acacia confusa*). Fruit trees such as Lychee (*Litchi chinensis*) and Longan (*Dimocarpus longan*) were commonly recorded in habitats including orchard/developed area. Only one flora species of conservation importance, Incense Tree was reported from the reviewed literature, recorded in secondary woodland near Ma Tso Lung (**Table 8.6** refers).

Fauna

8.5.9 Faunal diversity (particularly in wetland habitats) was relatively high in the Assessment Area. A total of 217 faunal species of conservation importance were noted from the reviewed literature, comprising 169 bird species (**Table 8.5** refers), 12 mammal species, 11 herpetofauna species, 14 butterfly species, 9 odonate species and two aquatic fauna species (**Table 8.6** and **60722948/A10/FIGURE 8.4** refers).

Avifauna

8.5.10 The Assessment Area comprises a wide array of wetland habitats which were considered to be significant for avifauna and supported a high diversity of resident and migrant bird species. The literature review illustrates the importance of wetland habitats in the area with 151 of the 169 bird species of conservation importance reported being wetland dependent to some extent, including notable species such as Black-faced Spoonbill, and Eastern Marsh Harrier (*Circus spilonotus*). The pond areas in HHW are covered by the Mai Po Inner Deep Bay Ramsar Site Waterbird Monitoring Programme (HKBWS, 2018-2023), with Little Egret, Great Cormorant and Chinese Pond Heron commonly recorded from pond and occasionally recorded from marsh habitats.

8.5.11 A small seasonal active ardeid night roost was recorded from a strip a mixed woodland at the northern end of Ma Tso Lung Road from January to April 2023, as

well as in December 2023, as shown in **60722948/A10/FIGURE 8.8**. A maximum count of 47 ardeids, comprising Chinese Pond Heron and Little Egret, were recorded in February 2023 (AECOM, unpublished). The summary of the number of roosting ardeids recorded is presented in **Table 8.4** below.

Table 8.4 Number of Ardeids Recorded at Ma Tso Lung Night Roost

Species	2023											
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Chinese Pond Heron	-	33	11	10	0	0	0	0	0	0	0	3
Little Egret	-	14	0	0	0	0	0	0	0	0	0	0
Total:	**-**	**47**	**11**	**10**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**3**

8.5.12 The Loop area and its vicinity are an important flight corridor linking HHW with the broader Northwestern New Territories (NWNT) wetland area to the southwest. Morning surveys recorded over 80% of ardeids flying northwards towards HHW and LMC, with lower usage of 10% toward the Ma Tso Lung. Most flights occurred at low altitudes (0-10m). Evening surveys showed a similar pattern, with nearly all ardeids returning from HHW to the roost (AECOM, unpublished).

8.5.13 Ove Arup (2013a) recorded a total of 9,155 bird flight records during surveys conducted for the LMC Loop EIA, with a clear seasonal pattern observed, whereby more bird flights were recorded during dry season. The majority of birds in flight were observed following a main corridor along Ha Wan Tsuen, the LMC Meander, and towards HHW.

8.5.14 Flight line surveys from areas to the south of LMC Spur Line and San Sham Road to Hoo Hok Wai were also conducted under the EM&A programme for the LMC Loop EIA (CEDD, 2023a). The majority of waterbirds notably Great Cormorant, Grey Heron, Great Egret, Black-faced Spoonbill continued to use this flight corridor (approximately 150m along LMC Meander and 50m along the EA and Shenzhen River). Overall, the majority of the flight surveys conducted between June 2021 to September 2023 observed more than 50% of monthly recorded birds utilising flight lines of 25m or above. Usage of flight lines between 15-25m height generally comprised less than 40% monthly records, and usage of flight lines between 10-15m is generally less than 5% usage. Seasonal and inter-year fluctuations were also observed, with usage of higher flight lines (25m or above) recorded during November to March of 2022 compared to 2021.

Nursery / Breeding Ground

8.5.15 The literature review recorded no notable nursery and breeding behaviour from the Project Boundary. However, Ove Arup (2013a and 2013b) reported that Eurasian Eagle Owl (*Bubo bubo*) is presumed to breed in the more extensive areas of grassland in the hills between Chua Tau and Ma Tso Lung villages, and utilise adjacent lowland, non-grassland habitats for foraging. Ove Arup (2013a) also reported that the pond of HHW was known to be an important foraging area for Chinese Pond Herons breeding at the Ho Sheung Heung egretry, although its significance in this respect appeared to have declined in recent years. Based on the egretry flight survey results (Ove Arup, 2013b) at Ho Sheng Heung, herons are not likely to fly towards HHW across the Project Boundary. Furthermore, AECOM

(unpublished) reported the presence of a Chinese Francolin (*Francolinus pintadeanus*) chick in the vicinity of Ma Tso Lung, and nesting behaviour of White-throated Kingfisher on a grassland margin at She Leng. These observations suggest that the hilly grassland and shrubland habitats south of the ECR Alignment potentially support breeding sites for these species.

Mammals

8.5.16 A total of 13 mammal species of conservation importance were noted from the reviewed literature (Ove Arup, 2013a and 2013b; AFCD, 2024). Foremost among these is Eurasian Otter, which was considered of regional concern by Fellowes *et al*. (2002) and was assessed as vulnerable and endangered in China by Wang (1998) and Jiang *et al*. (2016), respectively. It is largely restricted to Deep Bay wetlands where a very small breeding population remains (WWF, 2025). There are several records of otters at the LMC Meander, the EA, HHW, LMC EEA and OWCAs, as shown in **60722948/A10/FIGURE 8.7**. Based on the Territory Wide Long-term Monitoring Survey undertaken by AFCD, Eurasian Otters were observed through camera trapping between 2011-2014 near the ponds in Liu Pok, as indicated in **60722948/A10/FIGURE 8.7** as Area B (AFCD, unpublished data). In OWCA Area 2, otters were recorded by camera trap in 2021 (AECOM, 2021) and AFCD in 2024 (AFCD, 2024). Since 2020, Kadoorie Farm and Botanic Garden (KFBG) have partnered with WWF-Hong Kong to study otter distribution in Hong Kong, according to unpublished data by KFBG (2024), 10 otter spraints were found by the research team at a mitigation pond north of the Project Boundary. McMillan *et al*. (2022) studied the population of Eurasian Otter in Hong Kong and suggested that wetland areas within the HHW were part of their core range, but the exact locations of Eurasian Otters were not reported. These multiple records extending over a number of years indicates the HHW area is an existing habitat for Eurasian Otter in Hong Kong. Eurasian Otter and other mammals (i.e, Small Asian Mongoose (*Herpestes javanicus)*, Leopard Cat) were also previously recorded from marsh and/or grassland habitat within the Loop and the EA (Ove Arup, 2013a; AFCD, Unpublished data). The areas within the Loop have since been developed, and it is expected that these species would no longer occur in the newly developed habitats.

Herpetofauna

8.5.17 A total of 11 herpetofauna species of conservation importance were reported from the reviewed literature occurring in the Assessment Area**.**

8.5.18 A Three-banded Box Turtle was reported by Ove Arup (2013a) to have been recorded in a hill stream flowing towards Ma Tso Lung during surveys for the NENT NDA PES. The exact location was not specified in the Report to protect the species from potential poaching, but it potentially falls outside of the current Assessment Area (the hilly reaches of the Ma Tso Lung stream fall largely to the south of the Assessment Area). Ove Arup (2013a) also suggested that streams and riparian areas in Tse Koo Hang could provide potentially suitable habitat for this species. These habitats do fall within the Assessment Area, but outside of the Project Boundary. Three-banded Box Turtle was assessed as critically endangered in the China Red Data Book (Wang and Zhao, 1998) and has been assessed as a critically endangered species in the IUCN Red List of Threatened Species in 2018 (IUCN, 2022) and is also a Class II Protected Animal of China. This species is heavily poached for its unsubstantiated medicinal value globally, and has therefore been included in Appendix II of CITES and is listed under the Protection of Endangered Species of Animals and Plants Ordinance (Cap.

586). Furthermore, the Three-banded Box Turtle is listed in Schedule 2 of the Wild Animals Protection Ordinance (Cap. 170).

8.5.19 Chinese Soft-shelled Turtle (*Pelodiscus sinensis*) was not found during the ecological survey stage of the LMC Loop EIA. However, the EIA (Ove Arup, 2013a) notes it has been reported as widespread in Deep Bay ponds (Karsen *et al*., 1998). One individual was seen in a pond at HHW during survey for the Loop and its Peripheral Control Area Fishponds and Surroundings (Ove Arup, 2013b). Other herpetofauna species such as Chinese Cobra was also recorded in the EA within the current Project Boundary (AFCD, Unpublished data).

Butterflies

8.5.20 Previous studies have reported 14 butterfly species of conservation importance from the Assessment Area, although none were recorded from the current Project Boundary. These records were made from multiple habitat types, including concentrations of Grey Scrub Hopper (*Aeromachus jhora*), Metallic Cerulean (*Jamides alecto*) and Small Three-ring (*Ypthima norma*) from grassy habitats in the vicinity of Ma Tso Lung Stream.

Odonates

8.5.21 There were nine odonate species of conservation importance from the reviewed literature, with two species (i.e. Blue Sprite, Ruby Darter) falling within the current Project Boundary. The habitat associations of many of these records was not reported from the literature, although some species were noted around ponds (e.g., Sapphire Flutterer and Scarlet Basker) and Ma Tso Lung Stream (e.g., Blue Chaser (*Potamarcha congener*), Ruby Darter).

Aquatic Fauna

8.5.22 Across the reviewed literature, Small Snakehead (*Channa asiatica*) and the freshwater crab *Somanniathelphusa zanklon* were only two aquatic fauna species of conservation importance with records within the Assessment Area. They were both recorded at the same watercourse near Ma Tso Lung San Tsuen to the east of the Project Boundary.

Table 8.5 Avifauna Species of Conservation Importance in the Assessment Area Recorded from the Reviewed Literature

Number	Species	Rarity and Distribution in Hong Kong [1]	Conservation/Protection status [2,3,4,5,6,7,8,9,19]	Habitat Recorded	Source [10, 11,12, 13, 14, 15, 16, 17, 18, 19]
1	Amur Falcon *Falco amurensis*	Uncommon autumn passage migrant. Found in Lok Ma Chau, Mai Po.	List of Wild Animals under State Priority Conservation: Class II; Cap. 586	Pond	Ove Arup (2013a)
2	Amur Paradise Flycatcher *Terpsiphone incei*	Passage migrant, mainly autumn, and rare winter visitor. Found in Tai Po Kau, Lam Tsuen Country Park, Ng Tung Chai, Nam Chung, Chung Hom Kok, Ho Man Tin.	Fellowes *et al*. (2002): LC	Secondary Woodland / Plantation	Ove Arup (2013a)
3	Ashy Drongo *Dicrurus leucophaeus*	Uncommon winter visitor. Found in Shing Mun, Tai Po Kau.	Fellowes *et al*. (2002): LC	Secondary Woodland / Plantation, Shrubland	Mott Macdonald (2009) Ove Arup (2013a)
4	Asian Barred Owlet *Glaucidium cuculoides*	Locally common resident. Widely distributed in woodland of the north and central New Territories.	List of Wild Animals under State Priority Conservation: Class II; Cap. 586	Secondary Woodland / Plantation, Village, Compensatory Wetland	Ove Arup (2013a) AFCD Unpublished data
5	Asian Dowitcher *Limnodromus semipalmatus*	Common passage migrant. Found in Deep Bay area.	China Red Data Book: Rare; Fellowes *et al*. (2002): RC WWF: MR	Unspecified	Ove Arup (2013b) AFCD (2024) WWF (2025a)
6	Baer's Pochard *Aythya baeri*	Rare winter visitor. Found in Deep Bay area.	IUCN Red List of Threatened Species (2025): CR; Red List of China's Vertebrates: CR WWF: HR	Pond	Ove Arup (2013a) WWF (2025a)
7	Besra *Accipiter virgatus*	Common resident and migrant. Found in Tai Po Kau, Deep Bay area,	List of Wild Animals under State Priority Conservation: Class II	Secondary Woodland / Plantation, Shrubland	Ove Arup (2013a)

Number	Species	Rarity and Distribution in Hong Kong [1]	Conservation/Protection status [2,3,4,5,6,7,8,9,19]	Habitat Recorded	Source [10, 11,12, 13, 14, 15, 16, 17, 18, 19]
		Chek Lap Kok, Cheung Chau, Soko Islands.			
8	Black Bittern *Dupetor flavicollis*	Scarce passage migrant. Found in Deep Bay area, Tai Long Wan, Tap Mun, Long Valley, Sha Lo Tung, Mui Wo, Ma Tso Lung.	Fellowes *et al*. (2002): LC WWF: HR	Marsh	Ove Arup (2013a) Ove Arup (2013b) AFCD (2024) WWF (2025a)
9	Black Kite *Milvus migrans*	Common resident and winter visitor. Widely distributed in Hong Kong.	Fellowes *et al*. (2002): (RC)	Pond, Natural Watercourse, Grassland, Village, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b) AECOM (2021) Monthly EM&A Reports for EP-477/2013 and EP-477/2013/A - Development of LMCL AFCD (2024)
10	Black Stork *Ciconia nigra*	Rare autumn migrant and winter visitor. Found in Deep Bay area, Plover Cove Reservoir, Aberdeen Country Park.	List of Wild Animals under State Priority Conservation: Class I; China Red Data Book: EN; Fellowes *et al*. (2002): RC; CITES: Appendix II; Red List of China's Vertebrates: VU; Cap. 586	Pond	Ove Arup (2013a)
11	Black-capped Kingfisher *Halcyon pileata*	Uncommon passage migrant and winter visitor. Widely distributed in coastal areas throughout Hong Kong.	Fellowes *et al*. (2002): (LC) WWF: HR	Pond	Ove Arup (2013a) Black & Veatch (2019a) AFCD (2024) WWF (2025a)
12	Black-crowned Night Heron *Nycticorax nycticorax*	Common resident and winter visitor. Widely distributed in Hong Kong.	Fellowes *et al*. (2002): LC	Pond, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a)

Number	Species	Rarity and Distribution in Hong Kong [1]	Conservation/Protection status [2,3,4,5,6,7,8,9,19]	Habitat Recorded	Source [10, 11,12, 13, 14, 15, 16, 17, 18, 19]
					Ove Arup (2013b) AECOM (2021) Monthly EM&A Reports for EP-477/2013 and EP-477/2013/A - Development of LMCL
13	Black-faced Spoonbill *Platalea minor*	Common winter visitor. Found in Deep Bay area.	List of Wild Animals under State Priority Conservation: Class II; China Red Data Book: EN; Fellowes *et al*. (2002): PGC; IUCN Red List of Threatened Species (2025): VU; Red List of China's Vertebrates: EN WWF:MR	Pond, Natural Watercourse, Channelised Watercourse, Compensatory Wetland	Ove Arup (2013a) Ove Arup (2013b) AECOM (2021) Black & Veatch (2019b, 2019c, 2019d, 2019g) Monthly EM&A Reports for EP-477/2013 and EP-477/2013/A - Development of LMCL AFCD (2024) WWF (2025a)
14	Black-headed Gull *Chroicocephalus ridibundus*	Abundant winter visitor. Found in Deep Bay area and coastal waters.	Fellowes *et al*. (2002): PRC WWF: HR	Pond	Ove Arup (2013a) WWF (2025a)
15	Black-naped Oriole *Oriolus chinensis*	Common autumn passage migrant. Widely distributed in Hong Kong.	Fellowes *et al*. (2002): LC WWF:EX	Secondary Woodland / Plantation, Compensatory Wetland	Ove Arup (2013a) Ove Arup (2013b) AFCD Unpublished data WWF (2025a)
16	Black-tailed Godwit *Limosa limosa*	Abundant passage migrant and winter visitor. Found in Deep Bay area.	Fellowes *et al*. (2002): RC	Pond	Ove Arup (2013a)

Number	Species	Rarity and Distribution in Hong Kong [1]	Conservation/Protection status [2,3,4,5,6,7,8,9,19]	Habitat Recorded	Source [10, 11,12, 13, 14, 15, 16, 17, 18, 19]
17	Black-winged Kite *Elanus caeruleus*	Uncommon visitor. Found in Ha Tsuen, Deep Bay area.	China Red Data Book: VU; Fellowes et al. (2002): LC WWF: HR	Marsh, Pond	Ove Arup (2013a) AFCD (2024) WWF (2025a)
18	Black-winged Stilt *Himantopus himantopus*	Common passage migrant. Found in Deep Bay area, Long Valley, Kam Tin.	Fellowes et al. (2002): RC	Pond, Natural Watercourse, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b) AECOM (2021) Monthly EM&A Report for EP-477/2013 - Development of LMCL
19	Bluethroat *Luscinia svecica*	Locally common winter visitor. Widely distributed in wet agricultural areas throughout Hong Kong.	Fellowes et al. (2002): LC WWF: HR	Reed Marsh, Marsh, Compensatory Wetland	Ove Arup (2013a) Ove Arup (2013b) AFCD (2024) AFCD Unpublished data WWF (2025a)
20	Bridled Tern *Onychoprion anaethetus*	Common summer visitor and passage migrant. Found in Mirs Bay, Tolo Channel, Cheung Chau, Waglan Island, Cape D'Aguilar.	Fellowes et al. (2002): (LC)	Unspecified	Ove Arup (2013b)
21	Broad-billed Sandpiper *Limicola falcinellus*	Uncommon passage migrant. Found in Deep Bay area, Ho Chung, Shuen Wan.	Fellowes et al. (2002): LC WWF:MR	Unspecified	Ove Arup (2013b) WWF (2025a)
22	Bonelli's Eagle *Aquila fasciata*	Uncommon resident. Found in Deep Bay area, Hong Kong Island, Lamma Island, Lantau Island, Castle Peak, Sha Lo Tung.	China Red Data Book: Rare; Fellowes et al. (2002): (RC); Red List of China's Vertebrates: VU; Cap. 586 WWF: HR	Pond, Grassland	Ove Arup (2013a) AFCD (2024) WWF (2025a)

Number	Species	Rarity and Distribution in Hong Kong [1]	Conservation/Protection status [2,3,4,5,6,7,8,9,19]	Habitat Recorded	Source [10, 11,12, 13, 14, 15, 16, 17, 18, 19]
23	Brown-headed Thrush *Turdus chrysolaus*	Scarce winter visitor and migrant. Found in Tai Po Kau, Lam Tsuen, Shek Kong, Wu Kau Tang, Shing Mun, Kowloon Hills, Pok Fu Lam, Ngong Ping.	Fellowes et al. (2002): LC	Unspecified	Ove Arup (2013b)
24	Buff-bellied Pipit *Anthus rubescens*	Uncommon passage migrant and winter visitor. Widely distributed in areas of wet agriculture and drained fish ponds throughout Hong Kong.	Fellowes et al. (2002): LC	Pond	Ove Arup (2013a) Ove Arup (2013b)
25	Caspian Gull *Larus cachinnans*	Scarce winter visitor and passage migrant. Found in Deep Bay area.	Fellowes et al. (2002): LC	Pond	Ove Arup (2013a)
26	Chinese Grosbeak *Eophona migratoria*	Common winter visitor. Found in Kam Tin, Nam Chung, Shek Kong, Deep Bay area, Ho Chung, Lam Tsuen, Hok Tau, Island House, Kowloon Park.	Fellowes et al. (2002): LC	Secondary Woodland / Plantation, Village, Compensatory Wetland	Ove Arup (2013a) Ove Arup (2013b) AFCD (2024) AFCD Unpublished data
27	Chinese Hwamei *Garrulax canorus*	Common resident. Widely distributed in hillside shrubland throughout Hong Kong.	CITES: Appendix II; Cap. 586	Shrubland	Mott Macdonald (2009) Ove Arup (2013a)
28	Chinese Penduline Tit *Remiz consobrinus*	Common autumn migrant and winter visitor. Found in Deep Bay area, Tai O, Mui Wo, Long Valley, Luk Keng, Chek Lap Kok.	Fellowes et al. (2002): RC WWF:MR	Reed Marsh, Reedbed, Compensatory Wetland	Ove Arup (2013a) Black & Veatch (2019a) AFCD (2024) AFCD Unpublished data WWF (2025a)
29	Chinese Pond Heron *Ardeola bacchus*	Common resident. Widely distributed in Hong Kong.	Fellowes et al. (2002): PRC	Marsh, Pond, Channelised Watercourse, Natural Watercourse, Compensatory Wetland, Dry	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b) AECOM (2021)

Number	Species	Rarity and Distribution in Hong Kong [1]	Conservation/Protection status [2,3,4,5,6,7,8,9,19]	Habitat Recorded	Source [10, 11,12, 13, 14, 15, 16, 17, 18, 19]
				Agricultural Land, Seasonally Wet Grassland, Plantation, Reedbed	Black & Veatch (2019a, 2019c, 2019d, 2019e) Monthly EM&A Reports for EP-477/2013 and EP-477/2013/A - Development of LMCL AFCD (2024)
30	Chinese Sparrowhawk *Accipiter soloensis*	Common passage migrant. Found in Tsim Bei Tsui, Kadoorie Farm & Botanic Garden, Tai Po Kau.	List of Wild Animals under State Priority Conservation: Class II	Pond, Grassland and Grassland / Shrubland	Ove Arup (2013a)
31	Cinnamon Bittern *Ixobrychus cinnamomeus*	Scarce passage migrant. Found in Deep Bay area, Long Valley, Tai Yuen (Sheung Shui), Pui O.	Fellowes et al. (2002): LC WWF: HR	Reed Marsh, Marsh, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b) AECOM (2021) WWF (2025a)
32	Citrine Wagtail *Motacilla citreola*	Uncommon migrant and winter visitor. Found in Tsim Bei Tsui, Shuen Wan, Mai Po, Long Valley.	Fellowes et al. (2002): LC WWF: HR	Pond	Ove Arup (2013a) Ove Arup (2013b) AFCD (2024) WWF (2025a)
33	Collared Crow *Corvus torquatus*	Locally common resident. Found in Inner Deep Bay area, Nam Chung, Kei Ling Ha, Tai Mei Tuk, Pok Fu Lam, Chek lap Kok, Shuen Wan, Lam Tsuen.	Fellowes et al. (2002): LC; IUCN Red List of Threatened Species (2025): VU WWF:MR	Marsh, Pond, Reedbed, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) AECOM (2021) Black & Veatch (2019a, 2019b, 2019c, 2021h) Monthly EM&A Reports for EP-477/2013 and EP-477/2013/A - Development of LMCL

Number	Species	Rarity and Distribution in Hong Kong [1]	Conservation/Protection status [2,3,4,5,6,7,8,9,19]	Habitat Recorded	Source [10, 11,12, 13, 14, 15, 16, 17, 18, 19]
					AFCD (2024) WWF (2025a)
34	Common Emerald Dove *Chalcophaps indica*	Uncommon but widespread resident. Widely distributed in woodland throughout Hong Kong.	China Red Data Book: VU	Shrubland, Secondary Woodland / Plantation	Ove Arup (2013a)
35	Common Greenshank *Tringa nebularia Tringa nebularia*	Abundant winter visitor and migrant. Found in Deep Bay area.	Fellowes et al. (2002): RC WWF: HR	Pond, Compensatory Wetland	Ove Arup (2013a) AECOM (2021) AFCD (2024) WWF (2025a)
36	Common Kestrel *Falco tinnunculus*	Common autumn migrant and winter visitor. Widely distributed in Hong Kong.	List of Wild Animals under State Priority Conservation: Class II; Cap. 586 WWF: HR	Pond, Grassland and Grassland / Shrubland, Reedbed, Compensatory Wetland	Ove Arup (2013a) Black & Veatch (2019a) AFCD (2024) AFCD Unpublished data WWF (2025a)
37	Common Pochard *Aythya ferina*	Scarce winter visitor. Found in Deep Bay area, Lo Wu.	IUCN Red List of Threatened Species (2025): VU WWF: HR	Pond	Ove Arup (2013a) AFCD (2024) WWF (2025a)
38	Common Redshank *Tringa totanus*	Abundant passage migrant and winter visitor. Found in Deep Bay area.	Fellowes et al. (2002): RC	Pond	Ove Arup (2013a)
39	Common Rosefinch *Carpodacus erythrinus*	Rare winter visitor and migrant. Widely distributed in Hong Kong.	Fellowes et al. (2002): LC WWF: HR	Dry Agricultural Land	Ove Arup (2013b) WWF (2025a)
40	Common Starling *Sturnus vulgaris*	Scarce late autumn passage migrant and winter visitor. Found in Deep Bay area, Lok Ma Chau, Shatin, Mui Wo, Kam Tin.	Fellowes et al. (2002): LC	Unspecified	Ove Arup (2013b)

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41	Crested Goshawk *Accipiter trivirgatus*	Common resident. Widely distributed in woodlands and shrublands throughout Hong Kong.	List of Wild Animals under State Priority Conservation: Class II; China Red Data Book: Rare	Shrubland, Secondary Woodland / Plantation	Ove Arup (2013a) AFCD (2024)
42	Crested Honey Buzzard *Pernis ptilorhynchus*	Scarce autumn passage migrant. Found in Shek Kong Catchment, Tsim Bei Tsui, Lam Tsuen, Tai Po Kau, Mai Po, Ma Tso Lung, Wong Chuk Yeung.	China Red Data Book: VU; Fellowes et al. (2002): LC WWF:MR	Shrubland, Secondary Woodland / Plantation	Ove Arup (2013a) WWF (2025a)
43	Crested Serpent Eagle *Spilornis cheela*	Common resident. Widely distributed in shrublands on hillsides throughout Hong Kong.	China Red Data Book: VU; Fellowes et al. (2002): (LC)	Shrubland, Secondary Woodland / Plantation, Reedbed	Ove Arup (2013a) Ove Arup (2013b) Black & Veatch (2019a) AFCD (2024)
44	Curlew Sandpiper *Calidris ferruginea*	Abundant passage migrant. Found in Deep Bay area, Cape D'Aguilar.	Fellowes et al. (2002): RC	Pond	Ove Arup (2013a)
45	Daurian Starling *Agropsar sturninus*	Uncommon autumn passage migrant. Found in Mai Po, Long Valley, Kam Tin, Lam Tsuen, Tolo Harbour area, Kowloon Park, Mui Wo, Ho Chung.	Fellowes et al. (2002): LC	Pond, Grassland and Grassland / Shrubland, Village	Ove Arup (2013a)
46	Dunlin *Calidris alpina*	Abundant winter visitor and scarce passage migrant. Found in Deep Bay area.	Fellowes et al. (2002): RC	Pond	Ove Arup (2013a)
47	Dusky Thrush *Turdus eunomus*	Scarce winter visitor. Widely distributed in agricultural fields throughout Hong Kong.	Fellowes et al. (2002): LC WWF:MR	Marsh, Pond	Ove Arup (2013a) WWF (2025a)

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48	Eastern Buzzard *Buteo japonicus*	Common winter visitor. Widely distributed in Hong Kong.	Cap. 586	Reedbed, Compensatory Wetland	Black & Veatch (2019a) AFCD (2024) AFCD Unpublished data
49	Eastern Cattle Egret *Bubulcus coromandus*	Resident and common passage migrant. Widely distributed in Hong Kong.	Fellowes et al. (2002): LC WWF: HR	Marsh, Pond, Seasonally Wet Grassland, Dry Agricultural Land, Orchard, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b) AFCD Unpublished data WWF (2025a)
50	Eastern Imperial Eagle *Aquila heliaca*	Common winter visitor. Found in Deep Bay area, Ma Tso Lung.	List of Wild Animals under State Priority Conservation: Class I; China Red Data Book: VU; Fellowes et al. (2002): GC; IUCN Red List of Threatened Species (2025): VU; CITES: Appendix I; Red List of China's Vertebrates: EN; Cap. 586 WWF: HR	Pond, Compensatory Wetland	Ove Arup (2013a) AFCD (2024) AFCD Unpublished data WWF (2025a)
51	Eastern Marsh Harrier *Circus spilonotus*	Common winter visitor and passage migrant. Found in Deep Bay area, Starling Inlet area, Kadoorie Farm & Botanic Garden, Mount Austin.	Fellowes et al. (2002): LC WWF: HR	Reed marsh, Reedbed	Ove Arup (2013a) Black & Veatch (2019a) AFCD (2024) WWF (2025a)
52	Eurasian Bittern *Botaurus stellaris*	Uncommon winter visitor and spring migrant. Found in Deep Bay area, Shuen Wan, Pui O, Sandy Spur.	Fellowes et al. (2002): RC WWF: HR	Reed Marsh, Reedbed	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b) Black & Veatch (2019a)

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					WWF (2025a)
53	Eurasian Black Vulture *Aegypius monachus*	Previously rare winter visitor, last record in 1988. Found in Castle Peak, Mai Po.	China Red Data Book: VU; Cap. 586	Unspecified	Ove Arup (2013b)
54	Eurasian Coot *Fulica atra*	Uncommon winter visitor. Found in Deep Bay area, Plover Cove Reservoir, Shuen Wan.	Fellowes et al. (2002): RC WWF: HR	Pond, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) AECOM (2021) AFCD (2024) WWF (2025a)
55	Eurasian Curlew *Numenius arquata*	Abundant winter visitor. Found in Deep Bay area.	Fellowes et al. (2002): RC	Pond	Ove Arup (2013a)
56	Eurasian Eagle-Owl *Bubo bubo*	Scarce resident. Widely distributed in Hong Kong.	List of Wild Animals under State Priority Conservation: Class II; China Red Data Book: Rare; Fellowes et al. (2002): RC; Cap. 586 WWF: HR	Grassland and Grassland / Shrubland	Ove Arup (2013a) AFCD (2024) WWF (2025a)
57	Eurasian Hobby *Falco subbuteo*	Uncommon passage migrant. Widely distributed in marshes, agricultural land and lightly wooded hills throughout Hong Kong.	List of Wild Animals under State Priority Conservation: Class II; Fellowes et al. (2002): (LC); Cap. 586 WWF: HR	Pond, Grassland, Shrubland, Secondary Woodland, Reedbed	Ove Arup (2013a) Ove Arup (2013b) Black & Veatch (2019a) WWF (2025a)
58	Eurasian Jay *Garrulus glandarius*	Rare resident. Found in Yung Shue O, Plover Cove Reservoir, Shing Mun Reservoir, Tai Mo Shan, Kowloon Hills reservoirs, Pok Fu Lam	Fellowes et al. (2002): LC WWF: EX	Unspecified	Ove Arup (2013b) WWF (2025a)

Number	Species	Rarity and Distribution in Hong Kong [1]	Conservation/Protection status [2,3,4,5,6,7,8,9,19]	Habitat Recorded	Source [10, 11,12, 13, 14, 15, 16, 17, 18, 19]
		Reservoir, Crooked Island, King's Park.			
59	Eurasian Sparrowhawk *Accipiter nisus*	Scarce passage migrant. Found in Deep Bay area, Long Valley, Kadoorie Farm and Botanic Garden, Tai Po Kau, Lin Barn Tsuen, Ma Wan, Tung Ping Chau.	List of Wild Animals under State Priority Conservation: Class II	Pond, Grassland and Grassland / Shrubland	Ove Arup (2013a)
60	Eurasian Spoonbill *Platalea leucorodia*	Uncommon winter visitor. Found in Deep Bay area.	List of Wild Animals under State Priority Conservation: Class II; China Red Data Book: VU; Fellowes et al. (2002): LC; CITES: Appendix II; Cap. 586 WWF: HR	Pond, Compensatory Wetland	Ove Arup (2013a) Ove Arup (2013b) AFCD (2024) AFCD Unpublished data WWF (2025a)
61	Eurasian Teal *Anas crecca*	Common winter visitor. Found in Deep Bay area, Shuen Wan, Tai Lam Chung Reservoir, Victoria Harbour, urban parks.	Fellowes et al. (2002): RC WWF: HR	Marsh, Pond, Natural Watercourse, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b) AECOM (2021) AFCD (2024) WWF (2025a)
62	Eurasian Whimbrel *Numenius phaeopus*	Common passage migrant and scarce winter visitor. Found in Deep Bay area, Sai Kung, Tung Ping Chau, Ninepins, Cape D'Aguilar, Pok Fu Lam.	Fellowes et al. (2002): LC	Unspecified	Ove Arup (2013b)
63	Eurasian Wigeon *Mareca penelope*	Winter visitor. Found in Deep Bay area, Tai Lam Chung.	Fellowes et al. (2002): RC	Pond	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b)

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					AFCD (2024)
64	Falcated Duck *Mareca falcata*	Uncommon winter visitor. Found in Mai Po, Shuen Wan, Long Valley.	Fellowes et al. (2002): RC WWF: HR	Pond	Ove Arup (2013a) WWF (2025a)
65	Far Eastern Curlew *Numenius madagascariensis*	Uncommon passage migrant. Found in Deep Bay area.	Fellowes et al. (2002): LC; Red List of China's Vertebrates: VU WWF: HR	Unspecified	Ove Arup (2013b) WWF (2025a)
66	Ferruginous Duck *Aythya nyroca*	Rare visitor. Found in Mai Po.	WWF: HR	Pond	Ove Arup (2013a) WWF (2025a)
67	Golden-headed Cisticola *Cisticola exilis*	Locally common winter visitor. Widely distributed in grassland throughout Hong Kong.	Fellowes et al. (2002): LC WWF: HR	Reed Marsh, Grassland and Grassland / Shrubland, Compensatory Wetland	Ove Arup (2013a) Ove Arup (2013b) AFCD (2024) AFCD Unpublished data WWF (2025a)
68	Great Cormorant *Phalacrocorax carbo*	Common winter visitor. Widely distributed in coastal areas throughout Hong Kong.	Fellowes et al. (2002): PRC	Pond, Natural Watercourse, Channelised Watercourse, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b) AECOM (2021) Monthly EM&A Reports for EP-477/2013 and EP-477/2013/A - Development of LMCL AFCD (2024)
69	Greater Coucal *Centropus sinensis*	Common resident. Widely distributed in Hong Kong.	List of Wild Animals under State Priority Conservation: Class II; China Red Data Book: VU	Shrubland, Secondary Woodland / Plantation, Reedbed, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) AECOM (2021) Black & Veatch (2019a)

Number	Species	Rarity and Distribution in Hong Kong [1]	Conservation/Protection status [2,3,4,5,6,7,8,9,19]	Habitat Recorded	Source [10, 11,12, 13, 14, 15, 16, 17, 18, 19]
					AFCD (2024)
70	Great Crested Grebe *Podiceps cristatus*	Common winter visitor. Found in Tsim Bei Tsui, Starling Inlet.	Fellowes et al. (2002): RC	Pond	Mott Macdonald (2009) Ove Arup (2013a)
71	Great Egret *Egretta alba*	Common resident, migrant and winter visitor. Widely distributed in Hong Kong.	Fellowes et al. (2002): PRC	Marsh, Pond, Channelised Watercourse, Natural Watercourse, Reedbed, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b) AECOM (2021) Black & Veatch (2019a, 2019b, 2019c, 2019d, 2019e) Monthly EM&A Reports for EP-477/2013 and EP-477/2013/A - Development of LMCL AFCD (2024)
72	Greater Painted-snipe *Rostratula benghalensis*	Locally common resident. Found in Ha Tsuen, Lok Ma Chau, Kam Tin, Long Valley, Hong Kong Wetland Park.	China Red Data Book: VU; Fellowes et al. (2002): LC WWF:MR	Marsh, Reedbed, Compensatory Wetland	Ove Arup (2013a) Black & Veatch (2019a) AFCD Unpublished data WWF (2025a)
73	Greater Sand Plover *Charadrius leschenaultii*	Abundant passage migrant. Found in Deep Bay area.	Fellowes et al. (2002): RC	Pond	Ove Arup (2013a)
74	Greater Spotted Eagle *Clanga clanga*	Common winter visitor. Found in Deep Bay area.	China Red Data Book: Rare; Fellowes et al. (2002): GC; IUCN Red List of Threatened Species (2025): VU; Red List of China's Vertebrates: EN WWF:HR	Pond, Grassland and Grassland / Shrubland, Compensatory Wetland	Ove Arup (2013a) AECOM (2021) AFCD (2024) WWF (2025a)

Number	Species	Rarity and Distribution in Hong Kong [1]	Conservation/Protection status [2,3,4,5,6,7,8,9,19]	Habitat Recorded	Source [10, 11,12, 13, 14, 15, 16, 17, 18, 19]
75	Greater White-fronted Goose *Anser albifrons*	Vagrant. Found in Lok Ma Chau.	List of Wild Animals under State Priority Conservation: Class II	Pond	Ove Arup (2013a)
76	Grey Bush Chat *Saxicola ferreus*	Scarce passage migrant and winter visitor. Widely distributed in open cultivated fields throughout Hong Kong.	Fellowes et al. (2002): LC WWF:HR	Unspecified	Ove Arup (2013b) WWF (2025a)
77	Grey Heron *Ardea cinerea*	Common winter visitor. Found in Deep Bay area, Starling Inlet, Kowloon Park, Cape D'Aguilar.	Fellowes et al. (2002): PRC WWF:MR	Reed Marsh, Marsh, Pond, Channelised Watercourse, Natural Watercourse, Compensatory Wetland, Reedbed	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b) AECOM (2021) Black & Veatch (2019a, 2019b, 2019c, 2019d, 2019e) Monthly EM&A Reports for EP-477/2013 and EP-477/2013/A - Development of LMCL WWF (2025a)
78	Grey Nightjar *Caprimulgus jotaka*	Scarce passage migrant. Widely distributed in Hong Kong.	Fellowes et al. (2002): LC	Grassland	Ove Arup (2013b)
79	Grey Plover *Pluvialis squatarola*	Abundant winter visitor and scarce migrant. Found in Deep Bay area.	Fellowes et al. (2002): RC WWF:MR	Unspecified	Ove Arup (2013b) WWF (2025a)
80	Grey Treepie *Dendrocitta formosae*	Locally common resident. Found in Sai Kung, Luk Keng, Tai Po Kau, Tai Tam Reservoir, Ho Chung, Hong Kong University, Lam Tsuen, Hok Tau.	Fellowes et al. (2002): LC	Secondary Woodland / Plantation	Ove Arup (2013a)

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81	Grey-capped Greenfinch *Chloris sinica*	Scarce resident. Found in Shing Mun River, Lam Tsuen, Ping Shan, Lung Kwu Tang, Ho Man Tin, Tuen Mun.	Fellowes et al. (2002): LC	Plantation	Ove Arup (2013b)
82	Grey-headed Lapwing *Vanellus cinereus*	Locally common winter visitor and migrant. Found in Kam Tin, Tsim Bei Tsui, Lo Wu, Tai Long Wan, Shuen Wan, Castle Peak coast, Chek Lap Kok	Fellowes et al. (2002): LC WWF:HR	Pond	Ove Arup (2013a) WWF (2025a)
83	Grey-headed Woodpecker *Picus canus*	Occasional visitor. Found in Lam Tsuen, Fanling Golf Course, Liu Pok, Ho Sheung Heung.	Fellowes et al. (2002): LC	Unspecified	Ove Arup (2013b)
84	Grey-tailed Tattler *Tringa brevipes*	Common passage migrant. Found in Deep Bay area.	Fellowes et al. (2002): LC	Pond	Ove Arup (2013a)
85	Heuglin's Gull *Larus fuscus*	Common winter visitor and passage migrant. Found in Deep Bay area, Cape D'Aguilar.	Fellowes et al. (2002): LC	Pond	Ove Arup (2013a)
86	Indian Spot-billed Duck *Anas poecilorhyncha*	Rare visitor. Found in Deep Bay area.	Fellowes et al. (2002): RC WWF:HR	Pond	Ove Arup (2013a) WWF (2025a)
87	Intermediate Egret *Egretta intermedia*	Resident and passage migrant. Found in Deep Bay area, Tai Long Wan, Starling Inlet, Tai O, Cape D'Aguilar.	Fellowes et al. (2002): RC	Pond, Compensatory Wetland	Ove Arup (2013a) Ove Arup (2013b) AECOM (2021) AFCD (2024) AFCD Unpublished data
88	Japanese Paradise Flycatcher *Terpsiphone atrocaudata*	Uncommon passage migrant. Found in Tai Po Kau, Mai Po, Pok Fu Lam, Victoria Peak.	Fellowes et al. (2002): LC	Secondary Woodland / Plantation	Ove Arup (2013a)
89	Japanese Quail	Uncommon autumn passage migrant and rare winter visitor. Found in Long	Fellowes et al. (2002): LC	Grassland and Grassland / Shrubland	Ove Arup (2013a)

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	Coturnix japonica	Valley, Mai Po, Kam Tin, Lam Tsuen, Tin Shui Wai.	WWF:HR		Ove Arup (2013b) AFCD (2024) WWF (2025a)
90	Japanese Sparrowhawk *Accipiter gularis*	Uncommon passage migrant. Found in Chek Lap Kok, Mai Po, Tsim Bei Tsui, Mong Tseng, Cheung Chau, Po Toi, Tung Ping Chau.	List of Wild Animals under State Priority Conservation: Class II	Pond, Grassland and Grassland / Shrubland	Ove Arup (2013a)
91	Japanese Yellow Bunting *Emberiza sulphurata*	Scarce spring passage migrant. Found in Tung Ping Chau, Ho Chung, Cheung Chau, Tai Po Kau Tsuen, Shuen Wan, Long Valley, Mai Po, Tsim Bei Tsui.	Fellowes et al. (2002): GC; IUCN Red List of Threatened Species (2025): LC; Red List of China's Vertebrates: VU WWF:HR	Pond	Ove Arup (2013a) Ove Arup (2013b) WWF (2025a)
92	Kentish Plover *Charadrius alexandrinus*	Abundant winter visitor and scarce migrant. Found in Deep Bay area, Chek Lap Kok, Shuen Wan, Sai Kung, Lantau Island.	Fellowes et al. (2002): RC	Pond	Ove Arup (2013a) AFCD (2024)
93	Lesser Sand Plover *Charadrius mongolus*	Uncommon passage migrant and scarce winter visitor. Found in Deep Bay area.	Fellowes et al. (2002): LC	Pond	Ove Arup (2013a)
94	Lesser Whistling Duck *Dendrocygna javanica*	Occasional visitor. Found in Mai Po, Tin Shui Wai.	China Red Data Book: VU; Red List of China's Vertebrates: VU	Pond	Ove Arup (2013a)
95	Lesser White-fronted Goose *Anser erythropus*	Vagrant. Found in Lok Ma Chau.	IUCN Red List of Threatened Species (2025): VU; Red List of China's Vertebrates: VU	Pond	Ove Arup (2013a)
96	Long-toed Stint *Calidris subminuta*	Common passage migrant and scarce winter visitor. Found in Long	Fellowes et al. (2002): LC	Pond	Ove Arup (2013a) Ove Arup (2013b)

Number	Species	Rarity and Distribution in Hong Kong [1]	Conservation/Protection status [2,3,4,5,6,7,8,9,19]	Habitat Recorded	Source [10, 11,12, 13, 14, 15, 16, 17, 18, 19]
		Valley, Ma Tso Lung, Tsim Bei Tsui, Kam Tin, Pui O, Shuen Wan.			
97	Lesser Coucal *Centropus bengalensis*	Uncommon resident. Widely distributed in Hong Kong.	List of Wild Animals under State Priority Conservation: Class II; China Red Data Book: VU WWF:MR	Grassland and Grassland / Shrubland, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) AFCD (2024) AFCD Unpublished data WWF (2025a)
98	Little Curlew *Numenius minutus*	Rare passage migrant. Found in Kai Tak, Mai Po, Tsim Bei Tsui, Kam Tin, Long Valley, Shuen Wan.	Fellowes et al. (2002): LC WWF:HR	Pond	Ove Arup (2013a) WWF (2025a)
99	Little Egret *Egretta garzetta*	Common resident, migrant and winter visitor. Widely distributed in coastal area throughout Hong Kong.	Fellowes et al. (2002): PRC	Marsh, Pond, Channelised Watercourse, Natural Watercourse, Compensatory Wetland, Dry Agricultural Land, Plantation, Orchard, Reedbed	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b) AECOM (2021) Black & Veatch (2019a, 2019b, 2019c, 2019d, 2019e) Monthly EM&A Reports for EP-477/2013 and EP-477/2013/A - Development of LMCL AFCD (2024)
100	Little Grebe *Tachybaptus ruficollis*	Common resident. Found in Deep Bay area.	Fellowes et al. (2002): LC	Marsh, Pond, Natural Watercourse, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b) AECOM (2021) AFCD (2024) AFCD Unpublished data

Number	Species	Rarity and Distribution in Hong Kong [1]	Conservation/Protection status [2,3,4,5,6,7,8,9,19]	Habitat Recorded	Source [10, 11,12, 13, 14, 15, 16, 17, 18, 19]
101	Little Ringed Plover *Charadrius dubius*	Resident, common winter visitor and passage migrant. Widely distributed in freshwater areas throughout Hong Kong.	Fellowes et al. (2002): (LC)	Pond, Natural Watercourse, Channelised Watercourse, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b) AFCD (2024) AFCD Unpublished data
102	Little Tern *Sternula albifrons*	Uncommon passage migrant. Found in Mai Po, Ting Kau, Tsuen Wan Ferry Pier.	Fellowes et al. (2002): LC	Pond	Ove Arup (2013a)
103	Mallard *Anas platyrhynchos*	Scarce winter visitor. Found in Deep Bay area, Tai Lam Chung, Hok Tau Reservoirs, Tolo Harbour, Nam Chung, Long Valley, Kam Tin.	Fellowes et al. (2002): RC WWF:MR	Pond	Ove Arup (2013a) WWF (2025a)
104	Manchurian Reed Warbler *Acrocephalus tangorum*	Scarce autumn passage migrant. Found in Mai Po, Lin Barn Tsuen, Long Valley, Chek Lap Kok, Tin Shui Wai.	IUCN Red List of Threatened Species (2025): VU; Red List of China's Vertebrates: VU WWF:HR	Reed Marsh	Ove Arup (2013a) AFCD (2024) WWF (2025a)
105	Mandarin Duck *Aix galericulata*	Scarce winter visitor. Found in Mai Po, Tai Tam, Shing Mun.	List of Wild Animals under State Priority Conservation: Class II; China Red Data Book: VU WWF:HR	Unspecified	Ove Arup (2013a) WWF (2025a)
106	Marsh Sandpiper *Tringa stagnatilis*	Abundant winter visitor and migrant. Found in Deep Bay area, Shuen Wan, Long Valley, Kam Tin, Sai Kung.	Fellowes et al. (2002): RC WWF:HR	Pond, Compensatory Wetland	Ove Arup (2013a) Ove Arup (2013b) AECOM (2021) AFCD (2024) AFCD Unpublished data WWF (2025a)

Number	Species	Rarity and Distribution in Hong Kong [1]	Conservation/Protection status [2,3,4,5,6,7,8,9,19]	Habitat Recorded	Source [10, 11,12, 13, 14, 15, 16, 17, 18, 19]
107	Mountain Hawk-Eagle *Nisaetus nipalensis*	Vagrant. Found in Luk Keng, Ho Chung, Long Valley, Aberdeen.	Cap. 586	Unspecified	Ove Arup (2013b)
108	Naumann's Thrush *Turdus naumanni*	Rare winter visitor. Found in Chek Lap Kok.	Fellowes et al. (2002): LC	Marsh, Pond	Ove Arup (2013a)
109	Nordmann's Greenshank *Tringa guttifer*	Uncommon passage migrant and scarce winter visitor. Found in Deep Bay area.	List of Wild Animals under State Priority Conservation: Class II; Fellowes et al. (2002): GC; IUCN Red List of Threatened Species (2025): EN; CITES: Appendix I; Red List of China's Vertebrates: EN; Cap. 586 WWF:HR	Pond	Ove Arup (2013a) WWF (2025a)
110	Northern Lapwing *Vanellus vanellus*	Scarce winter visitor. Found in Mai Po, Long Valley, Chek Lap Kok, Ho Chung, Tai Long Wan, Tai Po, Castle Peak coast.	Fellowes et al. (2002): LC WWF:HR	Pond	Ove Arup (2013a) WWF (2025a)
111	Northern Pintail *Anas acuta*	Abundant winter visitor. Found in Deep Bay area, Shuen Wan, Long Valley, Kam Tin.	Fellowes et al. (2002): RC WWF:HR	Pond, Natural Watercourse	Mott Macdonald (2009) Ove Arup (2013a) WWF (2025a)
112	Northern Shoveler *Spatula clypeata*	Abundant winter visitor. Found in Deep Bay area.	Fellowes et al. (2002): RC	Pond, Compensatory Wetland	Ove Arup (2013a) AECOM (2021) AFCD (2024)
113	Eurasian Skylark *Alauda arvensis*	Uncommon autumn passage migrant and scarce winter visitor. Found in Mai Po, Tsim Bei Tsui, Kam Tin, Long Valley.	List of Wild Animals under State Priority Conservation: Class II WWF:HR	Compensatory Wetland	AFCD (2024) AFCD Unpublished data WWF (2025a)

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114	Oriental Skylark *Alauda gulgula*	Scarce passage migrant and winter visitor. Found in Chek Lap Kok, Deep Bay area.	Fellowes et al. (2002): LC WWF:EX	Pond	Ove Arup (2013a) AFCD (2024) WWF (2025a)
115	Oriental Stork *Ciconia boyciana*	Rare winter visitor. Found in Deep Bay area	China Red Data Book: EN; Fellowes et al. (2002): GC; IUCN Red List of Threatened Species (2025): EN; CITES: Appendix I; Red List of China's Vertebrates: EN; Cap. 586	Pond	Ove Arup (2013a)
116	Pacific Golden Plover *Pluvialis fulva*	Common migrant and winter visitor. Found in Deep Bay area, Chek Lap Kok, Long Valley.	Fellowes et al. (2002): LC	Pond	Ove Arup (2013a) Ove Arup (2013b)
117	Pacific Swift *Apus pacificus*	Uncommon spring migrant and summer visitor. Mainly found in Deep Bay area and islands.	Fellowes et al. (2002): (LC) WWF:HR	Pond, Grassland	Ove Arup (2013a) Ove Arup (2013b) WWF (2025a)
118	Pechora Pipit *Anthus gustavi*	Scarce passage migrant. Found in Tsim Bei Tsui, Mount Davis, Mai Po, Shuen Wan, Nam Sang Wai, Long Valley, Luk Keng, Tai O.	Fellowes et al. (2002): LC WWF:HR	Marsh, Seasonally Wet Grassland	Ove Arup (2013a) Ove Arup (2013b) WWF (2025a)
119	Pectoral Sandpiper *Calidris melanotos*	Scarce passage migrant. Found in Mai Po.	Fellowes et al. (2002): LC	Pond	Ove Arup (2013a)
120	Peregrine Falcon *Falco peregrinus*	Locally common resident and winter visitor. Widely distributed in Hong Kong.	List of Wild Animals under State Priority Conservation: Class II; Fellowes et al. (2002): (LC); CITES: Appendix I; Cap. 586	Marsh, Pond, Channelised Watercourse, Grassland, Reedbed, Compensatory Wetland	Ove Arup (2013a) Ove Arup (2013b) Black & Veatch (2019a) AFCD (2024) AFCD Unpublished data

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			WWF: HR		WWF (2025a)
121	Pheasant-tailed Jacana *Hydrophasianus chirurgus*	Uncommon migrant and rare winter visitor. Found in Deep Bay area, Kam Tin, Long Valley.	Fellowes et al. (2002): LC WWF: HR	Marsh, Pond	Ove Arup (2013a) Ove Arup (2013b) WWF (2025a)
122	Pied Avocet *Recurvirostra avosetta*	Abundant winter visitor. Found in Deep Bay area.	Fellowes et al. (2002): RC	Pond, Channelised Watercourse, Compensatory Wetland	Ove Arup (2013a) Ove Arup (2013b) AECOM (2021) AFCD (2024) AFCD Unpublished data
123	Pied Harrier *Circus melanoleucos*	Uncommon passage migrant. Found in Mai Po, Tsim Bei Tsui, Long Valley, Chek Lap Kok.	List of Wild Animals under State Priority Conservation: Class II; China Red Data Book: Rare WWF:HR	Reed marsh, Compensatory Wetland	Ove Arup (2013a) AFCD Unpublished data WWF (2025a)
124	Pied Kingfisher *Ceryle rudis*	Common resident. Widely distributed in lakes and ponds throughout Hong Kong.	Fellowes et al. (2002): (LC) WWF:HR	Pond, Channelised Watercourse, Compensatory Wetland	Ove Arup (2013a) Ove Arup (2013b) AECOM (2021) Monthly EM&A Report for EP-477/2013 - Development of LMCL AFCD (2024) WWF (2025a)
125	Pintail/Swinhoe's Snipe *Gallinago stenura/megala*	Pintail Snipe: Common passage migrant. Found in Long Valley, Chau Tau, Ha Tsuen.	Fellowes et al. (2002): LC for *Gallinago megala* WWF:MR	Pond, Compensatory Wetland	Ove Arup (2013a) Ove Arup (2013b) AFCD (2024) AFCD Unpublished data

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					WWF (2025a)
126	Purple Heron *Ardea purpurea*	Uncommon passage migrant. Found in Deep Bay area.	Fellowes et al. (2002): RC WWF:HR	Reed Marsh, Marsh, Natural Watercourse, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b) AECOM (2021) Monthly EM&A Reports for EP-477/2013 - Development of LMCL AFCD (2024) WWF (2025a)
127	Oriental Pratincole *Glareola maldivarum*	Passage migrant. Found in Mai Po, Tsim Bei Tsui.	Fellowes et al. (2002): LC WWF:MR	Pond	Ove Arup (2013a) Ove Arup (2013b) AFCD (2024) WWF (2025a)
128	Pallas's Grasshopper Warbler *Locustella certhiola*	Common autumn passage migrant. Found in wetland areas throughout Hong Kong.	Fellowes et al. (2002): LC WWF:MR	Reed Marsh, Seasonally Wet Grassland, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b) AFCD (2024) AFCD Unpublished data WWF (2025a)
129	Red-billed Starling *Spodiopsar sericeus*	Abundant winter visitor. Widely distributed in Hong Kong.	Fellowes et al. (2002): GC WWF:HR	Marsh, Pond, Channelised Watercourse, Natural Watercourse, Compensatory Wetland, Dry Agricultural Land, Grassland, Plantation, Village	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b) AECOM (2021) AFCD (2024) AFCD Unpublished data WWF (2025a)

Number	Species	Rarity and Distribution in Hong Kong [1]	Conservation/Protection status [2,3,4,5,6,7,8,9,19]	Habitat Recorded	Source [10, 11,12, 13, 14, 15, 16, 17, 18, 19]
130	Red-breasted Merganser *Mergus serrator*	Rare winter visitor and scarce spring passage migrant. Found in Deep Bay area, Plover Cove Reservoir, Shing Mun Reservoir.	Fellowes et al. (2002): LC WWF:HR	Pond	Ove Arup (2013a) WWF (2025a)
131	Red-necked Stint *Calidris ruficollis*	Abundant spring passage migrant. Found in Deep Bay area.	Fellowes et al. (2002): LC	Pond	Ove Arup (2013a) AFCD (2024)
132	Red-throated Pipit *Anthus cervinus*	Common passage migrant and winter visitor. Widely distributed in dry agricultural areas throughout Hong Kong.	Fellowes et al. (2002): LC WWF:MR	Marsh, Pond, Dry Agricultural Land, Channelised Watercourse, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b) AFCD (2024) AFCD Unpublished data WWF (2025a)
133	Ruddy-breasted Crake *Zapornia fusca*	Uncommon passage migrant and winter visitor. Found in Deep Bay area, Long Valley, Tin Shui Wai, Cheung Shu Tan, Fung Yuen and Ho Chung.	Fellowes et al. (2002): LC WWF:HR	Marsh, Compensatory Wetland	Ove Arup (2013a) Ove Arup (2013b) AFCD Unpublished data WWF (2025a)
134	Rose-ringed Parakeet *Psittacula krameri*	Locally common resident. Found in Kowloon Park, Hong Kong Park, Ocean Park, Aberdeen, Happy Valley.	List of Wild Animals under State Priority Conservation: Class II	Unspecified	Ove Arup (2013b)
135	Ruff *Philomachus pugnax*	Scarce passage migrant. Found in Deep Bay area, Kam Tin.	Fellowes et al. (2002): LC WWF:HR	Pond	Ove Arup (2013a) WWF (2025a)
136	Rufous-capped Babbler *Cyanoderma ruficeps*	Common resident. Found in Shing Mun, Tai Po Kau, Tai Mek Tuk, Ng Tung Chai, Fo Tan, Tai Mo Shan, The Peak, Kadoorie Agricultural Research Centre.	Fellowes et al. (2002): LC	Grassland	Ove Arup (2013b)

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137	Sanderling *Calidris alba*	Uncommon passage migrant. Found in Deep Bay area, Tai Long Wan, Sai Kung.	Fellowes et al. (2002): LC WWF:HR	Pond	Ove Arup (2013a) WWF (2025a)
138	Sharp-tailed Sandpiper *Calidris acuminata*	Common passage migrant. Found in Deep Bay area, Shuen Wan, Sai Kung, Long Valley, Tai Po, Lam Tsuen.	Fellowes et al. (2002): LC WWF:HR	Pond	Ove Arup (2013a) WWF (2025a)
139	Siberian Blue Robin *Larvivora cyane*	Scarce passage migrant. Widely distributed in woodland throughout Hong Kong.	Fellowes et al. (2002): LC	Shrubland	Ove Arup (2013a)
140	Slaty-breasted Rail *Lewinia striata*	Scarce resident and passage migrant. Found in Deep Bay area, Luk Keng, Chek Keng, Tung Chung, Tai O, Pui O.	China Red Data Book: Rare; Fellowes et al. (2002): RC WWF:HR	Marsh	Ove Arup (2013a) WWF (2025a)
141	Slaty-legged Crake *Rallina eurizonoides*	Locally common breeding season visitor, migrant and scarce winter visitor. Found in Shuen Wan, Tai Mei Tuk catchment, Kadoorie Agricultural Research Centre and Kadoorie Farm & Botanic Garden, Kau To Shan, Ting Kok, Mong Tseng, Lung Fu Shan, Shing Mun Country Park, Hok Tau and Tai Tam Country Park.	Red List of China's Vertebrates: VU	Grassland, Secondary Woodland	Ove Arup (2013b)
142	Spotted Redshank *Tringa erythropus*	Common spring passage migrant. Found in Deep Bay area.	Fellowes et al. (2002): RC WWF:HR	Pond	Ove Arup (2013a) WWF (2025a)
143	Steppe Eagle *Aquila nipalensis*	Vagrant. Found in Mai Po.	IUCN Red List of Threatened Species (2025): EN; Red List of China's Vertebrates: VU; Cap. 586	Pond	Ove Arup (2013a)

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144	Striated Heron *Butorides striata*	Common summer visitor. Widely distributed in Hong Kong.	Fellowes et al. (2002): LC	Pond	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b) AFCD (2024)
145	Styan's Grasshopper Warbler *Helopsaltes pleskei*	Scarce passage migrant and winter visitor. Found in Ma Tso Lung, Mai Po.	Fellowes et al. (2002): GC; IUCN Red List of Threatened Species (2025): LC; Red List of China's Vertebrates: VU WWF:HR	Marsh	Ove Arup (2013a) WWF (2025a)
146	Swinhoe's Egret *Egretta eulophotes*	Scarce spring passage migrant. Found in Mai Po, Lok Ma Chau, Nam Chung.	IUCN Red List of Threatened Species (2025): VU; Fellowes et al. (2002): GC; List of Wild Animals under State Priority Conservation: Class I; China Red Data Book: EN; Red List of China's Vertebrates: VU; China Red Data Book: EN	Unspecified	Ove Arup (2013b)
147	Swinhoe's Minivet *Pericrocotus cantonensis*	Scarce passage migrant. Found in Tsim Bei Tsui, Stanley Cemetery, Lam Tsuen, Tai Po Kau, Po Toi, Mai Po.	Fellowes et al. (2002): LC	Unspecified	Ove Arup (2013b)
148	Temminck's Stint *Calidris temminckii*	Uncommon winter visitor and migrant. Widely distributed in Deep Bay area ponds.	Fellowes et al. (2002): LC WWF:HR	Pond, Compensatory Wetland	Ove Arup (2013a) AFCD (2024) AFCD Unpublished data WWF (2025a)
149	Thick-billed Warbler	Scarce passage migrant. Found in Mai Po, Kadoorie Farm and Botanic	Fellowes et al. (2002): LC	Reed Marsh	Ove Arup (2013a)

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	Arundinax aedon	Garden, Lin Barn Tsuen, Kowloon Park, Wong Chuk Yeung, Sha Lo Tung, Nam Sang Wai, Mount Nicholson and Lok Ma Chau.	WWF:MR		WWF (2025a)
150	Tufted Duck *Aythya fuligula*	Abundant winter visitor. Found in Deep Bay area, Nam Chung, Starling Inlet.	Fellowes et al. (2002): LC	Marsh, Pond, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) AECOM (2021) AFCD (2024) AFCD Unpublished data
151	Von Schrenck's Bittern *Ixobrychus eurhythmus*	Scarce passage migrant. Found in Tai Long Wan, Luk Keng, Long Valley.	Fellowes et al. (2002): RC WWF:HR	Marsh	Ove Arup (2013a) Ove Arup (2013b) AFCD (2024) WWF (2025a)
152	Watercock *Gallicrex cinerea*	Scarce passage migrant. Found in Lantau Island, Tai Long Wan, Luk Keng, Nam Chung, Shuen Wan, Shek Kong, Sai Kung.	Fellowes et al. (2002): RC WWF:HR	Marsh	Ove Arup (2013a) AFCD (2024) WWF (2025a)
153	Western Osprey *Pandion haliaetus*	Common winter visitor. Widely distributed in coastal areas throughout Hong Kong.	China Red Data Book: Rare; Fellowes et al. (2002): RC; Cap. 586 WWF: MR	Pond, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) AFCD (2024) AFCD Unpublished data WWF (2025a)
154	Western Water Rail *Rallus aquaticus*	Vagrant. Found in Mai Po, Nam Chung, Wu Kau Tang, Mui Wo, Tai O.	Fellowes et al. (2002): LC	Marsh	Ove Arup (2013a)
155	White-bellied Sea Eagle *Haliaeetus leucogaster*	Locally common resident. Widely distributed in coastal areas throughout Hong Kong.	Fellowes et al. (2002): (RC); Red List of China's Vertebrates: VU	Pond	Mott Macdonald (2009) Ove Arup (2013a) WWF (2025a)

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			WWF:HR		
156	White-cheeked Starling *Spodiopsar cineraceus*	Locally common winter visitor. Found in Deep Bay area, Kam Tin, Long Valley.	Fellowes et al. (2002): PRC WWF:MR	Pond, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) AECOM (2021) AFCD (2024) AFCD Unpublished data WWF (2025a)
157	White-shouldered Starling *Sturnia sinensis*	Locally common passage migrant and uncommon winter visitor. Found in Kam Tin, Deep Bay area, Po Toi Island, Long Valley, Victoria Park, Ho Chung, Ma Tso Lung, Mui Wo, Lam Tsuen Valley.	Fellowes et al. (2002): (LC)	Pond, Secondary Woodland, Plantation, Reedbed, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b) Black & Veatch (2019a) AFCD (2024) AFCD Unpublished Data
158	White-throated Kingfisher *Halcyon smyrnensis*	Common resident. Widely distributed in coastal areas throughout Hong Kong.	Fellowes et al. (2002): (LC)	Reed Marsh, Marsh, Pond, Natural Watercourse, Compensatory Wetland, Reedbed	Mott Macdonald (2009) Ove Arup (2013a) AECOM (2021) Black & Veatch (2019a) AFCD (2024) AFCD Unpublished data
159	Wood Sandpiper *Tringa glareola*	Common migrant and winter visitor. Widely distributed in wetland area throughout Hong Kong.	Fellowes et al. (2002): LC WWF:HR	Pond, Channelised Watercourse, Natural Watercourse, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b) AFCD (2024) AFCD Unpublished data WWF (2025a)
160	Yellow Bittern	Uncommon summer visitor and common passage migrant. Found in	Fellowes et al. (2002): LC	Reed Marsh, Marsh, Pond, Reedbed,	Mott Macdonald (2009)

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	Ixobrychus sinensis	Deep Bay area, Chek Keng, Tai Long Wan.	WWF:HR	Compensatory Wetland	Ove Arup (2013a) Ove Arup (2013b) AECOM (2021) Black & Veatch (2019a) AFCD (2024) AFCD Unpublished data WWF (2025a)
161	Yellow-breasted Bunting *Emberiza aureola*	Scarce spring and rare winter passage migrant. Widely distributed in agriculture land and reedbeds throughout Hong Kong.	Fellowes et al. (2002): RC; IUCN Red List of Threatened Species (2025): CR; Red List of China's Vertebrates: EN WWF:HR	Reed Marsh, Marsh, Dry Agricultural Land, Compensatory Wetland	Ove Arup (2013a) Ove Arup (2013b) AFCD (2024) AFCD Unpublished data WWF (2025a)
162	Yellow-crested Cockatoo *Cacatua sulphurea*	Locally common resident. Found in Hong Kong Park, Pok Fu Lam, Happy Valley, Mong Tseng, Sai Kung, Ocean Park.	IUCN Red List of Threatened Species (2025): CR; CITES: Appendix I; Cap. 586 WWF:HR	Unspecified	Ove Arup (2013b) WWF (2025a)
163	Zitting Cisticola *Cisticola juncidis*	Common passage migrant and winter visitor. Widely distributed in grassland throughout Hong Kong.	Fellowes et al. (2002): LC WWF:HR	Reed Marsh, Grassland and Grassland / Shrubland, Dry Agricultural Land, Pond, Reedbed, Compensatory Wetland	Mott Macdonald (2009) Ove Arup (2013a) Ove Arup (2013b) Black & Veatch (2019a) AFCD (2024) AFCD Unpublished data WWF (2025a)
164	Siberian Rubythroat *Calliope calliope*	Common winter visitor and passage migrant. Widely distributed in Hong Kong.	List of Wild Animals under State Priority Conservation: Class II	Unspecified, Compensatory Wetland	AFCD (2024); AFCD Unpublished data WWF (2025a)

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			WWF:MR		
165	Chestnut-eared Bunting *Emberiza fucata*	Uncommon passage migrant. Found in Long Valley, Tai Mong Tsai, Luk Keng, Ho Chung, Kam Tin, Lantau, Sha Lo Tung.	Fellowes et al. (2002): LC WWF:MR	Unspecified, Compensatory Wetland	AFCD (2024); AFCD Unpublished data WWF (2025a)
166	Crested Kingfisher *Megaceryle lugubris*	Scarce resident and rare winter visitor. Found in Tai Tam Bay, Yi O, Tung Chung, Lai Chi Wo, Kuk Po, Tai Lam Chung Reservoir, Pak Tam, Tsing Lung Tau, Plover Cove, Bride's Pool.	Fellowes et al. (2002): RC WWF:EX	Unspecified	AFCD (2024) WWF (2025a)
167	Northern Boobook *Ninox japonica*	Uncommon passage migrant. Found in Stanley, Cheung Chau, Hong Kong University, Zoological and Botanical Gardens, Mount Nicholson, Magazine Gap Road, Barker Road, Tai Koo Shing, Shek Wu Wai, Cloudy Hill, Tung Chung, Mirs Bay.	Cap. 586; List of Wild Animals under State Priority Conservation: Class II; CITES: Appendix II	Unspecified	AFCD (2024)
168	White-bellied Green Pigeon *Treron sieboldii*	Vagrant. Found in Shek Kong Catchment, Tai Po Kau, Tung Ping Chau.	List of Wild Animals under State Priority Conservation: Class II	Unspecified	AFCD (2024)
169	Hen Harrier *Circus cyaneus*	Rare. Found in Mai Po Nature Reserve.	List of Wild Animals under State Priority Conservation: Class II; Red List of China's Vertebrates: EN ; IUCN Red List of Threatened Species (2025): LC	Compensatory Wetland	AFCD Unpublished data

Abbreviations:
• CR: Critically Endangered; EN: Endangered; GC: Global Concern; LC: Local Concern; PRC: Potential Regional Concern; RC: Regional Concern; VU: Vulnerable

Note:
* All birds in Hong Kong are protected under the Wild Animals Protection Ordinance (WAPO) (Cap. 170). However, their levels of conservation importance are unequal. As such, only bird species with conservation/protection status beyond the WAPO are considered of conservation importance.

1. AFCD (2022). Species Database of the Hong Kong Biodiversity Information Hub.
2. Cap. 170 Wild Animals Protection Ordinance
3. Cap. 586 Protection of Endangered Species of Animals and Plants Ordinance
4. Convention on International Trade in Endangered Species of Wild Fauna and Flora. (2022). Appendices I, II and III. Retrieved from: https://www.cites.org/eng/app/appendices.php
5. Fellowes et al. (2002). Wild animals to watch: Terrestrial and freshwater fauna of conservation concern in Hong Kong. For conservation status listed by Fellowes et al. (2002), letters in parentheses indicate that the assessment is on the basis of restrictedness in breeding and/or roosting sites rather than in general occurrence.
6. International Union of Conservation for Nature. (2025). The IUCN Red List of Threatened Species. Version 2025-2.
7. Jiang et al. (2016). Red list of China's Vertebrates.
8. State Forestry Administration & Ministry of Agriculture (1989). List of Wild Animals under State Protection.
9. Zheng, G. and Wang, Q. (1998). China Red Data Book of Endangered Animals: Aves.
10. AECOM (2021). Ecological Mitigation/Habitat Creation and Management Plan for Sub-package A & B (Final) – Issue 4 for the Agreement No. CE 5/2018 (CE) Development of the Loop: Main Works Package 1 – Design and Construction.
11. AECOM (2022). Report on Environmental Review (Draft) (Ref. G4-01) for the Agreement No. CE 19/2019 (CE) Development of Kwu Tung North Development area, Remaining Phase – Design and Construction: Compensation Event No. CE19/2019-01 – Planning and Engineering Review (KTN) [P&ER(KTN)].
12. AFCD (2024). Unpublished territory-wide long-term monitoring survey data on major groups.
13. Black and Veatch (2019a). Ecological Mitigation / Habitat Creation and Management Plan (184794/B&V/043/EP/Final/Issue 1 for the Agreement No. CE5/2014(CE) Development of the Loop: Land Decontamination and Advance Engineering Works – Design and Construction.
14. Wellab Limited/Nature & Technologies (HK) Limited. (2019-2023). Monthly EM&A Reports from January 2019 to March 2023 for EP-477/2013 and EP-477/2013/A - Development of the Loop.
15. Mott Macdonald. (2009). Approved EIA Report for the Construction of a Secondary Boundary Fence and new sections of Primary Boundary Fence and Boundary Patrol Road.
16. Ove Arup. (2013a). Approved EIA Report for the Agreement No. CE53/2008(CE) Planning and Engineering Study on Development of the Loop – Investigation.
17. Ove Arup. (2013b). Approved EIA Report for the Agreement No. CE61/2007(CE) North East New Territories New Development Areas Planning and Engineering Study – Investigation.
18. Unpublished AFCD Regular Monitoring Data at Lok Ma Chau Loop Ecological Area and Offsite Compensatory Wetland Area
19. WWF – Or, C.K.M. & Chan, B.P.L. (Eds.) (2025a). The State of Hong Kong Biodiversity 2025. WWF-Hong Kong. 65 pp.: EX = Extinct in Hong Kong under Hong Kong species status assessment; HR = High Risk of Extinction in Hong Kong species status assessment; MR = Moderate Risk of Extinction in Hong Kong species status assessment. Letters in parentheses indicate that the assessment is on the basis of breeding population.
20. Exact locations of recorded species were not available from previous studies.

Table 8.6 Non-avifauna Species of Conservation Importance Recorded within the Assessment Area from the Reviewed Literature

Species	Rarity and Distribution in Hong Kong (11 12)	Conservation/Protection status (12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29)	Habitat Recorded	Source (1 2 3 4 5 6 7 8 9 10 29)
Plant				
Incense Tree *Aquilaria sinensis*	Common. Lowland forests and fung shui woods.	IUCN Red List of Threatened Species (2025): VU; CITES: Appendix II; Threatened Species List of China's Higher Plants: VU; China Plant Red Data Book: VU; Included in Illustrations of Rare & Endangered Plant in Guangdong Province; Listed in Rare and Precious Plants of Hong Kong; Cap. 586; State Protection (Category II)	Secondary Woodland#	Ove Arup (2013b)
Mammal				
Red Muntjac *Muntiacus vaginalis*	Widely distributed in Hong Kong.	Fellowes et al. (2002): PRC	Grassland#, Compensatory Wetland#	Ove Arup (2013a) AFCD Unpublished data
Leopard Cat *Prionailurus bengalensis*	Widely distributed in Hong Kong.	China Red Data Book: VU; Cap. 170; Red List of China's Vertebrate: VU	Grassland#, Compensatory Wetland#	Ove Arup (2013a) AECOM (2021) AFCD (2024) AFCD Unpublished data
Eurasian Otter *Lutra lutra*	Records restricted to Mai Po Inner Deep Bay Ramsar Site, Hoo Hok Wai, and nearby areas.	State Protection (Category II); China Red Data Book: VU; Fellowes et al. (2002): RC; Cap. 170; Red List of China's Vertebrate: EN; Shek (2006) WWF:HR	Marsh#, Compensatory Wetland#, Pond#	Ove Arup (2013a) AECOM (2021) McMillan et al. (2022) AFCD (2024) Unpublished data from KFBG AFCD Unpublished data WWF (2025a)

Species	Rarity and Distribution in Hong Kong [11 12]	Conservation/Protection status [12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29]	Habitat Recorded	Source [1 2 3 4 5 6 7 8 9 10 29]
Small Asian Mongoose *Herpestes javanicus*	Fairly widely distributed in countryside areas in the New Territories.	Cap. 170; Red List of China's Vertebrate: VU	Reed Marsh#, Compensatory Wetland#	AECOM (2021) Monthly EM&A Reports for EP-477/2013 -Development of LMCL AFCD (2024)
Intermediate Horseshoe Bat *Rhinolophus affinis*	Widely distributed in countryside areas throughout Hong Kong.	Fellowes et al. (2002): (LC); Cap. 170	Developed Area#	AECOM (2022)
Chinese Noctule *Nyctalus plancyi*	Fairly widespread in Hong Kong.	Fellowes et al. (2002): PRC, (RC); Cap. 170	Marsh#	Ove Arup (2013a)
Japanese Pipistrelle *Pipistrellus abramus*	Widely distributed throughout Hong Kong.	Cap. 170	Developed Area#, Grassland#, Orchard#, Plantation#, Compensatory Wetland#	AECOM (2022) Black & Veatch (2019a) AFCD (2024) AFCD Unpublished data
Lesser Bamboo Bat *Tylonycteris pachypus*	Widely distributed in forested areas throughout Hong Kong.	China Red Data Book: Rare; Fellowes et al. (2002): (LC); Cap. 170	Developed Area, Plantation	AECOM (2022)
Unknown *Vespertilionidae* sp. 1	/	Cap. 170	Plantation#	AECOM (2022)
Small Indian Civet *Viverricula indica*	Widely distributed in Hong Kong.	List of Wild Animals under State Priority Conservation: Class II; Cap. 170; Red List of China's Vertebrate: VU	Grassland#, Compensatory Wetland#	Ove Arup (2013a) AECOM (2021) Monthly EM&A Report for EP-477/2013 -Development of LMCL AFCD (2024)
East Asian Porcupine *Hystrix brachyura*	Very widely distributed in countryside areas	Fellowes et al. (2002): PGC; Cap. 170	Unspecified#	AFCD (2024)

Species	Rarity and Distribution in Hong Kong [11 12]	Conservation/Protection status [12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29]	Habitat Recorded	Source [1 2 3 4 5 6 7 8 9 10 29]
	throughout Hong Kong, except for Lantau Island.			
Pallas's Squirrel *Callosciurus erythraeus*	Common. Fairly widely distributed, with the styani subspecies found in the New Territories (e.g. Tai Lam, Shing Mun and Tai Po Kau), and the thai subspecies found on the Hong Kong Island (e.g. Tai Tam and Pok Fu Lam).	Cap. 170	Compensatory Wetland#	AFCD Unpublished data
Short-nosed Fruit Bat *Cynopterus sphinx*	Very common. Very widely distributed in urban and countryside areas throughout Hong Kong.	Cap. 170	Compensatory Wetland#	AFCD Unpublished data
Herpetofauna				
Chinese Bullfrog *Hoplobatrachus rugulosus*	Quite widely distributed in Hong Kong.	List of Wild Animals under State Priority Conservation: Class II; Fellowes et al. (2002): PRC; Red List of China's Vertebrates: EN WWF:MR	Grassland#, Plantation#, Compensatory Wetland#	Ove Arup (2010) Ove Arup (2013a) Ove Arup (2013b) AFCD Unpublished data WWF (2025a)
Two-striped Grass Frog *Hylarana taipehensis*	Restricted in Hong Kong.	Fellowes et al. (2002): LC WWF:HR	Orchard#, Marsh#	Ove Arup (2013a) WWF (2025a)
Chinese Soft-shelled Turtle *Pelodiscus sinensis*	Locally found in reservoirs and ponds in Deep Bay area.	China Red Data Book: VU; Fellowes et al. (2002): GC; IUCN Red List of Threatened Species (2025): VU; Cap. 170; Red List of China's Vertebrates: EN WWF:MR	Unspecified^#	AFCD (2022); Karsen et al. (1998) WWF (2025a)

Species	Rarity and Distribution in Hong Kong (11 12)	Conservation/Protection status (12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29)	Habitat Recorded	Source (1 2 3 4 5 6 7 8 9 10 29)
Three-banded Box Turtle *Cuora trifasciata*	Widely distributed but a rare species in Hong Kong.	List of Wild Animals under State Priority Conservation: Class II; China Red Data Book: CR; Fellowes et al. (2002): GC; IUCN Red List of Threatened Species (2023): CR; CITES: Appendix II: Cap. 170; Cap. 586 WWF:HR	Modified Watercourse#	Ove Arup (2013b) WWF (2025a)
Chinese Cobra *Naja atra*	Widely distributed in Hong Kong.	China Red Data Book: VU; Fellowes et al. (2002): PRC: CITES: Appendix II; Red List of China's Vertebrates: VU	Secondary Woodland# Compensatory Wetland#	Ove Arup (2013a) AFCD Unpublished data
Common Rat Snake *Ptyas mucosus*	Widely distributed in Hong Kong.	China Red Data Book: EN; Fellowes et al. (2002): PRC; CITES: Appendix II; Red List of China's Vertebrates: EN	Marsh#, Plantation#, Compensatory Wetland#	Ove Arup (2010) Ove Arup (2013a) AFCD (2024) AFCD Unpublished data
Indo-Chinese Rat Snake *Ptyas korros*	Widely distributed in Hong Kong.	China Red Data Book: EN; Fellowes et al. (2002): PRC; Red List of China's Vertebrates: VU	Grassland#	Ove Arup (2013a) AFCD (2024)
Many-banded Krait *Bungarus multicinctus*	Widely distributed In Hong Kong.	China Red Data Book: VU; Fellowes et al. (2002): PRC; Red List of China's Vertebrates: EN	Orchard#	Ove Arup (2010) Ove Arup (2013b)
Burmese Python *Python bivittatus*	Widely distributed throughout Hong Kong.	List of Wild Animals under State Priority Conservation: Class I; China Red Data Book: CR; Fellowes et al. (2002): PRC; IUCN Red List of Threatened Species (2025): VU; CITES: Appendix II; Cap. 170; Cap. 586	Pond#	Mott Macdonald (2009) AFCD (2024)
Chinese Water Snake *Enhydris chinensis*	Distributed in freshwater or brackish wetlands in central and northern New Territories.	Red List of China's Vertebrates: VU WWF:MR	Semi-natural Watercourse#	Mott Macdonald (2009) AFCD (2024) WWF (2025a)

Species	Rarity and Distribution in Hong Kong [11 12]	Conservation/Protection status [12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29]	Habitat Recorded	Source [1 2 3 4 5 6 7 8 9 10 29]
Plumbeous Water Snake *Hypsiscopus plumbea*	Locally common and widely distributed in Hong Kong.	Red List of China's Vertebrates: VU	Unspecified#	AFCD (2024)
Butterflies				
Grey Scrub Hopper *Aeromachus jhora*	Rare. Yung Shue O, Kuk Po, Tai Lam, Sha Lo Tung.	-	Seasonally Wet Grassland#	AECOM (2022) AFCD (2024)
Colon Swift *Caltoris bromus*	Very rare. Yung Shue O, Pak Tam Chung, Chuen Lung, Hong Kong Wetland Park, Wu Kau Tang, Luk Keng, Kuk Po, Mai Po.	-	Unspecified#	AFCD (2024)
Dark Swift *Caltoris cahira*	Rare. Shing Mun, Sha Lo Tung, Mui Tsz Lam, Sai Kung, Lung Kwu Tan, Ho Sheung Heung, Muk Min Ha, Lin Ma Hang, Mai Po.	-	Unspecified#	AFCD (2024)
Oriental Straight Swift *Parnara bada*	Rare. Widely distributed throughout Hong Kong.	-	Unspecified#	AFCD (2024)
Large Branded Swift *Pelopidas subochraceus*	Very rare. Clear Water Bay Country Park	WWF:MR	Unspecified#	AFCD (2024) WWF (2025a)
Yellow Dart *Potanthus pava*	Very rare. Plover Cove.	-	Unspecified#	AFCD (2024)
Pale Palm Dart *Telicota colon*	Rare. Widely distributed in grassland and shrubland throughout Hong Kong.	Fellowes et al. (2002): LC	Marsh#	Ove Arup (2010) AFCD (2024)
Dark Palm Dart *Telicota ohara*	Rare. Widely distributed throughout Hong Kong.	-	Unspecified#	AFCD (2024)

Species	Rarity and Distribution in Hong Kong [11 12]	Conservation/Protection status [12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29]	Habitat Recorded	Source [1 2 3 4 5 6 7 8 9 10 29]
Grass Demon *Udaspes folus*	Rare. Widely distributed throughout Hong Kong.	-	Unspecified#	AFCD (2024)
Metallic Cerulean *Jamides alecto*	Very rare. Victoria Peak, Fung Yuen, Chuen Lung, Mui Wo.	-	Seasonally Wet Grassland#	AECOM (2022)
Malayan *Megisba malaya*	Very rare. North Lantau Island	Fellowes et al. (2002): LC	Unspecified#	AFCD (2024)
Small Three-ring *Ypthima norma*	Very rare/Species of conservation concern. Pak Tam Chung, Ngau Kwo Lo, Kai Shan.	Fellowes et al. (2002): LC WWF:HR	Grassland#	Ove Arup (2013a) Ove Arup (2013b) WWF (2025a)
Yellow Rajah *Charaxes marmax*	Uncommon. Cloudy Hill, Ma On Shan, Shing Mun, Yung Shue O, Fung Yuen, Ngong Ping	Fellowes et al. (2002): LC	Unspecified#	AFCD (2024)
Small Cabbage White *Pieris rapae*	Rare. Shep Mun Kap, Fan Lau, Ngong Ping, Kam Tin, Ho Chung, Luk Keng, Tuen Mun Ash Lagoon.	-	Compensatory Wetland#	AECOM (2021) AFCD (2024)
Odonates				
Blue Chaser *Potamarcha congener*	Common. Widely distribute in marsh throughout Hong Kong.	Fellowes et al. (2002): LC	Grassland#	AECOM (2022)
Blue Sprite *Pseudagrion microcephalum*	Common. Found in lowland streams and ponds; often perches on aquatic plants just above the water surface. Population scattered all over Hong	Fellowes et al. (2002): LC	Unspecified#, Compensatory Wetland#	AFCD (2024) AFCD Unpublished data

Species	Rarity and Distribution in Hong Kong [11 12]	Conservation/Protection status [12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29]	Habitat Recorded	Source [1 2 3 4 5 6 7 8 9 10 29]
	Kong and established in Hong Kong Wetland Park.			
Coastal Glider *Macrodiplax cora*	Common. Hong Kong Wetland Park, Kam Tin, Lai Chi Wo, Nim Wan and Luk Keng.	Fellowes et al. (2002): LC	Pond#, Developed Area#	Ove Arup (2013a) Black & Veatch (2019a)
Dancing Shadow-emerald *Idionyx victor*	Common. Found high in the forest canopy or over wooded streams. Widely distributed in wooded streams throughout Hong Kong.	Fellowes et al. (2002): LC	Unspecified#	AFCD (2024)
Eastern Lilysquatter *Paracercion melanotum*	Common. Found in ponds and streams with floating plants. Recorded in Hong Kong Wetland Park, Lai Chi Wo, Luk Keng and Shek Kwu Chau.	Fellowes et al. (2002): LC	Unspecified#	AFCD (2024)
Ruby Darter *Rhodothemis rufa*	Common. Widely distribute in ponds and marshes throughout Hong Kong.	Fellowes et al. (2002): LC	Modified Watercourse#, Village#, Developed Area#, Compensatory Wetland#	Ove Arup (2013a) Ove Arup (2013b) Black & Veatch (2019a) Reels (2019) AFCD (2024) AFCD Unpublished data
Sapphire Flutterer *Rhyothemis triangularis*	Common. Widely distribute in weedy ponds, sluggish rivers and marshes	Fellowes et al. (2002): LC	Pond#	Ove Arup (2013a) Reels (2019)
Scarlet Basker Urothemis signata	Common. Common in areas containing abandoned	Fellowes et al. (2002): LC	Pond#, Compensatory Wetland#	Ove Arup (2013a) AECOM (2018)

Species	Rarity and Distribution in Hong Kong [11 12]	Conservation/Protection status [12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29]	Habitat Recorded	Source [1 2 3 4 5 6 7 8 9 10 29]
	ponds throughout Hong Kong.			Reels (2019) AFCD (2024)
Tawny Hooktail *Paragomphus capricornis*	Uncommon. Found in woodland streams with fine to coarse sand beds. Recorded in Lion Rock Country Park, Lung Kwu Tang, Sham Tseng, Tai Lam Country Park, Tai Tong and Yeung Ka Tsuen.	Fellowes et al. (2002): RC	Unspecified#	AFCD (2024)
Aquatic Fauna				
Small Snakehead *Channa asiatica*	Uncommon in the wild.	Fellowes et al. (2002): LC	Semi-natural watercourse#	Ove Arup (2013a)
Somanniathelphusa zanklon	Endemic to Hong Kong, but quite widely distributed	Fellowes et al. (2002): GC IUCN Red List of Threatened Species (2025): EN Endemic to Hong Kong	Semi-natural watercourse #	Ove Arup (2013a) Ove Arup (2013b)

Abbreviations:
• CR: Critically Endangered; EN: Endangered; GC: Global Concern; LC: Local Concern; PRC: Potential Regional Concern; RC: Regional Concern; VU: Vulnerable

Note:
1. AECOM (2021). Ecological Mitigation/Habitat Creation and Management Plan for Sub-package A & B (Final) – Issue 4 for the Agreement No. CE 5/2018 (CE) Development of the Loop: Main Works Package 1 – Design and Construction.
2. AECOM (2022). Report on Environmental Review (Draft) (Ref. G4-01) for the Agreement No. CE 19/2019 (CE) Development of Kwu Tung North Development area, Remaining Phase – Design and Construction: Compensation Event No. CE19/2019-01 – Planning and Engineering Review (KTN) [P&ER(KTN)].
3. AFCD (2024). Unpublished territory-wide long-term monitoring survey data on major groups.
4. Black and Veatch (2019a). Ecological Mitigation / Habitat Creation and Management Plan (184794/B&V/043/EP/Final/Issue 1 for the Agreement No. CE5/2014(CE) Development of the Loop: Land Decontamination and Advance Engineering Works – Design and Construction.
5. Mott Macdonald. (2009). Approved EIA Report for the Construction of a Secondary Boundary Fence and new sections of Primary Boundary Fence and Boundary Patrol Road.
6. Ove Arup (2010). Final Report of the Agreement No. CE60/2005 (TP) Land Use Planning for the Closed Area – Feasibility Study.
7. Ove Arup. (2013a). Approved EIA Report for the Agreement No. CE53/2008(CE) Planning and Engineering Study on Development of the Loop – Investigation.
8. Ove Arup. (2013b). Approved EIA Report for the Agreement No. CE61/2007(CE) North East New Territories New Development Areas Planning and Engineering Study – Investigation.
9. McMillan et al. (2022). Spraints demonstrate small population size and reliance on fishponds for Eurasian otter (*Lutra lutra*) in Hong Kong.
10. Unpublished AFCD Regular Monitoring Data at Lok Ma Chau Loop Ecological Area and Offsite Compensatory Wetland Area
11. AFCD (2022). Species Database of the Hong Kong Biodiversity Information Hub.

12. Corlett *et al.* (2000). Hong Kong vascular plants: distribution and status.
13. Stanton & Leven (2016). Distribution, habitat, and conservation status of the freshwater crab, Somanniathelphusa zanklon Ng & Dudgeon, 1992 (Crustacea: Brachyura: Gecarcinucidae) endemic to Hong Kong.
14. Cap. 170 Wild Animals Protection Ordinance
15. Cap. 586 Protection of Endangered Species of Animals and Plants Ordinance
16. Convention on International Trade in Endangered Species of Wild Fauna and Flora. (2022). Appendices I, II and III. Retrieved from: https://www.cites.org/eng/app/appendices.php
17. Fellowes et al. (2002). Wild animals to watch: Terrestrial and freshwater fauna of conservation concern in Hong Kong. For conservation status listed by Fellowes et al. (2002), letters in parentheses indicate that the assessment is on the basis of restrictedness in breeding and/or roosting sites rather than in general occurrence.
18. Fu & Chin (1992). China Plant Red Data Book – Rare and Endangered Plants.
19. Hu et al. (2003). Rare and Precious Plants of Hong Kong.
20. International Union of Conservation for Nature. (2025). The IUCN Red List of Threatened Species. Version 2025-2.
21. Jiang et al. (2016). Red list of China's Vertebrates.
22. Karsen et al. (1998). Hong Kong Amphibians and Reptiles.
23. Qin et al. (2017). Threatened Species List of China's Higher Plants.
24. State Forestry Administration & Ministry of Agriculture (1989). List of Wild Animals under State Protection.
25. State Forestry Administration & Ministry of Agriculture (1999). List of Wild Plants under State Protection (Part 1).
26. Wang (1998). China Red Data Book of Endangered Animals: Mammalia.
27. Wang & Zhao (1998). China Red Data Book of Endangered Animals: Amphibia and Reptilia.
28. Wu et al. (1988). Illustration of Rare & Endangered Plant in Guangdong Province.
29. WWF – Or, C.K.M. & Chan, B.P.L. (Eds.) (2025a). The State of Hong Kong Biodiversity 2025. WWF-Hong Kong. 65 pp.: EX = Extinct in Hong Kong under Hong Kong species status assessment; HR = High Risk of Extinction in Hong Kong species status assessment; MR = Moderate Risk of Extinction in Hong Kong species status assessment. Letters in parentheses indicate that the assessment is on the basis of breeding population.
30. Habitat types: Habitat with ^ = recorded within Construction Area of the Project. Habitat with # = recorded outside of Construction Area of the Project but within Assessment Area

8.6 Ecological Baseline Information (Survey Findings)

Overview of Ecological Baseline Conditions

8.6.1 Ecological surveys were undertaken from July 2022 to December 2023. A total of 13 habitats were identified within the Assessment Area, comprising wetland habitats (compensatory wetland, pond, marsh, reed, modified watercourse, semi-natural / natural watercourse and seasonally wet grassland), and non-wetland habitats (woodland, shrubland / grassland, orchard, plantation, agricultural land and developed area, **60722948/A10/FIGURE 8.3** refers). A summary of all habitats identified within the Assessment Area is presented in **Table 8.7** below. Representative photos of habitat types and observed sites / species of conservation importance during ecological surveys are provided in **Appendices 8.2** and **8.3** respectively.

Table 8.7 Summary of Habitats Identified within Construction Area of the Project and Assessment Area

Habitat Types	Overall			
	Within Construction Area of the Project		Within the Assessment Area	
	ha	Percentage	ha (1)	Percentage
Compensatory Wetlands(2)	0.38	2.5	11.41	3.1
Pond	1.52	9.8	70.03	19.3
Marsh	0.33	2.2	15.88	4.4
Reedbed	0.00	0.0	8.24	2.3
Modified Watercourse	0.04	0.2	0.67	0.2
Semi-natural / Natural Watercourse	0.23	1.5	11.90	3.3
Seasonally Wet Grassland	1.61	10.4	12.31	3.4
Woodland	1.68	10.9	8.24	2.3
Grassland / Shrubland	2.88	18.6	124.67	34.4
Orchard	0.54	3.5	6.82	1.9
Plantation	1.09	7.1	13.86	3.8
Agricultural Land	0.00	0.0	1.27	0.4
Developed Area	5.14	33.3	77.45	21.4
Total(3)	**15.45**	**100.0**	**362.76**	**100.0**

Note:
1. Assessment Area includes areas within the Construction Area of the Project.
2. The portion of the EA included in the Project Boundary but outside of Construction Area of the Project would be properly fenced off and maintained by the project proponent during the construction phase, even when it would not be affected or occupied as temporary works/storage area.
3. Slight discrepancies in the totals due to rounding up.

Habitat and Vegetation

Compensatory Wetlands

Ecological Area

8.6.2 The entire EA is approximately 12.78 ha supporting reed marsh and marsh habitats, 6.92 ha of which falls within the Assessment Area. The marsh comprises a mix of floating, emergent and submerged plants such as Crested Floating Heart (*Nymphoides cristatum*), Water Lily (*Nymphaea spp.*) and Short-leaved Malacca Galingale (*Cyperus malaccensis*), while the reedbed is dominated by Common Reed (*Phragmites australis)*. Specific habitat features to promote usage by Eurasian Otter had been deployed, including artificial otter holts and otter resting platforms.

OWCA Area 2

8.6.3 Habitat creation works at OWCA Area 2 were completed in October 2022, comprising 6.61 ha pond, 1.66 ha marsh and 0.22 ha reed marsh (CEDD, 2023b). Of these, 4.49 ha fall within the Assessment Area. Specific habitats feature to promote usage by Eurasian Otter had been provided, including the establishment of otter holts, floating platforms, and rock platforms. The planting species from open pond bund in Area 2 including Brassica (*Brassica oleracea var. alboglabra*) and Couch Grass (*Cynodon dactylon*). Existing trees, shrubs and dense vegetation were retained, and emergent plants tolerant of some salinity, such as Short-leaved Malacca Galingale and Mexican Primrose-willow (*Ludwigia octovalvis*) were planted.

Pond

8.6.4 A mixture of active, inactive and abandoned aquaculture ponds was dominant across the HHW Area. Fish-farming devices (e.g., aerators, bird deterrent lines) were observed in some ponds. While most active ponds do not support submerged or emergent flora due to the pond profile and management regime, the associated narrow bunds support disturbance-associated herb and climber species, such as Couch Grass, Crow-foot Grass (*Dactyloctenium aegyptium*), Mikania (*Mikania micrantha*) and Passion Flower (*Passiflora foetida*). Fruit trees, such as Mango (*Mangifera indica*) and Guava (*Psidium guajava*), were occasionally cultivated by villagers on the bunds.

8.6.5 Abandoned ponds were scattered across the HHW Area in the Assessment Area. These ponds have been colonised by pioneer aquatic herb species in both the interior and pond bunds, including mono-dominant patches of Blunt Signal-grass (*Brachiaria mutica*) and Narrow-leaved Cat-tail (*Typha angustifolia*). Giant Alocasia (*Alocasia macrorrhiza*), White Popinac (*Leucaena leucocephala*) and Passion Flower were occasionally encountered. One small pond was recorded near the Ma Tso Lung Village that appeared to have been abandoned for some time and has become overgrown with Elephant Grass (*Pennisetum purpureum*), Knotweed (*Polygonum* sp.), and White Popinac. Another pond was recorded located at the junction of border road and Ma Tso Lung Road, which Winged Cassia (*Cassia alata*) and other ornamental species such as Copper Leaf (*Acalypha wilkesiana*) and Weeping Willow (*Salix babylonica*) were recorded at the fringe of pond. One small pond was found located within a private property at MTL Shun Yee San Tsuen and potentially used for amenity purpose. Further to the west, two ponds recorded at Hornhill, west to Ping Hang Stream, appeared to have been abandoned and colonised grassy vegetation, with Elephant's Ear (*Macaranga tanarius var. tomentosa*) and *Melastomataceae spp.*

recorded on bund. Three small ponds were identified at Tse Koo Hang and between the Ping Hang to Tai Law Hau area, however the current habitat condition of these ponds is unclear due to limited access, and potentially used for amenity purpose.

8.6.6 The ECR Alignment would pass through three ponds (Ponds 36-38) to the southeast to of the LMC Meander. During recent surveys, overgrown vegetation was observed along bunds and across much of the surface of Ponds 37-38, suggesting they have been inactive/abandoned for some time. Pond 36 was found to be recently reprofiled, with the northern portion of the pond partitioned into five smaller units. Another long abandoned small pond (Pond 39a) that falls partially within the Project Boundary was found to have been recently reprofiled, changing habitat type from seasonally wet grassland into a pond, and no vegetation was observed.

Marsh

8.6.7 Marsh areas were recorded as part of the broader HHW wetlands, with *Phragmites vallatorius* was very common in the marsh interiors. The edges of the marsh are colonised by exotic herbs such as Guinea Grass (*Panicum maximum*) and Narrow-leaved Cat-tail.

Reed

8.6.8 Two patches of reedbed were identified in HHW, dominated by *Phragmites vallatorius* and *P. australis*. Another patch of reedbed of about 3 ha was identified within the Loop, dominated by *Phragmites* spp.

Modified Watercourses

8.6.9 The Ma Tso Lung stream, located west of the Ma Tso Lung Shun Yee San Tsuen, falls partly within the Construction Area of the Project. Drainage improvement works undertaken in 2010 resulted in the stream being straightened with highly modified with concrete banks. Fine sediments were observed deposited in the downstream section of the channel.

8.6.10 A second modified watercourse occurs in the Assessment Area near the Lo Wu Range. This watercourse is subject to disturbance, modification and pollution discharges.

Semi-natural / Natural Watercourse

8.6.11 The LMC Meander is largest natural watercourse recorded in the western portion of the Assessment Area. Riparian vegetation was dominated by common grass and herb species such as Giant Alocasia and Gahnia (*Gahnia tristis*), along with common tree species including Opposite-leaved Fig (*Ficus hispid*a), Hairy Fig (*Ficus hirta*), Common Red-stem Fig (*Ficus variegata*).

8.6.12 Ping Hang Stream is a small, highly seasonal semi-natural watercourse, originating from the hillside at Ping Hang, flowing from south to north beneath the Border Road, through the pond area south of the Loop before discharging into the LMC Meander. The upper section of the stream course is largely natural, while the lower section has been straightened and channelised between the surrounding ponds. The lower section largely retains natural substrates, and bankside vegetation is dominated by common grassy vegetation such as Para Grass (*Brachiaria mutica*) and Guinea Grass.

8.6.13 Another water channel near the upstream of the Ma Tso Lung stream retains a natural riverbed with gabion channel banks. Aquatic herbaceous species, in particular Ramie (*Boehmeria nivea*) and Whorled Pennywort (*Hydrocotyle verticillata*), formed dense mats in areas with growing space and appropriate substrate. Several other small streams originating from Crest Hill, Tit Hang Shan and Ma Tso Lung were recorded in the Assessment Area. The upstream sections of these watercourses are located on hillsides and are largely natural and unmodified. Riparian vegetation was observed along the along these watercourses as previously described.

Seasonally Wet Grassland

8.6.14 An extensive area of seasonally wet grassland was identified south and southwest of Ma Tso Lung village, with smaller patches found interspersed between pond habitats in the HHW Area. The seasonally wet grassland was dominated by herb species, including Freshy Lady-fern (*Callipteris esculenta*), Taro (*Colocasia esculenta*), Morning Glory (*Ipomoea cairica*), Mikania and Guinea Grass. Scarce individuals of Lance-leaved Sterculia (*Sterculia lanceolata*) and White Popinac were also recorded from this habitat.

Woodland

8.6.15 Woodland belts at Horn Hill bounding the west of Lok Ma Chau Operational Base of the Hong Kong Police Force and woodland stands near Tse Koo Hang were identified within the Assessment Area. Native and exotic species were found in this habitat type, with native plants being more diverse and relatively more abundant. The canopy of the woodland was dominated by common native tree species and naturally dispersed fruit tree species in Hong Kong, such as Chinese Hackberry (*Celtis sinensis*), Camphor Tree (*Cinnamomum camphora*) and Longan. The understorey comprised native tree (e.g. Opposite-leaved Fig, Microcos (*Microcos nervosa*) and *Tetradium glabrifolium*), shrub, climber (e.g. *Desmos chinensis*) and herb species (e.g. *Liriope spicata*). Some exotic plantation species, like Taiwan Acacia and Slash Pine (*Pinus elliottii*), were also recorded. Due to the proximity of the woodland habitats to villages, some agricultural crop species, for instance, Winged Yam (*Dioscorea alata*), were found on the edge of the woodland. The at-grade road section of the ECR Alignment will pass through both recorded woodland areas.

Grassland / Shrubland

8.6.16 Grassland / shrubland within the Assessment Area dominated the hillsides of Crest Hill, She Leng and Tit Hang Shan, along with unnamed small hills in the southeast. The grassland / shrubland within the Assessment Area has a simple in floristic composition, being dominated by Dichotomy Forked Fern (*Dicranopteris pedata*) thickets along with common native shrub species Oblong-leaved Litsea (*Litsea rotundifolia* var. *oblongifolia*), Scansorial Climbing Fern (*Lygodium scandens*), Common Melastoma (*Melastoma malabathricum*), Chinese Fevervine (*Paederia scandens*) and Rose Myrtle (*Rhodomyrtus tomentosa*). This habitat is disturbed by frequent hillfires, with evidence of recent fires observed during the surveys.

Orchard

8.6.17 Scattered orchard patches were recorded at Tse Koo Hang, to the north of Ma Tso Lung San Tsuen and west of Ma Tso Lung Stream. The orchards were dominated by Carambola (*Averrhoa carambola*), Pomelo (*Citrus maxima*), Wampi (*Clausena lansium*), Lychee and Banana (*Musa* x *paradisiaca*). The at-grade road section of the

ECR Alignment will pass through a small area of orchard to the west of Ma Tso Lung Stream.

Plantation

8.6.18 Plantations belts / stands were generally found in Ma Tso Lung, near the foot of Crest Hill and Tit Hang Shan and to the east of Lok Ma Chau Operational Base. Plantation belts and stands were established for afforestation, amenity, landscaping and screening purposes, primarily utilising imported pioneer tree species. The plantations are dominated by exotic trees and fruit trees (e.g. Taiwan Acacia*,* Cook Pine (*Araucaria columnaris*)*,* Brisbane Box (*Lophostemon confertus*)*,* Lychee and Slash Pine), interspersed with native pioneer tree species (e.g., Elephant's Ear*,* Ivy Tree (*Schefflera heptaphylla*)*,* Aporosa (*Aporosa dioica*)*,* Chinese Banyan (*Ficus macrocarpa*) *and* Camphor Tree). Naturally recruited native small tree and shrub species, such as *Brucea javanica*, were likely dispersed from nearby woodlands by birds. Exotic herbaceous species such as *Bidens alba* and Mission Grass (*Pennisetum polystachion*) were abundant along the edges of these plantation stands.

Agricultural Land

8.6.19 Small patches of agricultural land were observed near villages and scattered in the Ma Tso Lung area. Food crop (e.g. Lettuce, *Lactuca sativa*) and fruit tree species (e.g. Jackfruit, *Artocarpus heterophyllus;* Longan and *Musa acuminata* 'Dwarf Cavendish') were cultivated by villagers in these habitats.

Developed Area

8.6.20 Developed area within the Assessment Area consists of the Loop, roads (e.g. Ma Tso Lung Road), villages and wasteland (e.g. Ma Tso Lung Shun Yee San Tsuen, Ma Tso Lung San Tsuen and Ma Tso Lung), government facilities (e.g. Police post, Lok Ma Chau Operational Base belonging to the Hong Kong Police Force and Hong Kong Police Lo Wu Range), schools (Lung Kai Public School), and nurseries (King Fung Nurseries). Areas that are currently being modified or occupied by temporary structures and construction machinery, or have been previously modified by human activities but now colonised by exotic vegetation are also classified as Developed Area. The limited growing space in concrete-paved areas only allows ruderal herbaceous species and exotic, cultivated landscape, ornamental and fruit tree species (e.g. *Bombax ceiba,* Longan*, Sterculia monosperma* and White Popinac) to grow in this habitat. The works within the Loop for its development into the Hong Kong-Shenzhen Innovation and Technology Park (the Park) commenced in 2018 and is currently on-going. Amid the COVID-19 epidemic, community isolation and treatment facilities were constructed on the Loop in early 2022.

Vegetation

8.6.21 A total of 310 plant species were recorded within the Assessment Area during surveys conducted for this EIA, comprising 192, 113 and 5 are native, exotic and of unknown origin to Hong Kong respectively (**Appendix 8.4** refers). Among the plant species recorded, 110 were found within the Project Boundary. Plant species and their relative abundance within each habitat of the Project Boundary and Assessment Area are listed in **Appendix 8.4**. Three plant species of conservation importance were recorded within the Assessment Area, namely Incense Tree, *Aralia chinensis* and Tongue Habenaria (*Habenaria linguella*). Several individuals of Incense Tree were recorded in the woodland at Tse Koo Hang as well as from a strip of shrubland along

the Border Road and woodland adjacent ponds near the LMC Operational Base. In total, four individuals were found within the Construction Area of the Project. Single individuals of *Aralia chinensis* and Tongue Habenaria were record in grassland / shrubland habitat at Tse Koo Hang, outside of the Construction Area of the Project.

Fauna

Avifauna

8.6.22 140 bird species were recorded within the Assessment Area, 81 of which are of conservation importance. Among the species recorded, 42 were recorded within the Construction Area of the Project, 15 of which are of conservation importance. Most of the recorded species are common and widespread in Hong Kong. Details of the recorded avifauna species recorded in each habitat are presented in **Appendix 8.5** and **60722948/A10/FIGURE 8.5**.

8.6.23 Pond habitats in HHW Area support a high diversity and abundance of waterbirds and wetland-dependent species, with a total of 105 species and 54 wetland dependant species recorded from this habitat type. Abundances and diversity were generally higher in the winter migratory season. Reflecting their generally poor condition, Ponds 36-38 were found to support a low diversity and abundance of wetland avifauna, dominated by common species such as White-breasted Waterhen (*Amaurornis phoenicurus*), Chinese Pond Heron and Black-crowned Night Heron (*Nycticorax nycticorax*). The marsh habitat in HHW area support moderate abundance and diversity of wetland dependent bird species, including White-breasted Waterhen and White-throated Kingfisher. Only occasional records of larger wetland birds (e.g., Grey Heron and Great Egret) were made from this habitat compared to open water habitat such as pond. Oriental Turtle Dove (*Streptopelia orientalis*) was also recorded from the habitat.

8.6.24 Aside from wetland habitats, relatively few records of species of conservation importance were recorded from other habitats in the Assessment Area. Notably, Asian Barred Owlet (*Glaucidium cuculoides*), Chinese Grosbeak (*Eophona migratoria*) and Grey Treepie (*Dendrocitta formosae*) were recorded from woodland habitats, and several raptor species, including Black Kite, Common Kestrel; Crested Goshawk (*Accipiter trivirgatus*); and Eastern Buzzard were recorded flying over various habitats including grassland / shrubland.

8.6.25 No breeding or nesting activity was recorded during ecological survey, and no records of Eurasian Eagle Owl or Chinese Francolin (species for which confirmed or suspected breeding in the Assessment Area was reported in the literature review) were made. White-throated Kingfisher was recorded from various habitats, including grassland / shrubland, suggesting it might be breeding in the Assessment Area.

Major Flight Paths

8.6.26 A total of five major flight lines were observed from vantage points adjacent to the LMC Meander. 12 waterbirds and/or larger avifauna species of conservation importance, namely Black Kite, Black-crowned Night Heron, Black-faced Spoonbill, Eastern Cattle Egret, Chinese Pond Heron, Eurasian Spoonbill, Great Cormorant, Great Egret, Grey Heron, Little Egret, White-throated Kingfisher and Pied Kingfisher were recorded during the coordinated bird flightline surveys (**Table 8.8** refers).

8.6.27 The HHW flight corridor, which lies above pond habitats between LMC Meander and the Border Road (**60722948/A10/FIGURE 8.8** refers), accounted for the largest proportion of the flights recorded (~53.4%) and was adopted by the highest diversity of waterbirds and/or larger avifauna species of conservation importance, which flew at a height of 15-25m (**Table 8.8** refers).

8.6.28 Major flight path No. 2 refers to the airspace directly above LMC Meander (**60722948/A10/FIGURE 8.8** refers), and was adopted by the second largest number of the observed waterbirds and/or larger avifauna species of conservation importance (~33.54% of all flightlines recorded) flying at a height of 15-25m (**Table 8.8** refers).

8.6.29 Major flight path no. 3 was only adopted by Great Cormorants flying at 25m or above from the direction of grassland/shrubland near Ping Hang towards ponds in Hoo Hok Wai (**60722948/A10/FIGURE 8.8** refers).

8.6.30 The remaining two identified flight paths (i.e. major flight path nos. 1 and 4) were rarely used by waterbirds and/or larger avifauna species of conservation importance (~1.25% of the total flightlines in total). Major Flight Path no. 1 was located along the EA, whereas Major Flight Path No. 4 is roughly aligned along the existing Border Road (**60722948/A10/FIGURE 8.8** refers).

Table 8.8 Summary of Coordinated Bird Flightline Surveys

Major flight path	Number of flight records	Percentage of total flight records (%)	Waterbirds and larger avifauna species for conservation importance	Number of bird species recorded	Modal flight height class (m)
1	3	0.32	Great Cormorant	1	25+
2	323	33.54	Black Kite Black-faced Spoonbill Eurasian Spoonbill Great Cormorant Great Egret Grey Heron Little Egret White-throated Kingfisher	8	15-25
HHW flight corridor	514	53.37	Black-crowned Night Heron Black-faced Spoonbill Chinese Pond Heron Eastern Cattle Egret Eurasian Spoonbill Great Cormorant Great Egret Grey Heron Little Egret Pied Kingfisher	10	15-25
3	114	11.84	Great Cormorant	1	25+
4	9	0.93	Great Cormorant Grey Heron	2	25+
Total	963	100	N.A.	12	N.A.

Mammals

8.6.31 21 mammal species were recorded within the Assessment Area, 16 of which are considered of conservation importance. Among the species recorded, eight were recorded within the Construction Area of the Project. The bat species Japanese Pipistrelle (*Pipistrellus abramus)* was recorded at various habitats, including wetland habitats (e.g. pond, compensatory wetland (the EA), marsh, modified watercourse, semi-natural / natural watercourse and seasonally wet grassland) as well as non-wetland habitats (e.g. orchard, plantation and developed area). The Short-nosed Fruit Bat (*Cynopterus sphinx)* was recorded exclusively in non-wetland habitats. The Lesser Yellow Bat (*Scotophilus kuhlii*) was recorded in upper section of the Ma Tso Lung Stream, and both the Horsfield's Myotis (*Myotis horsfieldii*) and Unknown Vespertilionidae Species 1 were recorded from developed area and modified watercourse. In addition, Pallas's Squirrel (*Callosciurus erythraeus*) was observed within village / orchard at Ma Tso Lung San Tsuen. Both Small Indian Civet *(Viverricula indica*) and Leopard Cat, were captured by camera traps in woodland habitat. Details of the recorded mammal species recorded in each habitat are presented in **60722948/A10/FIGURE 8.6** and **Appendix 8.5B** to **C**. Mammals species were captured by camera traps were presented in **Appendix 8.5A**.

8.6.32 While particular attention was given to Eurasian Otter during the ecological surveys (considering the significance of wetland habitats in HHW to this species), no direct sightings (during day and night surveys), indirect sightings (from camera traps), or signs of this species (spraints, footprints) were recorded. Long term and recent records have shown the presence of Eurasian Otters in the HHW, LMC Meander and its vicinity as described in **Section 8.5.16** and such records have been taken into account when conducting the impact assessment on the Eurasian Otter.

Herpetofauna

8.6.33 26 herpetofauna species were recorded within the Assessment Area, comprising 11 amphibian and 15 reptile species. Most of the recorded species are widely distributed throughout Hong Kong, and were predominately observed in orchard and developed areas. Three of the recorded species are considered of conservation importance: Chinese Bullfrog, Four-clawed Gecko (*Gehyra mutilata*), Indian Forest Skink (*Sphenomorphus indicus*). Only Chinese Bullfrog and Four-clawed Gecko were recorded within the Construction Area of the Project, from semi-natural and natural watercourse and developed area respectively. Details of the recorded herpetofauna species recorded in each habitat are presented in **60722948/A10/FIGURE 8.6** and **Appendix 8.5F** to **G**.

Butterfly

8.6.34 75 butterfly species were recorded within the Assessment Area, most of which are common and widespread species. A total of 12 recorded species are considered of conservation importance. Most records of the species of conservation importance such as Danaid Eggfly *Hypolimnas misippus*), Pale Palm Dart (*Telicota colon*), Peacock Royal (*Tajuria cippus*), Plain Hedge Blue (*Celastrina lavendularis*), Small Three-ring, Swallowtail (*Papilio xuthus*) and White Royal (*Pratapa deva*) were from upland shrubland / grassland habitats. Red-breasted Jezebel (*Delias acalis*), Metallic Cerulean and Paintbrush Swift (*Baoris farri*) were recorded in woodland. Common Onyx (*Horaga onyx*) was recorded from seasonally wet grassland, and Grass Demon (*Udaspes folus*) was sighted in the low-lying orchard habitat. Three butterfly species of conservation importance (Metallic Cerulean, Red-breast Jezebel and Paintbrush

Swift) were recorded within woodland habitat within the Construction Area of the Project. Details of the recorded butterfly species recorded in each habitat are presented in **60722948/A10/FIGURE 8.6** and **Appendix 8.5H** to **I**.

Odonate

8.6.35 32 odonate species were recorded within the Assessment Area, including five odonate species of conservation importance: Blue Chaser, Coastal Glider (*Macrodiplax cora*), Dingy Dusk-hawker (*Gynacantha subinterrupta*), Sapphire Flutterer and Scarlet Basker. Most of the recorded odonate species are common/abundant and widespread in Hong Kong. Eight odonate species were recorded within the Construction Area of the Project, none of which are considered species of conservation importance. Details of the recorded odonate species recorded in each habitat are presented in **60722948/A10/FIGURE 8.6** and **Appendix 8.5J** to **K**.

Firefly

8.6.36 Three firefly species were recorded within the Assessment Area, one of which, Motschulsky's Starworm (*Rhagophthalmus motschulskyi*), is considered of conservation importance. Motschulsky's starworm was recorded from seasonally wet grassland in the Ma Tso Lung area. While this species is currently considered endemic to Hong Kong, it is much more commonly encountered in villages and farmlands, even in urban parks, than in wild places. (Yiu, 2023). No firefly species of conservation importance were recorded within the Construction Area of the Project. Details of the recorded firefly species recorded in each habitat are presented in **60722948/A10/FIGURE 8.6** and **Appendix 8.5L** to **M**.

Freshwater Communities

8.6.37 24 freshwater aquatic faunal species were recorded within the Assessment Area, including the freshwater crab *Somanniathelphusa zanklon* and the fish *Gobiopterus macrolepis*, which are both species of conservation importance. Three individuals of *Somanniathelphusa zanklon* were recorded in the upstream of Ping Hang Stream. Two individuals of *Gobiopterus macrolepis* were recorded in the riparian zone of the LMC Meander. Notably, one individual of freshwater fish Blotched Snakehead (*Channa maculata*), classified as high risk of extinction in Hong Kong by the WWF HK State of Hong Kong Biodiversity report (WWF, 2025), was recorded at modified watercourse outside of the Project Boundary. The recorded individual is likely to be escaped cultivated individual from nearby ponds based on the location of discovery. No species of conservation importance were recorded within the Construction Area of the Project. Details of the recorded freshwater fauna species recorded in each habitat are presented in **60722948/A10/FIGURE 8.6** and **Appendix 8.5N** to **O**.

8.7 Evaluation of Ecological Value

8.7.1 The ecological importance of habitats within the Assessment Area was evaluated with reference to the criteria stipulated in Annex 8 of EIAO-TM, as summarised in **Table 8.9** to **Table 8.21** below.

Habitats within the Assessment Area

Table 8.9 Ecological Evaluation of Compensatory Wetland within the Assessment Area

Criterion	Description
Naturalness	Low, man-made for ecological purposes
Size	OWCA Area 2: 4.49ha and the EA: 6.92 ha totalling 11.41ha
Diversity	Low floral and medium faunal diversity
Rarity	Uncommon habitat in Hong Kong 37 fauna species of conservation importance were recorded during the ecological surveys: 33 Avifauna species: Black Drongo, Black-faced Bunting, Black Kite, Black-crowned Night Heron, Black-winged Stilt, Buff-bellied Pipit, Chestnut-eared Bunting, Chinese Penduline Tit, Chinese Pond Heron, Cinnamon Bittern, Common Moorhen, Common Snipe, Eurasian Spoonbill, Eurasian Wryneck, Great Egret, Greater Coucal, Grey Heron, Green Sandpiper, House Swift, Indian Cuckoo, Japanese Quail, Little Egret, Little Grebe, Little Ringed Plover, Long-tailed Shrike, Oriental Stork, Oriental Turtle Dove, Pied Kingfisher, Richard's Pipit, White-shouldered Starling, White-throated Kingfisher, Yellow Bittern and Zitting Cisticola 1 Herpetofauna species: Chinese Bullfrog 3 Mammal species: Japanese Pipistrelle, Leopard Cat, Small Indian Civet Literature Review: 74 fauna species of conservation importance were recorded in the Literature Review: 59 Avifauna species: Asian Barred Owlet, Black Kite, Black-crowned Night Heron, Black-faced Spoonbill, Black-naped Oriole, Black-winged Stilt, Bluethroat, Chinese Grosbeak, Chinese Penduline Tit, Chestnut-eared Bunting, Chinese Pond Heron, Cinnamon Bittern, Collared Crow, Common Greenshank, Common Kestrel, Eastern Buzzard, Eastern Cattle Egret, Eastern Imperial Eagle, Eurasian Coot, Eurasian Skylark, Eurasian Spoonbill, Eurasian Teal, Golden-headed Cisticola, Great Cormorant, Greater Coucal, Great Egret, Greater Painted-snipe, Greater Spotted Eagle, Grey Heron, Hen Harrier, Intermediate Egret, Japanese Quail, Lesser Coucal, Little Egret, Little Grebe, Little Ringed Plover, Marsh Sandpiper, Northern Shoveler, Pallas's Grasshopper Warbler, Peregrine Falcon, Pied Avocet, Pied Harrier, Pied Kingfisher, Pintail Snipe, Purple Heron, Red-billed Starling, Red-throated Pipit, Ruddy-breasted Crake, Siberian Rubythroat, Temminck's Stint, Tufted Duck, Western Osprey, White-cheeked Starling, White-shouldered Starling, White-throated Kingfisher, Wood Sandpiper, Yellow Bittern, Yellow-breasted Bunting, Zitting Cisticola 8 Mammal species: Eurasian Otter, Japanese Pipistrelle, Leopard Cat, Red Muntjac, Palla's Squirrel, Small Asian Mongoose, Small Indian Civet, Short-nosed Fruit Bat 1 Butterfly species: Small Cabbage White 3 Herpetofauna species: Chinese Bullfrog, Chinese Cobra, Common Rat Snake 3 Odonate species: Blue Sprite, Ruby Darter, Scarlet Basker
Re-creatability	Re-creatable with suitable hydrological regime

Criterion	Description
Fragmentation	None observed
Ecological linkage	Ecologically connected to the adjacent LMC meander and other wetland habitats
Potential value	Medium to high upon maturation and if actively managed under the Loop to fulfil specific mitigation targets
Nursery/breeding ground	No significant nursery or breeding ground known or observed during the ecological surveys. However, the EA and OWCA Area 2 are potential nursery and breeding grounds for wetland dependent species.
Age	Ecologically enhancement works have been completed in recent years
Abundance/richness of wildlife	Moderate
Overall ecological value	**EA – Moderate to High** **OWCA Area 2 – Moderate to High**

Note:

1. Please note that unpublished AFCD regular monitoring data of the LMCL EA and OWCA Area 2 is reviewed and duly incorporated under Literature Review.

Table 8.10 Ecological Evaluation of Pond within the Assessment Area

Criterion	Description	
	Hoo Hok Wai	**Other Ponds**
Naturalness	Low. Majority of the pond habitat is for fishery production.	Low.
Size	69.23 ha	0.80 ha
Diversity	Low diversity of flora, herpetofauna, butterfly, odonate and freshwater fauna, medium diversity of mammal and high diversity of birds recorded during the ecological surveys.	Low flora and fauna diversity
Rarity	Uncommon habitat. Mainly restricted to Northwestern New Territories Ponds within the Assessment Area are known to be part of the range of Eurasian Otter in Hong Kong (McMillan et al. 2022). <u>Current Survey:</u> A total of 77 fauna species of conservation importance were recorded during the ecological surveys: 66 Avifauna species: Black Drongo, Black-faced Bunting, Besra, Black Kite, Black-capped Kingfisher, Black-crowned Night Heron, Black-faced Spoonbill, Black-winged Stilt, Bluethroat, Buff-bellied Pipit,	Uncommon habitat. Mainly restricted to Northwestern New Territories <u>Current Survey:</u> 1 Avifauna species: Little Grebe <u>Literature Review:</u> No species of conservation importance recorded

Criterion	Description	
	Hoo Hok Wai	**Other Ponds**
	Chestnut-eared Bunting, Chinese Grosbeak, Chinese Pond Heron, Collared Crow, Common Greenshank, Common Moorhen, Common Redshank, Common Snipe, Crested Serpent Eagle, Eastern Buzzard, Eastern Cattle Egret, Eurasian Coot, Eurasian Spoonbill, Eurasian Teal, Eurasian Wigeon, Eurasian Wryneck, Great Cormorant, Great Egret, Greater Coucal, Grey Heron, Grey-headed Lapwing, Green Sandpiper, House Swift, Indian Cuckoo, Intermediate Egret, Japanese Quail, Lesser Coucal, Little Bunting, Little Egret, Little Grebe, Little Ringed Plover, Long-tailed Shrike, Long-toed Stint, Marsh Sandpiper, Northern Shoveler, Oriental Magpie, Oriental Pratincole, Oriental Turtle Dove, Pied Avocet, Pied Kingfisher, Purple Heron, Red-billed Starling, Red-necked Stint, Red-rumped Swallow, Red-throated Pipit, Richard's Pipit, Sharp-tailed Sandpiper, Sooty-headed Bulbul, Temminck's Stint, White-cheeked Starling, White-shouldered Starling, White-throated Kingfisher, Wood Sandpiper, Yellow Bittern, Yellow-breasted Bunting, Zitting Cisticola 6 Mammal species: Horsfield's Myotis, Japanese Pipistrelle, Lesser Yellow Bat, Unknown *Vespertilionidae* Species 1, Unknown *Vespertilionidae* Species 2, Small Asian Mongoose 1 Butterfly species: Swallowtail 2 Odonate species: Coastal Glider and Scarlet Basker <u>Literature Review:</u> 103 fauna species of conservation importance were recorded in the Literature Review: 98 Avifauna species: Amur Falcon, Baer's Pochard, Black Kite, Black Stork, Black-capped Kingfisher, Black-crowned Night Heron, Black-faced Spoonbill, Black-headed Gull, Black-tailed Godwit, Black-winged Kite, Black-winged Stilt, Bonelli's Eagle, Buff-bellied Pipit, Caspian	

Criterion	Description	
	Hoo Hok Wai	**Other Ponds**
	Gull, Chinese Pond Heron, Citrine Wagtail, Collared Crow, Common Greenshank, Common Kestrel, Common Pochard, Common Redshank, Curlew Sandpiper, Daurian Starling, Dunlin, Dusky Thrush, Eastern Cattle Egret, Eastern Imperial Eagle, Eurasian Coot, Eurasian Curlew, Eurasian Hobby, Eurasian Sparrowhawk, Eurasian Spoonbill, Eurasian Teal, Eurasian Wigeon, Falcated Duck, Ferruginous Duck, Great Cormorant, Great Crested Grebe, Great Egret, Greater Sand Plover, Greater Spotted Eagle, Greater White-fronted, Grey Heron, Grey-headed Lapwing, Grey-tailed Tattler, Heuglin's Gull, Indian Spot-billed Duck, Intermediate Egret, Japanese Sparrowhawk, Japanese Yellow Bunting, Kentish Plover, Lesser Sand Plover, Lesser Whistling Duck, Lesser White-fronted, Long-toed Stint, Little Curlew, Little Egret, Little Grebe, Little Ringed Plover, Little Tern, Mallard, Marsh Sandpiper, Naumann's Thrush, Normann's Greenshank, Northern Lapwing, Northern Pintail, Northern Shoveler, Oriental Skylark, Oriental Stork, Pacific Golden Plover, Pacific Swift, Pectoral Sandpiper, Peregrine Falcon, Pheasant-tailed Jacana, Pied Avocet, Pied Kingfisher, Pintail/Swinhoe's snipe, Oriental Pratincole, Red-billed Starling, Red-breasted Merganser, Red-necked Stint, Red-throated Pipit, Ruff, Sanderling, Sharp-tailed Sandpiper, Spotted Redshank, Steppe Eagle, Striated Heron, Temminck's Stint, Tufted Duck, Western Osprey, White-bellied Sea Eagle, White-cheeked Starling, White-shouldered Starling, White-throated Kingfisher, Wood Sandpiper, Yellow Bittern and Zitting Cisticola 1 Mammal species: Eurasian Otter 1 Herpetofauna species: Burmese Python	

Criterion	Description	
	Hoo Hok Wai	**Other Ponds**
	3 Odonate species: Coastal Glider, Sapphire Flutterer and Scarlet Basker	
Re-creatability	Readily re-created with suitable hydrological regime	Readily re-created with suitable hydrological regime
Fragmentation	Low	High
Ecological linkage	Form part of the wetland ecosystem in Hoo Hok Wai and the flight-line corridor for birds between Hoo Hok Wai and the rest of Deep Bay	Fragmented and not linked to adjacent wetland habitats
Potential value	Medium to high if they are converted to ecologically managed wetlands	Medium to high if they are converted to ecologically managed wetlands
Nursery/breeding ground	Reported as a foraging ground for breeding ardeids, especially from Ho Sheung Heung Egretry in the LMC Loop EIA. Not observed as a significant nursery or breeding ground during the ecological surveys	No notable nursery and breeding behaviour
Age	Ecologically non-applicable	N/A
Abundance/richness of Wildlife	Low abundance of herpetofauna, butterfly and odonate. Moderate abundance of mammals and high abundance of birds.	Low abundance of fauna and only recorded occasionally.
Overall ecological value	**High**	**Low**

Table 8.11 Ecological Evaluation of Marsh within the Assessment Area

Criterion	Description
Naturalness	Formed from abandoned artificial (rice-paddy) habitats
Size	15.88 ha
Diversity	Low floral and medium faunal diversity
Rarity	Uncommon habitat in Hong Kong 25 fauna species of conservation importance were recorded during the ecological surveys: 23 Avifauna species: Black Drongo, Black Kite, Black-crowned Night Heron, Black-winged Kite, Black-winged Stilt, Chestnut-eared Bunting, Chinese Pond Heron, Common Kestrel, Eastern Buzzard, Eastern Cattle Egret, Eurasian Wryneck, Great Egret, Greater Coucal, Grey Heron, Indian Cuckoo, Intermediate Egret, Little Bunting, Little Egret, Long-tailed Shrike, Oriental Turtle Dove, Sooty-headed Bulbul, White-shouldered Starling, Yellow Bittern 1 Mammal species: Japanese Pipistrelle 1 Odonate species: Sapphire Flutterer <u>Literature Review:</u>

Criterion	Description
	36 fauna species of conservation importance were recorded in the Literature Review: 31 Avifauna species: Black Bittern, Black-winged Kite, Bluethroat, Chinese Pond Heron, Cinnamon Bittern, Collared Crow, Dusky Thrush, Eastern Cattle Egret, Eurasian Teal, Great Egret, Greater Painted-snipe, Grey Heron, Little Egret, Little Grebe, Naumann's Thrush, Pechora Pipit, Peregrine Falcon, Pheasant-tailed Jacana, Purple Heron, Red-billed Starling, Red-throated Pipit, Ruddy-breasted Crake, Slaty-breasted Rail, Styan's Grasshopper, Tufted Duck, Von Schrenck's Bittern, Watercock, Western Water Rail, White-throated Kingfisher, Yellow Bittern, Yellow-breasted Bunting 2 Herpetofauna species: Two-striped Grass Frog, Common Rat Snake 2 Mammal species: Chinese Noctule, Eurasian Otter 1 Butterfly species: Pale Palm Dart
Re-creatability	Re-creatable with suitable hydrological regime
Fragmentation	Low
Ecological linkage	Part of the wetland ecosystem in Hoo Hok Wai
Potential value	Moderate
Nursery/breeding ground	No significant nursery/breeding ground known or observed
Age	Ecologically non-applicable
Abundance/richness of wildlife	Moderate
Overall ecological value	**Moderate**

Table 8.12 Ecological Evaluation of Reedbed within the Assessment Area

Criterion	Description
Naturalness	Natural
Size	8.24 ha
Diversity	Low floral and terrestrial faunal diversity
Rarity	Uncommon habitat in Hong Kong 5 fauna species of conservation importance were recorded during the ecological surveys: 5 Avifauna species: Black Drongo, Chinese Penduline Tit, Greater Coucal, Oriental Magpie and Yellow-breasted Bunting <u>Literature Review:</u> 30 fauna species of conservation importance were recorded in the Literature Review: 27 Avifauna species: Bluethroat, Chinese Penduline Tit, Chinese Pond Heron, Cinnamon Bittern, Collared Crow, Common Kestrel, Crested Serpent Eagle, Eastern Buzzard, Eastern Marsh Harrier, Eurasian Bittern, Eurasian Hobby, Golden-headed Cisticola, Great Egret, Greater Coucal, Greater painted-snipe, Grey Heron, Little Egret, Manchurian Reed Warbler, Pallas's Grasshopper Warbler, Peregrine Falcon, Pied Harrier,

Criterion	Description
	Purple Heron, Thick-billed Warbler, White-shouldered Starling, White-throated Kingfisher, Yellow-breasted Bunting and Zitting Cisticola 1 Herpetofauna species: Common Rat Snake 2 Mammal species: Japanese Pipistrelle, Small Asian Mongoose
Re-creatability	Re-creatable with suitable hydrological regime
Fragmentation	Low
Ecological linkage	Part of the wetland ecosystem in Hoo Hok Wai
Potential value	Low-medium
Nursery/breeding ground	No significant nursery or breeding ground known or observed during the ecological surveys
Age	Ecologically non-applicable
Abundance/richness of wildlife	Moderate
Overall ecological value	**Moderate**

Table 8.13 Ecological Evaluation of Modified Watercourse within the Assessment Area

Criterion	Description
Naturalness	Man-made
Size	About 0.6km (0.67 ha)
Diversity	Low floral and faunal diversity
Rarity	Common habitat in Hong Kong 16 fauna species of conservation importance were recorded during the ecological surveys: 9 Avifauna species: Black-crowned Night Heron, Chinese Pond Heron, Great Cormorant, Great Egret, Greater Coucal, Grey Heron, Intermediate Egret, Little Egret and Red-billed Starling 6 Mammal species: Horsfield's Myotis, Japanese Pipistrelle, Lesser Yellow Bat, Rickett's Big-footed Myotis, Unknown Vespertilionidae Species 1, Short-nosed Fruit Bat 1 Odonate species: Blue Chaser <u>Literature Review:</u> A total of 10 species of conservation importance were recorded in the literature review: 8 Avifauna species: Chinese Pond Heron, Great Cormorant, Great Egret, Greater Coucal, Grey Heron, Intermediate Egret, Little Egret, and Red-billed Starling 1 Mammal species: Japanese Pipistrelle 1 Odonate species: Ruby Darter
Re-creatability	Readily re-created
Fragmentation	None observed
Ecological linkage	Hydrologically connected to the more natural upstream watercourse in Ma Tso Lung and Hoo Hok Wai area downstream

Criterion	Description
Potential value	Low given its current engineering design, unless more diverse and natural substrate and banks are provided and ecological enhancement features, such as natural substrate and fish ladder, are applied
Nursery/breeding ground	No significant nursery or breeding ground known or observed during the ecological surveys
Age	Channelised in 2010 but ecologically non-applicable
Abundance/richness of wildlife	Low
Overall ecological value	**Low to Moderate** for the lower reaches of Ma Tso Lung Stream and its tributaries, which have been channelised

Table 8.14 Ecological Evaluation of Semi-natural / Natural Watercourse within the Assessment Area

Criterion	Description		
	The upper section of Ma Tso Lung Stream	**Section of the LMC Meander falling within the Assessment Area**	**Remaining watercourses within the Assessment Area**
Naturalness	The stream is largely natural	Semi-natural, subject to straightening, widening and deepening in the Shenzhen River Regulation Project. A substantial portion of the original riparian vegetation was cleared in the process.	Semi-natural, those in vicinity of village settlements had been subject to modification and effluent discharge
Size	About 11.9ha		
Diversity	Low floral and faunal diversity	Low floral and medium faunal diversity	Low floral and faunal diversity
Rarity	Common habitat in Hong Kong 3 fauna species of conservation importance recorded during the ecological surveys: 3 Mammal species: Japanese Pipistrelle, Lesser Bamboo Bat, Unknown Vespertilionidae Species 1	Common habitat in Hong Kong 21 fauna species of conservation importance were recorded during the ecological surveys: 20 Avifauna species: Black Drongo, Black Kite, Black-crowned Night Heron, Black-faced Spoonbill, Chinese Pond Heron, Collared Crow, Eastern Cattle Egret, Great Cormorant, Great Egret, Greater Coucal, Grey Heron, Little Egret, Little Grebe, Long-tailed Shrike, Pied Kingfisher, Straited Heron, Whimbrel, White-shouldered Starling, White-throated Kingfisher, Yellow Bittern 1 Aquatic Communities species: *Gobiopterus macrolepis*	Common habitat in Hong Kong 4 fauna species of conservation importance were recorded during the ecological surveys: 1 Butterfly species: Metallic Cerulean 1 Odonate species: Dingy Dusk-hawker 1 Herpetofauna species: Chinese Bullfrog 1 Aquatic Communities species: *Somanniathelphusa zanklon*
	24 fauna species of conservation importance were recorded in the Literature Review: 20 Avifauna species: Black Kite, Black-faced Spoonbill, Black-winged Stilt, Chinese Pond Heron, Eurasian Teal, Great Cormorant, Great Egret, Grey Heron, Little Egret, Little Grebe, Little Ringed Plover, Northern Pintail, Peregrine Falcon, Pied Avocet, Pied Kingfisher, Purple Heron, Red-billed Starling, Red-throated Pipit, White-throated Kingfisher and Wood Sandpiper		

Criterion	Description		
	The upper section of Ma Tso Lung Stream	**Section of the LMC Meander falling within the Assessment Area**	**Remaining watercourses within the Assessment Area**
	2 Herpetofauna species: Chinese Water Snake and Three-banded Box Turtle (was once recorded in the upper reaches of Ma Tso Lung Stream) 2 Aquatic Communities species: Small Snakehead and *Somanniathelphusa zanklon*		
Re-creatability	Difficult to re-create	Difficult to re-create	Natural sections are difficult to re-create
Fragmentation	None observed	None observed	None observed
Ecological linkage	Hydrologically linked to adjacent marsh or seasonally wet grassland	Hydrologically connected to Shenzhen River	Hydrologically connected to Hoo Hok Wai area downstream; Ping Hang Stream is hydrologically connected to riparian vegetation and LMC Meander
Potential value	Low to moderate for the upper sections of Ma Tso Lung Stream where Three-banded Box Terrapin was recorded	Low to Moderate	Low to moderate, no obvious sign suggesting that a higher diversity of freshwater-associated fauna, including amphibian, odonate and freshwater fauna species, can be supported
Nursery/breeding ground	No significant nursery or breeding ground known or observed	No significant nursery or breeding ground known or observed	No significant nursery or breeding ground known or observed
Age	Ecologically non-applicable	Ecologically non-applicable	Ecologically non-applicable
Abundance/richness of wildlife	Low	Moderate	Low
Overall ecological value	**High** for the upper reaches of Ma Tso Lung Stream and its tributaries, where Three-banded Box Terrapin was recorded	**High** due to reported usage by Eurasian Otter and as an important habitat of flight-line corridor for waterbirds and larger avifauna species of conservation importance	**Low to moderate**

Table 8.15 Ecological Evaluation of Seasonally Wet Grassland within the Assessment Area

Criterion	Description
Naturalness	Man-made
Size	12.31 ha
Diversity	Low floral and terrestrial faunal diversity
Rarity	Common habitat in Hong Kong 21 species of conservation importance were recorded during the ecological surveys: 1 Flora species: Incense tree 10 Avifauna species Black-crowned Night Heron, Black-winged Kite, Chinese Pond Heron, Eastern Buzzard, Great Cormorant, Greater Coucal, Little Egret, Long-tailed Shrike, Red-rumped Swallow and Sooty-headed Bulbul 6 Mammal species: Japanese Pipistrelle, Chinese Noctule, Lesser Yellow Bat, Unknown Vespertilionidae Species 1 and 2, and Small Indian Civet 1 Herpetofauna species: Indian Forest Skink 2 Butterfly species: Metallic Cerulean and Common Onyx 1 Firefly species: Motschulsky's Starworm <u>Literature Review:</u> 6 fauna species of conservation importance were recorded in the Literature Review: 4 Avifauna species: Chinese Pond Heron, Eastern Cattle Egret, Pechora Pipit, Pallas's Grasshopper Warbler 2 Butterfly species: Grey Scrub Hopper and Metallic Cerulean
Re-creatability	Re-creatable with suitable hydrological regime
Fragmentation	Fragmented by border road and lowland development
Ecological linkage	Linked to adjacent marsh, ponds and watercourse
Potential value	Low to medium
Nursery/breeding ground	No significant nursery or breeding ground known or observed during the ecological surveys
Age	Ecologically non-applicable
Abundance/richness of wildlife	Low terrestrial faunal abundance recorded
Overall ecological value	**Low to moderate**

Table 8.16 Ecological Evaluation of Woodland within the Assessment Area

Criterion	Description
Naturalness	Largely natural
Size	8.24 ha
Diversity	Low floral and terrestrial faunal diversity

Criterion	Description
Rarity	Common habitat in Hong Kong 17 species of conservation importance were recorded during the ecological surveys: 1 Flora species: Incense tree 8 Avifauna species: Asian Barred Owlet, Black Drongo, Chinese Grosbeak, Eastern Buzzard, Greater Coucal and Grey Treepie, House Swift and Oriental Turtle Dove 5 Mammal species: Leopard Cat, Intermediate Horseshoe Bat, Unknown Vespertilionidae Species 1, Pallas's Squirrel and Small Indian Civet 3 Butterfly species: Metallic Cerulean, Paintbrush Swift and Red-breast Jezebel <u>Literature Review:</u> 12 species of conservation importance were recorded in the Literature Review: 1 Flora species: Incense Tree 10 Avifauna species: Common Emerald Dove, Crested Goshawk, Crested Honey Buzzard, Crested Serpent Eagle, Eurasian Hobby, Greater Coucal, Grey Treepie, Japanese Paradise Flycatcher, Slaty-legged Crake, White-shouldered Starling 1 Herpetofauna species: Chinese Cobra
Re-creatability	Re-creatable over several decades
Fragmentation	Fragmented by the border road
Ecological linkage	Ecologically connected to shrubland/grassland upslope
Potential value	Low to medium due to lack of signs of further succession (e.g. colonization of shade-tolerant tree species typical of other more mature woodlands in Hong Kong) to become more mature woodland
Nursery/breeding ground	No significant nursery or breeding ground known or observed during the ecological surveys
Age	Over 30 years of age
Abundance/richness of wildlife	Moderate terrestrial faunal abundance recorded
Overall ecological value	**Moderate**

Table 8.17 Ecological Evaluation of Grassland / Shrubland within the Assessment Area

Criterion	Description
Naturalness	Created and disturbed by hillfires
Size	124.67 ha
Diversity	Low floral, bird, mammal, herpetofauna and odonate diversity, and low to medium butterfly diversity recorded during the ecological surveys
Rarity	Common habitat in Hong Kong 30 species of conservation importance were recorded during the ecological surveys:

Criterion	Description
	3 Flora species: *Aralia chinensis,* Tongue Habenaria and Incense Tree, 18 Avifauna species: Black Drongo, Black-faced Bunting, Black Kite, Chinese Hwamei; Chinese Francolin, Collared Scops Owl, Common Kestrel, Crested Goshawk, Crested Serpent Eagle, Eastern Buzzard, Greater Coucal, House Swift, Indian Cuckoo, Lesser Coucal, Oriental Turtle Dove, Red-rumped Swallow, Sooty-headed Bulbul, White-throated Kingfisher 1 Mammal species: Short-nosed Fruit Bat 1 Herpetofauna: Indian Forest Skink 7 Butterfly species: Danaid Eggfly, Pale Palm Dart, Plain Hedge Blue, Peacock Royal, Swallowtail, Small Three-ring, and White Royal <u>Literature Review:</u> 38 fauna species of conservation importance were recorded in the Literature Review: 28 Avifauna species: Ashy Drongo, Besra, Bonelli's Eagle, Chinese Hwamei, Chinese Sparrowhawk, Common Emerald Dove, Common Kestrel, Crested Goshawk, Crested Honey Buzzard, Crested Serpent Eagle, Daurian Starling, Eurasian Eagle-Owl, Eurasian Hobby, Eurasian Sparrowhawk, Golden-headed Cisticola, Greater Coucal, Greater Spotted Eagle, Grey Nightjar, Japanese Quail, Japanese Sparrowhawk, Lesser Coucal, Pacific Swift, Peregrine Falcon, Red-billed Starling, Rufous-capped Babbler, Siberian Blue Robin, Slaty-legged Crake and Zitting Cisticola 4 Mammal species: Red Muntjac, Leopard Cat, Japanese Pipistrelle, Small Indian Civet 2 Herpetofauna species: Chinese Bullfrog, Indo-Chinese Rat Snake 3 Butterfly species: Small Three-ring, Grey Scrub Hopper and Metallic Cerulean 1 Odonate species: Blue Chaser
Re-creatability	Readily re-created
Fragmentation	Fragmented by lowland development (e.g. villages in Ma Tso Lung area)
Ecological linkage	Ecologically linked to plantation and woodland
Potential value	Low as natural succession has apparently been arrested, as the extent and condition have largely remained unchanged at least over the last 30 years
Nursery/breeding ground	Potential Eurasian Eagle Owl, Chinese Francolin and White-throated Kingfisher nursery / breeding ground. No significant nursery or breeding ground known or observed during the ecological surveys
Age	Extent and condition have largely remained unchanged at least over the last 30 years
Abundance/richness of wildlife	Low faunal abundance recorded
Overall ecological value	**Low to moderate**

Table 8.18 Ecological Evaluation of Orchard within the Assessment Area

Criterion	Description
Naturalness	Man-made
Size	6.82 ha
Diversity	Low floral and faunal diversity
Rarity	Common habitat in Hong Kong 11 fauna species of conservation importance were recorded during the ecological surveys: 6 Avifauna species: Collared Scops Owl, Greater Coucal, White-bellied Erpornis, White-throated Kingfisher, Grey Treepie, Indian Cuckoo 3 Mammal species: Leschenault's Rousette, Pallas's Squirrel, Short-nosed Fruit Bat 1 Bufferfly Species: Grass Demon 1 Herpetofauna species: Indian Forest Skink <u>Literature Review:</u> 5 fauna species of conservation importance were recorded in the Literature Review: 2 Avifauna species: Eastern Cattle Egret and Little Egret 1 Mammal species: Japanese Pipistrelle 2 Herpetofauna species: Two-striped Grass Frog and Many-banded Krait
Re-creatability	Readily re-created
Fragmentation	Occurred as patches adjacent to village settlements
Ecological linkage	No significant ecological linkage
Potential value	Low
Nursery/breeding ground	No significant nursery or breeding ground known or observed
Age	Ecologically non-applicable
Abundance/richness of wildlife	Low
Overall ecological value	**Low**

Table 8.19 Ecological Evaluation of Plantation within the Assessment Area

Criterion	Description
Naturalness	Semi-natural with a mix of cultivated exotic trees and native plant species dispersed from habitats nearby, especially woodland.
Size	13.86 ha
Diversity	Low floral and terrestrial faunal diversity
Rarity	Common habitat in Hong Kong 7 fauna species of conservation importance were recorded during the ecological surveys:

Criterion	Description
	5 Avifauna species: Black-faced Bunting, Black Kite, Indian Cuckoo, Little Egret and Oriental Turtle Dove 2 Mammal species: Japanese Pipistrelle, Pallas's Squirrel Literature Review: 23 fauna species of conservation importance were recorded in the Literature Review: 18 Avifauna species: Amur Paradise Flycatcher, Ashy Drongo, Asian Barred Owlet, Besra, Black-naped Oriole, Chinese Grosbeak, Chinese Pond Heron, Common Emerald Dove, Crested Goshawk, Crested Honey Buzzard, Crested Serpent Eagle, Greater Coucal, Grey Treepie, Grey-capped Greenfinch, Japanese Paradise Flycatcher, Little Egret, Red-billed Starling, White-shouldered Starling 3 Mammal species: Japanese Pipistrelle, Lesser Bamboo Bat and Unknown Vespertilionidae sp. 1 2 Herpetofauna species: Chinese Bullfrog and Common Rat Snake
Re-creatability	Readily re-created
Fragmentation	Fragmented by developed area
Ecological linkage	No significant ecological linkage known or observed, although native plant species were possibly dispersed from nearby habitats, especially woodland
Potential value	Low. Potential may increase if progressive and selective thinning of exotic tree species is executed and time is given for the establishment of native tree species.
Nursery/breeding ground	Not known or observed as a significant nursery or breeding ground
Age	Unknown
Abundance/richness of wildlife	Low
Overall ecological value	**Low**

Table 8.20 Ecological Evaluation of Agricultural Land within the Assessment Area

Criterion	Description
Naturalness	Man-made
Size	1.27 ha
Diversity	Low floral and terrestrial faunal diversity
Rarity	Common habitat in Hong Kong 1 fauna species of conservation importance were recorded during the ecological surveys: 1 Avifauna species: Chinese Pond Heron Literature Review:

	8 fauna species of conservation importance were recorded in the Literature Review: 8 Avifauna species: Chinese Pond Heron, Common Rosefinch, Eastern Cattle Egret, Little Egret, Red-billed Starling, Red-throated Pipit, Yellow-breasted Bunting and Zitting Cisticola
Re-creatability	Readily re-created
Fragmentation	Disconnected agricultural land occurred as distinct patches
Ecological linkage	No significant ecological linkage with the remaining habitats within the Assessment Area
Potential value	Low under the current farming practices
Nursery/breeding ground	No significant nursery or breeding ground known or observed
Age	Ecologically non-applicable
Abundance/richness of wildlife	Low
Overall ecological value	**Low**

Table 8.21 Ecological Evaluation of Developed Area within the Assessment Area

Criterion	Description
Naturalness	Man-made
Size	77.45 ha
Diversity	Low floral and terrestrial faunal diversity
Rarity	Common habitat in Hong Kong 18 species of conservation importance were recorded during the ecological surveys: 1 Flora species: Incense tree 6 Avifauna species: Black Drongo, Chinese Pond Heron, Common Emerald Dove, Eastern Cattle Egret and Grey Treepie and Oriental Magpie 9 Mammal species: Horsfield's Myotis, Japanese Pipistrelle, Lesser Yellow Bat, Lesser Bamboo Bat, Short-nosed Fruit Bat, Unidentified Myotis, Unknown Vespertilionidae Species 1, Unknown Vespertilionidae Species 2, Pallas's Squirrel 1 Herpetofauna species: Four-clawed Gecko 1 Butterfly species: Metallic Cerulean <u>Literature Review:</u> 9 fauna species of conservation importance were recorded in the Literature Review: 4 Avifauna species: Asian Barred Owlet, Chinese Grosbeak, Daurian Starling and Red-billed Starling, 3 Mammal species: Intermediate Horseshoe Bat, Japanese Pipistrelle and Lesser Bamboo Bat 2 Odonate species: Coastal Glider and Ruby Darter
Re-creatability	Readily re-created

Criterion	Description
Fragmentation	None observed
Ecological linkage	None observed
Potential value	Very low, given the intensive and incessant anthropogenic disturbance
Nursery/breeding ground	No significant nursery or breeding ground known or observed
Age	Ecologically non-applicable
Abundance/richness of wildlife	Low
Overall ecological value	**Very low**

Overall ecological value

8.7.2 The overall ecological valuation of habitats recorded in the Assessment Area under this Study is generally consistent with that from the LMC Loop EIA, with ponds in HHW remaining of high ecological value; reedbeds of moderate ecological value; modified watercourses of low to moderate ecological value; semi-natural and natural watercourses ranging from low to moderate, moderate to high and high ecological value depending on locations, seasonally wet grassland and grassland and shrubland habitats in the Ma Tso Lung area of low to moderate ecological value, orchard, agricultural land and developed areas retaining ecological values from very low to low.

8.7.3 Notable differences between the two assessments reflect changes in site conditions, the establishment of new ecological features, and the fact that habitat assessments under the LMC Loop EIA covered a larger geographic area including other habitat patches. The EA and OWCAs, which were not established during preparation of the LMC Loop EIA, are newly assessed as having moderate to high ecological value. Reedbed in the HHW is reassessed as moderate ecological value as it is only seasonally wet and supports moderate fauna species richness. Scattered ponds recorded within the Assessment Area, which were not individually assessed in the LMC Loop EIA, are considered of low ecological value under this study. Woodland habitats, formerly rated as low to moderate ecological value, are reassessed as being of moderate ecological value given that they were found to support moderate faunal species richness and to have connectivity to nearby natural habitats. Plantation habitats, formerly rated low to moderate ecological value, are considered as low ecological value due to their fragmented distribution and limited plant diversity.

Species of Conservation Importance

8.7.4 With reference to Annex 8 of EIAO-TM, the ecological value of species recorded within the Assessment Area was assessed in **Table 8.22**, in terms of protection / conservation status (e.g. fauna protected under local legislation (except birds)), status under regional / global legislation / conventions, species distribution (e.g. endemic), and rarity (e.g. rare or restricted).

Table 8.22 Species of Conservation Importance Recorded within the Assessment Area during the Ecological Survey

Number	Species	Location (1,2,3,4,5,6,7)		Rarity and Distribution in Hong Kong (1, 2, 3, 4, 5, 6, 7)	Conservation/Protection status (5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21)
		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
	Vegetation				
1.	Incense Tree *Aquilaria sinensis*	Woodland, Developed Area	Woodland, Developed Area, Grassland / Shrubland, Seasonally Wet Grassland	Common. Lowland forests and fung shui woods.	IUCN Red List of Threatened Species (2025): VU; CITES: Appendix II; Threatened Species List of China's Higher Plants: VU; China Plant Red Data Book: VU; Included in Illustrations of Rare & Endangered Plant in Guangdong Province; Listed in Rare and Precious Plants of Hong Kong; Cap. 586; State Protection (Category II)
2.	*Aralia chinensis*	/	Grassland / Shrubland	Restricted. Found in forest margins.	IUCN Red List of Threatened Species (2025): VU
3.	Tongue Habenaria *Habenaria linguella*	/	Grassland / Shrubland	Restricted. Found in grassland.	Cap. 96A; Cap. 586; CITES: Appendix II
	Mammals				
1.	Small Asian Mongoose *Herpestes javanicus*	/	Pond	Uncommon. Fairly widely distributed in countryside areas in the New Territories.	Cap. 170; Red List of China's Vertebrate: VU

Number	Species	Location (1,2,3,4,5,6,7)		Rarity and Distribution in Hong Kong (1, 2, 3, 4, 5, 6, 7)	Conservation/Protection status (5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21)
		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
2.	Leschenault's Rousette *Rousettus leschenaultii*	/	Orchard	Common. Fairly widely distributed in countryside areas throughout Hong Kong.	Fellowes et al. (2002): (LC); Cap. 170
3.	Short-nosed Fruit Bat *Cynopterus sphinx*	Developed Area	Developed Area, Grassland / Shrubland, Orchard	Very common. Very widely distributed in urban and countryside areas throughout Hong Kong.	Cap. 170
4.	Intermediate Horseshoe Bat *Rhinolophus affinis*	/	Woodland	Uncommon. Widely distributed in countryside areas throughout Hong Kong	Fellowes et al. (2002): (LC); Cap. 170
5.	Pallas's Squirrel *Callosciurus erythraeus*	Woodland, Developed Area	Orchard, Plantation	Common. Fairly widely distributed, with the styani subspecies found in the New Territories (e.g. Tai Lam, Shing Mun and Tai Po Kau), and the thai subspecies found on the Hong Kong Island (e.g. Tai Tam and Pok Fu Lam).	Cap. 170
6.	Leopard Cat *Prionailurus bengalensis*	Woodland	Compensatory Wetland	Widely distributed in countryside areas throughout Hong Kong, except for Lantau Island.	List of Wild Animals under State Priority Conservation: Class II; IUCN Red List of Threatened Species (2025): LC; Red List of China's Vertebrate: VU; Cap. 170; Cap. 586

Number	Species	Location (1,2,3,4,5,6,7)		Rarity and Distribution in Hong Kong (1, 2, 3, 4, 5, 6, 7)	Conservation/Protection status (5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21)
		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
7.	Unknown *Vespertilionidae* Species *2*	/	Pond, Developed area, Seasonally Wet Grassland	Rare. Only several records in the countryside areas at Ting Kau, Ma On Shan and Lin Ma Hang, and several records of stray individuals inside buildings.	Fellowes et al. (2002): (LC); Cap. 170
8.	Horsfield's Myotis *Myotis horsfieldii*	Developed Area, Modified Watercourse	Pond; Marsh; Seasonally Wet Grassland; Modified Watercourse; Developed Area	Rare/Species of Conservation Concern. Found in Shek Kong, Pak Tam Chung, Fung Yuen, Plover Cove, Pat Sin Leng and Shing Mun Country Parks.	Fellowes et al. (2002): PRC; Cap. 170
9.	Rickett's Big-footed Myotis *Myotis ricketti*	/	Modified Watercourse	Common. Fairly widely distributed in countryside areas throughout Hong Kong.	IUCN Red List of Threatened Species (2025): VU; Fellowes et al. (2002): (LC); Cap. 170
10.	Chinese Noctule *Nyctalus plancyi*	/	Seasonally Wet Grassland	Common. Fairly widely distributed in countryside areas throughout Hong Kong.	Fellowes et al. (2002): PRC; Cap. 170
11.	Japanese Pipistrelle *Pipistrellus abramus*	Developed area	Pond, Compensatory Wetland, Marsh, Seasonally Wet Grassland, Modified Watercourse, Semi-natural / Natural Watercourse, Plantation, Developed Area	Very common. Widely distributed throughout Hong Kong.	Cap. 170

Number	Species	Location (1,2,3,4,5,6,7)		Rarity and Distribution in Hong Kong (1, 2, 3, 4, 5, 6, 7)	Conservation/Protection status (5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21)
		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
12	Unknown Vespertilionidae Species 1	Developed Area, Modified Watercourse	Pond, Woodland, Developed Area, Seasonally Wet Grassland, Modified Watercourse	Uncommon. Ten-something records found in Nam Chung, Sheung Wo Hang, Lin Ma Hang, Plover Cove Country Park, Yuen Long, Shek Pik, Deep Water Bay, Ho Pui and Ho Chung.	Cap. 170
13.	Lesser Yellow Bat *Scotophilus kuhlii*	Modified Watercourse	Pond, Marsh; Seasonally Wet Grassland; Semi-natural / Natural Watercourse; Developed Area	Uncommon. Fairly widely distributed in countryside areas throughout Hong Kong.	Fellowes et al. (2002): (LC); Cap. 170
14.	Small Indian Civet *Viverricula indica*	Seasonally Wet Grassland; Woodland	Compensatory Wetland, Woodland	Common. Very widely distributed in countryside areas throughout Hong Kong, except for Lantau Island.	List of Wild Animals under State Priority Conservation: Class I; Cap. 170; Red List of China's Vertebrate: VU
15.	Lesser Bamboo Bat *Tylonycteris pachypus*	/	Developed area, Semi-natural / Natural Watercourse	Very common. Fairly widely distributed in countryside areas throughout Hong Kong.	China Red Data Book: Rare; Fellowes et al. (2002): (LC); Cap. 170
16.	Unidentified Myotis *Myotis sp.*	/	Developed area	/	Cap. 170
	Avifauna				
1.	Asian Barred Owlet *Glaucidium cuculoides*	/	Woodland	Locally common resident. Widely distributed in woodland of the north and central New Territories.	Cap. 586; List of Wild Animals under State Priority Conservation:

Number	Species	Location (1,2,3,4,5,6,7)		Rarity and Distribution in Hong Kong (1, 2, 3, 4, 5, 6, 7)	Conservation/Protection status (5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21)
		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
					Class II; CITES: Appendix II
2.	Besra *Accipiter virgatus*	/	Pond	Common resident and migrant. Found in Tai Po Kau, Deep Bay area, Chek Lap Kok, Cheung Chau, Soko Islands.	Cap. 586; List of Wild Animals under State Priority Conservation: Class II; CITES: Appendix II
3.	Black Drongo *Dicrurus macrocercus*	Developed Area; Woodland	Pond, Compensatory Wetland, Marsh, Seasonally Wet Grassland, Semi-natural / Natural Watercourse, Modified Watercourse	Common summer visitor. Widely distributed in open area throughout Hong Kong.	WWF: HR
4.	Black Kite *Milvus migrans*	/	Pond, Compensatory Wetland, Marsh, Semi-natural / Natural Watercourse, Grassland/ Shrubland, Plantation	Common resident and winter visitor. Widely distributed in Hong Kong.	Fellowes et al. (2002): (RC); Cap. 586; List of Wild Animals under State Priority Conservation: Class II; CITES: Appendix II
5.	Black-capped Kingfisher *Halcyon pileata*	/	Pond	Uncommon passage migrant and winter visitor. Widely distributed in coastal areas throughout Hong Kong.	Fellowes et al. (2002): (LC); IUCN Red List of Threatened Species (2025): VU
6.	Black-crowned Night Heron *Nycticorax nycticorax*	Pond; Seasonally Wet Grassland	Pond, Compensatory Wetland, Marsh, Seasonally Wet Grassland, Semi-natural / Natural	Common resident and migrant. Widely distributed in Hong Kong.	Fellowes et al. (2002): LC

Number	Species	Location (1,2,3,4,5,6,7)		Rarity and Distribution in Hong Kong (1, 2, 3, 4, 5, 6, 7)	Conservation/Protection status (5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21)
		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
			Watercourse, Modified Watercourse		
7.	Black-faced Bunting *Emberiza spodocephala*	/	Pond, Compensatory Wetland	Common winter visitor and passage migrant. Widely distributed in Hong Kong.	WWF: MR
8.	Black-faced Spoonbill *Platalea minor*	/	Pond, Semi-natural / Natural Watercourse,	Common winter visitor. Found in Deep Bay area.	IUCN Red List of Threatened Species (2025): VU; Fellowes et al. (2002): PGC; China Red Data Book: EN; List of Wild Animals under State Priority Conservation: Class I; Red List of China's Vertebrates: EN WWF: MR
9.	Black-winged Stilt *Himantopus himantopus*	Pond	Pond; Compensatory Wetland; Marsh	Common migrant and winter visitor. Found in Deep Bay area, Long Valley, Kam Tin.	Fellowes et al. (2002): RC
10.	Black-winged Kite *Elanus caeruleus*	/	Marsh; Seasonally Wet Grassland	Uncommon visitor. Found in Ha Tsuen, Deep Bay area.	Fellowes et al. (2002): LC; Cap. 586; List of Wild Animals under State Priority Conservation: Class II; China Red Data Book: VU; CITES: Appendix II WWF: HR
11.	Bluethroat	/	Pond	Locally common winter visitor. Widely distributed in wet	Fellowes et al. (2002): LC; List of Wild Animals under

Number	Species	Location (1,2,3,4,5,6,7)		Rarity and Distribution in Hong Kong (1, 2, 3, 4, 5, 6, 7)	Conservation/Protection status (5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21)
		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
	Luscinia svecica			agricultural areas throughout Hong Kong.	State Priority Conservation: Class II WWF: HR
12.	Buff-bellied Pipit *Anthus rubescens*	/	Compensatory Wetland	Uncommon passage migrant and winter visitor. Widely distributed in areas of wet agriculture and drained fish ponds throughout Hong Kong.	Fellowes et al. (2002): LC
13.	Chestnut-eared Bunting *Emberiza fucata*	Pond	Pond; Compensatory Wetland; Marsh	Uncommon passage migrant. Found in Long Valley, Tai Mong Tsai, Luk Keng, Ho Chung, Kam Tin, Lantau, Sha Lo Tung.	Fellowes et al. (2002): LC WWF: MR
14.	Chinese Francolin *Francolinus pintadeanus*	/	Grassland / Shrubland	Common resident. Widely distributed in grassland throughout Hong Kong.	IUCN Red List of Threatened Species (2025): LC; Red List of China's Vertebrate: NT WWF: MR
15.	Chinese Grosbeak *Eophona migratoria*	/	Pond; Woodland	Common winter visitor. Found in Kam Tin, Nam Chung, Shek Kong, Deep Bay area, Ho Chung, Lam Tsuen, Hok Tau, Island House and Kowloon Park.	Fellowes et al. (2002): LC
16.	Chinese Penduline Tit *Remiz consobrinus*	/	Compensatory Wetland, Reedbed	Common autumn migrant and winter visitor. Found in Deep Bay area, Tai O, Mui Wo,	Fellowes et al. (2002): RC WWF: MR

Number	Species	Location (1,2,3,4,5,6,7)		Rarity and Distribution in Hong Kong (1, 2, 3, 4, 5, 6, 7)	Conservation/Protection status (5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21)
		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
				Long Valley, Luk Keng, Chek Lap Kok.	
17.	Chinese Pond Heron *Ardeola bacchus*	Pond; Seasonally Wet Grassland	Pond, Compensatory Wetland, Marsh, Seasonally Wet Grassland, Modified Watercourse, Semi-natural / Natural Watercourse; Agricultural Land Developed Area	Common resident. Widely distributed in Hong Kong.	Fellowes et al. (2002): PRC
18.	Cinnamon Bittern *Ixobrychus cinnamomeus*	/	Compensatory Wetland	Uncommon passage migrant and scarce summer visitor. Found in Deep Bay area, Long Valley, Tai Yuen (Sheung Shui), Pui O.	Fellowes et al. (2002): LC WWF: HR
19.	Collared Crow *Corvus torquatus*	Pond	Pond; Semi-natural / Natural Watercourse	Locally common resident. Found in Inner Deep Bay area, Nam Chung, Kei Ling Ha, Tai Mei Tuk, Pok Fu Lam, Chek lap Kok, Shuen Wan, Lam Tsuen.	Fellowes et al. (2002): LC WWF: MR
20.	Collared Scops Owl *Otus lettia*	/	Grassland / Shrubland; Orchard	Common resident. Widely distributed in shrubland throughout Hong Kong.	Cap. 586; List of Wild Animals under State Priority Conservation: Class II; CITES: Appendix II
21.	Common Emerald Dove *Chalcophaps indica*	Developed Area	Developed Area	Uncommon but widespread resident. Widely distributed in	China Red Data Book: VU

Number	Species	Location (1,2,3,4,5,6,7)		Rarity and Distribution in Hong Kong (1, 2, 3, 4, 5, 6, 7)	Conservation/Protection status (5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21)
		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
				woodland throughout Hong Kong.	
22.	Common Greenshank *Tringa nebularia*	/	Pond	Abundant passage migrant and winter visitor. Found in Deep Bay area.	Fellowes et al. (2002): RC WWF: HR
23.	Common Kestrel *Falco tinnunculus*	/	Pond, Marsh, Grassland / Shrubland	Common autumn migrant and winter visitor. Widely distributed in Hong Kong.	Cap. 586; List of Wild Animals under State Priority Conservation: Class II; CITES: Appendix II WWF: HR
24.	Common Moorhen *Gallinula chloropus*	/	Pond, Compensatory Wetland	Common winter visitor, resident and migrant. Found in Deep Bay area, Shuen Wan, Starling Inlet.	WWF: MR
25.	Common Redshank *Tringa totanus*	/	Pond	Abundant passage migrant and winter visitor. Found in Deep Bay area.	Fellowes et al. (2002): RC
26.	Common Snipe *Gallinago gallinago*	/	Pond	Common passage migrant and winter visitor. Found in Long Valley, Chau Tau, Sai Kung.	WWF: HR
27.	Chinese Hwamei *Garrulax canorus*	/	Grassland / Shrubland	Common resident	Cap. 170; NT
28.	Crested Goshawk *Accipiter trivirgatus*	/	Grassland / Shrubland	Common resident. Widely distributed in woodlands and	Cap. 586; List of Wild Animals under State Priority Conservation:

Number	Species	Location (1,2,3,4,5,6,7)		Rarity and Distribution in Hong Kong (1, 2, 3, 4, 5, 6, 7)	Conservation/Protection status (5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21)
		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
				shrublands throughout Hong Kong.	Class II; China Red Data Book: Rare; CITES Appendix II
29.	Crested Serpent Eagle *Spilornis cheela*	/	Pond, Grassland / Shrubland	Common resident. Widely distributed in shrublands on hillsides throughout Hong Kong.	Fellowes et al. (2002): (LC); Cap. 586; List of Wild Animals under State Priority Conservation: Class II; China Red Data Book: VU; CITES: Appendix II
30.	Eastern Buzzard *Buteo japonicus*	Pond; Seasonally Wet Grassland	Pond, Marsh, Seasonally Wet Grassland, Woodland, Grassland/Shrubland	Common winter visitor. Widely distributed in Hong Kong.	Cap. 586; List of Wild Animals under State Priority Conservation: Class II; CITES: Appendix II
31.	Eastern Cattle Egret *Bubulcus coromandus*	/	Pond, Marsh, Semi-natural / Natural Watercourse, Developed Area	Resident and common passage migrant. Widely distributed in Hong Kong.	Fellowes et al. (2002): LC WWF: HR
32.	Eurasian Coot *Fulica atra*	/	Pond	Uncommon winter visitor. Found in Deep Bay area, Plover Cove Reservoir, Shuen Wan.	Fellowes et al. (2002): RC WWF: HR
33.	Eurasian Spoonbill Platalea leucorodia	/	Pond, Compensatory Wetland	Uncommon winter visitor. Found in Deep Bay area.	Fellowes et al. (2002): LC; Cap. 586; List of Wild Animals under State Priority Conservation: Class II; China Red Data

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		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
					Book: VU; CITES: Appendix I WWF: HR
34.	Eurasian Teal *Anas crecca*	/	Pond	Common winter visitor. Found in Deep Bay area, Shuen Wan, Tai Lam Chung Reservoir, Victoria Harbour, urban parks.	Fellowes et al. (2002): RC WWF: HR
35.	Eurasian Wigeon *Mareca penelope*	/	Pond	Winter visitor. Found in Deep Bay area, Tai Lam Chung.	Fellowes et al. (2002): RC
36.	Eurasian Wryneck *Jynx torquilla*	Pond	Pond, Compensatory Wetland, Marsh	Uncommon passage migrant and winter visitor. Widely distributed in Hong Kong.	WWF: HR
37.	Great Cormorant *Phalacrocorax carbo*	/	Pond, Semi-natural / Natural Watercourse, Modified Watercourse, Seasonally Wet Grassland	Common winter visitor. Widely distributed in coastal areas throughout Hong Kong.	Fellowes et al. (2002): PRC
38.	Great Egret *Ardea alba*	/	Pond, Compensatory Wetland, Marsh, Semi-natural / Natural Watercourse, Modified Watercourse	Common resident, migrant and winter visitor. Widely distributed in Hong Kong.	Fellowes et al. (2002): PRC
39.	Greater Coucal *Centropus sinensis*	Pond; Modified Watercourse	Pond, Compensatory Wetland, Marsh, Reedbed, Seasonally Wet Grassland, Semi-natural / Natural Watercourse, Woodland,	Common resident. Widely distributed in Hong Kong.	List of Wild Animals under State Priority Conservation: Class II; China Red Data Book: VU

Number	Species	Location (1,2,3,4,5,6,7)		Rarity and Distribution in Hong Kong (1, 2, 3, 4, 5, 6, 7)	Conservation/Protection status (5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21)
		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
			Grassland/Shrubland, Orchard		
40.	Green Sandpiper *Tringa ochropus*	/	Pond, Compensatory Wetland	Common migrant and winter visitor. Found in Deep Bay area, Shuen Wan, Long Valley, Kam Tin, Shek Kong, Ho Chung.	WWF: HR
41.	Grey Heron *Ardea cinerea*	/	Pond, Compensatory Wetland, Marsh, Semi-natural / Natural Watercourse, Modified Watercourse	Common winter visitor. Found in Deep Bay area, Starling Inlet, Kowloon Park, Cape D'Aguilar.	Fellowes et al. (2002): PRC WWF: MR
42.	Grey Treepie *Dendrocitta formosae*	Woodland	Woodland, Orchard	Locally common resident. Found in Sai Kung, Luk Keng, Tai Po Kau, Tai Tam Reservoir, Ho Chung, Hong Kong University, Lam Tsuen, Hok Tau.	Fellowes et al. (2002): LC
43.	Grey-headed Lapwing *Vanellus cinereus*	/	Pond	Locally common winter visitor and migrant. Found in Kam Tin, Tsim Bei Tsui, Lo Wu, Tai Long Wan, Shuen Wan, Castle Peak coast, Chek Lap Kok.	Fellowes et al. (2002): LC WWF: HR
44.	House Swift *Apus nipalensis*	/	Pond, Grassland/Shrubland, Woodland	Abundant spring migrant and common resident. Widely distributed in Hong Kong.	WWF: HR

Number	Species	Location (1,2,3,4,5,6,7)		Rarity and Distribution in Hong Kong (1, 2, 3, 4, 5, 6, 7)	Conservation/Protection status (5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21)
		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
45.	Indian Cuckoo *Cuculus micropterus*	/	Pond, Compensatory Wetland, Marsh, Grassland/Shrubland	Locally common spring and summer visitor. Widely distributed in Hong Kong.	WWF: HR
46.	Intermediate Egret *Ardea intermedia*	/	Pond; Marsh, Modified Watercourse	Resident and passage migrant. Found in Deep Bay area, Tai Long Wan, Starling Inlet, Tai O, Cape D'Aguilar.	Fellowes et al. (2002): RC
47.	Japanese Quail *Coturnix japonica*	/	Compensatory Wetland,Pond	Uncommon autumn passage migrant and rare winter visitor. Found in Long Valley, Mai Po, Kam Tin, Lam Tsuen, Tin Shui Wai.	Fellowes et al. (2002): LC WWF: HR
48.	Lesser Coucal *Centropus bengalensis*	/	Pond, Grassland / Shrubland	Uncommon resident. Widely distributed in Hong Kong.	List of Wild Animals under State Priority Conservation: Class II; China Red Data Book: VU WWF: MR
49.	Little Bunting *Emberiza pusilla*	/	Pond, Marsh	Common passage migrant and winter visitor. Widely distributed in open area throughout Hong Kong.	WWF: HR
50.	Little Egret *Egretta garzetta*	/	Pond, Compensatory Wetland, Marsh, Seasonally Wet Grassland, Semi-natural / Natural Watercourse, Plantation, Modified Watercourse	Common resident, migrant and winter visitor. Widely distributed in coastal area throughout Hong Kong.	Fellowes et al. (2002): PRC

Number	Species	Location (1,2,3,4,5,6,7)		Rarity and Distribution in Hong Kong (1, 2, 3, 4, 5, 6, 7)	Conservation/Protection status (5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21)
		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
51.	Little Grebe *Tachybaptus ruficollis*	/	Pond, Compensatory Wetland, Marsh, Semi-natural / Natural Watercourse	Common resident. Found in Deep Bay Area.	Fellowes et al. (2002): LC
52.	Little Ringed Plover *Charadrius dubius*	/	Pond, Compensatory Wetland	Resident, common winter visitor and passage migrant. Widely distributed in freshwater areas throughout Hong Kong.	Fellowes et al. (2002): (LC)
53.	Long-tailed Shrike *Lanius schach*	/	Pond, Compensatory Wetland, Marsh, Semi-natural / Natural Watercourse	Common resident. Widely distributed in open areas throughout Hong Kong.	WWF: MR
54.	Long-toed Stint *Calidris subminuta*	/	Pond	Common passage migrant and scarce winter visitor. Found in Long Valley, Ma Tso Lung, Tsim Bei Tsui, Kam Tin, Pui O, Shuen Wan.	Fellowes et al. (2002): LC WWF: HR
55.	Marsh Sandpiper *Tringa stagnatilis*	/	Pond	Abundant winter visitor and migrant. Found in Deep Bay area, Shuen Wan, Long Valley, Kam Tin, Sai Kung.	Fellowes et al. (2002): RC WWF: HR
56.	Northern Shoveler *Spatula clypeata*	/	Pond	Abundant winter visitor. Found in Deep Bay area.	Fellowes et al. (2002): RC
57.	Oriental Magpie *Pica serica*	/	Pond	Common resident. Widely distributed in Hong Kong	WWF: HR

Number	Species	Location (1,2,3,4,5,6,7)		Rarity and Distribution in Hong Kong (1, 2, 3, 4, 5, 6, 7)	Conservation/Protection status (5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21)
		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
58.	Oriental Pratincole *Glareola maldivarum*	/	Pond	Passage migrant. Found in Mai Po, Tsim Bei Tsui.	Fellowes et al. (2002): LC WWF: MR
59.	Oriental Stork *Ciconia boyciana*	/	Compensatory Wetland	Rare winter visitor. Found in Deep Bay area.	IUCN Red List of Threatened Species (2025): EN; Fellowes et al. (2002): GC; Cap. 586; List of Wild Animals under State Priority Conservation: Class I; China Red Data Book: EN; Red List of China's Vertebrates: EN; CITES: Appendix I
60.	Oriental Turtle Dove *Streptopelia orientalis*	Pond	Pond, Compensatory Wetland, Marsh, Seasonally Wet Grassland, Semi-natural / Natural Watercourse, Plantation, Modified Watercourse	Common winter visitor. Widely distributed in Hong Kong.	WWF: HR
61.	Pied Avocet *Recurvirostra avosetta*	/	Pond	Abundant winter visitor. Found in Deep Bay area.	Fellowes et al. (2002): RC WWF: HR
62.	Pied Kingfisher *Ceryle rudis*	Compensatory Wetland; Semi-natural / Natural Watercourse	Pond, Compensatory Wetland, Semi-natural / Natural Watercourse	Common resident. Widely distributed in lakes and ponds throughout Hong Kong.	Fellowes et al. (2002): (LC)
63.	Purple Heron	/	Pond	Uncommon passage migrant. Found in Deep Bay area.	Fellowes et al. (2002): RC

Number	Species	Location (1,2,3,4,5,6,7)		Rarity and Distribution in Hong Kong (1, 2, 3, 4, 5, 6, 7)	Conservation/Protection status (5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21)
		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
	Ardea purpurea				WWF: HR
64.	Red-billed Starling *Spodiopsar sericeus*	/	Pond, Modified Watercourse	Abundant winter visitor. Widely distributed in Hong Kong.	Fellowes et al. (2002): GC WWF: HR
65.	Red-necked Stint *Calidris ruficollis*	/	Pond	Abundant spring passage migrant. Found in Deep Bay area.	Fellowes et al. (2002): LC
66.	Red-rumped Swallow *Cecropis daurica*	Seasonally Wet Grassland	Pond, Seasonally Wet Grassland, Grassland/Shrubland	Locally common passage migrant and winter visitor. Widely distributed in Hong Kong.	WWF: MR
67.	Red-throated Pipit *Anthus cervinus*	/	Pond	Common passage migrant and winter visitor. Widely distributed in dry agricultural areas throughout Hong Kong.	Fellowes et al. (2002): LC WWF: MR
68.	Richard's Pipit *Anthus richardi*	/	Pond, Compensatory Wetland	Common passage migrant, winter visitor and locally common resident. Widely distributed in Hong Kong.	WWF: HR
69.	Sharp-tailed Sandpiper *Calidris acuminata*	/	Pond	Common passage migrant. Found in Deep Bay area, Shuen Wan, Sai Kung, Long Valley, Tai Po, Lam Tsuen.	Fellowes et al. (2002): LC WWF: HR
70.	Sooty-headed Bulbul *Pycnonotus aurigaster*	/	Pond, Marsh, Seasonally Wet Grassland, Grassland/Shrubland	Common resident. Widely distributed in open areas throughout Hong Kong.	WWF: HR

Number	Species	Location (1,2,3,4,5,6,7)		Rarity and Distribution in Hong Kong (1, 2, 3, 4, 5, 6, 7)	Conservation/Protection status (5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21)
		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
71.	Striated Heron *Butorides striata*	/	Semi-natural / Natural Watercourse	Common summer visitor. Widely distributed in Hong Kong.	Fellowes et al. (2002): LC
72.	Temminck's Stint *Calidris temminckii*	/	Pond	Uncommon winter visitor and migrant. Widely distributed in Deep Bay area ponds.	Fellowes et al. (2002): LC WWF: HR
73.	Whimbrel *Numenius phaeopus*	/	Semi-natural / Natural Watercourse	Common passage migrant and scarce winter visitor. Found in Deep Bay area, Sai Kung, Tung Ping Chau, Ninepins, Cape D'Aguilar, Pok Fu Lam.	Fellowes et al. (2002): LC
74.	White-bellied Erpornis *Erpornis zantholeuca*	/	Orchard	Uncommon resident. Found in Tai Po Kau, Shing Mun, Ho Chung, Kowloon Hills, Ng Tung Chai, Wu Kau Tang, Sha Tau Kok, A Ma Wat, Kop Tong, Lau Shui Heung.	Fellowes et al. (2002): LC WWF: MR
75.	White-cheeked Starling *Spodiopsar cineraceus*	/	Pond	Locally common winter visitor. Found in Deep Bay area, Kam Tin, Long Valley.	Fellowes et al. (2002): PRC
76.	White-shouldered Starling *Sturnia sinensis*	/	Pond, Compensatory Wetland, Marsh, Semi-natural / Natural Watercourse	Locally common passage migrant and uncommon winter visitor. Found in Kam Tin, Deep Bay area, Po Toi Island, Long Valley, Victoria Park, Ho	Fellowes et al. (2002): (LC)

Number	Species	Location (1,2,3,4,5,6,7)		Rarity and Distribution in Hong Kong (1, 2, 3, 4, 5, 6, 7)	Conservation/Protection status (5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21)
		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
				Chung, Ma Tso Lung, Mui Wo, Lam Tsuen Valley.	
77.	White-throated Kingfisher *Halcyon smyrnensis*	Compensatory Wetland	Pond, Compensatory Wetland, Semi-natural / Natural Watercourse, Grassland / Shrubland, Orchard	Common resident. Widely distributed in coastal areas throughout Hong Kong	Fellowes et al. (2002): (LC); List of Wild Animals under State Priority Conservation: Class II
78.	Wood Sandpiper *Tringa glareola*	/	Pond	Common migrant and winter visitor. Widely distributed in wetland area throughout Hong Kong.	Fellowes et al. (2002): LC WWF: HR
79.	Yellow Bittern *Ixobrychus sinensis*	/	Pond, Compensatory Wetland, Marsh, Semi-natural / Natural Watercourse	Uncommon summer visitor and common passage migrant. Found in Deep Bay area, Chek Keng, Tai Long Wan	Fellowes et al (2002): LC WWF: HR
80.	Yellow-breasted Bunting *Emberiza aureola*	/	Pond; Reedbed	Scarce spring and rare winter passage migrant. Widely distributed in agriculture land and reedbeds throughout Hong Kong.	Fellowes et al. (2002): RC; List of Wild Animals under State Priority Conservation: Class I WWF: HR
81.	Zitting Cisticola *Cisticola juncidis*	/	Pond; Compensatory Wetland	Common passage migrant and winter visitor. Widely distributed in grassland throughout Hong Kong.	Fellowes et al. (2002): LC WWF: HR
	Herpetofauna				

Number	Species	Location (1,2,3,4,5,6,7)		Rarity and Distribution in Hong Kong (1, 2, 3, 4, 5, 6, 7)	Conservation/Protection status (5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21)
		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
1.	Chinese Bullfrog *Hoplobatrachus chinensis*	Semi-natural / Natural Watercourse	Compensatory Wetland	Widely distributed in Lantau Island and New Territories.	List of Wild Animals under State Priority Conservation: Class II; Fellowes et al. (2002): PRC; Red List of China's Vertebrates: EN WWF: MR
2.	Four-clawed Gecko *Gehyra mutilata*	Developed Area	Developed Area	Widely but thinly distributed throughout Hong Kong.	Red List of China's Vertebrates: VU
3.	Indian Forest Skink *Sphenomorphus indicus*	/	Seasonally Wet Grassland, Grassland / Shrubland, Orchard	Distributed in woodlands in eastern and central New Territories.	Fellowes et al. (2002): LC
	Butterfly				
1.	Common Onyx *Horaga onyx*	/	Seasonally Wet Grassland	Rare. Widely distributed throughout Hong Kong.	Fellowes et al. (2002): LC
2.	Danaid Eggfly *Hypolimnas misippus*	/	Grassland / Shrubland	Uncommon. Ngau Ngak Shan, Lung Kwu Tan, Hong Kong Wetland Park, Mount Parker, Cloudy Hill, Lin Ma Hang.	Fellowes et al. (2002): LC
3.	Grass Demon *Udaspes folus*	/	Orchard	Rare. Widely distributed throughout Hong Kong.	-

Number	Species	Location (1,2,3,4,5,6,7)		Rarity and Distribution in Hong Kong (1, 2, 3, 4, 5, 6, 7)	Conservation/Protection status (5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21)
		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
4.	Metallic Cerulean *Jamides alecto*	Woodland	Seasonally Wet grassland, Semi-natural / Natural Watercourse	Very rare. Victoria Peak, Fung Yuen, Chuen Lung, Mui Wo.	-
5.	Paintbrush Swift *Baoris farri*	Woodland	/	Rare. Deep Water Bay, Shing Mun, Lam Tsuen, Fung Yuen, Wu Kau Tang, Lai Chi Wo.	-
6.	Pale Palm Dart *Telicota colon*	/	Grassland / Shrubland	Rare. Widely distributed throughout Hong Kong.	Fellowes et al. (2002): LC
7.	Peacock Royal *Tajuria cippus malcolmi*	/	Grassland / Shrubland	Rare. Ma On Shan, Wu Kau Tang, Tai Po Kau, Mount Nicholson, Victoria Peak, Cloudy Hill	LC
8.	Plain Hedge Blue *Celastrina lavendularis*	/	Grassland / Shrubland	Very rare. Chuen Lung, Kap Lung, Tai Po Kau, Shing Mun Country Park, Tai Lam Country Park, Kadoorie Farm and Botanic Garden, Ngau Ngak Shan	LC
9.	Red-breast Jezebel *Delias acalis*	Woodland	/	Rare. Tai Tam, Tai Mo Shan, Ngau Ngak Shan, Pat Sin Leng, Tai Po Kau, Wu Kau Tang, Wong Nai Chung, Fung Yuen, Plover Cove.	Fellowes et al. (2002): LC
10.	Small Three-ring *Ypthima norma*	/	Grassland / Shrubland	Very rare. Pak Tam Chung, Ngau Kwo Lo, Kai Shan.	Fellowes et al. (2002): LC WWF: HR

Number	Species	Location (1,2,3,4,5,6,7)		Rarity and Distribution in Hong Kong (1, 2, 3, 4, 5, 6, 7)	Conservation/Protection status (5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21)
		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
11.	Swallowtail *Papilio xuthus*	/	Pond, Grassland / Shrubland	Rare. Kap Lung, Ma On Shan, Tai Tam, Sha Lo Wan, Kat O, Lung Kwu Tan, Wu Kau Tang, Lung Kwu Chau.	-
12.	White Royal *Pratapa deva*	/	Grassland / Shrubland	Very rare. Tai Po Kau, Pokfulam, Kuk Po, Pak Sha O, Victoria Peak, Wu Kau Tang, Fung Yuen.	-
	Odonate				
1.	Dingy Dusk-hawker *Gynacantha subinterrupta*	/	Semi-natural / Natural Watercourse	Common. Found in well-shaded woodlands. Recorded in Hong Kong Wetland Park, Lantau Island, Ping Shan Chai, Sha Lo Tung and Tai Mo Shan.	Fellowes et al. (2002): LC
2.	Coastal Glider *Macrodiplax cora*	/	Pond	Common. Frequents marshes and ponds with dense vegetation, especially adjacent to coastal areas.	Fellowes et al. (2002): LC
3.	Blue Chaser *Potamarcha congener*	/	Modified Watercourse	Common. Found in small weedy ponds, puddles and marshes. Widely distributed in the New Territories.	Fellowes et al. (2002): LC
4.	Sapphire Flutterer *Rhyothemis triangularis*	/	Marsh	Common. Widely distributed in weedy ponds, sluggish rivers and marshes.	Fellowes et al. (2002): LC

Number	Species	Location (1,2,3,4,5,6,7)		Rarity and Distribution in Hong Kong (1, 2, 3, 4, 5, 6, 7)	Conservation/Protection status (5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21)
		Within the Construction Area of the Project	Outside the Construction Area of the Project but within the Assessment Area		
5.	Scarlet Basker *Urothemis signata*	/	Pond; Grassland / Shrubland	Common. Common in areas with abandoned fish ponds throughout Hong Kong.	Fellowes et al. (2002): LC
	Firefly				
1.	Motschulsky's Starworm *Rhagophthalmus motschulskyi*	/	Seasonally Wet Grassland	Tai Po Kau, Kam Tin, Tin Shui Wai, Ma Shi Po, Fu Tei Au, Ma On Shan, Lam Tsuen, Wu Kau Tang, Diamond Hill, Tai Tong, Mount Parker Road, Mui Wo.	Endemic to Hong Kong
	Freshwater Fauna				
1.	*Gobiopterus macrolepis*	/	Semi-natural / Natural Watercourse	Recorded in Deep Bay.	Red List of China's Vertebrate: VU
2.	*Somanniathelphusa zanklon*	Semi-natural / Natural Watercourse	Semi-natural / Natural Watercourse	Widely distributed within Hong Kong, recorded from the northwestern and northeastern New Territories and Lantau Island.	Fellowes et al. (2002): GC IUCN Red List of Threatened Species (2025): EN Endemic to Hong Kong

Abbreviation

• Protection or conservation status: EN = Endangered; GC = Global Concern; LC = Local Concern; PGC = Potential Global Concern; PRC = Potential Regional Concern; RC = Regional Concern; VU = Vulnerable

Note:

1. Agriculture, Fisheries and Conservation Department. (2024). Species Database of the Hong Kong Biodiversity Information Hub.
2. Chan et al. (2011). A review of the local restrictedness of Hong Kong Butterflies.
3. Corlett et al. (2000). Hong Kong vascular plants: distribution and status.
4. Shek (2006). A Field Guide to the Terrestrial Mammals of Hong Kong.
5. Stanton & Leven (2016). Distribution, habitat utilisation and conservation status of the freshwater crab, *Somanniathelphusa zanklon* Ng & Dudgeon, 1992 (Crustacea: Brachyura: Gecarcinucidae) endemic to Hong Kong.

6. Tam et al. (2010). The Dragonflies of Hong Kong.
7. Yiu (2023). Hong Kong Fireflies.
8. Convention on International Trade in Endangered Species of Wild Fauna and Flora (2023). Appendices I, II and III.
9. Fellowes et al. (2002). Wild animals to watch: Terrestrial and freshwater fauna of conservation concern in Hong Kong.
• Letters in parentheses indicate that the assessment is on the basis of restrictedness in breeding and/or roosting sites rather than in general occurrence
10. Forests and Countryside Ordinance (Cap. 96)
11. Fu & Chin (1992). China Plant Red Data Book – Rare and Endangered Plants.
12. Hu et al. (2003). Rare and Precious Plants of Hong Kong.
13. International Union of Conservation for Nature. (2025). The IUCN Red List of Threatened Species. Version 2025-2.
14. Jiang et al. (2016). Red List of China's Vertebrates.
15. National Forestry and Grassland Administration and the Ministry of Agricultural and Rural Affairs. (2021). List of Wild Plants under the State Priority Protection.
16. National Forestry and Grassland Administration and the Ministry of Agricultural and Rural Affairs. (2021). List of Wild Animals under State Priority Conservation.
17. Protection of Endangered Species of Animals and Plants Ordinance (Cap. 586)
18. Qin et al. (2017). Threatened Species List of China's Higher Plants.
19. Wang (1998). China Red Data Book of Endangered Animals: Mammalia.
20. Wild Animals Protection Ordinance (Cap. 170)
21. Wu et al. (1988). Illustration of Rare & endangered plant in Guangdong Province
22. WWF – Or, C.K.M. & Chan, B.P.L. (Eds.) (2025a). The State of Hong Kong Biodiversity 2025. WWF-Hong Kong. 65 pp.: EX = Extinct in Hong Kong under Hong Kong species status assessment; HR = High Risk of Extinction in Hong Kong species status assessment; MR = Moderate Risk of Extinction in Hong Kong species status assessment. Letters in parentheses indicate that the assessment is on the basis of breeding population.

8.8 Identification of Potential Environmental Impacts

Construction Phase – Direct Impacts

Direct Impact on Recognised Sites of Conservation Importance and other Ecologically Sensitive Receivers

8.8.1 Within the Assessment Area, five recognised sites of conservation importance (proposed HHW WCP, WCA, WBA, "CA" zone and Priority Site for Enhanced Conservation), and other ecologically sensitive receivers including IBA, the EA and OWCA Area 2, off-site compensation woodland, and key dragonfly site in Ma Tso Lung) were identified. Part of the Assessment Area is also recognised as an Important Bird Area (IBA) and regarded as an ecologically sensitive resource under this project. The potentially affected recognised sites of conservation importance and other ecologically sensitive receivers are located in the HHW area as well as area to the southeast of the Loop.

8.8.2 A total of 13 habitats were identified within the Assessment Area, including both wetland and non-wetland habitats (**Section 8.6** refers). The Construction Area of the Project will encroach on all identified habitat types except reedbed and agricultural land. The potentially affected wetland habitats are mainly in the EA and WCA to the southeast of the Loop.

8.8.3 The ECR Alignment has been shifted northward to minimise the slope works in Tse Koo Hang, which would lead to an unavoidable encroachment into the area within the proposed HHW WCP as indicated in **60722948/A10/FIGURE 8.1.** Nevertheless, the encroachment is located at the fringe of the proposed HHW WCP boundary. The ECR layout is expected to be finalised prior to the detailed studies for the HHW WCP, therefore potential interface issues will be further reviewed and addressed at the detailed studies stage as indicated in the WCPs System FS (AFCD, 2024a).

8.8.4 Areas inside the Construction Area of the Project would be affected by construction works, resulting in direct impacts on wetland habitats within the sites of conservation importance and other ecological sensitive receivers, mainly within the WCA, WBA, "CA" zone, Priority Site for Enhanced Conservation, the EA, as well as the IBA. The portion of the EA included in the Project Boundary but outside the 0.38ha Construction area of the Project would not be affected or occupied as temporary works/storage areas. To avoid construction activities and to deter other unwanted human access, such area would be properly fenced off and maintained by the project proponent during the construction phase. Fences to be installed during the construction phase will incorporate a 30cm gap at the bottom to allow passage of otters and other mammal, reducing the potential impacts of wildlife on accessing the EA, subject to the Contractor's design. Although the boundary of Key Dragonfly Site in Ma Tso Lung has not been delineated, it generally refers to the mosaic of freshwater ponds, abandoned ponds and abandoned paddies lying in the former Frontier Closed Area to the east of Lok Ma Chau. For conservative assessment purposes, this ecological sensitive resource is considered to be directly impacted by the Construction Area of the Project. Considering the importance of these sites affected, the loss of these wetlands would be considered of moderate to high significance, in the absence of mitigation measures. The woodland habitat within the aforementioned areas is connected to natural habitats nearby (e.g. hillside grassland) supporting moderately diverse and abundant flora and fauna communities. As such, the loss of this habitat is considered to be of moderate significance. Relatively low

flora and fauna species are recorded in the other terrestrial habitat (e.g. grassland / shrubland, orchard, plantation and developed area), therefore no adverse impact is anticipated on loss of these habitats. No direct impacts are anticipated to the remaining portion of the EA (outside of the Construction Area of the Project), OWCA Area 2, and the off-site compensation woodland area. Identified habitat losses within Recognised Sites of Conservation Importance and other Ecologically Sensitive Receivers are summarised in **Table 8.23.**

Table 8.23 Direct Impacts to Habitats within Recognised Sites of Conservation Importance and Other Ecologically Sensitive Receivers

Sites	**Permanent Loss (ha)**			**Temporary Loss (ha)**		
	Wetland habitats(1)	Woodland habitats	Other habitats(2)	Wetland habitats(1)	Woodland habitats	Other habitats(2)
Proposed HHW WCP	1.11	0.12	0.04	1.49	-	-
WCA	0.74	0.00	0.17	1.11	-	-
WBA	1.35	1.68	6.09	0.02	-	-
"CA" zone	0.73	0.00	0.17	1.31	-	-
Priority Sites for Enhanced Conservation	0.25	-	-	0.88	-	-
IBA	2.09	1.68	9.66	1.74	-	-
The EA	-			0.38	-	-
OWCA Area 2	-					
Off-site Compensation Woodland	-					
The Key dragonfly site in Ma Tso Lung	-(3)					

Note:
1. The wetland habitat consists of compensatory wetland, pond, marsh, semi-natural and natural watercourses and seasonally wetland grassland
2. The other habitat consists of grassland/ shrubland, orchard, plantation and developed area.
3. Boundary of the Key Dragonfly Site in Ma Tso Lung is not delineated, therefore it is not possible to provide a breakdown of impacted habitats within this area.
4. Since the Sites may overlap with each other, the size of habitat loss in total should not be added up directly. For the total habitat loss, please refer to **Table 8.24** for details

Direct Impacts to Habitat and Vegetation

Overview of Direct Impact to Habitats

8.8.5 Considerable thought went into selecting the most appropriate ECR alignment and design from environmental and engineering perspectives, with a detailed option review process considered the base scheme and preferred location (see **Section 2.5** for details). In summary, the ECR will comprise a single 4-lane two-way carriage way, variously in the form of at-grade road, depressed road, underpass and viaduct, of a total length of approximately 2.5km. The preferred ECR Alignment avoids and minimises impacts to the Recognised Sites of Conservation Importance / Other Ecologically Sensitive Receivers and wetland habitats as far as practicable, with the section (of about 1.2 km long) that will traverse the EA, LMC Meander and three Ponds 36-38 being constructed as a shallow underpass (approximately 12m below ground). In order to minimise the extent of excavation so as to conserve adjacent wetland habitats, excavation and lateral support of this section would be used for the "cut and cover" construction of the underpass below ground as detailed **Section 2.6**. The ECR Alignment then follows the course of the existing Border Road northwest either as a viaduct or at-grade, before looping to the south and extending to the planned KTN NDA behind (to the east of) Ma Tso Lung Shun Yee San Tsuen. The LMC Meander, marsh and Ponds 36-38 included in the Project Boundary but outside of Construction Area of the Project would not be fenced off (except the EA as discussed in S.8.8.4) and would not be occupied as temporary works/ storage area. In addition, Ponds 36-38 would be maintained by the project proponent during the construction phase (S. 8.11.18 refers)

8.8.6 A fully enclosed electrical mechanical plant room (hereinafter referred to as E&M plant room) and drainage pumping room will be provided to support the operation of the depressed road and underpass. The dimensions and size of the proposed E&M plan room and drainage pumping room will be determined during the detailed design stage. The indicative location of the proposed E&M plant room and drainage pumping room is presented in **60722948/A10/FIGURE 8.10**. For conservative impact estimation, habitats across the entire potential area where these facilities will be located are assumed to be directly and permanently lost.

8.8.7 Development of the Project will result in temporary and permanent direct loss of habitats and vegetation within the Construction Area of the Project. Permanent habitat loss will result from construction works, whereby the habitat will be developed and no longer available for wildlife use, while temporary habitat loss under the Project will mainly arise from habitats that fall within the Construction Area, but will be reinstated / revitalised once the construction works are completed. The identified habitats losses are summarised in **Table 8.24.**

Table 8.24 Direct Impacts to Habitats within Construction Area of the Project

Habitats	Permanent Loss (ha)(4)	Temporary Loss (ha) (4)	Ecological Value
Compensatory Wetlands(1)	-	0.38	Ecological Area: Moderate to High
Pond (2)	0.25	1.26	Hoo Hok Wai: High
Marsh	0.23	0.10	Hoo Hok Wai: Moderate
Reedbed	-	-	No loss of habitat

Habitats	Permanent Loss (ha)(4)	Temporary Loss (ha) (4)	Ecological Value
Modified Watercourse(3)	-	-	No loss of habitat
Semi-natural / Natural Watercourse(3)	-	0.21	LMC Meander: High Ping Hang Stream: Low to Moderate
Seasonally Wet Grassland	1.61	-	Low to Moderate
Woodland	1.68	-	Moderate
Grassland/ Shrubland	2.88	-	Low to Moderate
Orchard	0.54	-	Low
Plantation	1.09	-	Low
Agricultural Land	-	-	No loss of habitat
Developed Area	5.15	-	Very Low
Total(4)	**13.44**	**1.96**	**-**

Notes:
1. The portion of the EA included in the Project Boundary but outside of Construction Area of the Project would be properly fenced off and maintained by the project proponent during the construction phase, even when it would not be affected or occupied as temporary works/storage area.
2. The habitat loss of pond comprises 1.13 ha of pond loss due to the Construction Area of the Project, about 0.18 ha of pond bunds associated with Ponds 39, 39a, and 93 would be temporarily occupied, 0.2 ha of Ponds 36–38 located outside the Construction Area of the Project but conservatively included in the assessment considering the overall integrity of the wetland habitats system. Among the total area, approximately 0.25 ha would be permanently lost due to the footprint of the ventilation opening and maintenance access required for the underpass road of the ECR Alignment. The remaining pond areas would be retained upon completion of the Project.
3. No permanent loss of this habitat is expected: the construction of the viaduct supporting columns would not divert or modify the existing watercourse (~0.06ha) within the Construction Area of the Project.
4. Slight discrepancies in the totals due to rounding up.

8.8.8 Areas subject to temporary loss as described in **Table 8.24** mainly comprise wetland habitats (including compensatory wetland, pond, marsh and semi-natural / natural watercourse) that would be temporarily affected by cut and cover construction of the sunken section of the ECR Alignment. Reinstatement of the EA Zone following completion of the depressed road / underpass sections of the ECR Alignment will be reinstated and established by the Project Proponent / the Contractor with reference to the approved Ecological Mitigation / Habitat Creation and Management Plan (CEDD, 2022) and the existing as-built conditions of the EA after the completion of the aboveground works. For underground reinstatement, all existing engineering and associated structures removed during construction will be reinstated upon completion. After construction of the underpass, the excavation cofferdam will be backfilled with compacted materials (compacted fill or rockfill, subject to the Contractor's design) in accordance with the requirements of the General Specification for Civil Engineering Works published by CEDD (CEDD, 2020). As the stone columns within the construction area will no longer be required to support the EA and its earth bund, the relevant geotextile layers and provisions will be reinstated to ensure that the original level of impermeability is maintained, and native plant species will be used for reinstatement. The ecological characteristics and function of the EA will be enhanced and maintained to a condition no less than the existing conditions, after the reinstatement. A reinstatement and monitoring plan for the EA would be prepared by

the Contractor and submitted to AFCD and EPD for agreement at least three months prior to commencement of construction works.

Direct Impacts to Compensatory Wetlands

8.8.9 The EA along the southeastern edge of the Loop has been established since December 2022. It is considered of moderate to high ecological value, with species of conservation importance including Little Egret, Grey Heron, and Pied Kingfisher recorded during recent surveys (**60722948/A10/FIGURE 8.5** refers). The ECR Alignment will traverse the EA as an underpass. Cut and cover construction will result in the temporary loss of 0.38 ha of the EA, which would be reinstated following the completion of construction works. The areas of the EA adjacent to 0.38 ha the works areas will be fenced off and maintained as wetland habitat during the construction phase, and not used for temporary access, works/storage area and other construction-related activities. Details of construction works near / in the EA (including but not limited to working schedule/period, construction method, reinstatement method, use of equipment, type and route of vehicle entry and water supply arrangements) will be carefully considered and incorporated during the detailed design stage to minimise impacts to the EA. Given that the EA was established as a compensatory wetland, according to the ongoing monitoring data the EA after its establishment, it supports moderate flora and moderate to high fauna diversity, temporary ecological impacts would be considered to be moderate to high in scale, in the absence of mitigation measures.

Direct Impacts to Ponds

8.8.10 The ECR Alignment would traverse pond habitats to the south of the Loop as an underpass. The construction of shallow underpass would primarily cause temporary loss of habitat at three ponds (Ponds 36 to 38). As discussed in **Sections 8.6.6**, the condition of these ponds is relatively poor, although there are some records of wetland species or species of conservation importance were recorded during recent surveys or from other available data (i.e., HKBWS, 2018-2023; AFCD, 2024b). It is also recognised that these ponds form part of the larger wetland system in LMC/HHW area and have significant potential for restoration.

8.8.11 The affected pond habitats would total 1.51 ha during construction phase. Despite poor condition of the ponds, and considering their ecological function as part of the larger wetland system of HHW, temporary direct ecological impacts to these habitats would be moderate to high in scale, in the absence of mitigation measures. As the affected ponds will largely be reinstated and enhanced after the construction works are completed (detailed below in **Section 8.11.14**), permanent operation phase impacts are considered minor in scale.

Direct Impacts to Marsh

8.8.12 Construction Area of the Project will result in the permanent loss of 0.23 ha of freshwater marsh along the Border Road and temporary loss of 0.10 ha of freshwater marsh immediately to the south of the LMC Meander. Marsh habitat is considered of moderate ecological value, with species of conservation significance recorded from the habitat including Japanese Pipistrelle and wetland-dependent species such as White-breasted Waterhen and White-throated Kingfisher. Considering the existing quality and associated common species of this habitat, the loss of this habitat is therefore considered to be moderate in scale.

Direct Impacts to Modified Watercourse

8.8.13 The ECR Alignment would avoid direct impacts to the lower sections of Ma Tso Lung Stream, crossing the stream as a viaduct. The viaduct piers would be located outside of the watercourse and its riparian zone, and no construction activities would be conducted within the watercourse or riparian zone. The affected section of the watercourse does not support high densities of disturbance-sensitive species and given that the viaduct section of the ECR Alignment would be elevated approximately 20m above the affected watercourses, the riparian vegetation along or adjacent to the watercourse would not be significantly affected by the shadow effect of the viaduct. Hence, no loss of this habitat is anticipated.

Direct Impacts to Semi-natural / Natural Watercourse

LMC Meander

8.8.14 The ECR Alignment will traverse the LMC Meander as an underpass, avoiding permanent impacts. However, cut and cover construction will result in temporary loss of the water channel and associated riparian vegetation, totalling approximately 0.21ha (**60722948/A10/FIGURE 8.9** refers). The LMC Meander is considered of high ecological value and was found to support species of conservation importance such as Eurasian Otter and Chinese Bullfrog, as well being utilised as a flight path by avifauna species including Black-faced Spoonbill, Great Egret, Great Cormorant. These temporary ecological impacts would be considered to be moderate to high in scale, in the absence of mitigation measures.

Ping Hang Stream and Other Watercourses

8.8.15 The ECR Alignment will avoid permanent direct impacts to Ping Hang stream. However, cut and cover construction will require that the Stream (approximately 0.4km) is temporarily diverted. The diversion works in themselves are considered a relatively low scale impact, as the diverted channel would be similar to the existing semi-natural stream that runs through the pond area. Ecological impacts would be considered to be minor in the absence of mitigation measures. Aside from Ping Hang Stream, a small section of watercourse (0.02 ha) located to the east of Ma Tso Lung Village will also fall within the Construction Area of the Project. However, the ECR Alignment would cross this stream as a viaduct, no construction activities would therefore be conducted within the watercourse or its riparian zone, with no encroachment on the watercourse and its riparian zone by the viaduct piers. The affected section of the watercourse does not support high densities of disturbance-sensitive species, and given that the viaduct section of the ECR Alignment would be elevated approximately 25m above the affected watercourses, the riparian vegetation along or adjacent to the watercourse would not be significantly affected by the shadow effect of the viaduct. Hence, no loss of this habitat is anticipated

Direct Impacts to Seasonally Wet Grassland Habitats

8.8.16 Several patches of seasonally wet grassland totalling 1.61 ha would be permanently lost under the Project. The habitat was found to support only low flora and fauna diversity, with an overall low-to-moderate ecological value. Given its ecological value and small size, the impact on the loss of this habitat is considered to be low to moderate in scale only.

Direct Impacts to Woodland Habitat

8.8.17 The ECR Alignment will pass through the woodland habitats alongside the existing Border Road. A total of 1.68 ha of woodland would be permanently lost. It is expected some associated flora species of conservation importance (e.g. Incense Tree) within the affected woodlands would also be directly impacted by the project construction. The significance of this impact to woodland is considered to moderate in scale, given the moderately diverse and abundant flora and fauna communities supported by woodland habitats, and the ecological linkage of woodlands with large areas of adjacent natural habitats (e.g. hillside grassland).

Direct Impacts to Other Habitats

8.8.18 Approximately 2.88 ha grassland / shrubland would be permanently lost due to development of the Project. The affected low-lying grasslands were found to supported limited diversity and abundance of wildlife, and impacted by disturbance from nearby by human activities, therefore the direct impact on this habitat is considered of low scale if no mitigation measures are adopted. Other habitats directly impacted by the Project (Orchard, Plantation) are of low ecological value with relatively small areas affected. Considering the low ecological value and low wildlife diversity and abundance supported, the loss of these habitats is considered a negligible to minor ecological impact. 5.15ha of existing Developed Area would be directly impacted by the project, and there would be no direct habitat loss of reedbed, modified watercourse or agricultural land habitats.

Habitat Fragmentation

8.8.19 Fragmentation refers to a discontinuity in habitat, decreasing its carrying capacity, hindering movement, and rendering it less attractive to flora and fauna (Haddad et al., 2015), or the isolation of individuals of a species' population resulting in their lower viability (Schlaepfer et al., 2018). The ECR Alignment will pass through lower value habitats (including developed area, small patches of grassland/shrubland and plantation) adjacent to Ma Tso Lung Shun Yee San Tsuen and Ma Tso Lung San Tsuen mainly on a viaduct, therefore habitat fragmentation impacts in these areas are anticipated to be minor. Fragmentation is anticipated to be more significant where the ECR Alignment passes through/adjacent to wetland and woodland habitats.

Fragmentation of Wetlands and Associated Fauna

8.8.20 Habitat fragmentation is expected to occur where the ECR Alignment crosses pond habitats, the EA and LMC Meander. This is of concern as the affected habitats supports major flight paths between the wetland habitats in the LMC/HHW area (on the east) and the San Tin area (on the west). As such, it is of importance particularly for Eurasian Otters which potentially use the area to transit between HHW and ST / SPS, as well as large wetland avifaunal species, which use the corridor as a flight path. These fragmentation impacts would be mostly temporary in nature, restricted to the construction phase, as the ECR will cross the area as a shallow underpass, with only two reinforced concrete ventilation openings (each approximately 0.13ha) remaining at grade following the completion of works. Nevertheless, the scale of these impacts during the construction phase are anticipated to be moderate if no mitigation measures are implemented.

8.8.21 Other potential wetland fragmentation impacts are expected to be lower in scale. The seasonally wet grassland habitat to the north of Tse Koo Hang is already fragmented

and disturbed by the existing Border Road, therefore additional impact to this habitat resulting from construction of at-grade sections of the ECR are anticipated to be of low scale only.

Fragmentation of Woodland Habitats

8.8.22 The woodland habitat to the north of Tse Koo Hang straddles the existing Border Road and is fragmented by this road and other developed areas. Similar to the fragmentation impact at the nearby seasonally wet grassland habitat, additional fragmentation impacts to these woodland habitats are expected with the widening of existing road for the ECR Project. The scale of this impact is anticipated to be low, considering the area is already subject to fragmentation and disturbance from regular human activities and traffic. Furthermore, major construction works will not be conducted along the whole alignment simultaneously, limiting the scale and extent of any fragmentation impacts. The other woodland habitats recorded would not be affected by fragmentation impacts.

Direct Impact on Species of Conservation Importance and Associated Habitats

Flora

8.8.23 Only one flora species of conservation importance, Incense Tree, was recorded within the Construction Area of the Project during previous and current surveys. Four individual incense trees were recorded in the developed area and woodland habitat to the north of Tse Koo Hang during recent surveys, inside the Construction Area of the Project, which would be subject to potential direct loss. Impacts to this species are regarded as a minor to moderate in significance if no mitigation measures are implemented. Suitable measures such as transplantation are discussed in **Section 8.11.29.**

Avifauna

8.8.24 A total of 15 avifauna species of conservation importance were recorded within the Construction Area of the Project during recent surveys. The majority of these species were wetland-dependent species recorded from ponds, with some species recorded from other wetland habitats (i.e. seasonally wet grassland, modified watercourse, semi-natural / natural watercourse and marsh) and non-wetland habitat (i.e. woodland). Considering the high mobility of avifauna species, it is anticipated that these species would be displaced and utilise nearby similar habitats during construction, therefore no major direct injury/mortality impacts are anticipated. Considering the general low avifauna usage at all habitats within the Construction Area of the Project but high ecological linkage to adjacent wetland habitats, the impact of loss of foraging ground is considered a minor to moderate impact with no mitigation in place.

8.8.25 Breeding adult avifauna, chicks and eggs are more vulnerable to construction works and higher risk of direct injury / mortality. Although no breeding activities were observed during the current survey, precautionary measures such as pre-construction survey should be carried out within and adjacent to the Construction Area of the Project, as well as proposed enhancement areas, to avoid any potential direct injuries to breeding pairs, chicks or eggs of avifauna species of conservation importance.

Mammals

8.8.26 A total of eight mammal species of conservation importance were recorded within the Construction Area of the Project during recent surveys. The bat species Japanese Pipistrelle was recorded from various habitats, including wetlands (e.g. pond, compensatory wetland (the EA), marsh, modified watercourse, semi-natural / natural watercourse and seasonally wet grassland) as well as non-wetland habitats (e.g. orchard, plantation and developed area). The Short-nosed Fruit Bat was recorded exclusively in non-wetland habitats. Other bat species such as Horsfield's Myotis, Lesser Bamboo Bat and Unknown Vespertilionidae Species 1 were recorded in developed area and modified watercourse. In addition, Pallas's Squirrel was observed within developed area and village / orchard at Ma Tso Lung San Tsuen. Both Small Indian Civet and Leopard Cat were captured by camera traps from woodland habitats and the compensatory wetland (e.g. the EA). No notable roosting sites of bats were recorded within the Assessment Area during the current survey. Considering that mammal species are highly mobile, direct impacts to these species are not anticipated, but associated impacts including the loss and fragmentation of habitat and foraging ground are considered to be of moderate significance.

8.8.27 Eurasian Otter was not recorded during recent surveys, but there are recent records from the Assessment Area and vicinity (AECOM, 2021; Ove Arup, 2013a) and McMillan *et al*. (2022) suggested that wetland areas within the Assessment Area were part of the core range for Hong Kongs otter population. Recent unpublished surveys by KFBG (2024), recorded an otter spraint from ponds in HHW within the current Assessment Area. Further recent records (from unpublished AFCD territory-wide long term monitoring surveys) include camera trap records of otters in the Liu Pok Area from 2011-2014 and OWCA Area 2 in 2024 (indicated in **60722948/A10/FIGURE 8.7** as Areas A and B respectively). In this context, it can be assumed that Eurasian Otters are potentially using parts of the Construction Area of the Project (i.e., the area to the south of the Loop) as a movement corridor and/or foraging habitat. As with other mammal species, otters are highly mobile, so direct impacts during the construction phase are not considered a major concern, unless there are any future occurrences of Eurasian Otter breeding within the Construction Area of the Project. Other direct impacts in the construction phase, particularly habitat fragmentation, are considered of moderate to high significance. Mitigation measures such as pre-construction survey for otter holts and natal dens will be conducted prior to the commencement of construction works to avoid any direct impacts on the Eurasian Otter. Details of mitigation measures are provided in **Sections 8.11.26** and **8.11.32.**

Herpetofauna

8.8.28 One reptile species of conservation importance (Four-clawed Gecko) was recorded from developed area habitat within the Construction Area the Project. As this species is known to be adapted to a wide range of habitat types (woodland, rocky, and built environments) in Hong Kong (AFCD, 2006), and considering the high mobility of this species and availability of simar habitats, direct impact to Four-clawed Gecko are considered to be of minor significance. One amphibian species of conservation importance (Chinese Bullfrog) was recorded from the upper section of Ping Hang Stream near the ECR Alignment and adjacent to Project Boundary in the EA, with both of these locations subject to direct impacts. Given that this species is highly dependent on wetlands throughout its lifecycle, and considering its relatively low mobility, the diversion of Ping Hang stream and construction works at the EA may have direct impact on this species. The potential impact on Chinese Bullfrog is therefore considered to be of minor to moderate significance in the absence of

mitigation measures. Mitigation measures such as pre-construction survey and translocation will be conducted to avoid direct impacts to Chinese Bullfrog. Details of these measures are provided in **Section 8.11.33**.

Butterfly

8.8.29 A total of three butterfly species of conservation importance (Red-breast Jezebel, and Metallic Cerulean and Paintbrush Swift) were recorded from woodland habitat within the Construction Area of the Project. Considering similar habitats are available in the vicinity (i.e. woodland habitat to the south of the Project Alignment) and butterflies are highly mobile, direct injury and mortality is not anticipated, while direct loss of associated habitat for Red-breast Jezebel, Metallic Cerulean and Paintbrush Swift would be of minor impact significance.

Odonate

8.8.30 A total of six odonate species of conservation importance were recorded within / adjacent to the Construction Area of the Project from literature review, with none recorded during the recent surveys. Although two odonate species of conservation importance (i.e., Blue Sprite, Ruby Darter) were recorded in the EA within the Project Boundary, and the wetland habitats within and adjacent to the Construction Area of the Project are considered potentially breeding / nursery ground for these species, no breeding and nursing behaviors (no nymph of the species) were recorded from recent surveys and the literature review. Given the small, localised and temporary nature of the construction works compared to the overall available wetland habitat in the Assessment Area, and that the low diversity and abundance of odonate species were recorded, as well as the high mobility of adult odonates, direct injury and mortality on these species is not anticipated. The direct loss of associated habitat or potential breeding ground for odonate would be of minor impact significance.

Firefly

8.8.31 No firefly species of conservation importance were recorded within the Construction Area of the Project, therefore no direct injury and mortality to firefly species of conservation importance is anticipated. Firefly species Motschulsky's Starworm, was recorded at seasonally wet grassland close to Ma Tso Lung Village. This species is also known to utilise human modified habitats such as developed areas and plantation (Yiu, 2023). There would be direct loss of these types of habitats under the current Project, but given their common occurrence and otherwise low ecological value, the loss of these potential habitats for fireflies is considered of minor impact significance

Freshwater Fauna

8.8.32 A total of two aquatic faunal species of conservation importance were recorded from recent surveys, including three individuals of *Somanniathelphusa zanklon* were recorded in upstream Ping Hang Stream and two individuals of *Gobiopterus macrolepis* were recorded in the riparian zone of the LMC Meander, which one record of *S. zanklon* was recorded within Construction Area of the Project. The construction works in the LMC Meander and diversion of Ping Hang stream may have direct impacts on these species given the high dependent on the wetlands, and considered of low mobility, it is therefore considered of minor to moderate impact if no mitigation measures are implemented. Measures such as pre-construction survey and

translocation will be conducted prior to avoid any direct impacts to these species. Details of mitigation measures are provided in **Section 8.11.36**.

Direct Injury / Mortality on Other Wildlife

8.8.33 Aside from the species described in the previous sections, construction activities may cause potential direct injury/mortality to other wildlife species found within the Construction Area of the Project. Recorded species that are highly mobile are anticipated to migrate to similar nearby habitats progressively during construction phase. Whereas other slow-moving or wetland-dependent species recorded within the Construction Area of the Project were mostly common and widespread in Hong Kong. Predicted impacts in the absence of mitigation measures are considered to be minor to moderate in significance.

Bird Collision

8.8.34 Construction activities may cause potential injury / mortality impacts to birds in flight, particularly for waterbirds using the HHW Flight Corridor and other flight paths along the EA and HHW area. Construction structures, heavy machinery (e.g. cranes), materials that are difficult to see (e.g. cables, wires) and night-time lighting would be of particular concern. Records of birds using the major flight corridors were generally 15 - 25m above ground level or higher (**Sections 8.5.14**, **8.6.27** to **8.6.29** refers). No construction activities would be undertaken at 15m above existing ground level or higher along the flightpaths, so general construction activities (as presented in **Appendix 2.1**) are unlikely to pose a significant mortality/injury risk to avifauna. Nevertheless, a small proportion of birds (less than 5% usage) were recorded flying below 15m near LMC Meander and the EA according to the EM&A programme under the LMC Loop EIA (CEDD, 2023a). Therefore, the potential impact is considered of minor to moderate in significance only.

8.8.35 The majority of ardeids utilising the seasonal active night roost near Ma Tso Lung Road as illustrated in **60722948/A10/FIGURE 8.8** were observed flying northwards to ponds in the HHW Area, and therefore no significant collision risks are expected during the construction phase.

Construction Phase – Indirect Impacts

Disturbance Impact to Recognised Sites of Conservation Importance

8.8.36 Temporary disturbance from noise, glare, dust, vibration and general human activity could result in indirect impact to areas adjacent to the Project works areas during the construction phase, including recognised sites of conservation importance and ecologically sensitive areas.

8.8.37 The Construction Area of the Project will occur in the vicinity of recognised sites of conservation importance, including proposed HHW WCP, WCA, WBA, “CA” zone and Priority Site for Enhanced Conservation, which largely comprise wetland habitats. Although these sites are already subject to disturbance such as traffic along the existing Border Road and existing pond fish culture production; construction activities of the ECR Project would result in additional disturbance impacts. This is considered a moderate to high impact if unmitigated. Particularly, wetland dependent avifauna and other wildlife using habitats adjacent to the Construction Area of the Project would likely be displaced by disturbance, decreasing the carrying capacity of habitats

in the area. Mitigation measures would therefore be required to minimise and compensate for potential disturbance impacts.

Disturbance Impact to Ecologically Sensitive Resources

8.8.38 The other ecologically sensitive areas in the vicinity of the Construction Area of the Project include the IBA, the EA, the OWCA Area 2, the off-site compensation woodland and key dragonfly site in Ma Tso Lung. Since the IBA largely overlaps with the "CA" zones, WCA and WBA, the disturbance impact is considered similar to the sites of conservation importance discussed in **Sections 8.8.36** to **8.8.37**. A portion of the EA falls within Construction Area of the Project, and disturbance impacts from construction works could affect adjacent areas of the EA. As the EA is known to support various avifauna species of conservation importance and nocturnal wildlife, this is considered a moderate to high scale impact if unmitigated. The OWCA Area 2 is located more than 180m from the Construction Area of the Project, and is separated by existing developed area within the Loop. Hence, the disturbance impacts to these Areas are anticipated to be minor. While the specific boundary of key dragonfly site in Ma Tso Lung is not delineated, odonates are not considered particularly disturbance sensitive, therefore indirect impacts to the key dragonfly site in Ma Tso Lung are considered minor in scale. Construction activities would be more than 130 m from the off-site compensation woodland, and this area is already subject to disturbance from the existing border road, therefore, no additional disturbance impacts to the off-site compensation woodland are anticipated.

Disturbance and Obstruction of Flight Path

8.8.39 Construction activities and tall machinery could potentially obstruct/disrupt the major flight paths in the area to the east of the Loop, including the HHW Flight Corridor. Movement of waterbirds between their roosting / breeding sites and foraging grounds could be hindered if the birds adopt a higher flight path, or adjust the flight path alignment, to avoid disturbance from construction activities. This would result in increased energy expenditure and reduced foraging efficiency. This potential disturbance impact is anticipated to be moderate without mitigation. Mitigation measures such as scheduling construction activities to avoid simultaneous works within sensitive wetland areas (i.e. Zone 1, Zone 2 and Zone 3 as illustrated in **60722948/A10/FIGURE 3.2b**) and construction time restriction, would be implemented as far as practicable to minimise this impact.

Disturbance Impact to Night Roost

8.8.40 The seasonally active night roost recorded from plantation habitat at the northern end of Ma Tso Lung Road will potentially be indirectly impacted by construction disturbance. The roost is within 100m northeast of the Construction Area of the Project. Although roosting birds can be sensitive to disturbance, the roost location is adjacent to Ma Tso Lung Road, suggesting some degree of disturbance tolerance, therefore the potential disturbance impact is anticipated to minor in significance. The adoption of precautionary measures such as construction time restriction on above-ground construction works, potential disturbance to the roosting ardeids (described in **Section 8.11.40**) will further minimise potential impacts.

Disturbance to Habitats and Associated Wildlife

8.8.41 During the construction phase, disturbance from traffic, noise, glare, for dust, vibration and general human activities would increase, which may degrade the quality of

nearby habitats and result in the displacement of fauna. Construction activities carried out at or above existing ground level (such as silent piling by press-in method, cofferdam construction, movement/using of construction machinery) will be visible and audible from the surrounding environment and therefore more likely to disturb waterbirds and other wetland-dependent species, particularly sensitive habitats in the EA, LMC Meander and Ponds 36 - 38 adjacent to the Construction Area of the Project as illustrated in **60722948/A10/FIGURE 8.9**. Underground construction of the depressed road and underpass, which will be undertaken using a cut-and-cover method, will occur largely below ground once construction of cofferdams is completed and the excavation has reached the final depth, with all designed struts and supports installed. Excavation, strut installation, and structural works will be undertaken within fully enclosed conditions, which significantly reducing the potential disturbance to nearby wetlands and wildlife. Moreover, the noise disturbance would be managed through established noise and vibration control measures, including use of QPME and moveable noise barriers/enclosures.

Extent of Disturbance Impacts

8.8.42 During construction of the Project, disturbance to surrounding habitats is anticipated mainly in the form of noise from silent piling by press-in method, excavation and machinery, as well as visual disturbance. Such activities are expected to influence the use of adjacent habitats by disturbance-sensitive fauna, particularly large waterbirds and mammals.

8.8.43 The severity of impacts will vary across species groups. Large waterbirds such as herons, egrets and cormorants are considered particularly sensitive and are likely to avoid disturbed areas (Yang *et al*, 2021; Ove, 2013a). Long term monitoring data of waterbirds from Mai Po Inner Deep Bay Ramsar Site Waterbird Monitoring Programme submitted by HKBWS to AFCD and EIA field survey results were reviewed; records indicate that several large water bird species have been observed using Ponds 36-41 (**60722948/A10/FIGURE 8.2**). The ECR Alignment would traverse Ponds 36-38 directly, while Ponds 39-41 would fall within the disturbance zone associated with the Construction Area of the Project.

8.8.44 Considering there are a number of disturbance sensitive species recorded from habitats in the vicinity of the Construction Area of the Project, a quantitative estimate of the potential extent of disturbance impacts has been developed adopting the following criteria:

- Exclusion Zone (EZ): Close to the source, disturbance levels are high enough to prevent disturbance sensitive species from using habitats in the zone, even if they would provide suitable habitat for the species in the absence of disturbance.

- Reduced Density Zone (RDZ): Further from the source, disturbance sensitive species would use suitable habitats but likely at a reduced density compared to undisturbed habitats. In this zone, it is assumed 0% usage of the habitat closest to the disturbance source, and 100% usage furthest from the source. Overall, usage in the reduced density zone is therefore considered to be 50% of that in undisturbed habitats.

8.8.45 Previous EIA reports in the NWNT including the LMC Loop EIA (AEIAR-176/2013), LMC Spur Line EIA (AEIAR-052/2002) and STLMC DN EIA (AEIAR-261/2024) have adopted similar approaches in assessing disturbance impacts, assuming EZ and

RDZ will be created around the disturbance source (KCRC, 2002; CEDD & PlanD, 2013a; CEDD & PlanD, 2024). Calculating the EZs and RDZs for larger, and commonly occurring disturbance sensitive waterbirds species (such as Great Cormorant, Great Egret, Grey Heron and Black-faced Spoonbill) will effectively delineate the maximum extent of ecological disturbance resulting from the Project (i.e. it can be assumed that indirect impacts to other less disturbance sensitive species would not extend beyond these zones). Raptor species such as Greater Spotted Eagle, Imperial Eagle, Osprey and Crested Serpent Eagle were also recorded occurring within the Assessment Area from literature review and recent surveys. While these species are considered very sensitive to disturbance, the abundance of these raptors in the Assessment Area is extremely low (varying from 0 to 3 from recent surveys). Moreover, records of these species are often closely associated with prey availability rather than habitat area *per se*. For example, monitoring data from the LMC EEA has indicated presence of multiple Imperial and Greater Spotted Eagles within a relatively small area of enhanced wetland (MTR, 2024), indicating that prey availability is a key driver for the distribution and abundance of these species. Therefore, adopting these raptor species (that occur in very low abundance and often distributed according to prey availability) as indicators for calculating EZs and RDZs is not representative of disturbance impacts to wetland communities more broadly.

8.8.46 The extent of EZ and RDZ adopted in previous EIA's differs from Project to Project. For example, the San Tin Technopole EIA (CEDD & PlanD, 2024) adopts various zoning distances, up to a 0-200m EZ and 200-400m RDZ in relation to the construction of high-rise buildings. This reflects the extensive disturbance expected from high-rise construction works, as well as the very large scale of the Project and high value of the affected wetlands. In contrast, the Loop adopts an EZ of 0-50m and a RDZ of 50-100m for at-grade roads. These zones were considered suitable given the lower overall disturbance expected from at-grade works. For the current project, an EZ of 0-50m and a RDZ of 50-100m will also be adopted for wetland habitats potentially supporting disturbance sensitive species. This is considered appropriate given that the majority of the ECR alignment running through/adjacent to wetland habitats will be either at-grade or underground. Indeed, this approach to disturbance zoning is considered conservative given that construction activities within the wetland areas would primarily occur below existing ground level, and most of the working areas would be covered within excavation cofferdam during the construction phase as described in **Section 8.8.41**. Therefore, the potential disturbance to avifauna from construction (i.e. noise, vibration, lighting and human presence) is expected to be lower than that associated with typical surface construction works. The use of hoarding, quieter piling methods (e.g. silent piling by press-in method), restrictions on tall plant machinery, and non-percussive construction methods will further reduce the disturbance magnitude. As also noted in the LMC Loop EIA, animals and birds generally habituate more readily to constant, non-explosive noise and are less affected by non-visual forms of disturbance.

8.8.47 The proposed ECR Alignment comprises depressed road, underpass, at-grade carriageway, and viaduct sections, connecting the Loop and the Kwu Tung North New Development Area (KTN NDA) via the Ma Tso Lung area, as shown in **60722948/A10/FIGURE 8.13**. The extent of the disturbance zone for the construction of the viaduct section at Ma Tso Lung is considered comparable to that of the at-grade section, as:

- The adjacent habitats that support potentially disturbance sensitive species (e.g. Pond 93) are already subject to disturbance from existing human activities and traffic volume. According to long-term monitoring data from the HKBWS and

recent ecological survey, no large waterbirds have been recorded in the Pond 93.

- The viaduct section of the ECR Alignment is located along the periphery of the core wetland areas of the HHW wetland system.

- The height of the viaduct section is limited. The viaduct section rises only approximately 8m above ground level at its staring point from the at-grade road and does not exceed 35m in height as it extends behind the Ma Tso Lung San Tsuen to join the adjacent hillside.

8.8.48 The extent of disturbance to habitats has been assessed based on these maximum distances, as summarised in **Table 8.25** and illustrated in **60722948/A10/FIGURE 8.9**. The EZ and RDZ is anticipated to support 0% and 50% of disturbance sensitive species respectively during the construction phase.

Table 8.25 Extent of Disturbance Impact on Disturbance Sensitive Wetland Species during Construction Phase

Phase / Construction Type	Extent of Disturbance Impact	
	Exclusion Zone	Reduced Density Zone
Construction Phase		
Vehicular Road	0 to 50m	50 to 100m

Notes:
1. Exclusion Zone is close to the disturbance source, which has high disturbance level to prevent species from using wetland habitats, even with the presence of suitable habitat (assuming 0% usage)
2. Reduced Density Zone is further from the disturbance source, therefore species would use suitable wetland habitats in this zone, but likely at a lower density compared to undisturbed wetland habitat (assuming 50% usage)

Disturbance to Pond Habitat

8.8.49 Pond habitats and associated wildlife would be subject to disturbance impacts from construction activities. Considering the high ecological value of the ponds and the importance of this habitat as a feeding resource and ecological corridor for wetland avifauna and Eurasian Otters, the disturbance impact on pond habitats in the area to the east of the Loop and in the HHW area, as well as the associated wildlife, is anticipated to be moderate to high in significance without mitigation, while the carrying capacity of the pond habitat is also anticipated to decrease during construction phase. The single pond at Ma Tso Lung is surrounded by village habitat and supports limited wildlife with only occasional records of individual wetland-dependent birds (e.g. Little Grebe). The pond near Ping Hang Stream is isolated from other wetland habitats in HHW and is observed be covered with vegetation and support limited wildlife use. The other ponds are isolated from other natural habitat, being surrounded by the adjacent village setting and are subject to disturbance from nearby human activities. Therefore, disturbance impacts to these ponds are considered to be of minor significance.

Disturbance to Compensatory Wetlands

8.8.50 Construction works at the EA would lead to disturbance of sensitive species including Greater Painted-snipe, Eurasian Otter, Leopard Cat, and other nocturnal species of conservation importance within man-made reed marsh at the eastern portion of the EA. ECR Construction works near the EA are tentatively scheduled to commence in Q1 / Q2 2027. The disturbance would primarily arise from construction works

including traffic, lighting, noise, dust, vibration, and human presence. Without mitigation, the disturbance impacts on the partition of the EA to the east of the ECR Alignment are anticipated to be moderate to high in significance. The detail of construction works near / in the EA (including but not limited to working schedule/period, construction method, reinstatement method, use of equipment, type and route of vehicle entry and water supply arrangements) will be carefully designed and incorporated during the detailed design stage to minimise the disturbance impacts arising from the construction works. Standard precautionary measures and other environmental mitigation measures for noise, air and glare as presented in **Sections 8.11.42** to **8.11.49** will be in place to address the potential disturbance impacts on the EA during both the construction and operational phases. The remaining marsh habitat within the EA (i.e. western marsh habitat outside of RDZ) would not subject to significant disturbance by the works. Reinstatement of the EA Zone following completion of the depressed road / underpass sections of the ECR Alignment will be reinstated and established by the Project Proponent / the Contractor with reference to the approved Ecological Mitigation / Habitat Creation and Management Plan (CEDD, 2022) and the existing as-built conditions of the EA after the completion of the aboveground works. For underground reinstatement, all existing engineering and associated structures removed during construction will be reinstated upon completion. After construction of the underpass, the excavation cofferdam will be backfilled with compacted materials (compacted fill or rockfill, subject to the Contractor's design) in accordance with the requirements of the General Specification for Civil Engineering Works published by CEDD (CEDD, 2020). As the stone columns within the construction area will no longer be required to support the EA and its earth bund, the relevant geotextile layers and provisions will be reinstated to ensure that the original level of impermeability is maintained, and native plant species will be used for reinstatement. The ecological characteristics and function of the EA will be enhanced and maintained to a condition no less than the existing conditions, after the reinstatement. A reinstatement and monitoring plan for the EA would be prepared by the Contractor and submitted to AFCD and EPD for agreement at least three months prior to commencement of construction works.

8.8.51 Other compensatory wetland in the Assessment Area (i.e., OWCA Area 2) are distant (>180 m) from the Construction Area of the Project, and separated by existing developed areas within the Loop. Hence, the disturbance impacts to these habitats are anticipated to be minor in significance.

Disturbance to Watercourse Habitat (LMC Meander)

8.8.52 Disturbance impacts to the LMC Meander are anticipated to be of moderate to high significance considering the LMC meander had previous records of Eurasian Otter, and it is also an ecological corridor for waterbirds and mammals.

Disturbance to Other Wetland Habitats

8.8.53 Apart from the LMC Meander, the remaining watercourses in the Assessment Area are relatively small and are subject to some level of disturbance already, whereas the section of Ma Tso Lung Stream within the Construction Area of the Project is channelised and located in a disturbed area. Only very low occurrence of disturbance sensitive species was observed from these habitats during recent surveys. Given that the viaduct section of the ECR Alignment would be elevated approximately 20m above the affected watercourses, the riparian vegetation along or adjacent to the watercourse would not be significantly affected by the shadow effect of the viaduct. Therefore, the potential disturbance impact is considered to be insignificant.

8.8.54 Other wetland habitats including marsh and seasonally wet grassland in the HHW area are located along the Construction Area of the Project and would also be subject to construction disturbance. The affected marsh and seasonally wet grassland areas support a relatively low diversity of species, which are generally tolerant to disturbance (CEDD & PlanD, 2024) with wetland avifauna recorded during recent surveys typically being less disturbance sensitive species such as White-breasted Waterhen and White-throated Kingfisher. Only a few records of other large, disturbance sensitive species (e.g. Grey Heron and Great Egret) were made from these affected habitats compared to open water habitat such as pond. According to long-term avifaunal monitoring results of EM&A programme for the WCR (CEDD, 2023a), Pond 12 indicated in **60722948/A10/FIGURE 8.8** has changed from pond to marsh as observed before commencement of the WCR works. Under the updated habitat condition, the avifaunal assemblage recorded from 62 survey events from March 2022 to August 2023 indicates that the pond is now supports reed marsh-dependent species (e.g. Yellow Bittern, White-breasted Waterhen), along with other common and widespread terrestrial species. No significant impact on the bird usage of Pond 12 was observed arising from the WCR construction activities and its proximity based on EM&A monitoring data. The avifaunal species recorded within the marsh and seasonally wet grassland habitat from the ecological surveys of the ECR Project are consistent with the species composition observed in Pond 12. Therefore, disturbance impacts on seasonally wet grassland and marsh are considered to be of low significance.

Disturbance to Other Non-Wetland Habitats

8.8.55 Non-wetland habitats recorded in the vicinity of the Construction Area of the Project, mostly comprise shrubland / grassland, plantation and woodland. These habitats are located on both sides of the existing Border Road, with a large patch of woodland habitat located to the north of Tse Koo Hang. Although these habitats are generally of low or low to moderate ecological value (apart from the moderate ecological value woodland), several individuals of Incense Trees, and some mammal and butterfly species of conservation importance were recorded in the woodland habitat to the north of Tse Koo Hang. Dust and pollutants generated during construction phase could deposit on leaves and affect growth of vegetation, in turn degrading the habitats. Construction activities would also bring disturbance to these habitats and associated wildlife. However, given these habitats are already subject to disturbance from traffic and other human activities, and that the fauna species recorded are generally mobile and recorded in other areas further from the ECR Alignment. The woodland peripheries are expected to function as natural buffers that help shield the core woodland areas from construction-related disturbance. Good site practices and standard mitigation measures, including the erection of hoardings around active works areas, will be implemented to further reduce disturbance to adjacent non-wetland habitats. Therefore, the potential construction disturbance impact to non-wetland habitats is anticipated to be of minor significance.

8.8.56 Breeding and nesting birds may be more sensitive to disturbance than at other times. Eurasian Eagle Owl is known to occur in the grassland/shrubland habitats in the hilly areas between HHW and Tse Koo Hang, and potentially breeds in these habitats. However, overseas studies indicate the home range of this species is very large, up to 25km^2 (Kang C *et al*, 2013), therefore potential disturbance impacts are considered to be of low significance as affected habitats are likely to form only a very small fringe of the species total range. Chinese Francolin and White-throated Kingfisher have also been observed exhibited breeding or nesting behaviours within the hillside grassland / shrubland margin habitats. Construction disturbance could displace breeding pairs

or reduce habitat suitability in adjacent areas. However, considering these habitats are subject to some level of disturbance from traffic and other human activities, the construction disturbance impact is anticipated to be minor. Nevertheless, as a precautionary measure, pre-construction nesting surveys for avifaunal species of conservation importance would be implemented to minimise impacts to breeding birds.

8.8.57 Disturbance impacts to other habitats that lie adjacent to the Construction Area of the Project such as orchard, plantation and developed area are anticipated to be of minor significance. These habitats do not support disturbance sensitive species and are already subject to existing disturbance.

Night-time Disturbance

8.8.58 Uncontrolled night-time lighting and glare from the construction sites has the potential to impact the light-sensitive cycles of roosting avifauna and activity of nocturnal fauna (e.g. Eurasian Otters, bats, herpetofauna and firefly) in the vicinity, potentially affecting migration, foraging and breeding success of these species. Overall, these species are already subject to minor nocturnal disturbance from nearby traffic and human activities at the nearby the Loop, intense artificial lights from construction sites would pose greater disturbance to these species and associated habitats, which is of moderate impact if unmitigated. Appropriate measures would be adopted to minimise night-time glare impacts.

8.8.59 Firefly species Motschulsky's Starworm, which was recorded at seasonally wet grassland close to Ma Tso Lung Village, breeding behaviour may potentially be impacted by the glare generated from the Construction Area of the Project along the hillside. Considering that there is a considerable distance (120 m) from the Construction Area of the Project, and that the species is more commonly encountered in disturbed areas such as village (or even in urban parks) compared to the wild (Yiu, 2023), it suggested that the species is tolerant to some level of disturbance and night-time lighting. It is therefore considered to have minor significance of indirect impacts on this species of conservation importance from glare/lighting generated from the Project.

Water Quality and Hydrological Impact

8.8.60 An underpass road will be constructed beneath the EA, LMC Meander and Pond 36-38 using the cut-and-cover method, requiring excavation works in these areas during construction. These works could result in the release of sediments into adjacent waterbodies. Sediment particles may cause physical injury to aquatic fauna and potentially clog the respiratory systems of fish and invertebrates. In addition, the potential discharge of pollutants or chemicals could lead to bioaccumulation and biomagnification within the Deep Bay ecosystem, thereby posing risks to waterbirds and other aquatic communities. Potential impacts on water quality are considered moderate in significance if unmitigated.

8.8.61 Prior to the commencement of works, all water in ponds and wetland areas inside the Construction Area of the Project Construction would need to be drained. This drained water could contain sediments and/or other pollutants (i.e., nutrients), so direct discharge of these drained waters to nearby watercourses will therefore be prohibited. The drained waters will be temporarily stored for on-site reuse as far as practicable, with any surplus tankered to a Sewage Treatment Works for proper disposal in compliance with TM-DSS. Moreover, dewatering will be carried out in the dry season

(i.e. Nov to Mar) to minimise water volume. Excavated materials and sediments will be managed and disposed of according to recommended mitigation measures to control water quality impacts, and with these measures implemented, no adverse water quality is anticipated.

8.8.62 To separate the construction works from the EA, LMC Meander and Ponds 36 - 38, cofferdams will be used to enclose the excavation works as described in **Chapter 2**. The construction of underpass road across the EA, LMC Meander and Pond 36-38 will be conducted by phases with only a single cofferdam installed at each stage. Restricting works to within the cofferdam will minimise potential water quality impacts. Since the cross-section area of the LMC Meander and Pond 36-38 to be enclosed by the cofferdams is anticipated to small and temporary, the change in flow regime would be localised and any resultant secondary impacts to water quality would be transient. As a result, the potential impact on hydrology of the LMC Meander and Ponds 36-38 would be insignificant.

8.8.63 In addition, as the Loop is currently under development and is tentatively scheduled for completion by 2035, the main water sources to feed the EA will comprise direct rainfall and water pumped from LMC Meander during construction phase of the ECR Project. According to CEDD (2022), LMC Meander provides a reliable water source sufficient to maintain water levels within the five cells of the EA during the dry season, (i.e. Nov to Mar), therefore the unmitigated impact on the remaining partitions would therefore be insignificant. However, a small portion (0.38 ha) of the easternmost partition of the EA will be temporarily affected by excavation works under the Project. This would interrupt the gravity-fed flow from the eastern partition to the larger portion of the EA, resulting in an unmitigated impact of moderate significance to the remaining parts within the affected eastern partition but outside the Construction Area. According to the construction programme provided in **Appendix 2.2**, works within Zone 1 (i.e. the EA zone and the LMC Meander) will be carried out first. Reinstatement works within Zone 1 will be completed prior to the commencement of works in Zone 2 (i.e. the remaining sections of the LMC Meander and ponds areas). As such, water flow within the LMC Meander will be maintained during the construction phase, ensuring that water of suitable quality will remain available for supplemental supply to the EA when required. During the dry season, (i.e. Nov to Mar), regular inspection of water depths and water quality of marsh and reed marsh habitats within the EA will be carried out by the Contractor. If the water level of the EA reaches the action water level as specified in Table 5.1 of CEDD (2022), the Contractor will use portable pumps to provide supplemental water supply from LMC Meander to maintain adequate water levels in the remaining parts of the EA, thereby sustaining the reedbed and freshwater marsh habitats. In case of non-compliance of water quality within the EA specified in Table 8.1 of CEDD (2022), appropriate water pollution control measures (e.g., emergency bypass mechanism) shall be carried out by the contractor to prevent polluted water entering the EA. Before pumping the water from LMC Meander to the EA, the Contractor will test water quality of LMC Meander, ensuring that the supplementary water quality is not inferior to the action levels of water quality specified for the EA, as referenced of the HCMP (CEDD, 2022). If the water quality of the LMC Meander fail to meet the relevant action levels, the Contractor shall take appropriate measures, such as water treatment, to achieve compliance before pumping water into the EA, and / or identify alternative water sources including the addition of rainwater harvesting systems, to ensure that the water to be supplied to the EA meets the required quality standards

8.8.64 Other watercourses (e.g. downstream channelised section of Ma Tso Lung stream and small portion of stream near Lo Wu Range) as well as ponds were identified

within/adjacent to works areas that could potentially be affected by pollution due to the release of construction site-runoff, accidental discharge and spillage. The potential impact on water quality is considered to be minor to moderate in significance if unmitigated. Given these impacts could be controlled by implementing standard good site practices and mitigation measures, no significant adverse water quality impact during construction phase is anticipated, as described in **Chapter 5**. Regular site inspections and water quality monitoring should be undertaken routinely to inspect the construction activities and works area to ensure the recommended measures are proper implemented.

8.8.65 Temporary diversion of a short section of Ping Hang Stream will be required during cut-and-cover construction of underpass sections. The Stream would be reinstated following the completion of construction works, with revitalisation works (e.g. landscaping along riparian zone) considered during the detailed design stage. The riparian vegetation removed along the LMC Meander would also be reinstated after the completion of construction works. Overall, no adverse hydrodynamic impacts (i.e., impacts to water flow hydraulics or flow regime) to the Ping Hang Stream is anticipated.

Impacts Due to Groundwater Drawdown

8.8.66 Cut-and-cover construction method is considered as a reliable and controllable approach and recommended to be adopted as described in **Sections 2.6.3** to **2.6.6**. Sheet pile walls, which have a good performance in water cut-off, would be used for the construction of the depressed road and underpass. Groundwater monitoring outside of the excavation cofferdam would be implemented to provide an indicator of potentially excessive groundwater drawdown or unexpected level changes. A recharge well(s) for groundwater recharge would also be implemented if excessive groundwater drawdown is identified. After the completion of the underpass, the area excavated for the cofferdam would be reinstated and backfilled with inert C&D materials. Therefore, the permanent groundwater drawdown is not expected and the risk of secondary impacts to adjacent wetlands are considered to be very low. With the implementation of the recommended groundwater control and mitigation measures described in **Chapter 5**, potential groundwater drawdown would be minimised and are considered as insignificant impact.

Operation Phase – Direct Impacts

Direct Impact on Recognised Sites of Conservation Importance

8.8.67 Along the ECR Alignment, there are five recognised sites of conservation importance (proposed HHW WCP, WCA, WBA, “CA” zone and Priority Site for Enhanced Conservation) and other ecologically sensitive resource including IBA and key dragonfly site in Ma Tso Lung. The scale of operation phase direct impact and permanent habitat loss to these recognised sites of conservation importance and sensitive receivers would be smaller than during the ECR construction, as habitats along the underpass sections and temporary works areas would be reinstated following completion of the Project, as presented in **Table 8.23**. No direct impact on the OWCA Area 2 and off-site compensation woodland is anticipated. Reinstatement of the EA Zone following completion of the depressed road / underpass sections of the ECR Alignment will be reinstated and established by the Project Proponent / the Contractor with reference to the approved Ecological Mitigation / Habitat Creation and Management Plan (CEDD, 2022) and the existing as-built conditions of the EA after the completion of the aboveground works. For underground reinstatement, all existing

engineering and associated structures removed during construction will be reinstated upon completion. After construction of the underpass, the excavation cofferdam will be backfilled with compacted materials (compacted fill or rockfill, subject to the Contractor's design) in accordance with the requirements of the General Specification for Civil Engineering Works published by CEDD (CEDD, 2020). As the stone columns within the construction area will no longer be required to support the EA and its earth bund, the relevant geotextile layers and provisions will be reinstated to ensure that the original level of impermeability is maintained, and native plant species will be used for reinstatement. The ecological characteristics and function of the EA will be enhanced and maintained to a condition no less than the existing conditions, after the reinstatement. A reinstatement and monitoring plan for the EA would be prepared by the Contractor and submitted to AFCD and EPD for agreement at least three months prior to commencement of construction works. Following reinstatement, no direct impact to the EA is anticipated. Upon completion of the ECR Alignment, no further additional habitat loss on the sites of conservation importance is expected, thus no significant ecological impact is anticipated during the operation of this Project.

Direct Impact on Wetlands and Other Habitats

8.8.68 The scale of direct impacts to habitat would be smaller in the operation phase as compared to the construction phase, as habitats along the underpass sections of the Project Alignment (along with temporary works areas) would be reinstated. Upon completion of the ECR Alignment, no additional habitat loss on wetlands and other habitats is anticipated, it is therefore considered of no significant ecological impact during the operation of the Project.

Habitat Fragmentation

8.8.69 Operation phase wetland habitat fragmentation impacts have largely been addressed through the Project design: where the ECR Alignment passes through the sensitive EA, LMC Meander and pond areas as an underpass, thereby avoiding potential fragmentation to animals moving through this area, including Eurasian Otter and flight paths utilised by avifauna species.

8.8.70 Fragmentation to other habitats such as woodland and seasonally wet grassland habitats are considered minor to moderate in scale, as the affected habitats are already fragmented by the existing Border Road and subject to ongoing traffic volume. The proposed viaduct sections and the areas adjacent to Ma Tso Lung Shun Yee San Tsuen and Ma Tso Lung San Tsuen comprise mainly developed area, small patches of grassland/shrubland and plantation. Given these areas are not regularly utilised by wetland avifaunal species, the habitat fragmentation impact is anticipated to be of minor significance. Furthermore, terrestrial animals should be able to move beneath the viaduct structure, and therefore fragmentation impacts are expected to be minimal.

Direct Injury / Mortality to Wildlife and Bird Collision

8.8.71 The main potential source of operation phase direct injury / mortality to wildlife will be from collisions with vehicles using the above-ground sections of the ECR, particularly at-grade sections. In wetland areas, this risk has largely been addressed through the ECR Alignment design: where the ECR Alignment passes through the sensitive EA, LMC Meander and pond areas, it will be as an underpass road, thereby avoiding potential impacts to animals moving through this area, including Eurasian Otter. Without mitigation, impacts to wildlife crossing at-grade sections of the road (e.g.,

Leopard Cat moving between woodland and shrubland habitats at Tse Koo Hang to the south and wetland habitats in HHW to the north) is anticipated to be minor to moderate in scale.

8.8.72 Fixed noise barriers will be provided at the proposed E&M plant room and drainage pumping room, subject to the detailed design. These will be designed with bird-friendly features such as patterned stickers to minimise the risk of bird collisions, ensuring environmental compatibility. Given the small and localised locations with fixed noise barriers, insignificant impacts associated with bird collisions to noise barriers are anticipated.

Operation Phase – Indirect Impacts

Disturbance to Recognised Sites of Conservation Importance, Wetland and Terrestrial Habitats, and Associated Fauna Species of Conservation Importance

8.8.73 The Project could result in operation phase disturbance impacts (e.g. traffic noise, vehicular emission, dust, surface runoff and glare) to the recognised sites of conservation importance, nearby habitats (including wetland habitats like ponds, wet grassland and marsh, and terrestrial habitats like grassland/shrubland, woodland and plantation), and associated fauna species of conservation importance. The scale of operation-phase impact is expected to be minor and localised for the following reasons:-

- The ECR Alignment would be in an underpass as it passes through key sensitive wetland habitats (the EA, LMC Meander and pond habitats). The only minor source of indirect disturbance in this area would be noise from two small (50m x 23m) ventilation openings. Screening planting (e.g. trees/tall shrubs/tall grass) would be established around the ventilation openings subject to detailed design.

- Habitats adjacent to the planned at-grade sections of the ECR Alignment are already subject to some disturbance from the existing Border Road. Furthermore, wildlife using areas close regular sources of disturbance such as vehicular traffic can habituate to some extent to the disturbance, for example, avifauna utilising marsh and seasonally wet grassland habitats along the at-grade road section of ECR Alignment are generally tolerant to disturbance. In contrast, disturbance sensitive avifauna were observed to occur at a lower abundance in these habitat types compared to preferred habitats (e.g. Pond). These two factors together indicate that disturbance along the at-grade sections of the ECR Alignment would not be dissimilar to existing conditions along the Border Road.

- Operation of the viaduct would generate disturbance impacts on nearby wetland habitats, primarily through visual disturbance and traffic noise. The wetland habitats (e.g. Pond 93) located adjacent to the viaduct section are already subject to disturbance from existing human activities and prevailing traffic levels. Low noise road surface will be adopted for viaduct section located adjacent to Pond 93 and traversing the Ma Tso Lung San Tsuen, further reducing potential operation disturbance. Additional screening planting (e.g. trees/tall shrubs/tall grass) would be established along the at-grade road subject to detailed design.

- The ECR Alignment would be in an underpass as it passes through key sensitive wetland habitats (the EA, LMC Meander and pond habitats). The

design largely avoids potential light impacts to disturbance sensitive waterbirds and wetland-dependent species, fireflies and nocturnal wildlife. A minimum number of streetlights should be constructed along the at-grade and viaduct sections of the ECR to provide a safe amount of night-time illumination. Lighting should use a narrower spectral wavelength rather than white light. Furthermore, light sources should be shielded to direct lighting onto the road and minimise light trespass.

- The proposed E&M plant room and drainage pumping room is not expected to cause significant disturbance to the surrounding environment. This is because a range of direct noise mitigation measures would be incorporated into the design, including the use of quieter fixed plants, enclosure of equipment within reinforced concrete structures or acoustic housings, installation of silencers and acoustic louvres and noise barriers if necessary.

8.8.74 With mitigation measures such as control of night-time lighting and provision of screen planting are implemented, the indirect impact on sites of conservation importance would be further minimised during operation phase.

Disturbance to Compensatory Wetland

8.8.75 Details of construction works near / in the EA (including but not limited to working schedule/period, construction method, reinstatement method, use of equipment, type and route of vehicle entry and water supply arrangements) will developed during the detailed design stage. Reinstatement of the EA Zone following completion of the depressed road / underpass sections of the ECR Alignment will be reinstated and established by the Project Proponent / the Contractor with reference to the approved Ecological Mitigation / Habitat Creation and Management Plan (CEDD, 2022) and the existing as-built conditions of the EA after the completion of the aboveground works. For underground reinstatement, all existing engineering and associated structures removed during construction will be reinstated upon completion. After construction of the underpass, the excavation cofferdam will be backfilled with compacted materials (compacted fill or rockfill, subject to the Contractor's design) in accordance with the requirements of the General Specification for Civil Engineering Works published by CEDD (CEDD, 2020). As the stone columns within the construction area will no longer be required to support the EA and its earth bund, the relevant geotextile layers and provisions will be reinstated to ensure that the original level of impermeability is maintained, and native plant species will be used for reinstatement. The ecological characteristics and function of the EA will be enhanced and maintained to a condition no less than the existing conditions, after the reinstatement. A reinstatement and monitoring plan for the EA would be prepared by the Contractor and submitted to AFCD and EPD for agreement at least three months prior to commencement of construction works. Furthermore, the at-grade section to the north of the ECR Alignment would be located within the developed area under the Loop. It is surrounded by other human activities and existing buildings or structure. As such, the operation of this section of at-grade road is not anticipated to result in any additional disturbance to the EA.

Disturbance and Obstruction of Flight Path

8.8.76 Operation phase disturbance and obstruction of flight paths, particularly the HHW Flight Corridor and other major flight paths recorded to the vicinity of the ECR, are considered of minor significance for the following reasons:-

- The ECR Alignment would be an underpass as it crosses the LMC Meander and major HHW Flight Corridor.
- The viaduct sections are located relatively distant from the recorded major flight paths. The viaduct section adjacent to the Pond 93 is approximately 12 - 20 m above existing ground level, which is lower than the majority of recorded bird flight paths, most of which observed at elevations above 25m aboveground.

Water Quality and Hydrological Impact

8.8.77 The key potential water quality issue of the operation phase is surface runoff generated from the new paved areas. However, as stated in **Chapter 5**, water quality impacts associated with the operation of the Project are expected to be minimal and acceptable with the proper implementation of the planned drainage system.

8.8.78 As described in **Section 8.8.75**, the EA will be reinstated during operation phase. Hence, no adverse hydrodynamic and hydrological impacts are anticipated.

8.9 Evaluation of Potential Environmental Impacts

8.9.1 Potential ecological impacts to identified habitats within the Assessment Area associated with the construction and operation of the ECR Alignment have been evaluated in accordance with the Annex 8 of the EIAO-TM, and are presented in **Table 8.26** to **Table 8.39.**

Table 8.26 Evaluation of Potential Ecological Impacts to Compensatory Wetland

Criterion	Compensatory Wetland	
	Ecological Area (the EA)	**OWCA Area 2**
Habitat Quality	Moderate to High	Moderate to High
Species/Ecological Resources	Current Study: 37 fauna species of conservation importance were recorded Literature Review: 73 species of conservation importance were recorded Breeding ground recorded: No significant nursery or breeding ground known or observed during the ecological surveys The total number of species of conservation importance recorded: 78	
Size/Abundance	0.38 ha would be temporarily lost(1)	Habitat would not be subject to direct loss
Duration	Direct impact (small area of the EA) would be temporary Construction phase disturbance (noise, dust, vibration, and glare) would be temporary Operation phase disturbance (noise, human disturbance, and glare) would be permanent	No impact is anticipated

Criterion	Compensatory Wetland	
	Ecological Area (the EA)	**OWCA Area 2**
Reversibility	Direct impact (small area of the EA) would be reversible Construction phase disturbance (noise, dust, vibration, and glare) would be reversible Operation phase disturbance (noise, human disturbance, and glare) would be irreversible	No impact is anticipated
Magnitude	Moderate	Low
Regional Significance	The impacted area is uncommon, man-made habitat of moderate to high value for ecological purpose and only contribute a small proportion of this habitat in Hong Kong. Recorded floral species are mostly common and widespread in Hong Kong. Recorded fauna species are common and widespread in Hong Kong, with most fauna species recorded being wetland-dependant species. No rare species recorded. Impact on ecological connectivity between habitats and species population is anticipated during construction phase only.	The impacted area is uncommon, man-made habitat of moderate to high value for ecological purpose and only contribute a small proportion of this habitat in Hong Kong. Recorded floral species are mostly common and widespread in Hong Kong. Recorded fauna species are common and widespread in Hong Kong, with most fauna species recorded being wetland-dependant species. Rare species such as Eurasian Otter and Greater Spotted Eagle were recorded in this habitat. Impact on ecological connectivity between habitats and species population is not anticipated.
Overall Impact Significance	Construction phase direct impact: Moderate to High Construction phase indirect impact: Moderate to High Operation phase direct impact: Nil Operation phase indirect impact: Minor	Construction phase direct impact: Nil Construction phase indirect impact: Minor Operation phase direct impact: Nil Operation phase indirect impact: Nil

Notes:
1. The portion of the EA included in the Project Boundary but outside of construction area of the Project would be properly fenced off and maintained by the project proponent during the construction phase, even when it would not be affected or occupied as temporary works/storage area.

Table 8.27 Evaluation of Potential Ecological Impacts to Pond

Criterion	Pond	
	Hoo Hok Wai	**Other Ponds**
Habitat Quality	High	Low
Species/Ecological Resources	Current Study: 77 fauna species of conservation importance were recorded Literature Review: 103 species of conservation importance were recorded Breeding ground recorded: No breeding record, nesting or roosting location or behaviour was exhibited by the observed bird species. The total number of species of conservation importance recorded: 136	Current Study: 1 fauna species of conservation importance were recorded Literature Review: No records of species of conservation importance were recorded Breeding ground recorded: No breeding record, nesting or roosting location or behaviour was exhibited by the observed bird species. The total number of species of conservation importance recorded: 1
Size/Abundance	0.25 ha would be permanently lost under the footprint of the ventilation opening and maintenance access in Pond 38, and an additional 1.26 ha would be temporarily lost due to construction works and conversion to freshwater wetland habitats during the construction phase.	Habitat would not be subject to direct loss
Duration	Construction phase direct impacts (0.25 ha) would be permanent. Construction phase disturbance (noise, dust, vibration, and glare) would be temporary Operation phase disturbance (noise, human disturbance, and glare) would be permanent	Construction phase disturbance (noise, dust, vibration, and glare) would be temporary Operation phase disturbance (noise, human disturbance, and glare) would be permanent
Reversibility	Direct impact (permanent habitat loss) will be irreversible Direct impact (temporary habitat loss) will be reversible Construction phase disturbance (noise, dust, vibration, and glare) would be reversible Operation phase disturbance (noise, human disturbance, and glare) would be irreversible	Construction phase disturbance (noise, dust, vibration, and glare) would be reversible Operation phase disturbance (noise, human disturbance, and glare) would be irreversible
Magnitude	High	Low
Regional Significance	The impacted area is uncommon, man-made habitat, and only contribute a small proportion of this habitat in Hong Kong.	The impacted area is uncommon, man-made habitat, and only contribute a small proportion of this habitat in Hong Kong.

Criterion	Pond	
	Hoo Hok Wai	**Other Ponds**
	Recorded floral species are mostly common and widespread in Hong Kong. Most of the recorded faunal species are common and widespread in Hong Kong, with most fauna species recorded being wetland-dependant species. Rare species Eurasian Otter was in previous surveys and pond habitat in Deep Bay is considered important habitat for the species. Impact on ecological connectivity between habitats and species and population is anticipated during construction phase only.	Recorded floral and faunal species are mostly common and widespread in Hong Kong, with very few fauna species recorded. No rare species recorded. Impact on ecological connectivity between habitats and species population is not anticipated.
Overall Impact Significance	Construction phase direct impact: Moderate to High Construction phase indirect impact: Moderate to High Operation phase direct impact: Nil Operation phase indirect impact: Minor	Construction phase direct impact: Nil Construction phase indirect impact: Minor Operation phase direct impact: Nil Operation phase indirect impact: Minor

Table 8.28 Evaluation of Potential Ecological Impacts to Marsh

Criterion	Marsh
Habitat Quality	Moderate
Species/Ecological Resources	Current Study: 25 fauna species of conservation importance were recorded during the ecological surveys Literature Review: 36 species of conservation importance were recorded Breeding ground recorded: No significant nursery or breeding ground known or observed The total number of species of conservation importance recorded: 54
Size/Abundance	0.23 ha would be permanently lost and 0.10 ha would be temporarily lost
Duration	Construction phase direct impact (0.23 ha) would be permanent Construction phase disturbance (noise, dust, vibration, and glare) would be temporary Operation phase disturbance (noise, human disturbance, and glare) would be permanent
Reversibility	Direct impact (permanent habitat loss) will be irreversible Direct impact (temporary habitat loss) will be reversible Construction phase disturbance (noise, dust, vibration, and glare) would be reversible Operation phase disturbance (noise, human disturbance, and glare) would be irreversible

Magnitude	Moderate
Regional Significance	The impacted area is uncommon habitat and contributes only a small proportion of this habitat in Hong Kong. Recorded floral species are mostly common and widespread in Hong Kong. Recorded faunal species of conservation importance are common and widespread in Hong Kong, with most fauna species recorded being wetland-dependant species. Impact on ecological connectivity between habitats and species population is anticipated during both construction and operation phase.
Overall Impact Significance	Construction phase direct impact: Moderate Construction phase indirect impact: Low Operation phase direct impact: Nil Operation phase indirect impact: Minor

Table 8.29 Evaluation of Potential Ecological Impacts to Reedbed

Criterion	Reedbed
Habitat Quality	Moderate
Species/Ecological Resources	Current Study: 5 fauna species of conservation importance were recorded during the ecological surveys Literature Review: 30 species of conservation importance were recorded Breeding ground recorded: No significant nursery or breeding ground known or observed The total number of species of conservation importance recorded: 32
Size/Abundance	Habitat would not be subject to direct loss
Duration	Indirect impact (disturbance) during construction phase would be temporary Indirect impact (disturbance) during operation phase would be permanent
Reversibility	Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be reversible
Magnitude	Moderate
Regional Significance	The impacted area is uncommon habitat and contributes only a small proportion of this habitat in Hong Kong. Recorded floral and faunal species are mostly common and widespread in Hong Kong, with most fauna species recorded being wetland-dependant species. Rare species recorded include Yellow-breasted Bunting. Impact on ecological connectivity between habitats and species population is not anticipated.
Overall Impact Significance	Construction phase direct impact: Nil Construction phase indirect impact: Nil Operation phase direct impact: Nil

Criterion	Reedbed
	Operation phase indirect impact: Nil

Table 8.30 Evaluation of Potential Ecological Impacts to Modified Watercourse

Criterion	Modified Watercourse
Habitat Quality	Low to moderate
Species/Ecological Resources	Current Study: 16 fauna species of conservation importance were recorded during the ecological surveys: Literature Review: A total of 10 species of conservation importance was recorded Breeding ground recorded: No significant nursery or breeding ground known or observed The total number of species of conservation importance recorded: 17
Size/Abundance	Habitat would not be subject to direct loss
Duration	Construction phase disturbance (noise, dust, vibration, and glare) would be temporary Operation phase disturbance (noise, human disturbance, and glare) would be permanent
Reversibility	Construction phase disturbance (noise, dust, vibration, and glare) would be reversible Operation phase disturbance (noise, human disturbance, and glare) would be irreversible
Magnitude	Low to moderate
Regional Significance	Impact area is artificial and common habitat and only contributes to a very small proportion of this habitat in Hong Kong. Recorded flora and fauna species are mostly common and widespread in Hong Kong, with common bat and avifauna species recorded. No rare species recorded. Impact on ecological connectivity between habitats and species population is not anticipated.
Overall Impact Significance	Construction phase direct impact: Nil Construction phase indirect impact: Minor Operation phase direct impact: Nil Operation phase indirect impact: Minor

Table 8.31 Evaluation of Potential Ecological Impacts to Semi-natural / Natural Watercourses

Criterion	Semi-natural / Natural Watercourses		
	The upper section of Ma Tso Lung Stream	**Section of the LMC Meander falling within the Assessment Area**	**Remaining watercourses within the Assessment Area**
Habitat Quality	Ma Tso Lung Stream and its tributaries: Moderate to High	LMC Meander: Moderate to High	Low to moderate
Species/Ecological Resources	Current Study: 3 fauna species of conservation importance was recorded during the ecological surveys	Current Study: 21 fauna species of conservation importance were recorded during the ecological surveys	Current Study: 4 fauna species of conservation importance were recorded during the ecological surveys
	Literature Review: A total of 24 species of conservation importance was recorded Breeding ground recorded: Breeding ground recorded: No significant nursery or breeding ground known or observed The total number of species of conservation importance recorded: 41		
Size/Abundance	Habitat would not be subject to direct loss	0.21 ha would be temporarily lost	Temporary diversion of Ping Hang Stream (approximately 0.4km) during construction phase
Duration	No direct and indirect impact anticipated	Construction phase direct impact would be temporary Indirect impact (disturbance) during construction phase would be temporary Indirect impact (disturbance) during operation phase would be permanent	Construction phase direct impact to Ping Hang Stream would be temporary Indirect impact (disturbance) during construction phase would be temporary Indirect impact (disturbance) during operation phase would be permanent
Reversibility	No direct and indirect impact anticipated	Direct impact during construction phase would be reversible Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be reversible	Direct impact to Ping Hang Stream during construction phase would be reversible Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be reversible

Criterion	Semi-natural / Natural Watercourses		
	The upper section of Ma Tso Lung Stream	**Section of the LMC Meander falling within the Assessment Area**	**Remaining watercourses within the Assessment Area**
Magnitude	Ma Tso Lung Stream and its tributaries: Low	LMC Meander: Moderate to High	Connected to riparian vegetation and LMC Meander: Low to moderate
Regional Significance	The impacted area is common habitat and contributes only a very small proportion of this habitat in Hong Kong. Recorded floral and faunal species are common and widespread in Hong Kong, with wetland dependent avifauna species recorded. Previous record of Three-banded Box Turtle, which wild individual is rare in Hong Kong. Impact on ecological connectivity between habitats and species population is not anticipated.	The impacted area is common habitat and contributes only a very small proportion of this habitat in Hong Kong. Recorded floral species are common and widespread in Hong Kong. Recorded faunal species are common and widespread in Hong, with wetland dependent avifauna species recorded mostly. Records of Eurasian Otter, which is considered rare and restricted to Deep Bay area in Hong Kong, utilise this habitat as movement corridor. Impact on ecological connectivity between habitats and species population is anticipated during construction phase only.	The impacted area is common habitat and contributes only a very small proportion of this habitat in Hong Kong. Recorded floral species are common and widespread in Hong Kong. Recorded faunal species are common and widespread in Hong, with wetland dependent avifauna species recorded mostly. Endemic freshwater crab *Somanniathelphusa zanklon* was recorded in Ping Hang Stream. Impact on ecological connectivity between habitats and species population is anticipated during construction phase only.
Overall Impact Significance	Construction phase direct impact: Nil Construction phase indirect impact: Nil Operation phase direct impact: Nil Operation phase indirect impact: Nil	Construction phase direct impact: Moderate to High Construction phase indirect impact: Moderate to High Operation phase direct impact: Nil Operation phase indirect impact: Nil	Construction phase direct impact: Ping Hang Stream: Minor Other watercourse: Nil Construction phase indirect impact: Ping Hang Stream: Minor Other watercourse: Nil Operation phase direct impact: Nil Operation phase indirect impact: Minor

Table 8.32 Evaluation of Potential Ecological Impacts to Seasonally Wet Grassland

Criterion	Seasonally Wet Grassland
Habitat Quality	Low to moderate
Species/Ecological Resources	Current Study: 1 flora and 20 fauna species of conservation importance were recorded during the ecological surveys Literature Review: 6 species of conservation importance were recorded Breeding ground recorded: No significant nursery or breeding ground known or observed The total number of species of conservation importance recorded: 24
Size/Abundance	1.61 ha would be permanently lost
Duration	Construction phase direct impacts (1.61 ha) would be permanent. Construction phase disturbance (noise, dust, vibration, and glare) would be temporary Operation phase disturbance would be permanent
Reversibility	Direct impact (permanent habitat loss) will be irreversible Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be reversible
Magnitude	Low to moderate
Regional Significance	The impacted area is common habitat and contributes only a very small proportion of this habitat in Hong Kong. Recorded floral and faunal species are common and widespread in Hong Kong, with wetland dependent avifauna species recorded mostly. Endemic but widespread firefly species Motschulsky's Starworm recorded. Impact on ecological connectivity between habitats and species population is anticipated during construction phase.
Overall Impact Significance	Construction phase direct impact: Low to Moderate Construction phase indirect impact: Low Operation phase direct impact: Nil Operation phase indirect impact: Minor

Table 8.33 Evaluation of Potential Ecological Impacts to Woodland

Criterion	Woodland
Habitat Quality	Moderate
Species/Ecological Resources	Current Study: 1 flora and 16 fauna species of conservation importance were recorded during the ecological surveys Literature Review: 1 flora and 11 species of conservation importance were recorded Breeding ground recorded: No significant nursery or breeding ground known or observed The total number of species of conservation importance recorded: 27

Criterion	Woodland
Size/Abundance	1.68 ha would be permanently lost
Duration	Construction phase direct impacts (1.68 ha) would be permanent. Indirect impact (disturbance) during construction phase would be temporary Indirect impact (disturbance) during operation phase would be permanent
Reversibility	Direct impact (permanent habitat loss) will be irreversible Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be reversible
Magnitude	Low to moderate
Regional Significance	The impacted area is common habitat and contributes only a very small proportion of this habitat in Hong Kong. Recorded floral and faunal species are common and widespread in Hong Kong. No rare species recorded. Impact on ecological connectivity between habitats and species population is anticipated during construction phase.
Overall Impact Significance	Construction phase direct impact: Moderate Construction phase indirect impact: Minor Operation phase direct impact: Nil Operation phase indirect impact: Minor

Table 8.34 Evaluation of Potential Ecological Impacts to Grassland / Shrubland

Criterion	Grassland / shrubland
Habitat Quality	Low to moderate
Species/Ecological Resources	Current Study: 3 flora and 27 fauna species of conservation importance were recorded during the ecological surveys Literature Review: 38 species of conservation importance were recorded Breeding ground recorded: No significant nursery or breeding ground known or observed The total number of species of conservation importance recorded: 61
Size/Abundance	2.88 ha would be permanently lost
Duration	Construction phase direct impacts (2.88 ha) would be permanent. Indirect impact (disturbance) during construction phase would be temporary Indirect impact (disturbance) during operation phase would be permanent
Reversibility	Direct impact (permanent habitat loss) will be irreversible Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be reversible
Magnitude	Moderate
Regional Significance	The impacted area is common habitat and contributes only a very small proportion of this habitat in Hong Kong.

Criterion	Grassland / shrubland
	Recorded floral and faunal species are common and widespread in Hong Kong. No rare species recorded. Impact on ecological connectivity between habitats and species population is anticipated during construction and operation phase.
Overall Impact Significance	Construction phase direct impact: Low Construction phase indirect impact: Minor Operation phase direct impact: Nil Operation phase indirect impact: Minor

Table 8.35 Evaluation of Potential Ecological Impacts to Orchard

Criterion	Orchard
Habitat Quality	Low
Species/Ecological Resources	Current Study: 11 fauna species of conservation importance were recorded during the ecological surveys Literature Review: 5 species of conservation importance were recorded Breeding ground recorded: No significant nursery or breeding ground known or observed The total number of species of conservation importance recorded: 16
Size/Abundance	0.54 ha would be permanently lost
Duration	Construction phase direct impacts (0.54 ha) would be permanent. Indirect impact (disturbance) during construction phase would be temporary Indirect impact (disturbance) during operation phase would be permanent
Reversibility	Direct impact (permanent habitat loss) will be irreversible Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be reversible
Magnitude	Low
Regional Significance	The impacted area is common habitat and contributes only a very small proportion of this habitat in Hong Kong. Recorded floral and faunal species are common and widespread in Hong Kong. No rare species recorded. Impact on ecological connectivity between habitats and species population is anticipated during construction and operation phase.
Overall Impact Significance	Construction phase direct impact: Minor Construction phase indirect impact: Minor Operation phase direct impact: Nil Operation phase indirect impact: Minor

Table 8.36 Evaluation of Potential Ecological Impacts to Plantation

Criterion	Plantation
Habitat Quality	Low
Species/Ecological Resources	Current Study: 7 fauna species of conservation importance were recorded during the ecological surveys Literature Review: 23 species of conservation importance were recorded Breeding ground recorded: No significant nursery or breeding ground known or observed The total number of species of conservation importance recorded: 28
Size/Abundance	1.10 ha would be permanently lost
Duration	Construction phase direct impacts (1.10 ha) would be permanent. Indirect impact (disturbance) during construction phase would be temporary Indirect impact (disturbance) during operation phase would be permanent
Reversibility	Direct impact (permanent habitat loss) will be irreversible Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be reversible
Magnitude	Low
Regional Significance	The impacted area is artificial and common habitat and contributes only a very small proportion of this habitat in Hong Kong. Recorded floral and faunal species are common and widespread in Hong Kong, with most floral species recorded being ornamental in nature. No rare species recorded. Impact on ecological connectivity between habitats and species population is anticipated during construction and operation phase.
Overall Impact Significance	Construction phase direct impact: Minor Construction phase indirect impact: Minor Operation phase direct impact: Nil Operation phase indirect impact: Minor

Table 8.37 Evaluation of Potential Ecological Impacts to Agricultural Land

Criterion	Agricultural Land
Habitat Quality	Low
Species/Ecological Resources	Current Study: 1 fauna species of conservation importance were recorded during the ecological surveys Literature Review: 8 species of conservation importance were recorded Breeding ground recorded: No significant nursery or breeding ground known or observed The total number of species of conservation importance recorded: 8
Size/Abundance	Habitat would not be subject to direct loss

Criterion	Agricultural Land
Duration	Indirect impact (disturbance) during construction phase would be temporary Indirect impact (disturbance) during operation phase would be permanent
Reversibility	Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be reversible
Magnitude	Low
Regional Significance	The impacted area is common habitat and contributes only a very small proportion of this habitat in Hong Kong. Recorded floral and faunal species are common and widespread in Hong Kong. Rare species Yellow-breasted Bunting was recorded in previous surveys. Impact on ecological connectivity between habitats and species population is not anticipated.
Overall Impact Significance	Construction phase direct impact: Nil Construction phase indirect impact: Minor Operation phase direct impact: Nil Operation phase indirect impact: Minor

Table 8.38 Evaluation of Potential Ecological Impacts to Developed Area

Criterion	Developed Area
Habitat Quality	Very low
Species/Ecological Resources	Current Study: 1 flora and 17 fauna species of conservation importance were recorded during the ecological surveys Literature Review: 9 species of conservation importance were recorded Breeding ground recorded: No significant nursery or breeding ground known or observed The total number of species of conservation importance recorded: 25
Size/Abundance	5.15 ha would be permanently lost
Duration	Construction phase direct impacts (5.15 ha) would be permanent. Indirect impact (disturbance) during construction phase would be temporary Indirect impact (disturbance) during operation phase would be permanent
Reversibility	Direct impact (permanent habitat loss) will be irreversible Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be reversible
Magnitude	Very low
Regional Significance	The impacted area is common habitat and contributes only a very small proportion of this habitat in Hong Kong. Recorded floral and faunal species are common and widespread in Hong Kong. No rare species recorded.

Criterion	Developed Area
	Impact on ecological connectivity between habitats and species population is not anticipated.
Overall Impact Significance	Construction phase direct impact: Negligible Construction phase indirect impact: Negligible Operation phase direct impact: Nil Operation phase indirect impact: Negligible

Table 8.39 Summary of Potential Ecological Impacts

Impact	Source	Receiver	Nature of Impact	Overall Impact Significance in the absence of Mitigation	Mitigation Required?	Residual Impact
Construction Phase – Direct Impact						
Loss of wetland habitats	Construction works	Sites of conservation importance (Proposed HHW WCP, WCA, WBA, "CA" zone, Priority Site for Enhanced Conservation), ecologically sensitive resources (IBA, the EA, OWCA Area 2, off-site compensation woodland and key dragonfly site in Ma Tso Lung)	Permanent habitat loss within sites of conservation importance and reduced ecological function / value	Proposed HHW WCP, WCA, WBA, "CA" zone, Priority Site for Enhanced Conservation, IBA, the EA[#] and Key dragonfly site in Ma Tso Lung: Moderate to High OWCA Area 2 and off-site compensation woodland: Nil	Yes (Wetland compensation and enhancement) *Sections 8.11.14 to 8.11.20*	Nil
Loss of wetland habitats	Construction works	Wetland habitats (compensatory wetland, pond, marsh, reedbed, modified watercourse, semi-natural and natural watercourse and seasonally wet grassland)	<u>Permanent loss of wetland habitats</u> Compensatory Wetland: Nil Pond: 0.25 ha Marsh: 0.23 ha Reedbed: Nil Modified Watercourse: Nil Semi-natural and natural Watercourse: Nil Seasonally Wet Grassland: 1.61 ha <u>Temporary loss of wetland habitats</u> Compensatory Wetland: 0.38 ha Pond: 1.26 ha	Compensatory Wetland (the EA): Moderate to High Pond: Moderate to High Marsh: Moderate Modified Watercourse: Nil Semi-natural and Natural Watercourse (the LMC Meander): Moderate to high Semi-natural and natural Watercourse (Ping Hang Stream): Minor Seasonally Wet Grassland: Low to Moderate	Yes (Wetland compensation and enhancement) *Sections 8.11.14 to 8.11.20*	Nil

Impact	Source	Receiver	Nature of Impact	Overall Impact Significance in the absence of Mitigation	Mitigation Required?	Residual Impact
			Marsh: 0.10 ha Reedbed: Nil Modified Watercourse: Nil Semi-natural and natural Watercourse: 0.21 ha Seasonally Wet Grassland: Nil			
Loss of terrestrial habitats	Construction works	Various terrestrial habitats	Permanent loss of terrestrial habitats Woodland: 1.68 ha Grassland / Shrubland: 2.88 ha Orchard: 0.54 ha Plantation: 1.09 ha Agricultural Land: Nil Developed Area: 5.15 ha	Woodland: Moderate Grassland / Shrubland: Low Orchard: Minor Plantation: Minor Agricultural Land: Nil Developed Area: Negligible	Yes (Woodland compensation) *Sections 8.11.21 to 8.11.25*	Minor
Habitat fragmentation	Construction works	Wetland habitats (compensatory wetland, pond, marsh, modified watercourse, semi-natural and natural watercourse and seasonally wet grassland) Non-wetland habitats (woodland, shrubland and grassland, plantation)	Fragmentation of wetland and terrestrial habitats	Wetland habitats (others): Moderate Wetland habitats (seasonally wet grassland): Low Terrestrial habitats: Low	Yes (Temporary Aboveground Wildlife Crossing) *Section 8.11.26*	Minor
Direct impact on flora species of conservation importance	Construction works	Incense Tree	Direct loss of Incense Tree	Minor to Moderate	Yes (Transplantation) *Section 8.11.29*	Nil

Impact	Source	Receiver	Nature of Impact	Overall Impact Significance in the absence of Mitigation	Mitigation Required?	Residual Impact
Direct impact on avifauna species of conservation importance	Construction works	Avifauna species of conservation importance Breeding grounds of avifauna species of conservation importance	Loss of habitat Potential injury / mortality of breeding pairs, chicks and eggs	Avifauna species: Minor to Moderate Breeding avifauna species of conservation importance: Minor to Moderate	Yes (Wetland and woodland compensation) *Sections 8.11.14 to 8.11.16 and 8.11.21 to 8.11.25* (Wetland enhancement) *Sections 8.11.17 to 8.11.20* (Pre-construction survey) *Section 8.11.30*	Nil
Direct impact on mammal species of conservation importance	Construction works	Mammal species of conservation importance (Horsfield's Myotis, Japanese Pipistrelle, Unknown Vespertilionidae Species 1, Lesser Yellow Bat, Short-nosed Fruit Bat, Pallas's Squirrel, Small Indian Civet, Leopard Cat and Eurasian Otter)	Loss of habitat and potential direct injury / mortality	Eurasian Otter: Moderate to High Other mammals: Moderate	Yes (Wetland and woodland compensation) *Sections 8.11.14 to 8.11.16 and 8.11.21 to 8.11.25* (Wetland enhancement) *Sections 8.11.17 to 8.11.20*	Nil

Impact	Source	Receiver	Nature of Impact	Overall Impact Significance in the absence of Mitigation	Mitigation Required?	Residual Impact
					(Pre-construction survey) *Section 8.11.32*	
Direct impact on herpetofauna species of conservation importance	Construction works	Herpetofauna (Four-clawed Gecko and Chinese Bullfrog)	Loss of habitat and potential direct injury / mortality	Four-clawed Gecko: Minor Chinese Bullfrog: Minor to Moderate	Yes (Translocation) *Section 8.11.33*	Nil
Direct impact on butterfly species of conservation importance	Construction works	Butterfly species of conservation importance (Red-breast Jezebel, Metallic Cerulean and Paintbrush Swift)	Loss of habitat and potential direct injury / mortality	Minor	No	Nil
Direct impact on odonate species of conservation importance	Construction works	Odonate species of conservation importance (Blue Sprite, Ruby Darter)	Loss of habitat and potential direct injury / mortality	Minor	No	Nil
Direct impact on firefly species of conservation importance	Construction works	No direct impact to firefly species of conservation importance is anticipated	Loss of habitat and potential direct injury / mortality	Minor	No	Nil
Direct impact on freshwater fauna species of conservation importance	Construction works	Freshwater fauna species of conservation importance (*Somanniathelphusa zanklon* and *Gobiopterus macrolepis)*)	Diversion of stream and potential direct injury / mortality	Minor to Moderate	Yes (Translocation) *Section 8.11.36*	Nil

Impact	Source	Receiver	Nature of Impact	Overall Impact Significance in the absence of Mitigation	Mitigation Required?	Residual Impact
Direct injury / mortality on other wildlife	Construction works	All wildlife other than species of conservation importance within the Construction Area of the Project	Loss of habitat and potential direct injury / mortality	Minor to Moderate	Yes (Temporary Aboveground Wildlife Crossing) *Section 8.11.26* *(*Mitigation measures*)* *Sections 8.11.30 to 8.11.31*	Nil
Bird collision	Construction structures, heavy machineries	Avifauna Seasonal Active Night Roost	Avifauna would potentially collide with construction structures and machinery	Avifauna: Minor to Moderate Seasonal Active Night Roost: Nil	Yes (Time control of construction activities) *Sections 8.11.31, 8.11.37 to 8.11.39*	Nil
Construction Phase – Indirect Impact						
Construction disturbance	Construction activities	Sites of conservation importance (Proposed HHW WCP, WCA, WBA, "CA" zone, Priority Site for Enhanced Conservation), ecologically sensitive resources (IBA, the EA, OWCA Area 2, off-site compensation woodland and key dragonfly site in Ma Tso Lung)	Construction disturbance including noise, glare, dust, vibration and other human activities	Proposed HHW WCP, WCA, WBA, "CA" zone, Priority Site for Enhanced Conservation, IBA and EA[#]: Moderate to High OWCA Area 2 and Key dragonfly site in Ma Tso Lung: Minor Off-site compensation woodland: Nil	Proposed HHW WCP, WCA, WBA, "CA" zone, Priority Site for Enhanced Conservation, IBA and the EA: Yes (Wetland compensation	Minor

Impact	Source	Receiver	Nature of Impact	Overall Impact Significance in the absence of Mitigation	Mitigation Required?	Residual Impact
					and enhancement) *Sections 8.11.14 to 8.11.20* (Mitigation measures to disturbance) *Sections 8.11.42 to 8.11.49*	
Construction disturbance	Construction activities	Wetland habitats (compensatory wetland, pond, marsh, reedbed, modified watercourse, semi-natural and natural watercourse and seasonally wet grassland)	Construction disturbance including noise, glare, dust, vibration and other human activities	Compensatory Wetland: Moderate to High Pond: Moderate to High Marsh: Low Reedbed: Nil Modified Watercourse: Minor Semi-natural and Natural Watercourse (the LMC Meander): Moderate to High Semi-natural and Natural Watercourse (Ping Hang Stream): Minor Seasonally Wet Grassland: Low	(Wetland compensation and enhancement) *Sections 8.11.14 to 8.11.20* (Mitigation measures to disturbance) *Sections 8.11.42 to 8.11.49*	Minor
Construction disturbance	Construction activities	Various terrestrial habitats	Construction disturbance including noise, glare, dust, vibration and other human activities	Woodland: Minor Grassland / Shrubland: Minor Orchard: Minor Plantation: Minor	Precautionary measures: (Mitigation measures to disturbance)	Minor

Impact	Source	Receiver	Nature of Impact	Overall Impact Significance in the absence of Mitigation	Mitigation Required?	Residual Impact
				Agricultural Land: Minor Developed Area: Negligible	*Sections 8.11.42, 8.11.46 to 8.11.49*	
Disturbance and obstruction of flight path	Construction activities	Avifauna usage of flight path	Construction activities and associated tall machinery	Moderate	Yes (Time restriction of construction activities) *Sections 8.11.31, 8.11.38 to 8.11.39* (Mitigation measures to disturbance) *Sections 8.11.42, and 8.11.47 to 8.11.49*	Minor
Disturbance to Night Roost	Construction activities	Seasonal active night roost	Construction disturbance including noise, glare, dust, vibration and other human activities	Minor	Precautionary measures: (Time restriction of construction activities) *Sections 8.11.31* (Mitigation measures to disturbance) *Sections 8.11.40, 8.11.42, and 8.11.47 to 8.11.49*	Minor

Impact	Source	Receiver	Nature of Impact	Overall Impact Significance in the absence of Mitigation	Mitigation Required?	Residual Impact
Night-time disturbance	Lighting and glare	Nocturnal animals	Uncontrolled lighting and glare from construction sites at night may affect nocturnal animals	Moderate	Yes (Night control of lightning) *Sections 8.11.32 and 8.11.48 to 8.11.49*	Minor
Water quality deterioration	Site runoff, discharge and accidental spillage	Compensatory wetland (the EA), pond and semi-natural watercourse (the Meander) and other wetland habitats and their associated wildlife	Potentially deterioration of water quality, increase in suspended solids and potential contaminants	Moderate	Yes (Minimise water quality impacts refer to ***Chapter 5***)	Nil
Hydrological impact	Removal, diversion and realignment of watercourses	Compensatory wetland (the EA), pond and semi-natural watercourse (LMC Meander)	Potential change in hydrodynamics of watercourses and ponds	Moderate	Yes (Minimise hydrological impact refer to *Section 8.11.44 to 8.11.45* and ***Chapter 5***)	Nil
Impacts Due to Groundwater Drawdown	Construction activities	Compensatory wetland (the EA), pond and semi-natural watercourse (LMC Meander) and other wetland habitats	Potential groundwater infiltration	Insignificant	Precautionary measures: (Groundwater infiltration control refer to ***Chapter 5***)	Nil
Operation Phase – Direct Impact						
Habitat loss	Footprint of ECR Alignment	Sites of conservation importance (Proposed HHW WCP, WCA, WBA, “CA” zone, Priority Site for Enhanced Conservation), ecologically sensitive resources	No further habitat loss during the operational phase	No	No	Nil

Impact	Source	Receiver	Nature of Impact	Overall Impact Significance in the absence of Mitigation	Mitigation Required?	Residual Impact
		(IBA, the EA, OWCA Area 2, off-site compensation woodland and key dragonfly site in Ma Tso Lung)				
Habitat loss	Footprint of ECR Alignment	Wetland habitats (compensatory wetland, pond, marsh, reedbed, modified watercourse, semi-natural and natural watercourse and seasonally wet grassland) Various terrestrial habitats	No further habitat loss during the operational phase	No	No	Nil
Habitat fragmentation	Footprint of ECR Alignment	Wetland habitats and terrestrial habitats	Disconnection	Minor to Moderate	Yes (Wildlife underpass with a standard solid parapet) *Sections 8.11.27 to 8.11.28*	Nil
Direct injury / mortality to wildlife	Footprint of ECR Alignment	Wildlife	Bird Collision Roadkill (non-flying wildlife)	Bird Collision: Insignificant Roadkill: Minor to Moderate	Yes (Wildlife underpass with a standard solid parapet) *Sections 8.11.27 to 8.11.28*	Nil
Operation Phase – Indirect Impact						
Operation disturbance	Operation of the ECR Alignment	Sites of conservation importance (Proposed HHW WCP, WCA, WBA, "CA" zone, Priority Site for Enhanced Conservation), ecologically sensitive resources	Increase in human, noise, air quality, noise, light pollution, traffic and visual disturbance	Minor	No	Minor

Impact	Source	Receiver	Nature of Impact	Overall Impact Significance in the absence of Mitigation	Mitigation Required?	Residual Impact
		(IBA, the EA, OWCA Area 2, off-site compensation woodland and key dragonfly site in Ma Tso Lung)				
Operation disturbance	Operation of the ECR Alignment	Wetland habitats (compensatory wetland, pond, marsh, reedbed, modified watercourse, semi-natural and natural watercourse and seasonally wet grassland)	Increase in human, noise, air quality, noise, light pollution, traffic and visual disturbance	Minor	Precautionary measures: (Noise and lighting management) *Sections 8.11.52 to 8.11.54*	Nil
Operation disturbance	Operation of the ECR Alignment	Various terrestrial habitats	Increase in human, noise, air quality, noise, light pollution, traffic and visual disturbance	Minor	No	Minor
Disturbance and obstruction of flight path	Operation of the ECR Alignment	Seasonal active night roost	Obstruction of flight path by establishment of viaduct section of ECR Alignment	Minor	No	Minor
Night-time disturbance	Operation of the ECR Alignment	Nocturnal animals	Uncontrolled lighting and glare at night may affect nocturnal animals	Minor	Precautionary measures: (Night control of lightning) *Section 8.11.54*	Nil
Water quality deterioration	Operation of the ECR Alignment	Pond and other wetland habitats and their associated wildlife	Potentially deterioration of water quality, increase in suspended solids and potential contaminants	Minor	Precautionary measures: (Water quality management) *Section 8.11.51 and refer to* ***Section 5***	Nil

Impact	Source	Receiver	Nature of Impact	Overall Impact Significance in the absence of Mitigation	Mitigation Required?	Residual Impact
Hydrological impact	Operation of the ECR Alignment	Compensatory wetland (the EA), pond and semi-natural and natural watercourse (LMC Menader)	Potential change in hydrodynamics of compensatory wetland (the EA), pond and semi-natural and natural watercourse (LMC Menader)	Minor	Precautionary measures: (Reinstatement plan) *Section 8.11.51*	Nil
Secondary Impact (Wetland Compensation)						
Habitat modification in HHW	Wetland enhancement	Ponds in HHW	Modification of the pond habitats	Wetland enhancement is anticipated to result in a positive ecological impact	No	Nil

Note:

refers to the portion of the EA that would be directly and indirectly affected by the Construction Area of the Project

8.10 Cumulative Ecological Impacts

8.10.1 The construction works of the Project will tentatively commence in Q1 / Q2 2027, and completed by Q4 2031. A full list of concurrent projects is detailed in **Chapter 2**. The cumulative impact assessment for ecology covers the vicinity of Hoo Hok Wai and Ma Tso Lung, where the Assessment Area would overlap with several major development projects which would be at various stages of construction/operation.

8.10.2 Potential cumulative impacts may arise from the combined effect of concurrent projects within the Assessment Area with potential ecological impacts, including:-

- Development of the Loop
- Development of the proposed Hoo Hok Wai Wetland Conservation Park under the Wetland Conservation Park System
- Ma Tso Lung Area and Other Sites in Kwu Tung North New Development Area and North District – Feasibility Study
- Northern Metropolis Highway (San Tin to Kwu Tung Section)

Cumulative Impacts on Wetland Habitats and Sites of Conservation Importance

8.10.3 Impacts to wetland habitats arising from the Development of the Loop (excluding ECR Development) includes habitat loss within the Loop, along the “Direct Link” (from the Loop to the LMC station), and along the WCR (along Ha Wan Tsuen Road East). Additional disturbance impacts were predicted in areas of wetland alongside construction works. The areas of directly and indirectly impacted wetlands affected by development within the Loop are partially included in the current Assessment Area, while wetlands affected by the proposed “Direct Link” and WCR are not within the current Project Assessment Area. Construction of buildings within the Loop commenced in 2024 and is expected to be fully completed by 2035, overlapping with the construction period of the current Project. However, cumulative impacts to wetlands and consequent effects on fauna species of conservation importance is anticipated to be low, as mitigation measures required under the LMC Loop EIA (including establishment of the EA and OWCAs) have already been implemented. Temporary impacts to a small portion of the EA resulting from development of the ECR Project would be mitigated by the provision and maintenance of compensatory wetlands in the OWCAs as described in **Section 8.11** of this report. The affected EA would be reinstated to the satisfaction of the future maintenance agent in accordance with a reinstatement and monitoring plan during the construction phase (**Section 8.8.8** refers). The footprint of the ECR alignment is not anticipated to result in adverse ecological impacts on wetland habitats upon completion by Q4 2031 as described in **Section 8.8.67**. Cumulative impacts on these ecological resources within the Assessment Area are also not anticipated during the operation phase.

8.10.4 The proposed WCPs System FS covers a large area of wetland habitats in the Northwest New Territories. Within the proposed HHW WCP, the existing network of pond habitats will largely be retained but zoned and managed to enhance the area’s ecological capacity while promoting the sustainable modernisation of aquaculture practices. Rural activities such as leisure farming may also be introduced to allow visitors to experience a rural lifestyle and enjoy the idyllic countryside landscape (AFCD, 2024a). The proposed HHW WCP overlaps with the current Project

Assessment Area, to the north of the ECR Alignment. The establishment of the WCP is anticipated to improve the overall quality of the wetland habitats WCPs System Area, including the wetland habitats at HHW WCP. The planning and design of the ECR would be expected to be completed prior detailed planning of the HHW WCP. Potential interface issues will be further reviewed and addressed at the detailed studies stage as indicated in WCPs System FS (AFCD, 2024a). Cumulative impacts on the HHW WCP within the Assessment Area are therefore not anticipated.

8.10.5 The "Ma Tso Lung Area and Other Sites in Kwu Tung North New Development Area and North District – Zone 1 Ma Tso Lung Area", will develop sites at Ma Tso Lung area for residential and employment uses, supporting G/IC facilities, open space and/or amenities. As the planning process for this NDA is still on-going, no definitive design information is available at the time of preparation of this EIA. Nevertheless, it is anticipated that the Ma Tso Lung Study would be planned to avoid recognised sites of conservation importance as far as practicable, and that any identified impacts would be mitigated through avoidance, minimisation and compensation.

8.10.6 A section of the Northern Metropolis Highway (NMH), running in a west–east direction from Tin Shui Wai to the NTN New Town, will traverse the Ma Tso Lung area (i.e. the Kwu Tung Section). The highway will consist of tunnel and viaduct / above-ground sections within the Ma Tso Lung area that will branch off at Hong Kong Police Lo Wu Range to another at-grade section aligned along the eastern portion of Ma Tso Lung area, connecting further north to the ECR Alignment. Given the available information at the time of preparation of this EIA, the wetlands at Hoo Hok Wai and Ma Tso Lung San Tsuen would not be subject to direct habitat loss or disturbance from these sections of the NMH alignment. On this basis, cumulative impacts on ecological resources within the Assessment Area are not anticipated.

Cumulative Impact on Woodland Habitats

8.10.7 As described in **Section 8.10.5**, the Ma Tso Lung NDA project may result in the loss of some woodland habitats. As the planning process for this NDA is still ongoing, no definitive design information is available at the time of preparation of this EIA. Nevertheless, it is anticipated that the Ma Tso Lung NDA studies will be planned to avoid recognised sites of conservation importance as far as practicable, and that any identified impacts will be mitigated through avoidance, minimisation, and compensation measures.

8.10.8 The impacts of the proposed NMH to other habitats would be limited to a small portion of woodland and mainly grassland / shrubland, located along the foothills of Crest Hill and the existing Ma Tso Lung Road within the Assessment Area of the Project. These habitats would be affected by at-grade section road construction and slope stabilisation works. However, considering the affected area only supported low diversity and abundance of fauna, and the appropriate mitigation measures would be implemented, the cumulative ecological impacts on ecological resources within the Assessment Area is expected to be low.

Cumulative Impact on Flight Paths

8.10.9 The east-west flight corridor along the LMC Meander and adjacent habitats support major flight paths between the wetland habitats in the LMC/HHW area (on the east) and the San Tin area (on the west). Development of the Loop was expected to have adverse impacts to this fight corridor, which were mitigated through the provision of the EA and associated buffer zones. As the construction of the current Project may

result in potential disturbance and obstruction of the LMC Meander and the EA, this could result in potential cumulative impacts to the flight path. Impacts to the flight corridor would therefore require mitigation.

8.11 Mitigation of Adverse Environmental Impacts

General Principles on Mitigation Measures

8.11.1 According to Annex 16 of EIAO-TM and EIAO Guidance Note. 3/2010, ecological impacts on important habitats and associated wildlife resulting from Projects should be mitigated by, in order of priority, avoidance, minimisation, and compensation approaches to the maximum practical extent. The proposed Project and associated mitigation measures have been designed such that it can satisfy the requirements of the EIAO-TM and the "no net loss in wetland" principle in Deep Bay as stipulated in TPB PG-No.12C.

8.11.2 Under the current ECR alignment and its design, key ecological impacts include the loss of wetland habitats (pond and other wetland habitats), loss of woodland habitats, and potential impacts on flora and fauna species of conservation importance, as discussed in **Section 8.8**. These impacts would require both temporary (during construction phase) and permanent (during operation phase) mitigation. Following the mitigation hierarchy, measures are proposed to address significant impacts to key ecological resources, as detailed in the following sections.

Impact Avoidance Through the Preferred Project Alignment

8.11.3 Considerable emphasis was placed on ecological considerations under the detailed option review process presented in **Section 2.5**. The selected alignment differs from the original ECR design presented in the LMC Loop EIA, with key permanent impacts to the pond habitats as well as fragmentation impacts at the western end of the alignment now largely avoided through adoption of an underpass rather than open sunken road design. Additional avoidance of wetland habitats would be achieved as follows:-

- Considerations for the selection of construction methods are detailed in **Section 2.6**. The preferred 'cut and cover' approach will allow for reinstatement of almost all wetland habitats including the LMC Meander, the EA and ponds habitats along underground sections of the ECR Alignment, following completion of construction works. While TBM tunnelling could further reduce construction phase ecological impacts, it also has considerable drawbacks, including high costs, environmental risks such as groundwater contamination from grouting and bentonite, extensive land requirements for launching and retrieval shafts, the need for a specialised workforce and equipment, and potential traffic disruptions during TBM delivery.

- Direct impacts to the EA could be avoided if the underpass section of ECR alignment is routed further to the north-east into the ponds of HHW, before turning west towards the Loop. This option would however have resulted in more extensive and prolonged overall impacts and disturbance to pond habitats during construction phase, increasing the overall direct and indirect impacts to wetlands. While it would affect the EA, the preferred alignment has a smaller overall impact to wetlands

- There would be no direct impacts the OWCA Area 2.

- The at-grade sections of the ECR Alignment largely follow the existing Border Road Alignment, largely avoiding further direct impacts to wetland habitats.

- At the eastern end of the ECR Alignment, the road sections would pass to the east of Ma Tso Lung Village, avoiding wetland habitats to the west of the Village.

8.11.4 By adopting this design, overall permanent impacts to wetland habitats requiring compensation would be reduced from 3.94 ha (comprising 1.99 ha pond, 1.76 ha freshwater marsh and 0.19 ha of seasonally wet grassland) as reported in the previous LMC Loop EIA, to 2.09 ha (comprising 0.25 ha pond, 0.23 ha marsh and 1.61 ha of seasonally wet grassland). For high value pond habitats, permanent direct impacts are almost wholly avoided, with the affected area would be reduced by over 87% (from 1.99 ha to 0.25 ha).

8.11.5 The potential extent of construction disturbance impacts is considered highly conservative, as it does not account for phased implementation of construction works under Phase 1, or that the majority of the construction activities will be undertaken underground as the Alignment passes through the EA, LMC Meander and Ponds 36-38. The area of wetland habitats temporarily impacted has increased from 7.46 ha to 9.04 ha during Phase 1 compared to the previous LMC Loop EIA. This increase is not a result of changes to design of the ECR Alignment, but from the following factors:

- In the LMC Loop EIA, the extent of disturbance was measured from the boundary of the permanent project alignment, whereas in the current assessment for the Project, the disturbance extent is measured from the Construction Area of the Project to provide a more conservative and robust evaluation of temporary construction impact area of wetland habitats that require compensation.

- The LMC Loop EIA did not account for temporary impacts and disturbance to the LMC Meander.

Although temporary losses would increase, reinstatement and further enhancement of affected areas will ensure ecological functional recovery and mitigation of residual impacts.

8.11.6 The ECR Alignment would pass beneath Ping Hang Stream by means of an underpass, thereby avoiding permanent direct impacts to this habitat.

8.11.7 The ECR Alignment was also designed to avoid direct impacts on the relatively undisturbed upper sections of the Ma Tso Lung Stream. Only the already channelised lower sections would fall within the Construction Area of the Project, and viaduct piers would not encroach upon the stream or result in any permanent impacts.

Impact Minimisation Through Construction Phasing

8.11.8 A phased construction approach would be adopted to minimise prolonged and concurrent direct and indirect impacts to wetland habitats. As described in **Section 2.9**, during construction, the ECR Alignment will be divided into three main sections: the first section extends from the depressed road at the Loop to the end of the underpass near the Lok Ma Chau Operational Base (Section 1); the second section covers the depressed road near Lok Ma Chau Operational Base, transitioning to an at-grade alignment along Broder Road before converting into a viaduct near Shun Yee San Tsuen (Section 2); and the third section comprises the remaining at-grade

alignment at Shun Yee Shan Tsuen, continuing to the viaduct near Ma Tso Lung San Tsuen (Section 3). For planning purposes, construction is assumed to proceed in two phases: Phase 1 will include Sections 1 and 3, while Phase 2 will include Sections 2 and 3. In general, Sections 1 and 2 will not be constructed concurrently, whereas Section 3 may be undertaken throughout the entire construction period. The indicative layout is shown in **60722948/A10/FIGURE 2.8.** Any construction activities (e.g. site clearance, excavation and lateral support (ELS), slope stabilisation works or use of powered machinery equipment (PME)) in Section 2 would only be permitted after habitat reinstatement and completion of all works in Section 1.

8.11.9 The portion of the EA included in the Project Boundary but outside the 0.38ha construction area of the Project would not be affected or occupied as temporary works/storage areas. To avoid construction activities and to deter other unwanted human access, this area would be properly fenced off and maintained by the project proponent during the construction phase. Reinstatement of the EA Zone following completion of the depressed road / underpass sections of the ECR Alignment will be undertaken by the Project Proponent / the Contractor with reference to the approved Ecological Mitigation / Habitat Creation and Management Plan (CEDD, 2022) and the existing as-built conditions of the EA after the completion of the aboveground works. For underground reinstatement, all existing engineering and associated structures removed during construction will be reinstated upon completion. After construction of the underpass, the excavation cofferdam will be backfilled with compacted materials (compacted fill or rockfill, subject to the Contractor's design) in accordance with the requirements of the General Specification for Civil Engineering Works published by CEDD (CEDD, 2020). As the stone columns within the construction area will no longer be required to support the EA and its earth bund, the relevant geotextile layers and provisions will be reinstated to ensure that the original level of impermeability is maintained, and native plant species will be used for reinstatement. The ecological characteristics and function of the EA will be enhanced and maintained to a condition no less than the existing conditions, after the reinstatement. A reinstatement and monitoring plan for the EA would be prepared by the Contractor and submitted to AFCD and EPD for agreement at least three months prior to commencement of construction works.

Impact on Wetland Habitats

8.11.10 The enhancement of existing wetlands is a strategy that has been adopted by several projects in the NWNT area (e.g. LMC Loop EIA, LMC Spur Line EIA and STLMC DN EIA) to compensate for the direct loss of wetland habitats and indirect disturbance impacts to the associated wildlife (especially disturbance sensitive bird species). Enhancing the ecological function and capacity of existing habitats would compensate for the loss or reduction of wetland areas and functional value resulting from the Project. For example, pond habitats can be enhanced with various features to increase function and capacity compared to typical commercially managed ponds, thereby compensating for the loss of the functional value by accommodating a higher abundance of wildlife. Enhancement measures could include:

- Physical modification of pond habitats to increase environmental capability;
- Managing and sequencing pond drain down across multiple ponds in the dry season (i.e. Nov to Mar);
- Providing fencing/controlling access to reduce disturbance from human activities and also prevent disturbance and predation of wildlife by feral dogs;

- Removal of existing bird scaring devices at actively managed ponds, where appropriate;

- Stocking ponds with suitable prey items (i.e., trash-fish) for target wildlife species

8.11.11 Using this approach, previous projects (e.g., LMC Spur Line) have assumed a 1:1 compensation ratio can be achieved, whereby the loss of one area of wetland habitat can be compensated for through the enhancement of another area of wetland of equal size. This approach has been largely validated through performance monitoring data from the LMC EEA (MTR, 2024).

8.11.12 A similar approach to compensation was adopted in the LMC Loop EIA. According to Table 2.3 to the HCMP developed under the LMC Loop EIA (AECOM, 2021), three OWCAs (Area 2, 7 and 9), **60722948/A10/FIGURE 8.11** refers) have been constructed to mitigate direct and indirect impacts to wetlands posed by the Development of the Loop, WCR, and Direct Link to LMC Station as summarised in **Table 8.40** below. These OWCAs would also partially address compensation requirements resulting from development of the ECR, as identified in the LMC Loop EIA.

Table 8.40 Actual Offsite Compensation Requirements for Development Components under the LMC Loop EIA

Off-site Compensation Area		Habitat Creation and Management Plan (HCMP)									Operation Manual for the OWCAs
		Lok Ma Chau Loop		Western Connection Road		Direct Link Road to LMC Station		Eastern Connection Road[1]		Total Minimum Required Area	Mitigation Area[2]
		Temp. Mitigation	Perm. Mitigation	Temp. Mitigation	Perm. Mitigation	Temp. Mitigation	Perm. Mitigation	Temp. Mitigation	Perm. Mitigation		
2	Marsh	-	-	-	0.07	-	-	-	1.95	2.02	1.66
	Reed Marsh	-	-	-	-	-	-	-	-	-	0.22
	Pond	-	4.51	-	0.92	-	-	-	-	5.43	6.61
7	Marsh	-	-	-	-	-	-	-	-	-	1.19
	Reed Marsh	-	-	-	-	0.032	-	-	-	0.032	-
	Pond	-	-	-	0.85	-	2.28	-	-	3.13	2.28
9	Marsh	-	-	-	-	-	-	-	-	-	1.03
	Pond	-	-	4.11	-	-	-	-	-	4.11	5.8
Total		-	4.51	4.11	1.84	0.032	2.28	-	1.95	14.722	18.79

Notes:
Temp. Mitigation refers to temporary mitigation and Perm Mitigation refers to permanent mitigation
1. Existing allocation for ECR wetland compensation designated under the LMC Loop EIA and associated HCMP
2. Actual areas constructed and managed under the Operation Manual for the OWCAs

8.11.13 Further to these compensation requirements, an additional to 0.11 ha of habitats within the San Tin Eastern Main Drainage Channel (STEMDC) wetlands would be temporarily impacted due to the construction of a pier of Direct Link Road on the eastern embankment of the STEMDC (CEDD, 2025). While the affected habitats are largely terrestrial, compensation for these temporary losses would also be accommodated within the OWCAs. As described in **Sections 8.4.15-8.4.18**, the OWCAs were completed in October 2022 (CEDD, 2023b), and the as-built areas of OWCA Area 2, 7 and 9 are summarised in **Table 8.2.** Taking into consideration areas of the OWCAs already allocated for compensation of other elements of the Loop development, 6.13ha of pond habitat and 3.89ha of marsh and reed marsh habitat are available to compensate for wetland loss and disturbance impacts associated with the ECR (**Table 8.41**). An updated baseline survey was conducted during 2018 to 2019 during preparation of the HCMP of the OWCAs under the "Development of Lok Ma Chau Loop" project (AECOM, 2021). This survey recorded a total of 96 fauna species, comprising 6 mammal species, 70 avifauna species, 7 herpetofauna species and 13 dragonfly species. Subsequently, CEDD carried out further monitoring during the establishment period of the OWCAs, commencing in mid-October 2022. Monitoring during the establishment period recorded a total of 152 fauna species comprising 10 mammal species, 106 avifauna species, 13 herpetofauna species and 23 dragonfly species (CEDD, 2023c). Following the official handover of the OWCAs to AFCD in January 2024 for habitat management (AFCD, unpublished), the species record at OWCAs continued to increase, with a total of 207 fauna species, including 15 mammal species, 149 avifauna species, 17 herpetofauna species and 26 dragonfly species. Based on the comparison of species recorded before and after the wetland enhancement works, the results suggest that the OWCAs were enhanced and properly maintained for compensation. In particular, the number of avifauna species increased from 70 in the updated baseline survey to 149 following OWCAs establishment. Overall, more than 100 additional fauna species, including 9 target species identified in the approved LMCL EIA, were newly recorded in the OWCAs when compared with the updated baseline survey presented in the HCMP.

Table 8.41 Summary of Areas of OWCAs Available for Wetland Compensation for the Project

Location	As-established OWCA for ECR Compensation (ha)	Remarks
OWCA Area 2, 7 and 9	Pond: 6.13 ha Marsh and Reed Marsh: 3.89 ha	Wetland compensation works at the OWCA were completed in October 2022 but are required to compensate for temporary construction phase impacts associated with the WCR. Upon the completion of WCR (estimated within 2026), 4.11 ha of Pond in Area 9 will be released and can be used to compensate for impacts associated with the current Project. (construction works of the ECR project will tentatively commence in Q1/Q2 2027).

Notes:
Areas based on Operation Manual for the Offsite Wetland Compensation Areas under the Development of Lok Ma Chau Loop: Main Works Package 1 (CEDD, 2023b) and differ slightly from areas stated in the LMC Loop EIA as habitats across the three OWCAs were reconfigured in response to updated site assessments. The numbers presented exclude areas already allocated for compensation under the WCR, Direct Road Link and the LMCL Project.

8.11.14 As discussed in **Sections 8.8.1 to 8.8.16**, the Construction Area of the Project would result in unavoidable loss and disturbance to wetland habitats of ecological significance, including habitats within some sites of conservation importance,

namely Proposed HHW WCP, WCA, WBA, "CA" zone, Priority Site for Enhanced Conservation, IBA and key dragonfly site in Ma Tso Lung. Furthermore, it would not be possible to fully mitigate these impacts with the avoidance or minimisation measures described in **Sections 8.11.8 to 8.11.18**. Compensation would therefore be required for habitats directly impacted by the construction works. In addition, wetland habitats including compensatory wetland (the EA), natural watercourse (the LMC Meander) and ponds that support disturbance sensitive species, would be subject to temporary disturbance during construction that would also require mitigation. To meet the "no net loss" compensation requirements under TPG-12C, all direct impacts to wetland habitats within the WCA/WBA would be compensated. In addition, significant indirect impact to disturbance-sensitive species and habitats would also be compensated. The compensation requirements during Phase 1 and Phase 2 are summarised below in **Table 8.42** and **Table 8.43**, and described below.

- During construction works in Phase 1, Ponds 36 to 38 will be partially drained. The bunds of Ponds 39, 39a will also fall partially within the Project Boundary during the construction phase. In total, 1.48 ha of pond habitat would be directly impacted. A permanent loss of 0.25 ha will accrue within the footprint of the ventilation shaft opening and maintenance access located in Pond 38, which cannot be reinstated upon completion of construction works (**60722948/A10/FIGURE 8.10** refers). In addition to these direct impacts, there would also be indirect disturbance impacts to pond habitats. With reference to the disturbance zoning described in **Section 8.8.48**, impacts to 2.85 ha of ponds in the EZ and 3.12 ha in the RDZ would occur during the construction phase (**60722948/A10/FIGURE 8.9** and **Table 8.42** refer). Regarding the loss and disturbance to other wetland habitats that require compensation, there would be no permanent loss of compensatory wetlands (i.e. the EA) or the LMC Meander. However, a temporary loss of 0.38 ha of the EA would occur, together with construction-phase disturbance impacts affecting 0.82 ha within the EZ and 0.57 ha within the RDZ. Similarly, a temporary loss of 0.21 ha of the LMC Meander would occur, with construction-phase disturbance impacts covering 0.55 ha in the EZ and 0.74 ha in the RDZ. In addition, the Project would result in a temporary loss of about 0.10 ha of marsh habitat, as well as a permanent loss of 0.42 ha of seasonally wet grassland.

- During construction works in Phase 2, only 0.03 ha pond bund of Pond 93 will fall partially within the Project Boundary during the construction phase. In addition to these direct impacts, there would also be indirect disturbance impacts. With reference to the disturbance zoning described in **Section 8.8.48**, impacts to 1.32 ha of ponds in the EZ and 2.41 ha in the RDZ would occur during the construction phase (**60722948/A10/FIGURE 8.9** and **Table 8.43** refer). In addition, the Project would result in a permanent loss of approximately 0.23 ha of marsh habitat, as well as an additional permanent loss of 1.19 ha of seasonally wet grassland together with the 0.07 ha of seasonally wet grassland to be impacted at the end of viaduct section at Section 3 of the Project Alignment. For conservative estimation, it is assumed directly impacted wetland habitats that would be reinstated under Phase 1 will require an establishment period to provide full ecological function. Therefore, the direct habitat loss of the EA, LMC Meander, Ponds 36 - 38 and marsh would also be considered as requiring compensation during phase 2 (i.e., after these habitats have been reinstated but are still within the establishment period). The direct, indirect, temporary and permanent impacts to wetland habitats that require compensation in Phase 2 are summarised in **Table 8.43**.

Table 8.42 Summary of Direct and Indirect Impacts (ha) to Wetland Habitats Requiring Compensation during Construction Phase (Phase 1)

Phase 1	Habitats	Direct Impacts	Indirect Disturbance Impacts		Total Compensation Requirement(1)
		Habitat Loss	EZ (0-50m)	RDZ (50-100m)	
Sections 1 and 3	Compensatory Wetland (the EA)(2)	0.38	0.82	0.57	**1.49**
	Pond	1.48	2.85	3.12	**5.89**
	Marsh	0.10	0.00	0.00	**0.10**
	Semi-natural / Natural Watercourse(3)	0.21	0.55	0.74	**1.14**
	Seasonally Wet Grassland	0.42	0.00	0.00	**0.42**
	Total(4)	**2.59**	**4.22**	**4.43**	**9.04**

Notes:
1. Compensation requirements are calculated at a 1:1 basis for habitat loss and Exclusion Zone (EZ) impacts, and 1:2 basis for the Reduced Density Zone (RDZ)
2. The portion of the EA included in the Project Boundary but outside of Construction Area of the Project would be properly fenced off and maintained by the project proponent during the construction phase, even when it would not be affected or occupied as temporary works/storage area.
3. The difference from Table 8.7 arises from the 0.02 ha of semi-natural/natural watercourse that will not be affected by habitat loss during the construction phase, as stated in Section 8.8.15.
4. Slight discrepancies in the totals due to rounding up.

Table 8.43 Summary of Direct and Indirect Impacts (ha) to Wetland Habitats Requiring Compensation during Construction Phase (Phase 2)

Phase 2	Habitats	Direct Impacts	Indirect Disturbance Impacts		Total Compensation Requirement(1)
		Habitat Loss(2)	EZ (0-50m)	RDZ (50-100m)	
Sections 2 and 3	Compensatory Wetland (the EA)	0.38	0.00	0.00	**0.38**
	Pond	1.51	1.32	2.41	**4.04**
	Marsh	0.33	0.00	0.00	**0.33**
	Semi-natural / Natural Watercourse	0.21	0.00	0.00	**0.21**
	Seasonally Wet Grassland	1.61	0.00	0.00	**1.61**
	Total(3)	**4.04**	**1.32**	**2.41**	**6.57**

Notes:
1. Compensation requirements are calculated at a 1:1 basis for habitat loss and Exclusion Zone (EZ) impacts, and 1:2 basis for the Reduced Density Zone (RDZ)
2. For conservative estimation, the directly impacted wetland habitats would be reinstated and require an establishment period to provide full ecological function. Therefore, the direct habitat loss of the EA, LMC Meander and Ponds 36 – 38 would also be considered as requiring compensation during Phase 2 (i.e., after these habitats have been reinstated but are still within the establishment period).
3. Slight discrepancies in the totals due to rounding up.

8.11.15 Unavoidable construction phase impacts to wetland habitats in Phase 1 (comprising 5.89 ha of pond, 1.49 ha of compensatory wetland within the EA, 0.10 ha of marsh, 1.14 ha of semi-natural and natural watercourse, and 0.42 ha of seasonally wet grassland as presented in **Table 8.42**) and Phase 2 (comprising 0.38 ha of compensatory wetland within the EA, 4.04 ha of pond, 0.33 ha of marsh, 0.21 ha of semi-natural and natural watercourse and 1.61 ha of seasonally wet grassland as presented in **Table 8.43**) can be compensated through OWCAs already developed under the LMC Loop EIA. The proposed compensation areas are summarised in **Table 8.44** below.

Table 8.44 Summary of Wetland Mitigation Requirements Arising from Each Phase of Construction Works of the Project

Phase	ECR Construction Phase Compensation Area Requirements (ha)	As-established OWCA for ECR Compensation (ha)
Phase 1 (Sections 1 and 3)	Pond: 5.89	Pond: 6.13
	Other Freshwater Wetland(1) : 3.15	Marsh and Reed Marsh: 3.89
Phase 2 (Sections 2 and 3)	Pond: 4.04	Pond: 6.13
	Other Freshwater Wetland(1) :2.53 ha	Marsh and Reed Marsh: 3.89

Notes:
1. Freshwater wetland habitats include Compensatory Wetland (the EA) ,Marsh, Semi-natural / Natural Watercourse and Seasonally Wet Grassland)

8.11.16 As construction phase impacts can be fully compensated through existing habitat enhancement at the OWCAs, no residual impacts are expected in the operation phase, where the extent of impacts to wetland habitats would be much smaller (**Section 8.11.14** refers). Detailed HCMP (AECOM, 2021) and the Operation Manual (CEDD, 2023b) for these OWCAs have already been developed and approved by relevant authorities. The HCMP should be updated and submitted to the relevant authorities (e.g. AFCD and EPD) at least three months prior to the commencement of enhancement works, to reflect the latest development of the Project and the corresponding mitigation requirements within the OWCAs. The HCMP should also include habitat enhancement works, management, monitoring requirements for Area 4 as described in **Section 8.11.18** to **8.11.19** below.

Ecological Enhancement of Ponds 36-38

8.11.17 Ponds 36-38 fall within the overall Project Boundary of the Project, but approximately 3.29 ha of these ponds are outside of the Construction Area of the Project, and would not be utilised for any construction related activities (i.e., site access, equipment/materials storage). Given the existing relatively poor condition of these habitats (**Section 8.6.6** refers), there is an opportunity to enhance the value of these ponds during the Phase 1 construction works.

8.11.18 Previous EIAs in the NWNT including the LMC Loop EIA, LMC Spur Line EIA and STLMC DN EIA have recommended various measures that can be adopted to increase the ecological function of pond habitats. These measures would be implemented at the 3.29 ha of Ponds 36-38 to be retained during the Phase 1 construction works, and would include:

- Ponds would be reprofiled to a maximum water depth of 2-2.5m, with 20-25% of overall pond area maintained as shallow water (<50cm) habitat. Pond base of all pond area would be reprofiled to have a gentle gradient, allowing progressive increase in feeding area for birds when ponds are drained;

- Creating habitat islands that provide refuge for avifauna and other wildlife, the island will have shallow-sloping muddy edges (gradient of 1:8) around the water edge to provide avifauna foraging habitat;

- On vegetated pond bunds, existing trees, shrubs and dense vegetation will be retained as they provide roosting habitat for birds as well as shelter for mammal species;

- On-going habitat management for other projects in the LMC Spur Line indicates that there is typically sufficient rainwater in the area to support the enhanced pond that will be created under this Project follows AEC NCSC (2016). In this regard, no external water supply system is required. However, some internal water management infrastructure is required to facilitate the management practices, subject to the detailed design of the Project;

- Providing fencing / controlling access to reduce disturbance from human activities and also prevent disturbance and predation of wildlife by feral dogs;

- Temporary stocking ponds with locally farmed native fish species for target wildlife species during construction phase, and would not comprise any highly invasive fish species; and

- Managing and sequencing pond drain down across multiple ponds in the dry season (i.e. Nov to Mar) to maximise feeding opportunities for avifauna and other wildlife. Studies from HKBWS (2021) recorded 83 water species, including critically endangered Spoon-billed Sandpiper and the endangered, Black-faced Spoonbill, were using drained pond. The number of waterbird species recorded during drain-down was found to be about 19 times higher than the number recorded before the pond drain-down. AEC NCSC (2016) recorded that pond drain-down is highly effective in attracting large numbers and a high diversity of waterbirds, with a marked increase in abundance and species richness. Furthermore, removing the actively managed pond from intensive fish farming, it can increase the pond ecological function and attract the target species.

8.11.19 Upon completion of Phase 1 of the ECR Alignment and reinstatement works, only a small area of 0.25 ha pond and 0.42 ha seasonally wet grassland would be permanently lost due to the footprint of the ventilation opening and maintenance access required for the underpass road. The provision of wetland compensation with the OWCAs will more than sufficiently offset wetland impacts arising from the Project, ensuring “no net loss” in the overall ecological function of the affected wetland habitats. The temporarily lost 0.89 ha of Pond 36–38 due to construction works will be reinstated, together with an additional strip of bund (Pond 36–38) of about 0.2 ha to be consolidated and converted into ecological enhanced pond (hereinafter referred as Area 4), totalling to 4.38 ha. The enhancement measures are outlined as **Section 8.11.18** above would be considered at Area 4. Located within the Hoo Hok Wai (HHW) flight corridor (**60722948/A10/FIGURE 8.8** refers), Area 4 will function as an important ecological linkage within the Deep Bay Wetland System, maintaining habitat connectivity and supporting the movement of wetland-dependent fauna across the corridor. These habitats can provide a continuous

movement pathway for avifauna using the major flight corridor. In addition, the enhanced corridor habitats would support the movement of Eurasian otters and other terrestrial mammals, allowing them to traverse between their established home ranges (from Mai Po to the Loop and HHW). Wetland associated avifauna species, non-avifauna species of conservation species as well as Eurasian Otters would benefit from better water quality management compared to existing pond, and more diverse and shallower sloping edge conditions to provide suitable foraging habitat for these species. The Contractor would be responsible for the reinstatement and enhancement works at Area 4 and will consult the relevant authorities by submitting an updated HCMP at least three months prior to commencing enhancement works. In addition, sufficient recurrent funding shall be provided by the Project Proponent to the authorities responsible for habitat management during the operation phase. Following establishment, the contractor shall seek written consent from related authorities (e.g. AFCD and EPD) for acceptance of the habitat management role of the ecologically enhanced ponds.

8.11.20 Upon the completion of the Project, the net loss of wetland habitats would be limited to 0.25 ha of pond, 0.23 ha of marsh and 1.61 ha of seasonally wet grassland as presented in **Table 8.45**. These direct losses will be fully compensated within the constructed OWCAs, where pond and other freshwater wetland habitats (i.e., marsh and reed marsh) of comparable ecological function are provided to ensure no reduction in overall wetland function. The total area of OWCAs will be more than sufficient to offset wetland area loss (by more than 20 times) and to maintain equivalent ecological function. In addition, the enhancement measures at Area 4 would provide a readily achievable increase in the wetland functioning compared to the existing ponds 36-38. During the operation phase, indirect impacts arising from the ECR alignment are expected to be insignificant. Viaduct sections of the ECR will adopt a low noise road surfacing, and at-grade sections would be screened by planting, minimising potential noise and visual disturbance to nearby wetland habitats. These design measures will maintain ecological functions of the adjacent wetlands. The Project is therefore anticipated to result in no net loss in wetland function within the WCA / WBA and will comply with the conservation intent of TPB Guideline No. 12C.

Table 8.45 Summary of Wetland Mitigation Requirements during the Operation Phase of the Project

Phase	ECR Operation Phase Compensation Area Requirements (ha)	As-established OWCA for ECR Compensation (ha)	Enhancement Works for ECR (ha)
Permanent loss of wetland habitats during operational phase	Pond: 0.25	Pond: 6.13	Area 4: 4.38
	Other Freshwater Wetland(1) : 1.84	Marsh and Reed Marsh: 3.89	-

Notes:
1. Freshwater wetland habitats include ,Marsh, and Seasonally Wet Grassland)

Impacts on Woodland Habitats

8.11.21 A 1.26 ha woodland and shrubland compensation area was identified under the LMC Loop EIA at Horn Hill, which included provision for impacts arising from the ECR construction (**60722948/A10/FIGURE 8.12 refers**). However, the Off-site

Woodland Compensation Proposal (CEDD, 2024) that has been implemented to date focuses on compensation for woodland loss associated with the WCR. This has been achieved through 0.4ha of woodland compensation area at the Horn Hill site. No compensation for woodland loss associated with the ECR Project has yet been implemented.

8.11.22 A site survey of the Horn Hill site was conducted in March 2021 during the preparation of the WCR Off-site Woodland Compensation Proposal. A patch of land (approximately 1500m2) was found to be inaccessible as the area is relatively steep. The remaining areas of the site (0.71ha) were comparatively flat with a maximum slope of 35°. Soils were predominantly composed of topsoil with medium-grained characteristics. This remaining area is therefore considered suitable for woodland compensation required to mitigate impacts associate the ECR. However, as described in **Section 8.8.17**, the total area of direct woodland impact associated with the current project is 1.68 ha, meaning an additional woodland compensation site of 0.97 ha will be required to fully address impacts resulting from the ECR Project construction.

8.11.23 As on-site mitigation is not possible given the narrow footprint of the ECR Alignment, an off-site area of 1.5 ha has been identified close to the impacted areas (**60722948/A10/FIGURE 8.12** refers). The area is currently a grassland /shrubland habitat and has been selected on a relatively shallow slope (less than 1:3) that provides a better physical environment for woodland establishment. Pre-construction surveys for nesting activities of avifaunal species of conservation importance that could potentially utilise grassland/shrubland habitats would be carried out before the commencement of construction works. The area to be provided will be on “no net loss’ and ‘like for like’ basis, at minimum, at a ratio of 1:1 in terms of area, with native tree species preferred for planting (**Table 8.46** refers). A vegetation survey should be conducted within the proposed woodland compensation area to identify the suitable locations to fulfill the minimum requirements of an additional 0.97 ha of woodland compensation. The survey should also determine the dominant native tree species in the nearby area, which should be considered as a reference for the species to be planted.

Table 8.46 Species Proposed for Woodland Compensation Planting at this EIA Stage

Flora Species	Growth Form
Acronychia pedunculata	Tree
Bischofia javanica	Tree
Bridelia tomentosa	Shrub or small tree
Canthium dicoccum	Tree or shrub
Cinnamomum camphora	Tree
Daphniphyllum calycinum	Tree
Elaeocarpus chinensis	Tree or small tree
Ficus microcarpa	Tree
Ficus variegata	Tree
Garcinia oblongifolia	Tree
Litsea glutinosa	Tree

Flora Species	Growth Form
Machilus pauhoi	Tree
Phyllanthus emblica	Tree or shrub
Schefflera heptaphylla	Tree
Schima superba	Tree
Sterculia lanceolata	Semi-deciduous tree
Viburnum odoratissimum	Shrub or small tree

8.11.24 Key measures for the establishment of the compensation woodland include:

- Survey and retain any the plant species of conservation importance growing on the site if any,
- Site clearance of grass and common fern should commence after the setting up of plant protection zones for recorded plant species of conservation importance;
- Any invasive plants detected during site preparation works should be uprooted, packed and removed from the work site;
- Select a diverse mix of planting species that the woodland's overall adaptability, to provide ecological benefits at different times of year, and to enhance wildlife use, and commonly used and maintenance are familiar by local practices, with preference given to stock that is readily available from the market or supplied by the local nursery.

8.11.25 To facilitate successful establishment of the compensatory woodland, a detailed Woodland Compensation Plan should be prepared by local ecologists / botanist with at least 5 years of relevant experience. The Woodland Compensation Proposal should include implementation details, management requirement and monitoring requirements (e.g., methodology, schedule, frequency of monitoring, and monitoring parameters). Details of the Woodland Compensation Proposal would be submitted to relevant authority(ies) for approval before commencement of the compensatory planting. Upon completion of planting, monitoring maintenance works (e.g. irrigation, weeding, pruning, control of pests and diseases, replacement planting and repair of damage) should be implemented and regular performed.

Habitat Fragmentation

8.11.26 Habitat fragmentation impacts would be most substantial where the ECR Alignment crosses the EA, LMC Meander and pond habitats in the southwest of the Project Boundary. Measures would be implemented during both the construction and operation phases to mitigate these impacts. Based on the ecological survey and literature review conducted under this Project, non-flying mammal species of conservation importance were recorded in the Assessment Area including Eurasian Otter, Red Muntjac, Leopard Cat, Pallas's Squirrel, Small Asian Mongoose and Small Indian Civet. They are susceptible to potential habitat fragmentation resulting from the Project. Thus, mitigation measures such as provision of wildlife crossing (i.e., temporary aboveground wildlife crossing, permanent wildlife underpass) are proposed accordingly. In order to maintain the movement corridor of non-flying mammals, during construction, works would be phased in this area to ensure some areas of intact habitat are maintained at all times, with only one cofferdam will be

installed in each phase and simultaneous work at two or more of the sections traversing this area is avoided. As described in **Section 8.11.14**, habitats in this area would be reinstated following completion of construction works, avoiding long-term fragmentation impacts associated with the ECR Alignment. As illustrated in **60722948/A10/FIGURE 3.2b**, the length of each construction zone may vary, and simultaneous construction works in Zone 3 and the adjoining eastern section will exceed 150 m in total length, therefore a temporary aboveground wildlife crossing would be provided above the construction area through the ponds near Ngau Kok Shan to mitigate the habitat fragmentation during construction (**60722948/A10/FIGURE 8.13** refers). The temporary aboveground wildlife crossing will be approximately 40 m wide and will be planted with shrubs. It is further recommended that dense shrub planting be incorporated along the crossing edges to function as soft fencing, thereby minimise disturbance from the surrounding environment and to facilitate wildlife movement of non-flying mammal species of conservation importance (i.e., Eurasian Otters). The temporary wildlife crossing will be designed and constructed by the Contractor during construction phase.

8.11.27 Non-flying wildlife (including mammal species of conservation importance such as Small Indian Civet and Leopard Cat) were also recorded (during recent surveys and previous studies) along other sections of the ECR Alignment, where the road follows the existing Border Road. Fragmentation impacts in this area are considered to be low in significance during the construction phase, as major construction works will not be conducted along the whole alignment simultaneously. During the operation phase, fragmentation impacts would be minimised through various design features. Approximately 100 m of the ECR Alignment would be constructed on a viaduct, allowing non-flying mammals to cross under the ECR unimpeded. The ECR would also cross the Ping Hang Stream as an underpass, avoiding fragmentation impacts to aquatic fauna, and allowing for habitat enhancement works (as described in **Section 8.11.41** below). In addition, two permanent wildlife underpasses would be provided along at grade sections of the ECR Alignment (**60722948/A10/FIGURE 8.13** refers). The wildlife underpasses would comprise shallow concrete box culvert (1.8-2m wide, 1.2-1.5m tall) under the proposed ECR Alignment, subject to detailed design. The design will include measures to avoid flooding and to facilitate potential wildlife movement, subject to detailed design.

8.11.28 Additionally, a standard 0.8-metre-high solid parapet will be installed along all depressed road, at-grade roads and viaducts as one of the standard road designs. The parapet will serve to limit access to the road area by terrestrial wildlife, guiding them towards the wildlife underpasses, and reducing potential vehicle collisions and incidence of injury/roadkill.

Impacts on Species of Conservation Importance

Flora

8.11.29 Pre-construction surveys of all works areas would be undertaken to locate and identify potentially affected Incense Trees. Individuals recorded in temporary works areas would be retained *in-situ* where practicable. Individuals that would be unavoidably impacted by the Project would be transplanted. A detailed transplantation plan should be prepared before the commencement of the construction phase.

Fauna

8.11.30 While direct mortality / injury impacts to birds during the construction and operation phases are expected to be minor, measures would be taken to minimise these risks further:-

- During the construction phase, site hoarding at least 3m high (painted matt green) will prevent collision with construction vehicles and/or structures within the works areas. Detailed design of any moveable noise barriers/enclosures should be agreed with the relevant authorities before the commencement of works.

- During the operation phase, dense vegetative screening (landscape planting) will also be provided along the at grade sections of the ECR Alignment, to minimise visual disturbance and potential collision impacts.

- The solid parapet along the ECR Alignment will serve to reduce mortality associated with vehicle collision by preventing wildlife from accessing the road.

- Pre-construction surveys during breeding season (February to July) within the construction area of the ECR Alignment for nesting activities of Eurasian Eagle Owl, Chinese Francolin and White-throated Kingfisher would be undertaken in any areas of potentially suitable habitat (HKBWS, n.d.). Should any of these species be encountered during the surveys or during construction, work in the area will cease and the site is examined by a suitably qualified Ecologist. An appropriate exclusion zone of will be established around the nest until the Ecologist confirms that works can safely resume. The extent of the exclusion zone should be determined by the Ecologist and factors such as the bird species and the nest location should be considered.

8.11.31 Additional measures to minimise disturbance impacts that could impact avifauna (particularly wetland avifauna) and other disturbance sensitive wildlife will include:-

- Quality power mechanical equipment (PME) and moveable barrier / disclosure would be used to limit noise emission and therefore reduce disturbance impact.

- No aboveground noisy construction activities (with the use of PME), as described in **Section 8.8.41** are to be carried out in the sensitive wetland areas that fall within the Construction Area of the Project (i.e. Zone 1, Zone 2 and Zone 3 as illustrated in **60722948/A10/FIGURE 3.2b** and **FIGURE 8.14)** before 0900h and after 1700h throughout the year, unless fully enclosed conditions are provided through the provision of an appropriate cover or enclosure. This restriction is intended to avoid disturbance during period when flight lines and otters are most active in the region.

- During the construction phase, night-time lighting will be restricted to the minimum necessary to ensure site safety. Directional lighting will be adopted to avoid excessive light spill to sensitive receivers (e.g. the EA, LMC Meander and nearby ponds).

Eurasian Otters and Other Mammals

8.11.32 Potential habitat loss and fragmentation impacts to Eurasian Otters and other non-flying mammals would be addressed through mitigation measures described in

Sections 8.11.26 to **8.11.28**. Additional measures to mitigate impacts to otters will include:

- The small areas of the EA (approximately 0.38ha) directly impacted would be reinstated after construction.

- No aboveground noisy construction activities (with the use of PME) as described in **Section 8.8.41** are to be carried out in the sensitive wetland area that fall within the Construction Area of the Project (i.e. Zone 1, Zone 2 and Zone 3 as illustrated in **60722948/A10/FIGURE 3.2b** and **FIGURE 8.14)** before 0900h and after 1700h throughout the year, unless fully enclosed conditions are provided through the provision of an appropriate cover or enclosure. This restriction is intended to avoid disturbance when otters are most active in the region.

- Construction activities will be scheduled to avoid simultaneous works within sensitive wetland areas (i.e. Zone 1, Zone 2 and Zone 3 as illustrated in **60722948/A10/FIGURE 3.2b**).

- Pre-construction surveys for otter holts or natal dens should be carried out in wetland habitats located within 150m of the Construction Area of the Project before the commencement of construction works. Should either be encountered during these surveys or during construction, work in the area will cease and the site is examined by a suitably qualified Ecologist. If the holt or natal den is in use for breeding (e.g. dependent young are present), works in the area will cease, and an access corridor and buffer area (as required), around the natal den or holt will be designated in which no human activity will be allowed until breeding is completed. In general, no works should be undertaken within 150m of any active natal den or within 30m of any active holt. Appropriate mitigation measures should also be activated to restrict vehicular/public access. Further measures will be taken, as required, in consultation with the Ecologist and AFCD.

- Given the sensitivity of otters to dogs, no dogs should be fed or kept by workers within construction areas. Any dogs that are found would be trapped and removed.

Herpetofauna

8.11.33 Two herpetofauna species of conservation importance were recorded in the Construction Area of the Project that would be subject to habitat loss, with one species (Chinese Bullfrog) at risk of potential direct injury / mortality. Accordingly, pre-construction surveys and translocation of any identified individuals of this species would be implemented from Ping Hang Stream, the EA and other suitable wetland habitats within the Construction Area of the Project. A pre-construction survey and a translocation proposal should be submitted to relevant authority(ies) for approval at least one month before the commencement of works.

Butterfly

8.11.34 Three butterfly species of conservation importance were recorded from woodland habitat within the Construction Area of the Project during recent surveys (i.e., Metallic Cerulean, Paintbrush Swift and Red-breast Jezebel). Habitat loss for these species would be mitigated through woodland and wetland compensation works.

Host plants (i.e., *Bambusa sp*. and Sweet Viburnum) and nectar species (i.e., *Abelmoschus moschatus* and *Atalantia buxifolia*) would be considered in the Woodland Compensation Plan planting species, as well as other landscaping works alongside the ECR alignment. Ginger lily (*Hedychium coronarium*, the host plant of Metallic Cerulean) would be incorporated into landscaping works around the Ping Hang Stream.

Odonate

8.11.35 Habitat loss for odonate species of conservation importance would be mitigated through wetland compensation / enhancement works. Odonate species would benefit from improve water quality resulting from better water quality monitoring and feeding regimes in enhanced ponds, as well as from the provision of increased riparian / emergent vegetation.

Freshwater Fauna

8.11.36 *Sommaniathelphusa zanklon* was recorded in the upstream section of Ping Hang Stream (within the Construction Area of the Project) during recent surveys. Permanent direct habitat loss for this species would be avoided as the ECR Alignment would cross through this area as an underpass. Moreover, where it passes under the existing Border Road, Ping Hang Stream is channelised and lined with concrete. This section would be revitalised (as described in **Section 8.11.41** below), providing additional habitat and improved connectivity for *S. zanklon* and other aquatic species. However, it is possible that individuals of *S. zanklon* could be impacted during the construction phase. Two individuals of *Gobiopterus macrolepis* were also recorded within the riparian zone of the LMC Meander, which could be impacted during the construction phase. A pre-construction survey and a translocation proposal for *S. zanklon* and *Gobiopterus macrolepis* should be submitted for approval from relevant government departments prior to the construction commencement.

Impacts on Flight Path

8.11.37 As described in **Section 8.8.39,** impacts of ECR construction and operation to identified flight paths are potentially significant, and would require mitigation.

8.11.38 Construction phase impacts would be addressed through measures to mitigate fragmentation and disturbance, as described in **Sections 8.11.26** to **8.11.28** and **8.11.30** to **8.11.31.** Operation phase impacts would primarily be addressed through the project design, with the ECR Alignment passing through the key flight path areas as an underpass (as described in **Section 8.11.3**).

8.11.39 During both the construction phase, no above-ground structures exceeding 15 m above the existing ground level are allowed in Zone 1, Zone 2, and Zone 3 (**as illustrated in 60722948/A10/Figure 3.2b**) during the period from October to March of the following year to avoid potential impacts on flight paths. The maximum existing ground level in these zones is +5.5mPD; therefore, no above-ground structure may exceed +20.5mPD (i.e., +5.5mPD + 15m). In addition, no aboveground noisy construction activities (with the use of PME described in **Section 8.8.41** are to be carried out in the sensitive wetland area that fall within the Construction Area of the Project (i.e. Zone 1, Zone 2 and Zone 3 as illustrated in **60722948/A10/FIGURE 3.2b** and **FIGURE 8.14)** before 0900h and after 1700h throughout the year, unless fully enclosed conditions are provided through the

provision of an appropriate cover or enclosure. This measure can minimise disturbance impacts on the flight lines.

Impacts on Night Roost

8.11.40 Direct encroachment onto the seasonal active night roost at the northern end of Ma Tso Lung Road would be avoided, and indirect impacts are considered minor given that there are existing sources of disturbance close to the roost, the distance of works areas from the roost, and the fact that works areas would not affect access from the roost to wetland habitats to the north. Nonetheless, a pre-construction night roost survey would to be undertaken to confirm location and extent of the roosting area. Any noisy construction activities (with the use of PME) within 100 meters of the existing seasonal active night roost (as presented in **60722948/A10/FIGURE 8.14)** would be subject to timing restriction from October to April, permitted only between 0900h to 1700h to avoid the period of night roost utilisation. Construction phase night-time lighting will be reduced to the minimum possible, and direct lighting towards the night roost will be avoided.

Impacts on Streams / Watercourses

8.11.41 Under the ECR Alignment Design, the Ping Hang Stream will be temporarily diverted during the construction phase. Upon completion of works, the stream would be reinstated and revitalised, with details of the revitalisation would be developed in the detailed design stage. Opportunities for ecological enhancement would be considered to improve its ecological value. Provision of natural substrate that would encourage colonisation of flora and fauna in the bottom and banks of the revitalised watercourses would be considered, subject to detailed design of the proposed revitalisation measures. Vegetation species to be planted along the riparian zone would be selected on the basis that it would benefit the wildlife recorded in the vicinity. Fauna species recorded from recent surveys and previous studies (e.g. *S. zanklon*) would be potentially benefit from the revitalised watercourse. A Stream Revitalisation Proposal should be prepared and submitted to the relevant authorities for approval prior to the commencement of revitalisation works.

Other Potential Impact from Construction Disturbance

Good Site Practice

8.11.42 Adverse disturbance impacts can be minimised to areas within and adjacent to the Construction Area of the Project by following good site practices. These will include:-

- Nomination of approved personnel, such as a site manager, to be responsible for good site practices, and making arrangements for collection of all waste generated at the site and effective disposal to an appropriate facility;
- Training of site personnel in proper waste management and chemical waste handling procedures;
- Provision of sufficient waste reception / disposal points, of a suitable vermin-proof design that minimises windblown litter;

- Arrangement for regular collection of waste for transport off-site and final disposal;

- Appropriate measures to minimise windblown litter and dust during transportation of waste by either covering trucks or by transporting wastes in enclosed containers;

- Regular cleaning and maintenance programme for drainage systems, sumps and oil interceptors;

- A recording system for the amount of wastes generated, recycled and disposed (including the disposal sites) should be proposed; and

- Preparation of a WMP as a part of the EMP in accordance with ETWB TCW No. 19/2005 Environmental Management on Construction Sites and submit to the Engineer / Architect for approval.

Minimising Water Quality and Hydrological Impacts

8.11.43 As stated in the Water Quality Impact Assessment (**Chapter 5** refers), control of potential pollution of nearby water bodies (i.e. the EA, LMC Meander) during the construction of the Project should be achieved by measures to:-

- Prevent or minimise the likelihood of pollutants (generated from construction activities) being in contact with rainfall or run-off.

- Abate pollutants in the stormwater surface run-off prior to the discharge of surface run-off to the nearby water bodies.

- Preventive measures include installation of silt traps, sedimentation tanks, and perimeter drainage channels to collect and treat site runoff prior to discharge.

- Temporary surface protection such as tarpaulin covers and hydroseeding will be applied to exposed slopes and stockpiles to minimise erosion during the wet season.

- Regular maintenance and silt removal will be undertaken, particularly after heavy rainfall, to maintain drainage efficiency; and

- Chemical and fuel storage will be bunded and located away from water bodies, and wastewater/discharge will be treated on-site before disposal.

8.11.44 To maintain the hydrological regime of the eastern partition of the EA during dry season (i.e., Nov to Mar) in construction phase, regular inspection of water depths and water quality within marsh and reed marsh habitats will be carried out by the Contractor. If the water level of the EA reaches the action water level specified in Table 5.1 of CEDD (2022), the Contractor will use portable pumps to provide supplemental water supply from LMC Meander, ensuring adequate water levels are maintained in the remaining parts of the EA and sustaining the reedbed and freshwater marsh habitats. In case of non-compliance of water quality within the EA specified in Table 8.1 of CEDD (2022), appropriate water pollution control measures (e.g., emergency bypass mechanism) shall be carried out by the contractor to prevent polluted water entering the EA. Before pumping the water from LMC

Meander to the EA, the Contractor will test water quality of LMC Meander to ensure that the supplementary water quality is not inferior to the action levels of water quality specified for the EA in the HCMP (CEDD, 2022). If the water quality of the LMC Meander fail to meet the relevant action levels, the Contractor shall take appropriate measures, such as water treatment, to achieve compliance before pumping water into the EA, and / or identify alternative water sources including the addition of rainwater harvesting systems, to ensure that the water to be supplied to the EA meets the required quality standards.

8.11.45 In the event of accidental chemical spillages or flooding of EA under heavy rainstorm, the following actions will be implemented:

- In the event of a chemical spillage within the EA and LMC Meander, appropriate measures outlined in Chapter 5 (**Section 5.7.26 – 5.7.28**) will be implemented to remove the spillage.

- Valves at the most eastern partition of the EA should be managed by the Contractors, particularly before and during extreme weather/flooding event, to adjust water to the targeted level as specified in CEDD (2022) and to control flow in between the meander and the EA. Inspections to the EA shall be arranged and any damage to the valves and habitats in the EA identified during the inspections will be repaired and reinstated.

Groundwater Infiltration Control

8.11.46 As stated in the Water Quality Impact Assessment (**Chapter 5** refers), control of groundwater infiltration during the construction of the Project should be achieved by:-

- Adopting cut-and-cover excavation that is fully enclosed by cofferdams; and

- Ensuring any groundwater from dewatering process will be pumped and discharged via silt traps and silt removal facilities before entering stormwater system.

- If excessive filtration is observed, grouting or equivalent Engineer-approved methods will be applied to minimise groundwater inflow

Dust Suppression

8.11.47 As described in **Chapter 3** Air Quality Impact Assessment, dust suppression measures (i.e., regular spraying of haul roads, proper storage of construction materials, covering trucks or transporting waste in enclosed containers, and environmental control measures as stipulated in the Air Pollution Ordinance (Construction Dust) Regulation) should be implemented to avoid and minimise emission and dispersal dust.

Noise and Lighting Management

8.11.48 In addition to the use of site hoarding described in the Noise Impact Assessment (**Chapter 4**), good site practice should be implemented to minimise noise disturbance. This should include shutdown of machines and plants that are in intermittent use, and the use of well-maintained and good quality PME to limit noise emissions at source. Machines and plant known to emit strong directional noise

should, wherever practicable, be orientated so that the noise is directed away from the nearby disturbance habitats, particularly wetland habitats. Quiet Mechanical Plant and other machines and plants should be covered by noise enclosure to further reduce noise impact.

8.11.49 During the construction phase, night-time lighting will be restricted to the minimum necessary to ensure site safety. Excessive light spill to sensitive receivers (e.g. the EA, LMC Meander and nearby ponds) will be avoided through the use of directional lighting.

Operation Phase Impacts

8.11.50 Major operation phase impacts such as fragmentation, impacts to species of conservation interest etc. are already addressed in the preceding sections. Further measures would also be adopted to address three specific operation phase impacts, water quality impacts, noise and light disturbance.

Water Quality and Hydrological Impact

8.11.51 Best Management Practices (BMPs) to reduce stormwater pollution from road surfaces should be implemented, as described in **Chapter 5**. Reinstatement of the EA Zone following completion of the depressed road / underpass sections of the ECR Alignment will be undertaken by the Project Proponent / the Contractor with reference to the approved Ecological Mitigation / Habitat Creation and Management Plan (CEDD, 2022) and the existing as-built conditions of the EA after the completion of the aboveground works. For underground reinstatement, all existing engineering and associated structures removed during construction will be reinstated upon completion. After construction of the underpass, the excavation cofferdam will be backfilled with compacted materials (compacted fill or rockfill, subject to the Contractor's design) in accordance with the requirements of the General Specification for Civil Engineering Works published by CEDD (CEDD, 2020). As the stone columns within the construction area will no longer be required to support the EA and its earth bund, the relevant geotextile layers and provisions will be reinstated to ensure that the original level of impermeability is maintained, and native plant species will be used for reinstatement. The ecological characteristics and function of the EA will be enhanced and maintained to a condition no less than the existing conditions, after the reinstatement. A reinstatement and monitoring plan for the EA would be prepared by the Contractor and submitted to AFCD and EPD for agreement at least three months prior to commencement of construction works.

Noise Management

8.11.52 During the operation phase, roadside screen planting will be implemented along the at-grade sections of the ECR Alignment and low noise road surfacing will be used in viaduct sections of the ECR Alignment to mitigate noise disturbance to the adjoining wetland habitats. Noise impact from the pumping station can be effectively mitigated through the implementation of source-based noise control measures at the detailed design stage. Mitigation measures include using quieter fixed plant, enclosing the fixed plant within reinforced concrete building or acoustic enclosure with openings directed away from NSRs, use of silencer, installation of acoustic louvre, installation of noise barrier, and installation of noise enclosure, etc

8.11.53 With the adoption of the proposed mitigation measures, no adverse noise impact from the proposed fixed sources of the Project is anticipated.

Lighting

8.11.54 Road lighting along the entire at-grade and viaduct sections of the ECR Alignment should be planned and designed to minimise impact to surrounding habitats. With reference to published literature (e.g., Pauwels *et al*., 2021), measures that should be adopted include:-

- The minimum number of streetlights should be constructed along the alignment to provide a safe amount of night-time illumination
- Lighting intensity should be reduced to the minimum as per safety requirements
- Adopt lighting with a narrower spectral wavelength rather than white light.
- Light sources should be shielded to direct lighting onto the road and minimise light trespass.

8.12 Evaluation of Residual Impacts

8.12.1 Direct impact from the permanent loss of habitats are identified and presented in **Section 8.11.4.** While potential ecological impacts may arise from the loss of 0.25 ha pond and 1.84 ha of other wetland habitats, as well as 1.68 ha woodland habitats, mitigation measure have been proposed according in the above sections, including wetland and woodland compensation. With the implementation of the proposed mitigation measures, no unacceptable residual ecological impacts from habitat loss are anticipated. Furthermore, with proposed enhanced measures in the Area 4, overall habitat quality in the affected area within Project Boundary is anticipated to be improved.

8.12.2 The construction and operation of the ECR Alignment would result in potential disturbance (i.e. lightning and glare, air, noise, traffic and visual), injuries / mortality to wildlife and habitat fragmentation on flight path. Mitigation measures including provision of underpass and control of lighting) were proposed to maintain the habitat connectivity. No unacceptable residual ecological impact is anticipated.

8.12.3 Species of conservation importance were recorded within the Construction Area of the Project. Mitigation measures were proposed accordingly in above sections (e.g. transplantation of vegetation species of conservation importance, translocation of amphibian and freshwater communities of conservation importance, pre-construction site check for nesting and breeding avifauna species of conservation importance, and incorporation of wildlife corridor for the non-flying mammal species recorded within the Construction Area of the Project as well as potential usage of Eurasian Otter). No unacceptable residual ecological impact is anticipated.

8.13 Environmental Monitoring and Audit

8.13.1 The implementation of mitigation measures described above shall be audited periodically during the construction phase. This will include implementation of the mitigation measures described in **Section 8.11** and in the Environmental Monitoring & Audit (EM&A) Manual (**Section 7** of this report).

8.13.2 To assess potential construction phase disturbance, quantitative avifauna monitoring of all wetlands within 100m of the Construction Area of the Project should be conducted using transect surveys. The monitoring frequency should be monthly for 12 months prior to commencement of construction and throughout the construction period until 12 months after completion of works. The monitoring should be carried out by qualified ecologist(s) with at least 5 years of relevant experience.

8.13.3 Reinstatement of the EA Zone following completion of the depressed road / underpass sections of the ECR Alignment will be reinstated and established by the Project Proponent / the Contractor with reference to the approved Ecological Mitigation / Habitat Creation and Management Plan (CEDD, 2022) and the existing as-built conditions of the EA after the completion of the aboveground works. Monitoring of water level and quality will be carried out by the Contractor during the construction phase. In case of non-compliance, suitable mitigation measures shall be undertaken to meet requirement specified in Table 5.1 and Table 8.1 of CEDD (2022). For underground reinstatement, all existing engineering and associated structures removed during construction will be reinstated upon completion. After construction of the underpass, the excavation cofferdam will be backfilled with compacted materials (compacted fill or rockfill, subject to the Contractor's design) in accordance with the requirements of the General Specification for Civil Engineering Works published by CEDD (CEDD, 2020). As the stone columns within the construction area will no longer be required to support the EA and its earth bund, the relevant geotextile layers and provisions will be reinstated to ensure that the original level of impermeability is maintained, and native plant species will be used for reinstatement. The ecological characteristics and function of the EA will be enhanced and maintained to a condition no less than the existing conditions, after the reinstatement. A reinstatement and monitoring plan for the EA would be prepared by the Contractor and submitted to AFCD and EPD for agreement at least three months prior to commencement of construction works.

8.13.4 OWCAs would be utilised as a mitigation measure for the direct and indirect impacts during both construction and operation phases on wetland habitats under the Project. The 3.29ha of Ponds 36-38 located outside of the Construction Area of the Project in Area 4 would not be utilised for any construction related activities. This provides an opportunity to enhance the ecological value of these ponds during the Phase 1 construction works. Pond habitat will be retained and, together with the remaining area of Pond 36-38, converted into 4.38 ha ecologically enhanced pond in Area 4 upon completion of construction works in Phase 1. An updated HCMP for the OWCAs and Area 4 would be prepared by the Contractor and submitted to AFCD and EPD for agreement at least three months prior to commencement of construction works.

8.13.5 For the unavoidable loss of woodland resulting from impacts within the Construction Area of the Project, compensatory planting has been proposed as a mitigation measure, with the proposed location(s) and detailed design subject to further agreement. A detailed Woodland Compensation Plan should be prepared by an ecologist / botanist with at least 5 years relevant experience to form the basis of the proposed compensatory planting. The mitigation plan should include implementation details, associated management and monitoring requirements (e.g. methodology, schedule, monitoring frequency and parameters) of the mitigation area. Upon the completion of planting, monitoring and maintenance works (e.g., irrigation, weeding, pruning, control of pests and diseases, replacement planting and repair of damage) of the compensatory woodland should be implemented. The Woodland Compensation Plan shall be submitted to the relevant authority(ies) for approval before commencement of the compensatory planting.

8.13.6 Given the importance of the flight path corridor and the significance of potential impacts, its use should be monitored to identify any effects of construction activities focusing on the composition of bird species and their abundance using the flight path. The flight path monitoring survey should be conducted monthly, commencing 30 minutes before sunrise and continued for at least 2 hours. The composition of bird species and their abundance, together with the flight height and pathways of all flight lines should be recorded.

8.13.7 A seasonal active night roost at the northern end of Ma Tso Lung Road was identified within the Assessment Area. Pre-construction survey is necessary to confirm the extent of night roost, as well as the 100m buffer area before commencement of the construction works. The pre-construction survey shall be conducted once per month between October to April by qualified ecologist with at least 5 years relevant experience.

8.13.8 Incense trees as a species of conservation importance, should be protected as far as practicable. As a mitigation measure, any unavoidably affected individuals should be conserved on site or, as a last resort, transplanted to nearby suitable habitat(s) prior to commencement of site clearance. A detailed Pre-Construction Vegetation Survey should be carried out by a suitably qualified botanist / ecologist to identify and record affected individuals prior to the commencement of site clearance works. A protection and transplantation proposal, including subsequent monitoring of the affected individuals, should be prepared and implemented by a suitably qualified ecologist / botanist with at least 5 years relevant experience. The proposal should be submitted to the relevant authority(ies) for approval at least one month prior to commencement of work. Upon the transplantation of the identified individuals, a post-transplantation monitoring should be implemented to monitor the health conditions and survival of the transplanted individuals. Whenever transplantation is not suitable (e.g. due to technical feasibility, poor plant health, form or structural condition), compensatory planting for this species should be proposed. Details of post-transplantation monitoring are provided in the standalone EM&A Manual.

8.13.9 Mammal species of conservation importance with high mobility (i.e. Eurasian Otter, Small Indian Civet, Leopard Cat) should be monitored by transect survey and infrared camera trapping. Transect count would be carried out monthly and the cameras are suggested to be checked monthly. Pre-construction surveys for otter holts or natal dens should be carried out in wetland habitats within 150m of the Construction Area of the Project before the commencement of construction works. Should either be encountered during these surveys or during construction, work in the area will cease until the site has been inspected by an experienced ecologist. Any other mammal species of conservation concern sighted will be reported.

8.13.10 A temporary aboveground wildlife crossing would be provided above the construction area through the Ponds near Ngau Kok Shan to minimise habitat fragmentation during construction phase. In addition, two permanent underpasses would be provided under the at-grade road section of the ECR Alignment. The detailed design of both aboveground wildlife crossing and underpasses would be provided in design stage. To ensure their efficiency, monitoring of the wildlife crossing shall be conducted during the construction phase, and underpasses shall be conducted during the first three years upon establishment The conditions of the constructed wildlife crossing / underpasses (e.g. structural integrity, vegetation overgrown, any observable usage) shall be monitored every two months, while any potential usage of the wildlife corridor by mammal species (e.g. Eurasian Otter and Small Indian Civet) should also be recorded (e.g. with the use of camera traps). Maintenance work such

as weeding, screening, and repairing broken structure should be conducted by the Contractor, where necessary, during the monitoring period for both wildlife crossing and underpasses.

8.13.11 Other fauna species of conservation importance (one herpetofauna species Chinese bullfrog, one freshwater crab species *Somanniathelphusa zanklon* and one freshwater fish species *Gobiopterus macrolepis*) should also be protected as far as practicable. The species would be translocated to nearby suitable habitat(s) prior to the commencement of site clearance. A detailed Pre-Construction Survey carried out by a suitably qualified ecologist to identify, and record affected individuals prior to the commencement of site clearance works. A translocation proposal, including subsequent monitoring of affected individuals, should be prepared and implemented by a suitably qualified ecologist with at least 5 years relevant experience. The proposal should be submitted to the relevant authority(ies) for approval at least one month prior to commencement of work.

8.13.12 Breeding / nesting behaviour of avifauna species of conservation importance such as White-throated Kingfisher, Chinese Francolin, Eurasian Eagle Owl were recorded or suspected in the Assessment Area, and they should be protected as far as practicable. Pre-construction site check for nesting activities is proposed to avoid direct injury to breeding pairs, chicks and eggs of avifauna species of conservation importance.

8.14 Conclusion

8.14.1 Ecological impact assessment for the Project has been conducted following the EIA Study Brief No. ESB-375/2025 and EIAO-TM Guidelines Annexes 8 and 16. A detailed literature review and ecological surveys (July 2022 to December 2023) were conducted to characterise the ecological baseline of the Assessment Area, which extended up to 500m from the Project Boundary. Several sites of conservation importance were identified within the Assessment Area, including the Proposed HHW WCP, WCA, WBA, “CA”, Priority Site for Enhanced Conservation, IBA, the EA, OWCA Area 2, off-site compensation woodland and Key Dragonfly Site in Ma Tso Lung.

8.14.2 A total of 13 habitat types were identified within the Assessment Area, with wetland habitats comprising compensatory wetland, pond, marsh, reedbed, modified watercourse, semi-natural and natural watercourse, seasonally wet grassland; and non-wetland habitats comprising woodland, grassland / shrubland, orchard, plantation, agricultural land and developed land. Wetland habitats were mostly concentrated in the northeastern portion of the Assessment Area around Hoo Hok Wai, comprising high ecological value pond habitats, moderate to high ecological value compensatory wetland (the EA) and semi-natural and natural watercourses (LMC Meander), moderate ecological value marsh habitats, and low to moderate ecological value seasonally wet grassland. Terrestrial woodland habitats were mainly found in Horn Hill, Tse Koo Hang and Crest Hill, and are considered to be of moderate ecological value. Flora species of conservation importance, including Incense Tree (*Aralia chinensis*) and Tongue Habenaria were recorded within the Assessment Area.

8.14.3 Sensitive wetland habitats (i.e., compensatory wetland and ponds) in the HHW area supported high diversity and abundance of waterbirds. Mammal species of conservation importance, such as Leopard Cat, Small Indian Civet, and several bat species were also recorded. While not recorded in surveys conducted specifically for this EIA, long-term and recent records confirm the presence of Eurasian Otter in wetland habitats of the HHW area. Other fauna of conservation importance recorded

within the Assessment Area include herpetofauna (i.e., Chinese Bullfrog, Four-clawed Gecko); common and widespread butterfly species of conservation concern (i.e., Metallic Cerulean, Red-breast Jezebel, Paintbrush Swift); odonate species (i.e., Blue Chaser, Coastal Glider); firefly species including Motschulsky's Starworm; and freshwater species such as the fish *Gobiopterus macrolepis* and freshwater crab *Somanniathelphusa zanklon* and .

8.14.4 Direct and indirect impacts to sites of conservation importance and associated wetland / woodland habitats and species of conservation importance would arise from the proposed project. Compared to the design of the ECR presented in the previously approved LMC Loop EIA (2013a), the latest design proposed in this EIA has been developed to avoid and minimise ecological impacts as far as practicable. In particular, key permanent impacts to the pond habitats as well as fragmentation impacts at the western end of the alignment now largely avoided through adoption of an underpass rather than open sunken road design. By adopting this design, the total affected wetland habitats would be reduced by over 46 % (from 3.94 ha to 2.09 ha). Permanent direct impacts on high value pond habitats are almost wholly avoided, with the affected area of high ecological value pond habitats reduced by over 87% (from 1.99 ha to 0.25 ha). The ecological characteristics and function of the portion of the EA directly affected by the construction works will be reinstated following the completion of works, and enhanced and maintained to a condition no less than the existing conditions.

8.14.5 The provision of wetland compensation at the OWCAs already constructed as part of the LMC Loop EIA will more than sufficiently offset unavoidable wetland impacts arising from the construction and operation of the Project, ensuring "no net loss" in the overall ecological function of the affected wetland habitats, in accordance with the requirements of the EIAO-TM. Additional enhancement works of 4.38 ha of pond habitats will be implemented at Area 4, over and above the required compensation to further strengthen ecological functionality upon the completion of the Project. The enhancement of Area 4 will serve to improve local ecological performance as well as strengthen ecological linkage within the Deep Bay Wetland System, maintaining habitat connectivity and supporting the movement of wetland-dependent fauna through the corridor. Wetland associated avifauna species, non-avifauna species of conservation species as well as Eurasian Otters would benefit from better water management compared to existing abandoned and poor condition of ponds, and more diverse and shallower sloping edge conditions to provide suitable foraging habitat for these species. The enhanced ponds will also improve connectivity by reinforcing the major avifaunal flight corridor traversing from Mai Po to the Loop and HHW. In addition, the enhanced habitats would support the movement of Eurasian Otters and other terrestrial mammals between these areas.

8.14.6 Aside from wetland habitats, approximately 1.68 ha of woodland will be directly affected within the Construction Area of the Project. To comply with the requirements of the EIAO-TM, equivalent woodland compensation will be provided on a 1:1 basis. The location of the proposed compensatory woodland was preliminarily identified to the southwest of the ECR Alignment. A woodland compensation plan will be formulated and submitted prior to the commencement of construction at the woodland habitat. With the proposed compensation measures, no unacceptable ecological impact is anticipated to arise from the loss of habitats under the proposed development.

8.14.7 There would be no direct encroachment on a seasonally active night roost identified at a strip a mixed woodland at the northern end of Ma Tso Lung Road, and only minor

indirect impacts anticipated. Pre-construction site checks for the night roost are also proposed to update the status of the roost and minimise potential impacts. Time and seasonal restriction will be adopted for any noisy construction activities (with the use of PME) within 100 meters of the existing seasonal active night roost between October to April, permitted only between 0900h to 1700h to avoid period of night roost utilisation.

8.14.8 Major flight paths for wetland avifauna were observed along the Hoo Hok Wai Flight Corridor, which lies above pond habitats between LMC Meander and the Border Road (**60722948/A10/FIGURE 8.8** refers). Construction activities within sensitive areas used by Eurasian otters or intersecting the bird flight line will be largely confined to restricted areas (i.e. Zones 1–3), and with the implementation of appropriate mitigation and precautionary measures, no overlap with major flight lines or core otter activity areas is anticipated outside the restricted zones. In addition, impacts to these flight paths during the construction phase are expected to be minimal considering no above-ground structures exceeding 15m above the existing ground level be established in this area. No operation phase impacts are expected. Time and seasonal restriction will be adopted for any aboveground noisy construction activities (with the use of PME) within sensitive wetland area that fall within the Construction Area of the Project, permitted only between 0900h to 1700h throughout the year, unless a cover is provided to create the fully enclosed conditions. This restriction is intended to avoid the time of day when flight lines and otters are most active in the region.

8.14.9 Several individuals of a plant species of conservation importance (Incense Tree) were recorded within the Construction Area of the Project. A detailed vegetation survey should be conducted prior to the commencement of construction works to identify potentially affected plant individuals. All identified individuals would be labelled and fenced off on-site for better preservation, or in the case of unavoidable direct impacts, for transplantation or seedling planting according to the Protection and Transplantation / Seedling Planting Proposal.

8.14.10 Fauna species of conservation importance (comprising one herpetofauna species, Chinese bullfrog; one freshwater crab species, *Somanniathelphusa zanklon*; and one freshwater fish species, *Gobiopterus macrolepis*) were recorded within or close to Construction Area of the Project, as well as actual / suspected breeding of three avifauna species of conservation importance (White-throated Kingfisher, Chinese Francolin and Eurasian Eagle Owl). Due to the low mobility of these species and/or potential breeding individuals, translocation or pre-construction site check for nesting activities should be implemented to minimise impacts to these species. Detailed pre-construction surveys should be conducted for these species, followed by translocation of the low mobility species, and nest control measures for the potential breeding / nesting avifauna, including establishing an appropriate exclusion and minimising disturbance impacts from nearby construction activities.

8.14.11 While no records of Eurasian Otter were made during surveys conducted for this project, there are recent records of this species from within/adjacent to the Construction Area of the Project. Pre-construction surveys for otter holts or natal dens should be carried out in wetland habitats located within 150m of the Construction Area of the Project before the commencement of construction works to minimise potential impact on the species' holts and natal dens.

8.14.12 Movement corridors for wildlife (including non-flying mammal species) have also been considered and would be maintained during the project construction and

operation. During the construction phase, works would be divided into three large sections (Sections 1,2,3), to be completed in two phases (Sections 1 and 3 and Sections 2 and 3). All works would be completed at Section 1 before any works commence at Section 2, minimising disturbance and fragmentation impacts. Within the most sensitive wetland area (i.e., the EA, LMC Meander and ponds 36 - 38), works would be further phased to ensure some areas of intact habitat are maintained at all times, and a temporary aboveground wildlife crossing (approximately 40m) would be installed to ensure connectivity is maintained during construction phase. Habitats in this area would be reinstated following completion of construction works, avoiding long-term fragmentation impacts. Two permanent wildlife underpasses would be incorporated in the at-grade section of ECR Alignment, linking the wooded habitat at Tse Koo Hang to the south with wetland habitats at Hoo Hok Wai to the north. These wildlife underpasses would facilitate ecological linkage, allowing movement of non-flying mammals and terrestrial herpetofauna.

8.14.13 A range of mitigation measures as well as general good site practises have been proposed to reduce the disturbance impacts (e.g. noise, glare and dust) and water quality impacts on habitats, sites of conservation importance and wildlife during the construction phase. The operational impact would be largely avoided through the design of the ECR Alignment, and the proposed underpasses with solid parapet, as well as noise and lighting management as precautionary measures, which further reduce habitat fragmentation associated with the at-grade sections of the ECR and other potential indirect impacts during the operation phase.

8.14.14 With the implementation of the aforementioned mitigation measures, no unacceptable impacts are anticipated to arise from the construction or operation of the Project on ecological resources recorded within the Project Assessment Area.

8.15 References

Advisory Council on the Environment Nature Conservation Subcommittee (ACE NCSC) (2016). Nature Conservation Management Agreement Project – Progress of Implementation of the Fishpond Management Agreement (Committee Paper NCSC 1/2016).

AECOM. (2021). Development of the Loop: Main Works Package 1 – Design and Construction. Ecological Mitigation/ Habitat Creation and Management Plan for Sub-package A & B (Final) – Issue 4

AECOM. (Unpublished). Ecological Baseline Desktop Study and Survey Report for Zone 1 – Ma Tso Lung (Draft) (Ref. TR17a-01) under the Agreement No. CE 46/2022 (CE) Ma Tso Lung Area and Other Sites in Kwu Tung North New Development Area and North District – Feasibility Study.

Agriculture, Fisheries and Conservation Department (AFCD). (2004a). Deep Bay Wetland outside Ramsar Site. Retrieved from: https://www.afcd.gov.hk/english/conservation/con_nncp/con_nncp_list/files/09_DeepBayWetland_Public_18_E.pdf

Agriculture, Fisheries and Conservation Department (AFCD). (2004b). Long Valley and Ho Sheung Heung. Retrieved from: https://www.afcd.gov.hk/english/conservation/con_nncp/con_nncp_list/files/08_LongValley_Public_18_E.pdf

Agriculture, Fisheries and Conservation Department (AFCD). (2006). The Geckos of Hong Kong Retrieved from: https://www.afcd.gov.hk/english/publications/publications_con/files/hkbonewsletter13.pdf

Agriculture, Fisheries and Conservation Department (AFCD). (2021). Hong Kong Herbarium – Hong Kong Plant Database. Available at: https://www.herbarium.gov.hk/en/home/index.html. Accessed on 30 August 2025.

Agriculture, Fisheries and Conservation Department (AFCD). (2024a). Strategic Feasibility Study on the Development of Wetland Conservation Parks System Under the Northern Metropolis Development Strategy. Retrieved from: https://www.afcd.gov.hk/english/conservation/con_wet/WCPS_SYSTEM/wcps_system.html

Agriculture, Fisheries and Conservation Department. (2024b). Unpublished territory-wide long-term monitoring survey data on major groups.

Agriculture, Fisheries and Conservation Department (AFCD). (2024c). Hong Kong Biodiversity Information Hub. Available at: https://bih.gov.hk/en/species-database/index.html. Accessed on 30 August 2025.

Agriculture, Fisheries and Conservation Department (AFCD). (2024d). Management Agreement and Public-Private Partnership. Available at: https://www.afcd.gov.hk/english/conservation/con_nncp/con_nncp_new/con_nncp_new.html Accessed on 31 August 2025.

Anon. (2002). Summer 2002 Report: Egretry counts in Hong Kong with articular reference to the Mai Po Inner Deep Bay Ramsar Site. Report by The Hong Kong Bird Watching Society to the Agriculture, Fisheries and Conservation Department, Hong Kong Special Administrative Region Government.

Anon. (2022). Summer 2022 Report: Egretry counts in Hong Kong with particular reference to the Mai Po Inner Deep Bay Ramsar Site. Report by The Hong Kong Bird Watching Society to the Agriculture, Fisheries and Conservation Department, Hong Kong Special Administrative Region Government.

Anon. (2021). Summer 2021 Report: Egretry counts in Hong Kong with particular reference to the Mai Po Inner Deep Bay Ramsar Site. Report by The Hong Kong Bird Watching Society to the Agriculture, Fisheries and Conservation Department, Hong Kong Special Administrative Region Government.

ArchSD. (2009). Study for the Construction of a Secondary Boundary Fence and New Sections of Primary Boundary Fence and Boundary Patrol Road

Black and Veatch (2019a). Ecological Mitigation / Habitat Creation and Management Plan (184794/B&V/043/EP/Final/Issue 1 for the Agreement No. CE5/2014(CE) Development of Lok Ma Chau Loop: Land Decontamination and Advance Engineering Works – Design and Construction.

Black and Veatch. (2019b). Monthly Environmental Monitoring and Audit Report for January 2019 (version 1.0) for Development of Lok Ma Chau Loop: Land Decontamination and Advance Engineering Works – Environmental Team.

Black and Veatch. (2019c). Monthly Environmental Monitoring and Audit Report for February 2019 (version 2.0) for Development of Lok Ma Chau Loop: Land Decontamination and Advance Engineering Works – Environmental Team.

Black and Veatch. (2019d). Monthly Environmental Monitoring and Audit Report for March 2019 (version 2.0) for Development of Lok Ma Chau Loop: Land Decontamination and Advance Engineering Works – Environmental Team.

Black and Veatch. (2019e). Monthly Environmental Monitoring and Audit Report for April 2019 (version 1.0) for Development of Lok Ma Chau Loop: Land Decontamination and Advance Engineering Works – Environmental Team.

Civil Engineering and Development Department (CEDD) & Planning Department (PlanD), (2013). Agreement No. CE 13/2014(CE) Development of Kwu Tung North and Fanling North New Development Areas Phase 1 – Design and Construction. Report for New Territories North and West Development Office, Civil Engineering and Development Department.

Civil Engineering and Development Department (CEDD). (2019). Agreement No. CE 19/2019(CE) Development of Kwu Tung North New Development Area, Remaining Phase – Design and Construction Compensation Event No. CE/2019-01 – Planning and Engineering Review (KTN) [P&ER(KTN)]

Civil Engineering and Development Department (CEDD). (2020). General Specification for Civil Engineering Works, 2020 Edition. Available at: https://www.cedd.gov.hk/eng/publications/standards-spec-handbooks-cost/stan-gs-2020/index.html

Civil Engineering and Development Department (CEDD). (2022). Development of Lok Ma Chau Loop: Land Decontamination and Advance Engineering Works – Design and Construction - Ecological Mitigation / Habitat Creation and Management Plan

Civil Engineering and Development Department (CEDD). (2023a). Development of Lok Ma Chau Loop: Main Works Package 1. Monthly Environmental Monitoring and Audit Report from March 2022 -August 2023.

Civil Engineering and Development Department (CEDD). (2023b). Development of Lok Ma Chau Loop: Main Works Package 1. Operation Manual for the Offsite Wetland Compensation Areas.

Civil Engineering and Development Department (CEDD). (2023c). Development of Lok Ma Chau Loop: Main Works Package 1. Monthly Monitoring and Management Report for OWCAs from October 2022 -October 2023.

Civil Engineering and Development Department (CEDD) & Planning Department (PlanD), (2024). Agreement No. CE 20/2021 (CE). First Phase Development of the New Territories North – San Tin / Lok. Ma Chau Development Node – Investigation

Civil Engineering and Development Department (CEDD). (2024). Development of Lok Ma Chau Loop: Main Works Package 1- Design and Construction. Off-site Woodland Compensation Proposal. Prepared by AECOM for Civil Engineering and Development Department.

Civil Engineering and Development Department (CEDD). (2025). Development of Lok Ma Chau Loop: Main Works Package 1- Design and Construction. Ecological Review Report for the Construction of the Pier DRL-P08. Prepared by China Road and Bridge Corporation.

Chan, A. H. Y. et al. (2023) 'Reconstructing 34 Years of Fire History in the Wet, Subtropical Vegetation of Hong Kong Using Landsat', Remote Sensing 2023, Vol. 15, Page 1489. Multidisciplinary Digital Publishing Institute, 15(6), p. 1489. doi: 10.3390/RS15061489.

Chan, A. H. Y. and Coomes, D. A. (2024) 'Fire traps in the wet subtropics: New perspectives from Hong Kong', Journal of Applied Ecology. John Wiley & Sons, Ltd, 61(4), pp. 884–898. doi: 10.1111/1365-2664.14575.

Chan, S.K.F., Cheung, K.S., Ho, C.Y., Lam, F.N., Tang, W.S., Lau, M.W.N. & Bogadek, A. (2005). A Field Guide to the Amphibians of Hong Kong. Friends of the Country Parks.

Chan, K. F., Cheung, K. S., Ho, C. Y., Lam F. N. and Tang, W. S. (2006). The Geckos of Hong Kong. Hong Kong Biodiversity 13: 1-9.

Chan, A. Cheung, J., Sze, P., Wong, A., Wong, E. and Yau, E. (2011). A review of the local restrictedness of Hong Kong Butterflies. Hong Kong Biodiversity 21: 1-12.

Corlett, R. T., Xing, F. W., Ng, S. C., Chau, L. K. C., & Wong, L. M. Y. (2000). Hong Kong vascular plants: distribution and status. Memoirs of the Hong Kong Natural History Society 23:1-157.

Dudgeon, D. (1999). Tropical Asian Streams: Zoobenthos, Ecology and Conservation. Hong Kong University Press, Hong Kong.

Fellowes, J.R., Lau, M.W.N., Dudgeon, D., Reels, G.T., Ades, G.W.J., Carey, G.J., Chan, B.P.L., Kendrick, R.C., Lee, K.S., Leven, M.R., Wilson, K.D.P. and Yu, Y.T. (2002). Wild animals to watch: Terrestrial and freshwater fauna of conservation concern in Hong Kong. Memoirs of the Hong Kong Natural History Society No. 25, 123-160.

Fireflies.hk. (n.d.). *Rhagophthalmus motschulskyi* (Motschulsky's Starworm). Available at : https://fireflies.hk/species/species/Rh_motschulskyi.html

Fu, L. and Chin, C. M. (1992). China Plant Red Data Book. Science Press, Beijing.

Haddad, N. M., et al. (2015). "Habitat fragmentation and its lasting impact on Earth's ecosystems."

Hong Kong Bird Watching Society. (2018). Mai Po Inner Deep Bay Ramsar Site Waterbird Monitoring Programme, Monthly Waterbird Monitoring

Hong Kong Bird Watching Society (HKBWS). (2021). Available at: https://cms.hkbws.org.hk/cms/en/hkbws/work/monitor/other-monitor/knowing-deep-bay-en

Hong Kong Bird Watching Society (HKBWS) (2024). List of Hong Kong Birds. Available at: https://cms.hkbws.org.hk/cms/en/resource/bird-list . Accessed on 30 August 2025.

The Hong Kong Biodiversity Information Hub (HKBIH). Retrieved from: https://bih.gov.hk/en/home/index.html

Hong Kong Biodiversity – Newsletter of Agriculture, Fisheries and Conservation Department (AFCD)

"Territory-wide long-term monitoring survey results from the Agriculture, Fisheries and Conservation Department" (AFCD, 2024)

Hu, Q.M, Wu, T.L., Xia, N.H., Xing F.W., Lai, C.C.P., Yip, K.W. (2003). Rare and Precious Plants of Hong Kong. Agriculture, Fisheries and Conservation Department, The Government of the Hong Kong Special Administrative Region.

International Union of Conservation for Nature. (2023). The IUCN Red List of Threatened Species. Version 2022-2. Retrieved from: https://www.iucnredlist.org.

Jiang, Z.G., Jiang, J.P., Wang, Y.Z., Zhang, E., Zhang, Y.Y., Li, L.L., Xie, F., Cai, B., Cao, L., Zheng, G.M., Dong, L., Zhang, Z.W., Ding, P., Luo, Z.H., Ding, C.Q., Ma, Z.J., Tang, S.H., Cao, W.X., Li, C.W., Hu, H.J., Ma, Y., Wu, Y., Wang, Y.X., Zhou, K.Y., Liu, S.Y., Chen, Y.Y., Li, J.T., Feng, Z.J., Wang, Y., Wang, B., Li, C., Song, X.L., Cai, L., Zang, C.X., Zeng, Y., Meng, Z.B., Fang, H.X., and Ping, X.G., (2016). Red List of China's Vertebrates. Biodiversity Science, 24 (5), 500-551.

Kadoorie Farm and Botanic Garden (KFBG). (2022). In search of the last otters of Hong Kong. Retrieved from: https://www.kfbg.org/en/KFBG-blog/post/In-search-of-the-last-otters-of-Hong-Kong. Accessed on 14 March 2024.

Kaodorie Farm Botanic Garden (KFBG). (2022). Bird Survey of Selected Farmlands in the Proposed Northern Metropolis, Hong Kong SAR: Interim Report

Kang, C., Jin, S-D., Lee, W-S. & Rhim, S-J. (2013). Analysis of Home Range of Eurasian Eagle Owl (Bubo bubo) by WT-100. Korean Journal of Ornithology, 20(1), 1-8. Available at: https://www.researchgate.net/publication/264149073

Karsen, S., Lau, M. and Bogadek, A. (1998). Hong Kong Amphibians and Reptiles. 2nd edition. The Provisional Urban Council, Hong Kong.

Kowloon-Canton Railway Corporation (KCRC). (2002). Sheung Shui to Lok Ma Chau Spur Line. EIA Report (AEIAR-052/2002).

Lee, L. F., Lam, K. S., Ng, K. Y., Chan, K. T., & Young, L. C. (2004). Field guide to the freshwater fish of Hong Kong. Friends of the Country Parks and Cosmos Books Ltd: Hong Kong.

MTR Corporation Limited (MTR). (2024). Minutes of Meeting of the Environmental Committee for Sheung Shui to Lok Ma Chau Spur Line. Hong Kong: MTR Corporation Limited. Available at: https://www.mtr.com.hk/archive/corporate/en/env_report/2024_01_No.46_EC_Meeting.pdf McMillan, S. E., Wong, A. T. C., Tang, S. S. Y., Yau, E. Y. H., Gomersall, T., Wong, P. Y. H., Vu, A. K. H., Sin, S. Y. W., Hau, B. C. H. & Bonebrake, T. C. (2023). Spraints demonstrate small population size and reliance on fishponds for Eurasian otter (*Lutra lutra*) in Hong Kong. Conservation Science and Practice, 5(1), e12851.

Mott Macdonald. (2009). Approved EIA Report for the Construction of a Secondary Boundary Fence and new sections of Primary Boundary Fence and Boundary Patrol Road.

National Forestry and Grassland Administration and the Ministry of Agricultural and Rural Affairs. (2021). List of Wild Animals under State Priority Conservation. The State Council, Beijing. (promulgated on 5 February 2021).

National Forestry and Grassland Administration and the Ministry of Agricultural and Rural Affairs. (2021). List of Wild Plants under the State Priority Conservation. The State Council, Beijing (promulgated on 7 August 2021).

Ove Arup. (2010). Final Report of the Agreement No. CE60/2005 (TP) Land Use Planning for the Closed Area – Feasibility Study.

Ove Arup. (2013a). Approved EIA Report for the Development of Lok Ma Chau Loop Agreement

Ove Arup. (2013b). Approved EIA Report for the Agreement No. CE61/2007(CE) North East New Territories New Development Areas Planning and Engineering Study – Investigation.

Pauwels, J., Le Viol, I., Bas, Y., Valet, N., and Kerbiriou, C. (2021) Adapting street lighting to limit light pollution's impacts on bats, Global Ecology and Conservation (28)

Porcupine! Newsletter of Ecology & Biodiversity, The School of Biological Sciences, The University of Hong Kong

Qin, H. N., Yang, Y., Dong, S. Y., He, Q., Jia, Y., Zhao, L. N., Yu, S. X., Liu, H. Y., Liu, B., Yan, Y. H., Xiang, J. Y., Xia, N. H., Peng, H., Li, Z. Y., Zhang, Z. X., He, X. J., Yin, L. K., Lin, Y. L., Liu, Q. R., Hou, Y. T., Liu, Y., Liu, Q. X., Cao, W., Li, J. Q., Chen, S. L., Jin, X. H., Gao, T. G., Chen, W. L., Ma, H. Y., Geng, Y. Y., Jin, X. F., Chang, C. Y., Jiang, H., Cai, L., Zang, C. X., Wu, J. Y., Ye, J. F., Lai, Y. J., Liu, B., Lin, Q., W. & Xue, N. X. (2017). Threatened species list of China's higher plants. Biodiversity science, 25(7), 696-744.

Reels, G. T. (2019). An annotated check list of Hong Kong dragonflies and assessment of their local conservation significance. International Dragonfly Fund-Report, (30), 1-49.

Schlaepfer, D. R., Braschler, B., Rusterholz, H.-P., & Baur, B. (2018). "Genetic effects of anthropogenic habitat fragmentation on remnant animal and plant populations: a meta-analysis."

Shek, C.T. (2006) A Field Guide to the Terrestrial Mammals of Hong Kong. Agriculture, Fisheries and Conservation Department, The Government of the Hong Kong Special Administrative Region.

Stanton, D.J. & M.R. Leven (2016). Distribution, habitat utilisation and conservation status of the freshwater crab, *Somanniathelphusa zanklon* Ng & Dudgeon, 1992 (Crustacea: Brachyura: Gecarcinucidae) endemic to Hong Kong. Journal of Threatened Taxa 8(3): 8564–8574; http://dx.doi.org/10.11609/jott.2070.8.3.8564-8574

Tam, T.W., Leung, K.K., Kwan, B.S.P., Wu, K.K.Y., Tang, S.S.H., So, I.W.Y., Cheng, J.C.Y.,

The Avifauna of Hong Kong from HKBWS. (2023). Available at: https://avifauna.hkbws.org.hk/species/0030/003700#:~:text=According%20to%20Pereira%20,at%20any%20sign%20of%20alarm

Tong, C. P. (2016). Distribution and preference of landscape features and foraging sites of insectivorous bats in Hong Kong urban parks. HKU Theses Online (HKUTO).

Town Planning Board. (2014). Town Planning Board Guidelines for Application for Developments within Deep Bay Area under Section 16 of the Town Planning Ordinance. Retrieved from: https://www.info.gov.hk/tpb/en/forms/Guidelines/pg12c_e.pdf

Town Planning Board. (2018). Approved Lok Ma Chau Loop Outline Zoning Plan No. S/LMCL/2 Notes – Schedule of Uses – Explanatory Statement. Retrieved from: https://www.ozp.tpb.gov.hk/api/Plan/PlanNote?planNo=S%2fLMCL%2f2&lang=EN&ext=pdf&dType=in#nameddest=O

Wang, S. (1998). China Red Data Book of Endangered Animals: Mammalia. Science Press, Beijing.

Wang, S. and Zhao, E. M. (1998). China Red Data Book of Endangered Animals: Amphibia and Reptilia. Science Press, Beijing.

Wellab Limited/Nature & Technologies (HK) Limited. (2019-2023). Monthly EM&A Reports from January 2019 to March 2023 for EP-477/2013 and EP-477/2013/A - Development of Lok Ma Chau Loop.

Weston, M. A, McLeod, Emily M, Blumstein, Daniel T and Guay, Patrick-Jean (2012) A Review of Flight-Initiation Distances and their Application to Managing Disturbance to Australian Birds. Emu : Austral Ornithology, 112 (4). pp. 269-286. ISSN 0158-4197 (print), 1448-5540 (online)

World Wide Fund for Nature Hong Kong (WWF). (2025a and 2025b). State of Hong Kong Biodiversity 2025 and Hong Kong Terrestrial Biodiversity Hotspot Map 2025. Available at: https://www.wwf.org.hk/en/biodiversity/hkbiodiversity2025/

Wu, D. L. and Hu. C.X. (1988). Illustrations of Rare and Endangered Plants in Guangdong Province. China Environmental Science Press, Beijing.

YANG Ai-fang, LIU Zhao, WU Yin-huan, ZHANG Meng, JI Shuang, LIANG Xiang-ping, XU Zhi-wei, LI Min-qian, XIE Zong-ping, XIE Hui-chun. Analysis of Vigilance Distance of Wintering Waterbirds in Zhangye National Wetland Park[J]. WETLAND SCIENCE & MANAGEMENT, 2021, 17(1): 4-8.

Yip, J. Y., Yip, J. K. L., Liu, E. K. Y., Ngar, Y. N., & Lai, P. C. C. (2010). A floristic survey of marshes in Hong Kong. Hong Kong Biodiversity 19: 7-16.

Yiu, V. (2023). Hong Kong Fireflies. Retrieved from: http://fireflies.hk

Yuen, E.F.M., Tsang, Y.M., and Hui, W.L. (2011). The Hong Kong Dragonflies. Agriculture, Fisheries and Conservation Department, The Government of the Hong Kong Special Administrative Region.

Zheng, G. and Wang, Q. (1998). China Red Data Book of Endangered Animals: Aves. Science Press, Beijing.

Zheng, X., Chan, B.P.L., Au, T.K., Yang, J. & Lo, P.Y.F. (2023) The Hidden Oases of Hong Kong: High Ecological Value Farmlands in the Proposed Northern Metropolis. Kadoorie Farm and Botanic Garden Publication Series No. 19. Kadoorie Farm and Botanic Garden, Hong Kong SAR, China. 38pp. Available at: https://www.kfbg.org/images/download/kfbg%20northern%20metropolis%20bird%20report%20sept%202023%20eng.pdf